

SILLAH KANU SERIES

MASTERING STANDARD MATHEMATICS FOR JUNIOR SECONDARY SCHOOLS

$$3x^{\frac{3}{2}} + 5y^{\frac{5}{3}} = 250 \dots \text{equation(1)}$$

$$5x^{\frac{3}{2}} + 3y^{\frac{5}{3}} = 270 \dots \text{equation(2)}$$

Find x^3 and y^5 ?

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PREFACE

The Author has witnessed the fears exhibited by pupils learning mathematics. For that, he has prepared a simple and straight forward mathematics text that is very user friendly. There is no doubt whoever goes through the main texts, the worked examples and the exercises will not fear mathematics as it used to be.

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ACKNOWLEDGEMENT

**TO ALLAH,
MY FAMILY,
LECTURERS,
AND FRIENDS.**

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DEDICATION

This book is dedicated to my late elder sister, Kadijah Sillah Kanu.

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TABLE OF CONTENTS

CHAPTER 1 Page 1 – 8

NUMBER PATTERN, HCF AND LCM

SECTION 1: NUMBER PATTERN

Subsection 1: Prime numbers

Subsection 2: Even numbers

Subsection 3: Odd numbers

Subsection 4: Factors

Subsection 5: Multiples

SECTION 2: HIGHEST COMMON FACTOR (HCF)

SECTION 3: LOWEST COMMON MULTIPLE (LCM)

SECTION 4: SKS PAPER 1

CHAPTER 2 Page 9 – 22

APPROXIMATIONS

SECTION 1: ROUNDING UP DECIMAL NUMBERS TO A GIVEN NUMBER OF PLACE VALUE

SECTION 2: ROUNDING NUMBERS TO THE NEAREST TENTHS, HUNDREDTHS, THOUSANDTHS, ETC.

SECTION 3: ROUNDING NUMBERS TO THE NEAREST TEN, HUNDRED, THOUSAND, ETC.

SECTION 4: APPROXIMATING A DECIMAL NUMBER TO A GIVEN NUMBER OF SIGNIFICANT FIGURES

SECTION 5: SKS PAPER 2

CHAPTER 3 Page 23 – 34

STANDARD FORM

SECTION 1: CONVERTING FROM ORDINARY FORM TO STANDARD FORM

SECTION 2: CONVERTING FROM STANDARD FORM TO ORDINARY FORM

Subsection 1: When n is positive

Subsection 2: when n is negative

SECTION 2: BASIC OPERATIONS IN STANDARD FORM

Subsection 1: Addition

Subsection 2: Subtraction

Subsection 3: Multiplication

Subsection 4: Division

SECTION 3: APPROXIMATING A DECIMAL NUMBER TO A GIVEN NUMBER OF SIGNIFICANT FIGURES

SECTION 4: SKS PAPER 3

CHAPTER 4 Page 35 – 44

DECIMALS

SECTION 1: BASIC OPERATION ON DECIMALS

Subsection 1: Addition and Subtraction

Subsection 2: Multiplication and Division

SECTION 2: CONVERSION OF DECIMAL TO FRACTION

SECTION 3: CONVERSION OF DECIMAL TO PERCENTAGE

SECTION 4: SKS PAPER 4

CHAPTER 5 Page 45 – 66

RECIPROCAL AND SQUARES

SECTION 1: RECIPROCAL

SECTION 2: SQUARES

Subsection 1: Square of variables with square one (1) only

Subsection 2: Square of variables with powers other than one (1)

Subsection 3: Squares of numbers

Subsection 4: Squares of square roots

Subsection 4: Squares of Cube roots

SECTION 2: CONVERSION OF DECIMAL TO FRACTION

SECTION 3: CONVERSION OF DECIMAL TO PERCENTAGE

SECTION 4: SKS PAPER 5

CHAPTER 6 Page 67 – 86

NUMBER BASES

SECTION 1: INTRODUCTION

SECTION 2: CONVERSION OF NUMBER BASE

Subsection 1: Conversion from one base to base ten

Subsection 2: Conversion from number base ten to other base

Subsection 3: Conversion between non decimal bases

Subsection 4: Equations involving number bases

Subsection 4: Squares of Cube roots

SECTION 2: BASIC OPERATTION IN NUMBER BASE

Subsection 1: Addition

Subsection 2: Subtraction

Subsection 3: Finding missing numbers

Subsection 4: Multiplication

SECTION 3: SKS PAPER 6

CHAPTER 7 Page 87 – 98

SURDS

SECTION 1: INTRODUCTION

Subsection 1: Rational numbers

Subsection 2: Irrational numbers

SECTION 2: REDUCTION TO BASIC FORM

SECTION 3: ADDITION AND SUBTRACTION OF SURDS

SECTION 4: PROPERTIES OF SURDS

SECTION 5: MULTIPLICATION OF SURDS

SECTION 6: RATIONALISING THE DENOMINATOR OF SURDS

SECTION 7: CONJUGATE OF SURDS

SECTION 8: SKS PAPER 7

CHAPTER 8 Page 99 – 126

FRACTIONS AND PERCENTAGES

SECTION 1: FRACTIONS

Subsection 1: Definitions and worked examples

Subsection 2: Order of Magnitude

SECTION 2: PERCENTAGES

Subsection 1: Conversion of a fraction to a percentage

Subsection 2: Conversion of a decimal to a percentage

Subsection 3: Express a percentage as a decimal

Subsection 4: Percentage of a quantity

Subsection 5: Increasing a quantity

Subsection 6: Decreasing a quantity

Subsection 7: Faulty and not faulty in percentages

Subsection 8: Percentage profit and loss

Subsection 9: Finding the original quantity in percentages

SECTION 3: SKS PAPER 8

CHAPTER 9 Page 127 – 182

ALGEBRA

SECTION 1: ADDITION AND SUBTRACTION

Subsection 1: The same sign

Subsection 2: Different signs

SECTION 2: MULTIPLICATION AND DIVISION

SECTION 3: WORD PROBLEMS

Subsection 1: Word problems involving sum and difference

Subsection 2: Word problems involving product

SECTION 4: SUBSTITUTION

SECTION 5: LIKE AND UNLIKE TERMS

SECTION 6: BRACKETS AND EXPANSION

SECTION 7: FACTORISATION

Subsection 1: Common factors

Subsection 2: Factorisation by grouping

Subsection 3: Factorisation by splitting the middle term

SECTION 8: DIFFERENCE OF TWO SQUARES

SECTION 9: MISCELLANEOUS EXAMPLES

CHAPTER 10 Page 183 – 206

INDICES

SECTION 1: IDENTIFICATION OF BASES AND POWERS

SECTION 2: LAWS OF INDICES

Subsection 1: Multiplication Law

Subsection 2: Division Law

Subsection 3: Power Law

Subsection 4: Product power law

Subsection 5: Zero Index

Subsection 6: Negative Index

Subsection 7: Fractional Index

SECTION 3: SKS PAPER 3

CHAPTER 11 Page 207 – 212

BINARY OPERATION

SECTION 1: DEFINITION AND WORKED EXAMPLES

SECTION 2: SKS PAPER 11

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CHAPTER 12 Page 213 – 256

CHANGE OF SUBJECT OF FORMULA

SECTION 1: DEFINITION AND WORKED EXAMPLES

SECTION 2: WORD PROBLEMS

SECTION 3: SKS PAPER 12

SILLAH KANU SERIES @ 2022

CHAPTER 13 Page 257 – 312

EQUATIONS

SECTION 1: LINEAR EQUATIONS

Subsection 1: Simple linear equation

Subsection 2: Equations in fraction form

SECTION 2: SIMULTANEOUS EQUATIONS

Subsection 1: Using the elimination method

Subsection 2: Using the substitution method

SECTION 3: EQUATIONS INVOLVING WORD PROBLEMS

SECTION 4: QUADRATIC EQUATIONS

SECTION 5: SKS PAPER 13

CHAPTER 14 Page 313 – 326

GRAPH

SECTION 1: INTRODUCTION TO LINEAR GRAPHS

SECTION 2: GENERAL EQUATIONS OF A STRAIGHT LINE

SECTION 3: SKS PAPER 14

SILLAH KANU SERIES @ 2022

CHAPTER 15 Page 327 – 350

EVERYDAY ARITHMETIC

SECTION 1: SIMPLE INTEREST

SECTION 2: RATE

SECTION 3: TIME

SECTION 4: PRINCIPAL AND AMOUNT

SECTION 5: COMMISSION

SECTION 6: DISCOUNT

SECTION 7: TAXES AND BILLS

SECTION 8: RATIO

SECTION 9: PROPORTION

SECTION 10: SPEED, DISTANCE AND TIME

SECTION 11: DENSITY AND POPULATION DENSITY

SECTION 12: SKS PAPER 15

CHAPTER 16 Page 351 – 362

POLYGONS

SECTION 1: DEFINITION

SECTION 2: CLASSIFICATION OF POLYGONS

SECTION 3: THE SUM OF THE INTERIOR ANGLE OF A POLYGON

SECTION 4: FINDING THE NUMBER OF SIDES OF A REGULAR POLYGON WHEN THE SUM OF THE INTERIOR ANGLE IS GIVEN

SECTION 5: FINDING THE NUMBER OF SIDES OF A POLYGON

SECTION 6: FINDING THE NUMBER OF SIDES OF A REGULAR POLYGON WHEN THE INTERIOR ANGLE IS GIVEN

SECTION 7: FINDING THE EXTERIOR ANGLE OF A REGULAR POLYGON

SECTION 8: FINDING THE NUMBER OF SIDES WHEN THE EXTERIOR ANGLE IS GIVEN

SECTION 9: FINDING THE SUM OF THE INTERIOR ANGLES

WHEN THE EXTERIOR ANGLE IS GIVEN

SECTION 10: SKS PAPER 16

SILLAH KANU SERIES @ 2022

CHAPTER 17 Page 363 – 414

PLANE GEOMETRY

SECTION 1: INTRODUCTION AND DEFINITION OF ANGLES

SECTION 2: ANGLES AROUND A POINT

SECTION 3: ANGLES ON A STRAIGHT LINE

SECTION 4: COMPLEMENTARY ANGLES

SECTION 5: VERTICALLY OPPOSITE ANGLES

SECTION 6: USING TWO OR MORE ANGLE TYPE

SECTION 7: WORD PROBLEMS INVOLVING ANGLES

SECTION 8: ANGLE PROPERTIES OF PARALLEL LINES

SECTION 9: PARALLEL LINES IN A TRIANGLE

SECTION 10: TRIANGLES

Subsection : Angle properties of a triangle

SECTION 11: SKS PAPER 17

CHAPTER 18 Page 415 – 426

LOGARITHM

SECTION 1: DEFINITION

SECTION 2: FINDING LOGARITHM

SECTION 3: ADDITION IN LOGARITHM

SECTION 4: SUBTRACTION IN LOGARITHM

SECTION 5: MULTIPLICATION IN LOGARITHM

SECTION 6: USING LOGARITHM TO EVALUATE EXPRESSIONS

Subsection 1: Products

Subsection 2: Division

Subsection 3: Powers

Subsection 4: Roots

SECTION 7: SKS PAPER 18

CHAPTER 19 Page 427 – 464

STATISTICS

SECTION 1: INTRODUCTION

SECTION 2: MEASURE OF CENTRAL TENDENCY

Subsection 1: mode

Subsection 2: mean

Subsection 3: median

SECTION 3: THE USE OF AT LEAST

SECTION 4: THE USE OF AT MOST

SECTION 5: THE USE OF ABOVE AND BELOW

SECTION 6: FREQUENCY TABLE

SECTION 7: GRAPHICAL REPRESENTATION OF DATA

Subsection 1: pie chart

Subsection 2: bar chart

Subsection 3: median

SECTION 8: SKS PAPER 19

CHAPTER 20 Page 465 – 516

CIRCLE THEOREMS

SECTION 1: PROPERTIES OF ANGLES

Subsection 1: property one (1)

Subsection 1: property two (2)

Subsection 1: property three (3)

Subsection 1: property four (4)

Subsection 1: property five (5)

Subsection 1: property six (6)

Subsection 1: property seven (7)

Subsection 1: property eight (8)

SECTION 2: ANGLES AROUND A POINT OF A CYCLIC QUADRILATERAL

SECTION 3: ISOSCELES TRIANGLE

SECTION 4: OPPOSITE ANGLES

SECTION 5: EXAMPLES INVOLVING MORE THAN ONE THEOREM

SECTION 6: SKS PAPER 20

CHAPTER 21 Page 517 – 570

MEASURATION

SECTION 1: RECTANGLE

SECTION 2: PARALLELOGRAM

SECTION 3: TRIANGLES

SECTION 4: SQUARE

SECTION 5: TRAPEZIUM

SECTION 6: CIRCLE

SECTION 7: CUBE

SECTION 8: CUBOID

SECTION 9: CYLINDER

SECTION 10: CONE

SECTION 11: SPHERE

SECTION 12: HEMISPHERE

SECTION 13: SKS PAPER 21

CHAPTER 22 Page 571 – 596

TRIGONOMETRY

SECTION 1: TRIGONOMETRIC RATIOS

SECTION 2: WORD PROBLEMS

SECTION 3: ANGLE OF ELEVATION AND DEPRESSION

SECTION 4: SKS PAPER 22

CHAPTER 23 Page 597 – 612

BEARING AND DISTANCE

SECTION 1: BEARING OF A POINT FROM ANOTHER

SECTION 2: REVERSAL BEARING

SECTION 3: DISTANCE BEARING FORM

SECTION 4: SKS PAPER

SECTION 5: SKS PAPER 23

CHAPTER 24 Page 613 – 633

CONSTRUCTION

SECTION 1: TO BISECT A STRAIGHT LINE

SECTION 2: TO BISECT A GIVEN LINE

SECTION 3: TO CONSTRUCT AN ANGLE OF 90°

SECTION 4: TO CONSTRUCT AN ANGLE OF 45°

SECTION 5: TO CONSTRUCT AN ANGLE OF 60°

SECTION 6: TO CONSTRUCT AN ANGLE OF 30°

SECTION 7: CONSTRUCTION OF TRIANGLES

Subsection: Construction of triangles given three sides

SECTION 8: MORE WORKED EXAMPLES ON

CONSTRUCTION

SECTION 9: SKS PAPER 24

REFERENCES Page 634

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CHAPTER 1

NUMBER PATTERN, HCF AND LCM

SECTION 1: NUMBER PATTERN

This is a list of numbers that follow a certain sequence or pattern.

For example: even numbers, odd numbers, prime numbers, etc

Subsection 1: Prime numbers

A prime number is a number with two factors; one (1) and the number itself. Examples of prime numbers are: 2, 3, 5, 7, 11, 13, etc.

Subsection 2: Even numbers

These are numbers which are exactly divisible by two. Examples of even numbers are: 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, etc.

Subsection 3: Odd numbers

These are numbers which have remainder one(1) when divided by two(2). Examples of odd numbers are: 1, 3, 5, 7, 9, 11, etc.

Subsection 4: Factors

A factor is a number that will divide another number exactly without leaving a remainder. For example 24 has factors: 1, 2, 3, 4, 6, 8, 12 and 24.

Subsection 5: Multiples

The multiple of a number can be divided exactly by that number without leaving a remainder. There is an infinite number to the multiple of numbers.

E. g

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<i><u>Numbers</u></i>	<i><u>Multiples</u></i>
5	5, 10, 15, 20, <i>etc.</i>
10	10, 20, 30, 40, <i>etc.</i>

SECTION 2: HIGHEST COMMON FACTOR(HCF)

Factors can be used to find the highest common factor for a given set of numbers.

ILLUSTRATIVE EXAMPLES

Example 1. Find the HCF of 5, 15 and 45.

Solution

Step 1: List all the factors of 5, 15 and 45.

Numbers	Factors
5	1 and 5
15	1, 3, 5 and 15
45	1, 3, 5, 9, 15 and 45

Step 2: From the factors of 5, 15 and 45, we observed that 5 is the highest common factor.

$$\therefore HCF = 5$$

Example 2. Find the HCF of 4, 6, 8 and 12.

Solution

Step 1: List all the factors of 4, 6, 8 and 12.

Numbers	Factors
4	1, 2 and 4
6	1, 2, 3 and 6
8	1, 2, 4 and 8
12	1, 2, 3, 4, 6 and 12

Step 2: From the factors of 4, 6, 8 and 12, we observed that 2 is the

highest common factor.

$$\therefore HCF = 2$$

Example 3. Find the HCF of 4, 6 and 16.

Solution

Step 1: List all the factors of each given number.

<i>Numbers</i>	<i>Factors</i>
4	1, <u>2</u> and 4
6	1, <u>2</u> , 3 and 6
16	1, <u>2</u> , 4, 8 and 16

Step 2: From the factors of 4, 6 and 16 we observed that 2 is the highest common factor.

$$\therefore HCF = 2$$

Example 4. Find the HCF of 8, 16, 24 and 32.

Solution

Step 1: List all the factors of 8, 16, 32 and 24

<i>Numbers</i>	<i>Factors</i>
8	1, <u>2</u> , <u>4</u> and <u>8</u>
16	1, <u>2</u> , <u>4</u> , <u>8</u> and 16
24	1, <u>2</u> , 3, <u>4</u> , 6, <u>8</u> , 12 and 24
32	1, <u>2</u> , <u>4</u> , <u>8</u> , 16 and 32

Step 2: The underlined numbers are the common factors and the highest of these is 8

$\therefore HCF = 8$

Example 5. Find the HCF of 6, 12, 24 and 48.

Solution

Step 1: List all the factors of 6, 12, 24 and 48.

<u>Numbers</u>	<u>Factors</u>
6	1, <u>2</u> , <u>3</u> and <u>6</u> .
12	1, <u>2</u> , <u>3</u> , 4, <u>6</u> and 12.
24	1, <u>2</u> , <u>3</u> , 4, <u>6</u> , 8, 12 and 24.
48	1, <u>2</u> , <u>3</u> , 4, <u>6</u> , 8, 12, 16, 24 and 48.

Step 2: The underlined numbers are the common factors and the highest of these is 6.

$\therefore HCF = 6$

SECTION 3: LOWEST COMMON MULTIPLE(LCM)

LCM is the smallest of the multiples of a given set of numbers.

ILLUSTRATIVE EXAMPLES

Example 1. Find the LCM of 5, 15 and 45.

Solution

Step 1: Start from the lowest prime number of the given numbers

3	5	15	45
3	5	5	15
5	5	5	5
	1	1	1

$$\therefore LCM = 3 \times 3 \times 5 = 45$$

Example 2. Find the LCM of 2, 16, 40 and 60

Solution

Step 1: Start from the lowest prime number of the given numbers

2	2	16	40	60
2	1	8	20	30
2	1	4	10	15
2	1	2	5	15
3	1	1	5	15
5	1	1	5	5
	1	1	1	1

$$\therefore LCM = 2 \times 2 \times 2 \times 2 \times 3 \times 5 = 240$$

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SECTION 4: SKS PAPER 1

Answer all questions

Part 1

Question 1. Find the HCF of 25, 30, 35 and 45.

Answer[$HCF = 5$]

Question 2. Find the HCF of 8, 10, 12 and 14.

Answer[$HCF = 2$]

Question 3. Find the HCF of 16, 40 and 60.

Answer[$HCF = 4$]

Question 4. Find the HCF of 12, 24, 48 and 60.

Answer[$HCF = 12$]

Question 5. Find the HCF of 10, 20, 30 and 40.

Answer[$HCF = 10$]

Question 6. Find the HCF of 15, 30, 45 and 60.

Answer[$HCF = 15$]

Part 2

Question 1. Find the LCM of 5, 15, 45 and 60

Answer[$LCM = 180$]

Question 2. Find the LCM of 2, 6, 7 and 8

Answer[$LCM = 168$]

Question 3. Find the LCM of 2, 4, 6 and 8

Answer[$LCM = 24$]

CHAPTER 2

APPROXIMATIONS

It is a method of assuming precise values to figures. Approximations are often linked to terms such as decimal places, significant figures, standard forms and so on.

SECTION 1: ROUNDING UP DECIMAL NUMBERS TO A GIVEN NUMBER OF PLACE VALUE

The decimal places (d.p) in a number come after the decimal point.

For example, each of the following numbers has 3 decimal places: 0.456, 3.002, 84.010 and 123.000

To the number places (1^{st} , 2^{nd} , 3^{rd} , ...), start at the decimal point and come to the right.

For example, in the number 4.2613

2 is in the first decimal place

6 is in the second decimal place

1 is the third decimal place

3 is the fourth decimal place

A common way to approximate a decimal is to reduce the number of decimal places it has. The decimal is given 'correct to' a stated number of decimal places. For example:

$$0.4821 = 0.5 \text{ (correct to 1 d.p)}$$

$$27.063 = 27.06 \text{ (correct to 2 d.p)}$$

$$8.2865 = 8.287 \text{ (correct to 3 d.p.)}$$

To correct a decimal to a stated number of decimal places

- Find the last decimal place you want
- Then look at the figure in the next place (to the right)

If this figure is 5 or more than 5, add 1 to the figure in the last place you want.

Thus to correct a decimal

to 1 d.p., look at the 2nd d.p.

to 2 d.p., look at the 3rd d.p.

to 3 d.p., look at the 4th d.p. and so on ...

ILLUSTRATIVE EXAMPLES

Example 1. Round 33.3625 to

(i). 1 d.p (ii). 2 d.p (iii). 3 d.p

Solution

(i). The 1st d.p. is 3 but the figure after it (i.e. the 2nd d.p) is 6 which is more than 5. Therefore we add 1 to 3 to give 4 in the 1st d.p.

$$33.3625 = 33.4 \text{ (correct to 1 d.p.)}$$

(ii). The 2nd d.p. is 6 but the figure after it (i.e. the 3rd d.p) is 2 which is less than 5. Therefore we leave 6 as it is.

$$33.3625 = 33.36 \text{ (correct to 2 d.p.)}$$

(iii). The 3rd d.p. is 2 but the figure after it (i.e. the 4th d.p.) is 5.

Therefore we add 1 to 2 to give 3 in the 3rd decimal place)

$$33.3625 = 33.363 \text{ (correct to 3 d.p.)}$$

Example 2. Round 6.98 to 1 d.p.

Solution

$6.98 = 7.0$ (to 1 d.p.), because adding 1 to 9 we have

6.9

1

7.0

Do not forget the zero since without it you do not have 1 d.p.

Example 3. Round 2.3498 to 3 d.p.

Solution

$2.349 = 2.350$ (to 3 d.p.). Since the digit after the 3rd d.p. (i.e. 8) is more than 5, we add 1 to 9 to give

2.349

1

2.350

Example 3. Round 2.9998 to 3 d.p.

Solution

2.999

1

3.000

$\therefore 2.9998 = 3.000$ (to 3 d.p.)

Example 4. Round 0.2998 to 3 d.p.

Solution

0.299

$$\begin{array}{r} 1 \\ \hline 0.300 \\ \hline \end{array}$$

$$\therefore 0.2998 = 0.300 \text{ (to 3 d.p.)}$$

Example 5. Round 699.996 to 2 d.p.

Solution

$$\begin{array}{r} 699.99 \\ \hline 1 \\ \hline 700.00 \\ \hline \end{array}$$

SECTION 2: ROUNDING NUMBERS TO THE NEAREST TENTHS, HUNDREDTHS, THOUSANDTHS ETC.

For any number with a decimal part, the first decimal place is called the tenths, the second decimal place is called the hundredths, the third decimal place is called the thousandths, etc. For example, for the number 543.3456, 3 is the tenth, 4 is the hundredth, 5 is the thousandth etc.

Rounding decimals to the nearest tenths is the same as rounding numbers to 1 d.p. Also rounding to the nearest hundredth is the same as rounding numbers to 2 d.p and so on. For example, 543.3456 to the nearest

i). Whole number is 543

[Discard figures after decimal point if the figure immediately after the decimal point is less than 5. If the figure is 5 or more, add 1 to the whole number part. For example, 543.5456 to the nearest whole number is 544]

ii). tenth(i. e. 1 d.p) is 543.3

iii). hundredth(i. e. 2 d.p) is 543.35

iv). thousandth(i. e. 3 d.p) is 543.346

ILLUSTRATIVE EXAMPLES

Example 1. Write 463.1598 to the nearest

i). whole number ii). tenth iii). hundredth iv). thousandth

Solution

i). whole number is 463

ii). tenth(i. e. 1 d. p) is 463. 2

iii). hundredth(i. e. 2 d. p) is 463. 16

iv). thousandth(i. e. 3 d. p) is 463. 160

Example 2. Write 40. 023045 to the nearest

i). whole number ii). tenth iii). hundredth iv). thousandth

Solution

i). whole number is 40

ii). tenth(i. e. 1 d. p) is 40. 0

iii). hundredth(i. e. 2 d. p) is 40. 02

iv). thousandth(i. e. 3 d. p) is 40. 023

Example 3. Write 11. 11111 to the nearest

i). whole number ii). tenth iii). hundredth iv). thousandth

Solution

i). whole number is 11

ii). tenth(i. e. 1 d. p) is 11. 1

iii). hundredth(i. e. 2 d. p) is 11. 11

iv). thousandth(i. e. 3 d. p) is 11. 111

SECTION 3: ROUNDING UP NUMBERS TO THE NEAREST TEN, HUNDRED, THOUSAND ETC

For any whole number, a figure's place in the number is important. The first figure, counting from right is the units, the second figure is the tens, the third figure is the hundreds, the fourth figure is the thousands, etc. For example the number 8912765, we have 5 units, 6 tens, 7 hundreds, 2 thousands, 1 ten thousands, 9 hundred thousands and 8 millions.

For example, 8912765 to the nearest

i). ten is 8,912,770

ii). hundred is 8,912,800

iii). thousand is 8,913,000

SECTION 4: APPROXIMATING A DECIMAL NUMBER TO A GIVEN NUMBER OF SIGNIFICANT FIGURES

- *When zero(0) comes before a non – zero digit number, it is not significant. Thus:*

0.002834

- *When zero comes after a non – zero digit number, it is not significant. Thus:*

243.40

- *When zero lies between two non – zero digit, then that zero is consider to be significant. Thus:*

204.115

To correct a number to a stated number of significant figures:

- 1. Find the last significant figure you want*
- 2. Then look at the next significant figure(to the right)*

If this figure is less than 5 leave the last significant figure you want as it is. If this figure is 5 or more add 1 to the last significant figure you want. Always check that your approximated number is of the same order of size as the original number.

ILLUSTRATIVE EXAMPLES

Example 1.

- Round 0.008327 to 1 significant figure**
- Round 16.063 to 3 significant figure**
- Round 0.3662 to 2 significant figure**

Solution

a). the 1st significant figure is 8 but the figure after it(i. e. the 2nd significant figure) is 3 which is less than 5.

Therefore we leave 8 as it is.

$$0.008327 = 0.008 \text{ (to 1 significant figure)}$$

b). the 3rd significant figure is 0 but the figure after it(i. e. the 4th significant figure) is 6 which is more than 5.

Therefore we add 1 to 0 to give it 1 in the 3rd significant figure

$$16.063 = 16.1 \text{ (to 3 significant figure)}$$

c). the 2nd significant figures is 6 but the figure after it(i. e. the 3rd significant figure) is 6 which is more than 5.

Therefore we add 1 to 6 to give 7 in the 2nd significant figure.

$$0.3662 = 0.37 \text{ (to 2 significant figure)}$$

Example 2. Correct each of the following numbers to 2 significant figures.

a) 0.0597

b) 0.0998

Solution

a) $0.0597 = 0.060$ (to 2 significant figures). Since the digit after the 2nd s. f. (i. e. 7) is more than 5, we add 1 to 9 to give

$$0.059$$

$$\begin{array}{r} 1 \\ \hline \end{array}$$

$$\begin{array}{r} 0.060 \\ \hline \end{array}$$

b) $0.0998 = 0.100$ (to 2 significant figures). Since the digit after the 2nd s.f. (i.e. 8) is more than 5, we add 1 to 9 to give

0.099

$$\begin{array}{r} 1 \\ \hline 0.100 \end{array}$$

• **SOMETIMES YOU HAVE TO FILL SOME 'PLACES' WITH 0's TO KEEP THE PLACE VALUE OF EACH FIGURE CORRECT.**

ILLUSTRATIVE EXAMPLE

Example 1. Correct 396 640 to

(a). 4 s.f (b). 3 s.f (c). 1 s.f

Solution

a). $396\,640 = 396\,600$ (to 4 s.f)

b). $396\,640 = 397\,000$ (to 3 s.f)

c). $396\,640 = 400\,000$ (to 1 s.f)

SECTION 5: SKS PAPER 2

Answer all questions

Part 1

Question 1. Round 564.6569 to

(i). 1 d.p (ii). 2 d.p (iii). 3 d.p

Answers:

(i). $564.6567 = 564.7$ (to 1. d.p)

(ii). $564.6567 = 564.66$ (to 2. d.p)

(iii). $564.6567 = 564.657$ (to 3. d.p)

Question 2. Round 346.75746 to

(i). 1 d.p (ii). 2 d.p (iii). 3 d.p

Answers:

(i). $346.75746 = 346.8$ (to 1. d.p)

(ii). $346.75746 = 346.76$ (to 2. d.p)

(iii). $346.75746 = 346.757$ (to 3. d.p)

Question 3. Round 346.12564 to 3 d.p.

Answer:

$346.12564 = 346.126$ (to 3. d.p)

Question 4. Round 4536.993 to 1 d.p.

Answer:

$4536.993 = 4537.0$ (to 1. d.p)

Question 6. Round 214.6986 to 3 d.p.

Answer:

$$214.6986 = 214.699 \text{ (to 3. d.p)}$$

Question 7. Round 499.997 to 1 d.p.

Answer:

$$499.997 = 500.0 \text{ (to 1. d.p)}$$

Part 2

Question 1. Write 5789.5467 to the nearest

i). whole number ii). tenth iii). hundredth iv). thousandth

Answers:

i). whole number is 5790

ii). tenth(i. e. 1 d.p) is 5789.5

iii). hundredth(i. e. 2 d.p) is 5789.55

iv). thousandth(i. e. 3 d.p) is 5789.547

Question 2. Write 1234.1035 to the nearest

i). whole number ii). tenth iii). hundredth iv). thousandth

Answers:

i). whole number is 1234

ii). tenth(i. e. 1 d.p) is 1234.1

iii). hundredth(i. e. 2 d.p) is 1234.10

iv). thousandth(i. e. 3 d.p) is 1234.104

Question 3. Write 2368.7243 to the nearest

i). whole number ii). tenth iii). hundredth iv). thousandth

Answers:

i). whole number is 2369

ii). *tenth*(i. e. 1 d. p) is 2368.7

iii). *hundredth*(i. e. 2 d. p) is 2368.72

iv). *thousandth*(i. e. 3 d. p) is 2368.724

Part 3

Question 1.

a). *Correct 0.007865 to 1 significant figure*

b). *Give 45.0553 to 3 significant figure*

c). *Round 0.2576 to 2 significant figure*

Answers:

a). $0.007865 = 0.008$ (to 1 s. f)

b). $45.0553 = 45.1$ (to 3 s. f)

c). $0.2576 = 0.26$ (to 2 s. f)

Question 2. Correct each of the following numbers to 2 significant figures.

a) 0.06796

b) 0.05995

Answers:

a). $0.06796 = 0.068$ (to 2 s. f)

b). $0.05995 = 0.060$ (to 2 s. f)

Part 4

Question 1. Correct 426 950 to

(a). 4 s. f (b). 3 s. f (c). 1 s. f

Answers:

a). $426\,850 = 426\,900$ (*to 4 s.f*)

b). $426\,850 = 427\,000$ (*to 3 s.f*)

c). $426\,850 = 400\,000$ (*to 1 s.f*)

Question 2. Correct 297 860 to

(a). 4 s.f (b). 3 s.f (c). 1 s.f

Answers:

a). $297\,860 = 297\,900$ (*to 4 s.f*)

b). $297\,860 = 298\,000$ (*to 3 s.f*)

c). $297\,860 = 300\,000$ (*to 1 s.f*)

CHAPTER 3

STANDARD FORM

This is a number expressed in the form $A \times 10^n$, where A is a number between 1 and 9 and n is an integer. For example:

$3 \times 10^5, 2 \times 10^4, 8 \times 10^{-4}$, etc.

SECTION 1: CONVERTING FROM ORDINARY FORM TO STANDARD FORM

In converting from ordinary form to standard form, we move the decimal point and place it between the 1st and 2nd significant figure.

ILLUSTRATIVE EXAMPLES

Example 1. Express 10000.06 in standard form

Solution

Step 1: Move and place the decimal point between the 1st and 2nd significant figure

10000.06 now becomes:

1.000006

Step 2: Multiply the answer in 'step 1' by 10^n

1.000006×10^n

*Step 3: Since the decimal point is moved four(4) times to the left,
 $n = 4$*

1.000006×10^4

Example 2. Express 3947.0779 in standard form

Solution

Step 1: Move and place the decimal point between the 1st and 2nd significant figure

3947.0779 now becomes:

3.9470779

Step 2: Multiply the answer in 'step 1' by 10^n

3.9470779×10^n

*Step 3: Since the decimal is moved three(3) times to the left,
 $n = 3$.*

3.9470779×10^3

Example 3. Express 2770004 in standard form

Solution

Step 1: Add ".0" behind the last number if it is a whole number

2770004.0

Step 2: Move and place the decimal point between the 1st and 2nd significant figure

2770004.0 now becomes:

2.770004

Step 3: Multiply the answer in 'step 2' by 10^n

2.770004×10^n

*Step 4: Since the decimal point is moved six(6) times to the left,
 $n = 6$*

$$2.770004 \times 10^6$$

Example 4. Express 177283 in standard form

Solution

Step 1: Add ".0" behind the last number if it is a whole number.

$$177283.0$$

Step 2: Move and place the decimal between the 1st and 2nd significant figure

177283.0 now becomes:

$$1.77283$$

Step 3: Multiply the answer in 'step 2' by 10^n

$$1.77283 \times 10^n$$

Step 4: Since the decimal is moved five(5) times to the left, $n = 5$

$$1.77283 \times 10^5$$

Example 5. Express 0.045 in standard form

Solution

Step 1: Move and place the decimal point between the 1st and 2nd significant figure

0.045 now becomes:

$$4.5$$

Step 2: Multiply the answer in 'step 2' by 10^n

$$4.5 \times 10^n$$

Step 3: Since the decimal is moved two(2) times to the right,

$$n = -2$$

$$4.5 \times 10^{-2}$$

Example 6. Express 0.0004893 in standard form.

Solution

Step 1: Move and place the decimal point between the 1st and 2nd significant figure

0.0004893 now becomes:

$$4.893$$

Step 2: Multiply the answer in 'step 2' by 10^n

$$4.893 \times 10^n$$

*Step 3: since the decimal is moved four(4) times to the right,
 $n = -4$.*

$$4.893 \times 10^{-4}$$

Example 7. Express 0.00000756 in standard form

Solution

Step 1: Move and place the decimal between the 1st and 2nd significant figure

0.00000756 now becomes:

$$7.56$$

Step 2: Multiply the answer in 'step 2' by 10^n

$$7.56 \times 10^n$$

Step 3: Since the decimal is moved 6 times to the right, $n = -6$

$$7.56 \times 10^{-6}$$

SECTION 2: CONVERTING FROM STANDARD FORM TO ORDINARY

Numbers in ordinary form are quite opposite to numbers in standard form. The end result is not multiplied by 10.

Subsection 1: When n is positive

When n is positive, we move the decimal point to the right.

ILLUSTRATIVE EXAMPLES

Example 1. Express 6.324×10^2 in ordinary form

Solution

Step 1: 10^2 means to move and place the decimal point two(2) times to the right

632.4

Example 2. Express 1.000005×10^4

Solution

Step 1: 10^4 means to move and place the decimal point four(4) times to the right

10000.05

Subsection 2: When n is negative

When n is negative, we move the decimal point to the left.

ILLUSTRATIVE EXAMPLES

Example 1. Express 2.5005×10^{-3} in ordinary form

Solution

Step 1: 10^{-3} means to add three(3) zeroes in front of the first number

0002.50054

Step 2: Since 10^{-3} , move and place the decimal point three(3) times to the left

0.00250054

Example 2. Express 0.324×10^{-3} in ordinary form

Solution

Step 1: 10^{-3} means to add three(3) zeroes in front of the first number

0000.324

Step 2: Since 10^{-3} , move and place the decimal point three(3) times to the left

0.000324

Example 3. Express 0.324×10^{-5} in ordinary form

Solution

Step 1: 10^{-5} means to add five(5) zeroes in front of the first number

000000.324

Step 2: Since 10^{-5} , move and place the decimal point five(5) times to the left

0.00000324

SECTION 3: BASIC OPERATION IN STANDARD FORMS

Subsection 1: Addition

Example 1. Simplify $2 \times 10^2 + 3 \times 10^3$

Solution

Step 1: convert 2×10^2 and 3×10^3 to ordinary form

$$2 \times 10^2 = 200$$

$$3 \times 10^3 = 3000$$

Step 2: add the numbers (200 and 3000)

$$\begin{array}{r} 3000 \\ + 200 \\ \hline 3200 \end{array}$$

Example 2. Simplify $3.12 \times 10^2 + 2.11 \times 10^3$

Solution

Step 1: convert 3.12×10^2 and 2.11×10^3 to ordinary form

$$3.12 \times 10^2 = 312$$

$$2.11 \times 10^3 = 2110$$

Step 2: add the numbers (2110 and 312)

$$\begin{array}{r} 2110 \\ + 312 \\ \hline 2422 \end{array}$$

Subsection 2: Subtraction

Example 1. Simplify $3 \times 10^3 - 2 \times 10^2$

Solution

Step 1: convert 3×10^3 and 2×10^2 to ordinary form

$$\Rightarrow 3 \times 10^3 = 3000$$

$$\Rightarrow 2 \times 10^2 = 200$$

Step 2: subtract

$$\begin{array}{r} 3000 \\ - 200 \\ \hline 2800 \end{array}$$

Example 2. Simplify $2.11 \times 10^3 - 3.12 \times 10^2$

Solution

Step 1: convert 3.12×10^2 and 2.11×10^3 to ordinary form

$$\Rightarrow 3.12 \times 10^2 = 312$$

$$\Rightarrow 2.11 \times 10^3 = 2110$$

Step 2: subtract

$$\begin{array}{r} 2110 \\ - 312 \\ \hline 1798 \end{array}$$

Subsection 3: Multiplication

Example 1. Simplify $(3 \times 10^3) \times (2 \times 10^2)$

Solution

Step 1: convert 2×10^2 and 3×10^3 to ordinary form

$$\Rightarrow 2 \times 10^2 = 200$$

$$\Rightarrow 3 \times 10^3 = 3000$$

Step 2: multiply

$$\Rightarrow 3000 \times 200 = 600000$$

OR

$$= (3 \times 2) \times (10^{3+2})$$

$$= 6 \times 10^5$$

$$= 6 \times 100000$$

$$= 600000$$

Example 2. Simplify $(2.11 \times 10^3) \times (3.12 \times 10^2)$

Solution

Step 1: convert 3.12×10^2 and 2.11×10^3 to ordinary form

$$\Rightarrow 3.12 \times 10^2 = 312.0$$

$$\Rightarrow 2.11 \times 10^3 = 2110.0$$

Step 2: multiply

$$\Rightarrow 2110.0 \times 312.0 = 658320.0$$

OR

$$= (2.11 \times 3.12) \times (10^{3+2})$$

$$= 6.5823 \times 10^5$$

$$= 6.5823 \times 100000$$

$$= 658320.0$$

Subsection 4: Division

Example 1. Simplify: $(3 \times 10^3) \div (2 \times 10^2)$

Solution

Step 1: convert 2×10^2 and 3×10^3 to ordinary form

$$2 \times 10^2 = 200$$

$$3 \times 10^3 = 3000$$

Step 2: divide

$$\frac{3000}{200} = 15$$

OR

$$= \frac{3}{2} \times 10^{3-2}$$

$$= 1.5 \times 10^1$$

$$= 1.5 \times 10$$

$$= 15$$

Example 2. Simplify: $(5.6 \times 10^3) \div (7.0 \times 10^1)$

Solution

Step 1: convert 5.6×10^2 and 7.0×10^1 to ordinary form

$$5.6 \times 10^2 = 560$$

$$7.0 \times 10^1 = 70$$

Step 2: divide

$$\frac{560}{70} = 8$$

SECTION 4: SKS PAPER 3

Part 1

Question 1. Express 2.46×10^{-5} in ordinary form

Answer: 0.0000246

Question 2. Express 6.381×10^3 in ordinary form

Answer: 6381.0

Question 3. Express 1.000004×10^4 in ordinary form

Answer: 10000.04

Question 4. Express 0.00250067 in standard form

Answer: 2.50067×10^{-3}

Question 5. Express 45671.4905 in standard form

Answer: 4.56714905×10^4

Question 6. Express 0.000123 in standard form

Answer: 1.23×10^{-4}

Part 2

Question 1. Simplify $4 \times 10^3 + 7 \times 10^7$

Answer: 70004000

Question 2. Simplify $6.345 \times 10^2 + 7.234 \times 10^3$

Answer: 7868.5

Question 3. Simplify $4.56 \times 10^3 - 2.172 \times 10^2$

Answer: 4342.8

Question 4. Simplify $2.11 \times 10^3 - 3.12 \times 10^2$

Answer: 1798

Question 5. Simplify $(3.12 \times 10^3) \times (2.24 \times 10^2)$

Answer: 698880

Question 6. Simplify: $(3.36 \times 10^6) \times (3.12 \times 10^2)$

Answer: 1048320000

Question 7. Simplify: $(7 \times 10^5) \div (2 \times 10^4)$

Answer: 35

Question 8. Simplify: $75.0 \times 10^3 \div 25.0 \times 10^3$

Answer: 3

CHAPTER 4

DECIMALS

SECTION 1: BASIC OPERATION ON DECIMALS

There are four(4) basic operation on decimals which are addition (+), subtraction (−), multiplication (×) and division (÷).

Subsection 1: Addition and Subtraction of decimals

Example 1. Simplify $4.52 + 3.48$

Solution

Step 1: Arrange

$$\begin{array}{r} 4.52 \\ +3.48 \\ \hline 8.00 \end{array}$$

Example 2. Simplify $4.5234 + 3.48$

Solution

Step 1: Arrange

$$\begin{array}{r} 4.5234 \\ +3.48 \\ \hline 8.0034 \end{array}$$

Example 3. Simplify $4.52 - 3.48$

Solution

Step 1: Arrange

$$\begin{array}{r} 4.52 \\ - 3.48 \\ \hline 1.04 \end{array}$$

Example 4. Simplify $4.5234 - 3.48$

Solution

Step 1: Arrange

$$\begin{array}{r} 4.5234 \\ - 3.48 \\ \hline 1.0434 \end{array}$$

Subsection 2: Multiplication and Division of decimals

Example 1. Simplify 4.52×3.48

Solution

Step 1: Convert to standard form

$$4.52 = 452 \times 10^{-2}$$

$$3.48 = 348 \times 10^{-2}$$

Step 2:

$$= (452 \times 10^{-2}) \times (348 \times 10^{-2})$$

Step 3:

$$= (452 \times 348) \times 10^{-2+(-2)}$$

$$= (452 \times 348) \times 10^{-2-2}$$

$$= 157296 \times 10^{-4}$$

$$= 15.7296$$

OR

Step 1: Arrange the figure as whole numbers, while the multiplication is done

$$\begin{array}{r} 452 \\ \times 348 \\ \hline 3616 \\ 1808 \\ 1356 \\ \hline 157296 \end{array}$$

Step 2: Count the total number of places of decimal in the two numbers

$\Rightarrow$ There are 4 decimal places (.52 and .48)

Step 3: The answer should have 4 decimal places

$$\therefore 4.52 \times 3.48 = 15.7296$$

Example 2. Simplify 0.52×1.228

Solution

Step 1: Convert to standard form

$$0.52 = 52 \times 10^{-2}$$

$$1.228 = 1228 \times 10^{-3}$$

Step 2:

$$= (52 \times 10^{-2}) \times (1228 \times 10^{-3})$$

Step 3:

$$= (52 \times 1228) \times 10^{-2+(-3)}$$

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$$= (52 \times 1228) \times 10^{-2-3}$$

$$= 63856 \times 10^{-5}$$

$$= 0.63856$$

OR

Step 1: Arrange the figure as whole numbers, while the multiplication is done

$$\begin{array}{r} 1228 \\ \times 52 \\ \hline 2456 \\ 6140 \\ \hline 63856 \end{array}$$

Step 2: Count the total number of places of decimal in the two numbers

$\Rightarrow$ There are 5 decimal places (.52 and .228)

Step 3: The answer should have 5 decimal places

$$\therefore 1.228 \times 0.52 = 0.63856$$

Example 3. Simplify $6.4 \div 0.16$

Solution

Step 1: Convert to standard form

$$6.4 = 64 \times 10^{-1}$$

$$0.16 = 16 \times 10^{-2}$$

Step 2:

$$= (64 \times 10^{-1}) \div (16 \times 10^{-2})$$

Step 3:

$$= \left(\frac{64}{16}\right) \times 10^{-1-(-2)}$$

$$= 4 \times 10^{-1+2}$$

$$= 4 \times 10^1$$

$$= 4 \times 10$$

$$= 40$$

Example 4. Simplify $6.25 \div 0.25$

Solution

Step 1: Convert to standard form

$$6.25 = 625 \times 10^{-2}$$

$$0.25 = 25 \times 10^{-2}$$

Step 2:

$$= (625 \times 10^{-2}) \div (25 \times 10^{-2})$$

Step 3:

$$= \left(\frac{625}{25}\right) \times 10^{-2-(-2)}$$

$$= 25 \times 10^{-2+2}$$

$$= 25 \times 10^0$$

$$= 25 \times 1$$

$$= 25$$

SECTION 2: CONVERSION OF DECIMALS TO FRACTIONS

Example 1. Convert 0.1 to fraction

Solution

Step 1: assume the number to be a whole number

⇒ Assume 0.1 to be 1

Step 2: since the original number has one decimal place, we divide the whole number by 10

$$\Rightarrow \frac{1}{10}$$

Example 2. Convert 21.34 to fraction

Solution

Step 1: assume the number to be a whole number

⇒ Assume 21.34 to be 2134

Step 2: since the original number has two decimal places, we divide the whole number by 100

$$\Rightarrow \frac{2134}{100}$$

Step 3: reduce the fraction to its lowest form

$$\Rightarrow \frac{1067}{50} = 21\frac{17}{50}$$

Example 3. Convert 0.234 to fraction

Solution

Step 1: assume the number to be a whole number

⇒ Assume 0.234 to be 234

Step 2: since the original number has three decimal places, we divide the

whole number by 1000

$$\Rightarrow \frac{234}{1000}$$

Step 3: reduce the fraction to it lowest form

$$\Rightarrow \frac{117}{500}$$

Example 4. Convert 0.2345 to fraction

Solution

Step 1: assume the number to be a whole number

$\Rightarrow$ Assume 0.2345 to be 2345

Step 2: Since the original number has four decimal places, we divide the whole number by 10000

$$\Rightarrow \frac{2345}{10000}$$

Step 3: Reduce the fraction to it lowest form

$$\Rightarrow \frac{469}{2000}$$

SECTION 3: CONVERSION OF DECIMALS TO PERCENTAGES

Example 1. Convert 0.12 to a percentage

Solution

Step 1: Multiply the decimal number by 100

$$= 0.12 \times 100$$

$$= 12\%$$

Example 2. Convert 0.1234 to a percentage

Solution

Step 1: Multiply the decimal number by 100

$$= 0.1234 \times 100$$

$$= 12.34\%$$

SECTION 4 : SKS PAPER 4

Answer all questions

Part 1

Question 1. Simplify $3.457 + 3.43$

Answer: 6.887

Question 2. Simplify $4.5234 + 3.48$

Answer: 8.0034

Question 3. Simplify $4.52 + 3.48$

Answer: 8

Question 4. Simplify $4.5234 + 3.4833$

Answer: 8.0067

Part 2

Question 1. Simplify $7.78 - 5.54 - 3.456$

Answer: -1.216

Question 2. Simplify $23.65 - 3.45 - 5.23$

Answer: 14.97

Part 3

Question 1. Simplify 72.34×4.23

Answer: 305.9982

Question 2. Simplify 0.2045×1.786

Answer: 0.365237

Part 4

Question 1. Simplify $23.14 \div 11.57$

Answer: 2

Question 2. Simplify $7.65 \div 3.06$

Answer: 2.5

Part 5

Question 1. Convert 0.567 to fraction

Answer: $\frac{567}{1000}$

Question 2. Convert 42.84 to fraction

Answer: $42\frac{21}{25}$

Question 3. Convert 0.222 to fraction

Answer: $\frac{111}{500}$

Question 4. Convert 0.458 to fraction

Answer: $\frac{229}{500}$

Part 6

Question 1. Convert 0.16 to a percentage

Answer: 16%

Question 2. Convert 0.4356 to a percentage

Answer: 43.56%

CHAPTER 5

RECIPROCAL AND SQUARES

SECTION 1: RECIPROCAL

Is the result of interchanging the numerator and the denominator of a fraction.

ILLUSTRATIVE EXAMPLES

Example 1. Find the reciprocals of the following?

a) x

b) $\frac{x}{y}$

c) $\frac{1}{x}$

d) $\frac{1}{3}m$

e) $\frac{1}{xy}$

f) $\frac{1}{3m}$

g) $\frac{b-1}{b}$

h) $\frac{ab}{a+b}$

i) $\frac{2f-v}{fv}$

Solution

a) x

Step 1: Add denominator of one (1) to x, since x is not in fraction form.

$$\frac{x}{1}$$

Step 2: Interchange the numerator and the denominator

$$= \frac{1}{x}$$

b). $\frac{x}{y}$

Solution

Step 1: Interchange the numerator and the denominator

$$= \frac{y}{x}$$

c). $\frac{1}{x}$

Solution

Step 1: Interchange the numerator and the denominator

$$= \frac{x}{1}$$

$$= x$$

d). $\frac{1}{3}m$

Solution

Step 1: Multiply the numerators

$$\frac{1 \times m}{3} = \frac{m}{3}$$

Step 2: Interchange the numerator and the denominator

$$= \frac{3}{m}$$

$$e). \frac{1}{xy}$$

Solution

Step 1: Interchange the numerator and the denominator

$$= \frac{xy}{1}$$

$$= xy$$

$$f). \frac{1}{3m}$$

Solution

Step 1: Interchange the numerator and the denominator

$$= \frac{3m}{1}$$

$$= 3m$$

$$g). \frac{b-1}{b}$$

Solution

Step 1: Interchange the numerator and the denominator

$$= \frac{b}{b-1}$$

$$h). \frac{ab}{a+b}$$

Solution

Step 1: Interchange the numerator and the denominator

$$= \frac{a+b}{ab}$$

$$i). \frac{2f-v}{fv}$$

Solution

Step 1: Interchange the numerator and the denominator

$$= \frac{fv}{2f - v}$$

SECTION 2: SQUARES

In mathematics, a square is the result of multiplying a term by itself.

Subsection 1: Square of variables with power one(1) only

Example 1. Find the squares of the following?

a) p

b). xb

c). $\frac{a}{t}$

Solution

a).

Step 1: Give the expression a power of two(2)

$$= p^2$$

b).

Step 1: Put the expression inside a bracket

$$(xb)$$

Step 2: Raise the bracket to power two(2)

$$= (xb)^2$$

$$= x^2b^2$$

c).

Step 1: Put the expression inside a bracket

$$= \left(\frac{a}{t}\right)$$

Step 2: Raise the bracket to power two(2)

$$= \left(\frac{a}{t}\right)^2$$

$$= \frac{a^2}{t^2}$$

Subsection 2: Square of variables with powers other than one(1)

Example 1. Find the squares of the following?

a). a^2b^4

b). a^2b^3

c). $x^4y^2z^3$

d). $3t^3 + 4m^2$

e). $\frac{2x^3}{5y^2}$

Solution

a).

Step 1: Put the expression inside a bracket

$$= (a^2b^4)$$

Step 2: Raise the bracket to power two(2)

$$= (a^2b^4)^2$$

Step 3: Multiply the power outside the bracket by each of the powers inside the bracket

$$= a^{2 \times 2} b^{4 \times 2}$$

$$= a^4 b^8$$

b).

Step 1: Put the expression inside a bracket

$$= (a^2b^3)$$

Step 2: Raise the bracket to power two(2)

$$= (a^2b^3)^2$$

Step 3: Multiply the power outside the bracket by each of the powers inside the bracket

$$= a^{2 \times 2} b^{3 \times 2}$$

$$= a^4 b^6$$

c).

Step 1: Put the expression inside a bracket

$$= (x^4y^2z^3)$$

Step 2: Raise the bracket to power two(2)

$$= (x^4y^2z^3)^2$$

Step 3: Multiply the power outside the bracket by each of the powers inside the bracket

$$= x^{4 \times 2} y^{2 \times 2} \times z^{3 \times 2}$$

$$= x^8 y^4 z^6$$

d).

Step 1: Put the quantexpression inside a bracket

$$= (3t^2 \times 4m^3)$$

Step 2: Simplify

$$= (3 \times 4 \times t^2 \times m^3)$$

$$= (12t^2m^3)$$

Step 3: Raise the bracket to power two(2)

$$= (12t^2m^3)^2$$

Step 4: Multiply the power outside the bracket by each of the powers

inside the bracket

$$= 12^2 t^{2 \times 2} m^{3 \times 2}$$

$$= 144 t^4 m^6$$

e).

Step 1: Put the expression inside a bracket

$$= \left(\frac{2x^3}{5y^2} \right)$$

Step 2: Raise the bracket to power two(2)

$$= \left(\frac{2x^3}{5y^2} \right)^2$$

Step 3: Multiply the power outside the bracket by each of the powers inside the bracket

$$= \frac{2^{1 \times 2} x^{3 \times 2}}{5^{1 \times 2} y^{2 \times 2}}$$

$$= \frac{2^2 x^6}{5^2 y^4}$$

$$= \frac{2 \times 2 \times x^6}{5 \times 5 \times y^4}$$

$$= \frac{4x^6}{25y^4}$$

Subsection 3: Square of numbers

Example 1. Find the square of the following numbers

a). 1

b). -2

c). 25

d). -36

e). -100

Solution

a).

Step 1: Raise the number to power two(2)

$$= 1^2$$

Step 2: Multiply the number by itself

$$= 1 \times 1$$

$$= 1$$

b).

Step 1: Raise the number to power two(2)

$$= (-2)^2$$

Step 2: Multiply the number by itself

$$= -2 \times -2$$

$$= 4$$

c).

Step 1: Raise the number to power two(2)

$$25^2$$

Step 2: Multiply the number by itself

$$= 25 \times 25$$

$$= 625$$

d).

Step 1: Raise the number to power two(2)

$$= (-36)^2$$

Step 2: Multiply the number by itself

$$= -36 \times -36$$

$$= 1296$$

e).

Step 1: Raise the number to power two(2)

$$= (-100)^2$$

Step 2: Multiply the number by itself

$$= -100 \times -100$$

$$= 10000$$

Subsection 4: Square of square roots

Example 1. Find the square of the following:

(i) $\sqrt{a}$

(ii) $\sqrt{ab}$

(iii) $\sqrt{(a^3b)}$

(iv) $\sqrt{abc^3}$

(v) $\sqrt{(msk^5)}$

(vi) $\sqrt{\frac{d}{g}}$

(vii) $a\sqrt{m}$

(viii) $2\pi\sqrt{a}$

(ix) $\frac{3}{\sqrt{b}}$

$$(x) \frac{\pi}{3} \sqrt{2g}$$

Solution

(i).

Step 1: Put the square root inside a bracket

$$= (\sqrt{a})$$

Step 2: Square the bracket and simplify

$$= (\sqrt{a})^2$$

Step 3: The square will cancel the square root

$$(\sqrt{a})^2$$

$$= a$$

(ii).

Step 1: Put the square root inside a bracket

$$= (\sqrt{ab})$$

Step 2: Square the bracket and simplify

$$= (\sqrt{ab})^2$$

Step 3: The square will cancel the square root

$$= (\sqrt{ab})^2$$

$$= ab$$

(iii).

Step 1: Put the square root inside a bracket

$$= (\sqrt{a^3b})$$

Step 2: Square the bracket and simplify

$$= \left(\sqrt{a^3 b} \right)^2$$

Step 3: The square will cancel the square root

$$\left(\sqrt{a^3 b} \right)^2$$

$$= a^3 b$$

(iv).

Step 1: Put the square root inside a bracket

$$= \left(\sqrt{ac^3 b} \right)$$

Step 2: Square the bracket and simplify

$$= \left(\sqrt{ac^3 b} \right)^2$$

Step 3: The square will cancel the square root

$$= \left(\sqrt{ac^3 b} \right)^2$$

$$= ac^3 b$$

(v).

Step 1: Put the square root inside a bracket

$$= \left(\sqrt{msk^5} \right)$$

Step 2: Square the bracket

$$= \left(\sqrt{msk^5} \right)^2$$

Step 3: The square will cancel the square root

$$= \left(\sqrt{msk^5} \right)^2$$

$$= msk^5$$

(vi).

Step 1: Put the expression inside a bracket

$$= \left(\sqrt{\frac{d}{g}} \right)$$

Step 2: Square the bracket and simplify

$$= \left(\sqrt{\frac{d}{g}} \right)^2$$

Step 3: The square will cancel the square root

$$= \frac{d}{g}$$

(vii).

Step 1: Put the expression inside a bracket

$$= (a\sqrt{m})$$

Step 2: Square the bracket and simplify

$$= (a\sqrt{m})^2$$

$$= a^2(\sqrt{m})^2$$

$$= a^2m$$

(viii).

Step 1: Put the expression inside a bracket

$$= (2\pi\sqrt{a})$$

Step 2: Square the bracket and simplify

$$= (2\pi\sqrt{a})^2$$

$$= (2\pi)^2(\sqrt{a})^2$$

$$= 2^2\pi^2a$$

$$= 4\pi^2a$$

(ix).

Step 1: Put the expression inside a bracket

$$= \left(\frac{3}{\sqrt{b}}\right)$$

Step 2: Square the bracket and simplify

$$= \left(\frac{3}{\sqrt{b}}\right)^2$$

$$= \frac{3^2}{\sqrt{b}^2}$$

$$= \frac{9}{b}$$

(x).

Step 1: Put the expression inside a bracket

$$= \left(\frac{\pi}{3}\sqrt{2g}\right)$$

Step 2: Square the bracket and simplify

$$= \left(\frac{\pi}{3}\sqrt{2g}\right)^2$$

$$= \frac{\pi^2 \times 2g}{3^2}$$

$$= \frac{2\pi^2g}{9}$$

Example 2. Simplify the following

a) $\sqrt{9b^2}$

b) $\sqrt{\frac{16}{9}t^2m^2}$

Solution

a).

Step 1: $\sqrt{9b^2}$ can also be written as:

$$= \sqrt{9} \times \sqrt{b^2}$$

$$= 3 \times b$$

$$= 3b$$

Note: $\sqrt{9} = 3$ and $\sqrt{b^2} = b$

b).

Step 1: $\sqrt{\frac{16}{9}t^2m^2}$ can also be written as:

$$\sqrt{\frac{16}{9}} \times \sqrt{t^2} \times \sqrt{m^2}$$

Step 2: Simplify

$$= \frac{\sqrt{16}}{\sqrt{9}} \times t \times m$$

$$= \frac{4}{3}tm$$

Subsection 5: Square of cube roots

Example 1. Find the square of the following and simplify

(i) $\sqrt[3]{a}$

(ii) $\sqrt[3]{abc^3}$

Solution

(i).

Step 1: Put the square root inside a bracket

$$= (\sqrt[3]{a})$$

Step 2: Square the bracket

$$= (\sqrt[3]{a})^2$$

$$= a^{\frac{1}{3} \times 2}$$

$$= a^{\frac{2}{3}}$$

(ii).

Step 1: Put the square root inside a bracket

$$(\sqrt[3]{abc^3})$$

Step 2: Square the bracket and simplify

$$(\sqrt[3]{abc^3})^2$$

$$= \left[(abc^3)^{\frac{1}{3}} \right]^2$$

$$= \left[a^{\frac{1}{3}} b^{\frac{1}{3}} c^{\frac{1}{3} \times 3} \right]^2$$

$$= a^{\frac{1}{3} \times 2} b^{\frac{1}{3} \times 2} c^{\frac{1}{3} \times 3 \times 2}$$

$$= a^{\frac{2}{3}} b^{\frac{2}{3}} c^2$$

SECTION 3: SKS PAPER 5

Answer all questions

Part 1

Question 1. Find the reciprocal of the following?

a) y

b). $\frac{y}{x}$

c). $\frac{1}{b}$

d). $\frac{2}{3}h$

e). $\frac{2}{xyz}$

f). $\frac{2}{3mn}$

g). $\frac{3b-1}{b}$

h). $\frac{4ab}{a+b}$

i). $\frac{2f-3v}{fv}$

Answers:

(a). $\frac{1}{y}$ (b). $\frac{x}{y}$ (c). b (d). $\frac{3h}{2}$ (e). $\frac{xyz}{2}$ (f). $\frac{3mn}{2}$ (g). $\frac{b}{3b-1}$ (h). $\frac{a+b}{4ab}$ (i). $\frac{fv}{2f-3v}$

Part 2

Question 1. Find the squares of the following?

a). cvx

b). ay

c). $\frac{v}{t}$

Answers:

(a). $c^2v^2x^2$

(b). a^2y^2

(c). $\frac{v^2}{t^2}$

Part 3

Question 1. Find the squares of the following and simplify?

a). v^2z

b). m^2n^3

c). $x^4y^4z^3$

d). $7t^3 \times 4m^2$

e). $\frac{24}{5y^2}$

Answers:

(a). v^4z^2

(b). m^4n^6

(c). $x^8y^8z^6$

(d). $784t^6m^4$

(e). $\frac{576}{25y^4}$

Part 4

Question 1. Find the squares of the following numbers

a). 1000

b). -2000

c). 25000

d). -36000

e). -1001

Answers:

(a). 1000000

(b). 4000000

(c). 625000000

(d). 1296000000

(e). 1002001

Part 5

Question 1. Find the squares of the following and simplify:

(i) $\sqrt{acd}$

(ii) $\sqrt{abmn}$

(iii) $\sqrt{(a^3b)}$

(iv) $\sqrt{abc^3de}$

(v) $\sqrt{(msk^5ln)}$

Answers:

(i). acd (ii). $abmn$ (iii). a^3b (iv). abc^3de and (v). msk^5ln

Part 6

Question 1. Find the square of the following and simplify

(i). $\sqrt[3]{p}$

(ii). $\sqrt[3]{abc^3mn}$

Answers:

(i). $p^{\frac{2}{3}}$ and (ii). $a^{\frac{2}{3}}b^{\frac{2}{3}}c^2m^{\frac{2}{3}}n^{\frac{2}{3}}$

Question 2. Write down the squares of the following and simplify

a). dhp

b). $\frac{msk}{t}$

c). $\frac{x^2yz}{y}$

d). $abc + y$

e). $\sqrt{\frac{e}{g}}$

f). $a\sqrt{mn}$

g). $2\pi\sqrt{abcd}$

h). $\frac{3acd}{\sqrt{b}}$

i). $\frac{\pi}{3}\sqrt{2mpz}$

Answers:

(a). $d^2h^2p^2$

$$(b). \frac{m^2 s^2 k^2}{t^2}$$

$$(c). x^4 z^2$$

$$(d). a^2 b^2 c^2 + y^2$$

$$(e). \frac{e}{g}$$

$$(f). a^2 mn$$

$$(g). 4\pi^2 abcd$$

$$(h). \frac{9a^3 c^2 d^2}{b}$$

$$(i). \frac{2\pi^2 mpz}{9}$$

Question 3 .Simplify the following

$$a)\sqrt{900b^2}$$

$$b). \sqrt{\frac{160}{90} t^2 m^2}$$

Answers:

$$a). 30b$$

$$b). \frac{4}{3} tm$$

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CHAPTER 6

NUMBER BASES

SECTION 1: INTRODUCTION

This is a system of counting in which number makes up a bundle.

The bundle thus form the place value of the digit. The highest digit of any number base is always one less than the base. If the base is y , the highest digit ever is $y - 1$.

SECTION 2: CONVERSION OF NUMBER BASE

A number base can be changed from one base to another and it involves the following:

- 1. To change from other base to base ten.*
- 2. To change from base ten to another base.*
- 3. To change from other base to another base that is not in base ten*

Subsection 1: Conversion from one base to base ten

Example 1. Convert 210_{three} to number base 10.

Solution

Step 1: Use the format below

$$\begin{aligned} & (1^{\text{st}} \text{ number} \times \text{the base}) + (2^{\text{nd}} \text{ number} \times \text{the base}) \\ & \quad + (3^{\text{rd}} \text{ number} \times \text{the base}) \end{aligned}$$

$$= (2 \times 3) + (1 \times 3) + (0 \times 3)$$

Step 2: Power each base in ascending order starting from 0 on the right hand side

$$= (2 \times 3^2) + (1 \times 3^1) + (0 \times 3^0)$$

$$= (2 \times 3 \times 3) + (1 \times 3) + (0 \times 1)$$

$$= 18 + 3 + 0$$

$$= 21_{10} = 21$$

Note: Any number raise to power zero is 1.

Example 2. Convert 234_{five} to number base 10

Solution

Step 1: Use the format below

$$(1^{\text{st}} \text{ number} \times \text{the base}) + (2^{\text{nd}} \text{ number} \times \text{the base}) \\ + (3^{\text{rd}} \text{ number} \times \text{the base})$$

$$= (2 \times 5) + (3 \times 5) + (4 \times 5)$$

Step 2: Power each base in ascending order starting from "0" on the right hand side

$$= (2 \times 5^2) + (3 \times 5^1) + (4 \times 5^0)$$

$$= (2 \times 5 \times 5) + (3 \times 5) + (4 \times 1)$$

$$= 50 + 15 + 4$$

$$= 69_{10} = 69$$

Note: Any number raise to power zero is 1.

Example 3. Convert 124_{five} to number base 10

Solution

Step 1: Use the format below

$$(1^{\text{st}} \text{ number} \times \text{the base}) + 2^{\text{nd}} \text{ number} \times \text{the base}) \\ + (3^{\text{rd}} \text{ number} \times \text{the base})$$

$$= (1 \times 5) + (2 \times 5) + (4 \times 5) + (9 \times 5)$$

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Step 2: Power each base in ascending order starting from zero on the right hand side

$$= (1 \times 5^2) + (2 \times 5^1) + (4 \times 5^0)$$

$$= (1 \times 5 \times 5) + (2 \times 5) + (4 \times 1)$$

$$= 25 + 10 + 40$$

$$= 75_{10} = 75$$

Note: Any number raise to power zero is 1.

Example 4. Find in base ten the value of the 2 in 123_{four}

Solution

Step 1: Convert to base 10

$$= (1 \times 4^2) + (2 \times 4^1) + (3 \times 4^0)$$

$$= (1 \times 4 \times 4) + (2 \times 4) + (3 \times 1)$$

$$= 16 + 8 + 3$$

Step 2: by comparing 123_4 and $16 + 8 + 3$ it can be seen that the value of 2 is 8



Therefore the value of 2 is 8.

Subsection 2: Converting from number base ten to other bases

Example 1. Convert 48_{ten} to number base three

Solution

Step 1: Divide repeatedly by 3

3	48	
3	16	R 0
3	5	R 1
3	1	R 2
	0	R 1

$$\therefore 48_{10} = 1210_3$$

Example 2. Convert 21_{10} to number base two

Solution

Step 1: Divide repeatedly by 2

2	21	
2	10	R 1
2	5	R 0
2	2	R 1
2	1	R 0
	0	R 1

$$\therefore 21_{10} = 10101_2$$

Subsection 3: Conversion between non – decimal bases

To convert between non – decimal bases first convert to base ten and then convert the result to the required base.

Example 1. Convert 210_{three} to number base seven.

Solution

Step 1: Change 210_{three} to number base 10.

$$= (2 \times 3^2) + (1 \times 3^1) + (0 \times 3^0)$$

$$= (2 \times 3 \times 3) + (1 \times 3) + (0 \times 1)$$

$$= 18 + 3 + 0$$

$$= 21_{10}$$

Step 2: We then convert 21_{10} to number base seven

7	21	
7	3	R 0
	0	R 3



$$\therefore 210_3 = 30_7$$

Example 2. Convert 234_{five} to number base four

Solution

Step 1: Change 234_{five} to number base 10.

$$(1^{st} \text{ number} \times \text{the base}) + 2^{nd} \text{ number} \times \text{the base} \\ + (3^{rd} \text{ number} \times \text{the base})$$

$$= (2 \times 5) + (3 \times 5) + (4 \times 5)$$

$$= (2 \times 5^2) + (3 \times 5^1) + (4 \times 5^0)$$

$$= (2 \times 5 \times 5) + (3 \times 5) + (4 \times 1)$$

$$= 50 + 15 + 4$$

$$= 69_{10}$$

Step 2: We then convert 69_{ten} to number base four

4	69	
4	17	R 1
4	4	R 1
4	1	R 0
	0	R 1



$$\therefore 69_{10} = 1011_4$$

Subsection 4: Equations involving number bases

Example 1. If $42_x = 30$, find the value of x .

Solution

Step 1: Changing the left hand side to number base 10

$$(4 \times x^1) + (2 \times x^0) = 30$$

$$(4 \times x) + (2 \times 1) = 30$$

$$4x + 2 = 30$$

Step 2: Collect like terms

$$4x = 30 - 2$$

$$4x = 28$$

Step 3: Divide both sides by the coefficient of x

$$\frac{4x}{4} = \frac{28}{4}$$

$$x = 7$$

Note: The right hand side is already in base 10.

Example 2. If $21_x = 9_{10}$, find the value of x .

Solution

Step 1: Changing the left hand side to base 10

$$(2 \times x^1) + (1 \times x^0) = 9$$

$$(2 \times x) + (1 \times 1) = 9$$

$$2x + 1 = 9$$

Step 2: Collect like terms

$$2x = 9 - 1$$

$$2x = 8$$

Step 3: Divide both sides by the coefficient of x

$$\frac{2x}{2} = \frac{8}{2}$$

$$x = 4$$

Note: The right hand side is already in base 10.

Example 3. If $45_x = 131_4$, find the value of x .

Solution

Step 1: Changing both sides of the equation to number base 10

$$(4 \times x^1) + (5 \times x^0) = (1 \times 4^2) + (3 \times 4^1) + (1 \times 4^0)$$

$$(4 \times x) + (5 \times 1) = 16 + 12 + 1$$

$$4x + 5 = 29$$

Step 2: Collect like terms

$$4x = 29 - 5$$

$$4x = 24$$

Step 3: Divide both sides by the coefficient of x

$$\frac{4x}{4} = \frac{24}{4}$$

$$x = 6$$

Example 4. If $232_x = 92$, find the value of x .

Solution

Step 1: Change the LHS of the equation to number base 10

$$(2 \times x^2) + (3 \times x^1) + (2 \times x^0) = 92$$

$$2x^2 + 3x + 2 = 92$$

Step 2: Collect like terms

$$2x^2 + 3x + 2 - 92 = 0$$

$$2x^2 + 3x - 90 = 0$$

[This is a quadratic equation which can be solved by factorization]

Step 3: Find two numbers which when multiplied gives " $-180x^2$ " and when added gives " $+3x$ "

$$\text{Sum} = +15x - 12x$$

$$\text{Product} = +15x (\times) - 12x$$

Step 4: Replace the middle term by " $+15x - 12x$ "

$$2x^2 + 15x - 12x - 90 = 0$$

Step 5: For the LHS, group the first two terms and the last two as well

$$(2x^2 + 15x)(-12x - 90) = 0$$

Step 6: Factorize

$$x(2x + 15) - 6(2x + 15) = 0$$

$$(x - 6)(2x + 15) = 0$$

$$(i). x - 6 = 0$$

$$x = 6$$

$$(ii). 2x + 15 = 0$$

$$x = -\frac{15}{2}$$

Step 7: Since the equation is true for positive values of x ;

$$\Rightarrow x = 6 \dots \text{Answer}$$

Example 5. If $(3m_6)_7 = 188_{10}$, find the value of x .

Solution

Step 1: Converting the LHS to base 10

$$3 \times 7^2 + m \times 7^1 + 6 \times 7^0 = 188$$

$$3 \times 49 + m \times 7 + 6 \times 1 = 188$$

$$147 + 7m + 6 = 188$$

Step 2: Collect like terms

$$7m + 147 + 6 = 188$$

$$7m + 153 = 188$$

$$7m = 188 - 153$$

$$7m = 35$$

Step 3: Divide both sides by the coefficient of m

$$\frac{7m}{7} = \frac{35}{7}$$

$$m = 5$$

Example 6. Find the number x such that $35_6 + 14_5 + 15_x = 45_{10}$

Solution

Step 1: Change the L.H.S of the equation to base 10.

$$(3 \times 6^1) + (5 \times 6^0) + (1 \times 5^1) + (4 \times 5^0) + (1 \times x^1) + (5 \times x^0) = 45$$

$$(3 \times 6) + (5 \times 1) + (1 \times 5) + (4 \times 1) + (1 \times x) + (5 \times 1) = 45$$

$$18 + 5 + 5 + 4 + x + 5 = 45$$

$$x + 37 = 45$$

Step 2: Collect like terms

$$x = 45 - 37$$

$$x = 8$$

Example 7. If $x^2 = 20_8$, find ?

Solution

Step 1: Converting 20_8 to base 10

$$x^2 = (2 \times 8^1) + (0 \times 8^0)$$

$$x^2 = (2 \times 8) + (0 \times 1)$$

$$x^2 = 16 + 0$$

$$x^2 = 16 + 0$$

$$x^2 = 16$$

Step 2: Taking the square root of both sides

$$\sqrt{x^2} = \pm\sqrt{16}$$

$$x = \pm 4$$

SECTION 2: BASIC OPERATION IN NUMBER BASES**Subsection 1: Addition in number base****Example 1. Evaluate $1101_2 + 101_2 + 1011_2$** **Solution**

$$\begin{array}{r}
 1101_2 \\
 101_2 \\
 + 1011_2 \\
 \hline
 11101_2
 \end{array}$$

➤ Working**Starting from the right hand side****Column 1:** $1 + 1 + 1 = 3$. Since the number (i.e 3) is greater than the base (i.e 2), we divide the number by the base as shown below

$$\Rightarrow \frac{3}{2} = 1 \text{ time remainder } 1. \text{ Write the 1 and carry the 1.}$$

Column 2: $0 + 0 + 1 + 1(\text{number carried}) = 2$.**Since the number (i.e 2) is equal to the base (i.e 2), we divide the number by the base as shown below**

$$\Rightarrow \frac{2}{2} = 1 \text{ time remainder } 0. \text{ Write the 0 and carry the 1.}$$

Column 3: $1 + 1 + 0 + 1(\text{number carried}) = 3$.**Since the number (i.e 3) is greater than the base (i.e 2), we divide the number by the base as shown below**

$$\Rightarrow \frac{3}{2} = 1R1. \text{ Write the 1 and carry the 1}$$

Column 4: $1 + 1 + 1(\text{number carried}) = 3$. Write both the time and

the remainder, since it is the last column

Note: We can only divide the number by the base if the number is equal to or greater than the base.

Example 2. Evaluate $234_5 + 102_5 + 321_5$

$$\begin{array}{r} 234_5 \\ 102_5 \\ + 321_5 \\ \hline 1212_5 \end{array}$$

Solution

➤ Working

Starting from the right hand side

Column 1: $4 + 2 + 1 = 7$. Since the number (i.e 7) is greater than the base (i.e 5), we divide the number by the base as shown below

$$\Rightarrow \frac{7}{5} = 1R2. \text{ Write the 2 and carry the 1.}$$

Column 2: $3 + 0 + 2 + 1 = 6$. Since the number (i.e 6) is greater than the base (i.e 5), we divide the number by the base as shown below

$$\Rightarrow \frac{6}{5} = 1R1. \text{ Write the 1 and carry the 1.}$$

Column 3: $2 + 1 + 3 + 1 = 7$. Since the number (i.e 7) is greater than the base (i.e 5), we divide the number by the base as shown below

$$\Rightarrow \frac{7}{5} = 1R2. \text{ Write both the times and the remainder, since it is the}$$

last column

Note: We can only divide the number by the base if the number is equal to or greater than the base.

Example 3. Evaluate in base four

$$\begin{array}{r} 123_4 \\ + 32_4 \\ + 1_4 \\ \hline 222_4 \end{array}$$

Subsection 2: Subtraction in number base

Example 1. Evaluate in base five

$$\begin{array}{r} 313_5 \\ - 124_5 \\ \hline 134_5 \end{array}$$

➤ Working

Column 1: $3 - 4$, we can not. We borrow 1 and add it to 3 so that it becomes 8. We now have:

$$\Rightarrow 8 - 4 = 4. \text{ We write } 4.$$

Note: The number we borrow represents the base (i. e 5) and not 1.

Column 2: $0 - 2$, we can not. We borrow 1 and add it to 0 so that it becomes 5. We now have

$$\Rightarrow 5 - 2 = 3. \text{ We write } 3.$$

Note: The number we borrow represents the base (i. e 5) and not 1.

Column 3: $2 - 1 = 1$. We write 1

Example 2. Evaluate in base 9

$$\begin{array}{r} 482_9 \\ -364_9 \\ \hline 117_9 \end{array}$$

Column 1: $2 - 4$, we can not. We borrow 1 and add it to 2 so that it becomes 11. We now have:

$\Rightarrow 11 - 4 = 7$. We write 7.

Note: The number we borrow represents the base (i.e 9) and not 1.

Column 2: $7 - 6 = 1$. We write 1

Column 3: $4 - 3 = 1$. We write 1

Subsection 3: Finding missing numbers in number base

Example 1. The following addition was performed in base 2

$$\begin{array}{r} 1101_2 \\ \quad \text{XXX}_2 \\ + 1011_2 \\ \hline 11101_2 \end{array}$$

Solution

Step 1:

$$\begin{array}{r} 1101_2 \\ + 1011_2 \\ \hline 11000_2 \end{array}$$

And

$$\begin{array}{r} 11101_2 \\ -11000_2 \\ \hline 101_2 \end{array}$$

Example 2. Find the missing number if the following subtraction is in base 3

$$\begin{array}{r} 211_3 \\ -XXX_3 \\ \hline 111_3 \end{array}$$

Solution

Step 1:

$$\begin{array}{r} 211_3 \\ -111_3 \\ \hline 100_3 \end{array}$$

Subsection 4: Multiplication in number base

Example 1. Evaluate $20_3 \times 20_3$

Solution

Step 1:

$$\begin{array}{r} 20_3 \\ \times 20_3 \\ \hline 00 \\ +110 \\ \hline 1100_3 \end{array}$$

Example 2. Evaluate $(31_4)^2$

Solution

Step 1:

$$\begin{array}{r} 31_4 \\ \times 31_4 \\ \hline 31 \\ +213 \\ \hline 2221_4 \end{array}$$

SECTION 3: SKS PAPER 6

Answer all questions

Part 1

Question 1. Evaluate in base 3

$$\begin{array}{r} 111_3 \\ 111_3 \\ + 121_3 \\ \hline \\ \hline \end{array}$$

Answer: 10210_3

Question 2. Evaluate in base 6

$$\begin{array}{r} 234_6 \\ 542_6 \\ + 321_6 \\ \hline \\ \hline \end{array}$$

Answer: 1541_6

Question 3. Evaluate in base five

$$\begin{array}{r} 143_5 \\ 30_5 \\ + 1_5 \\ \hline \\ \hline \end{array}$$

Answer: 224_5

Part 2

Question 1. Evaluate in base six

$$\begin{array}{r} 315_6 \\ -124_6 \\ \hline \end{array}$$

Answer: 151_6

Question 2. Evaluate in base nine

$$\begin{array}{r} 572_9 \\ -364_9 \\ \hline \end{array}$$

Answer: 207_9

Part 3

Question 1. The following addition was performed in base four

$$\begin{array}{r} 1321_4 \\ YYY_4 \\ +1331_4 \\ \hline 13303_4 \end{array}$$

Answer: 3331_4

Question 2. Find the missing number if the following subtraction is in base 9

$$\begin{array}{r} 778_9 \\ -XXX_9 \\ \hline 110_9 \end{array}$$

Answer: 668_9

Part 4

Question 1. Evaluate $40_5 \times 40_5$

Answer: 3100_5

Question 2. Evaluate $(78_9)^2$

Answer: 6821_9

Part 5

Question 1. Convert 433_{five} in base ten.

Answer: 118_{10}

Question 2. Convert 834_{nine} in base ten

Answer: 679_{10}

Question 3. Convert 5154_{six} in base ten

Answer: 1150_{10}

Part 6

Question 1. Evaluate $75_x = 61_{\text{ten}}$

Answer: $x = 8$

Question 2. Evaluate $65_x = 47_{\text{ten}}$

Answer: $x = 7$

Question 3. Find in base ten the value of the 3 in 123_{four}

Answer: The value of 3 is 3

Part 7

Question 1. Convert 222_{ten} to base three

Answer: 22020_3

Question 2. Convert 1111_{10} to base two

Answer: 10001010111_2

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CHAPTER 7

SURDS

SECTION 1: INTRODUCTION

Subsection 1: Rational numbers

Taking only the positive square root, $\sqrt{1}$, $\sqrt{4}$, $\sqrt{100}$ etc. give exact values 1, 2 and 10 respectively are called rational numbers.

Subsection 2: Irrational numbers

*$\sqrt{3} = 1.732 \dots$ and $\sqrt{13} = 3.606 \dots$ etc. can not be obtained directly, and those roots are said to be irrational numbers, generally called **surds**.*

SECTION 2: REDUCTION TO BASIC FORM

If $\sqrt{x}$ is such that it cannot be reduced further then it is said to be in the basic form. therefore x does not contain a factor which is a perfect square. For example, $\sqrt{11}$, $\sqrt{17}$, $\sqrt{37}$ and $\sqrt{41}$ are in the basic form and can not be simplified any further.

ILLUSTRATIVE EXAMPLES

Example 1. Simplify:

(i). $\sqrt{27}$ (ii). $\sqrt{45}$ (iii). $\sqrt{12}$ (iv). $\sqrt{48}$ (v). $\sqrt{125}$

Solution

(i). $\sqrt{27}$

Step 1: Find two numbers which when multiplied gives $\sqrt{27}$, but one of them must be a rational number

$$\sqrt{27} = \sqrt{9 \times 3}$$

$$= \sqrt{9} \times \sqrt{3}$$

$$= 3 \times \sqrt{3}$$

$$= 3\sqrt{3}$$

Note: $\sqrt{9} = 3$... Rational number

(ii). $\sqrt{45}$

Step 1: Find two numbers which when multiplied gives $\sqrt{45}$, but one of them must be a rational number

$$\sqrt{45} = \sqrt{9 \times 5}$$

$$= \sqrt{9} \times \sqrt{5}$$

$$= 3 \times \sqrt{5}$$

$$= 3\sqrt{5}$$

(iii).

$$\sqrt{12} = \sqrt{4 \times 3}$$

$$= \sqrt{4} \times \sqrt{3}$$

$$= 2\sqrt{3}$$

Note: $\sqrt{4} = 2$... *Rational number*

(iv).

$$\sqrt{48} = \sqrt{16 \times 3}$$

$$= \sqrt{16} \times \sqrt{3}$$

$$= 4\sqrt{3}$$

Note: $\sqrt{16} = 4$... *Rational number*

(v).

$$\sqrt{125} = \sqrt{25 \times 5}$$

$$= \sqrt{25} \times \sqrt{5}$$

$$= 5\sqrt{5}$$

Note: $\sqrt{25} = 5$... *Rational number*

SECTION 3: ADDITION AND SUBTRACTION OF SURDS

We can only add or subtract surds which are alike or have the same form.

Note: reduce first to their basic form if they are not.

ILLUSTRATIVE EXAMPLES

Example 1. Simplify: $2\sqrt{3} + 4\sqrt{3}$

Solution

Step 1: Add the numbers in front of each root inside a bracket and place the common root outside the bracket

$$= (2 + 4)\sqrt{3}$$

$$= 6\sqrt{3}$$

Example 2. Simplify: $4\sqrt{2} + 6\sqrt{2}$

Solution

Step 1: Add the numbers in front of each root inside a bracket and place the common root outside the bracket

$$= (4 + 6)\sqrt{2}$$

$$= 10\sqrt{2}$$

Example 3. Simplify: $10\sqrt{3} - 4\sqrt{3}$

Solution

Step 1:

$$= (10 - 4)\sqrt{3}$$

$$= 6\sqrt{3}$$

Example 4. Simplify: $7\sqrt{6} - 4\sqrt{6}$

Solution

Step 1:

$$= (7 - 4)\sqrt{6}$$

$$= 3\sqrt{6}$$

Example 5. Simplify: $\sqrt{75} + \sqrt{27}$

Solution

Step 1: Reduce each root to it basic form and simplify

$$= \sqrt{75} + \sqrt{27}$$

$$= \sqrt{25 \times 3} + \sqrt{9 \times 3}$$

$$= (\sqrt{25} \times \sqrt{3}) + (\sqrt{9} \times \sqrt{3})$$

$$= 5\sqrt{3} + 3\sqrt{3}$$

$$= 8\sqrt{3}$$

Example 6. Simplify: $\sqrt{50} + \sqrt{32}$

Solution

Step 1: Reduce each root to it basic form and simplify

$$= \sqrt{50} + \sqrt{32}$$

$$= \sqrt{25 \times 2} + \sqrt{16 \times 2}$$

$$= (\sqrt{25} \times \sqrt{2}) + (\sqrt{16} \times \sqrt{2})$$

$$= 5\sqrt{2} + 4\sqrt{2}$$

$$= 9\sqrt{2}$$

SECTION 4: PROPERTIES OF SURDS

The following properties of surds should be noted carefully.

$$1. \sqrt{a} \times \sqrt{a} = (\sqrt{a})^2 = a$$

$$2. (\sqrt{a})^2 = a$$

$$3. \sqrt{a} \times \sqrt{b} = \sqrt{ab}$$

$$4. \frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\left(\frac{a}{b}\right)}$$

$$5. a\sqrt{b} \times \sqrt{c} = a\sqrt{bc}$$

$$6. a \times \sqrt{b} = a\sqrt{b}$$

$$7. a\sqrt{b} \times c\sqrt{d} = ac\sqrt{bd}$$

$$8. a \times b\sqrt{c} = ab\sqrt{c}$$

$$9. \sqrt{m+n} \neq \sqrt{m} + \sqrt{n}$$

$$10. \sqrt{m-n} \neq \sqrt{m} - \sqrt{n}$$

SECTION 5: MULTIPLICATION OF SURDS

When multiplying two surds in brackets, the rules of surds must be used.

ILLUSTRATIVE EXAMPLES

Example 1. Simplify: $(3 - 4\sqrt{2})(\sqrt{2} - 6)$

Solution

Step 1: Multiply the numbers in the first bracket by each of the numbers in the second bracket

$$= 3 \times \sqrt{2} - 3 \times 6 - 4\sqrt{2} \times \sqrt{2} + 4\sqrt{2} \times 6$$

$$= 3\sqrt{2} - 18 - 4(2) + 24\sqrt{2}$$

$$= 3\sqrt{2} - 18 - 8 + 24\sqrt{2}$$

Step 2: Collect like terms

$$= 3\sqrt{2} + 24\sqrt{2} - 18 - 8$$

$$= 27\sqrt{2} - 26$$

Example 2. Simplify: $(5 - 3\sqrt{2})(\sqrt{2} - 7)$

Solution

Step 1: Multiply the numbers in the first bracket by each of the numbers in the second bracket

$$= 5 \times \sqrt{2} - 5 \times 7 - 3\sqrt{2} \times \sqrt{2} + 3\sqrt{2} \times 7$$

$$= 5\sqrt{2} - 35 - 3(2) + 21\sqrt{2}$$

$$= 5\sqrt{2} - 35 - 6 + 21\sqrt{2}$$

Step 2: Collect like terms

$$5\sqrt{2} + 21\sqrt{2} - 35 - 6 = 26\sqrt{2} - 41$$

SECTION 6: RATIONALISING THE DENOMINATOR OF SURDS

This is the process of making the denominator of fractional surds whole or rational numbers.

Surds of the form $\frac{1}{\sqrt{5}}$ or $\frac{1}{\sqrt{7}}$ etc, can be rationalised by multiplying both the numerator and the denominator by the denominator.

ILLUSTRATIVE EXAMPLES

Example 1. Simplify: $\frac{2}{\sqrt{2}}$

Solution

Step 1: Multiply both the numerator and the denominator by $\sqrt{2}$

$$= \frac{2}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$= \frac{2 \times \sqrt{2}}{\sqrt{2} \times \sqrt{2}}$$

$$= \frac{2\sqrt{2}}{2}$$

$$= \sqrt{2}$$

Example 2. Simplify: $\frac{4}{\sqrt{6}}$

Solution

Step 1: Multiply both the numerator and the denominator by $\sqrt{6}$

$$= \frac{4}{\sqrt{6}} \times \frac{\sqrt{6}}{\sqrt{6}}$$

$$= \frac{4 \times \sqrt{6}}{\sqrt{6} \times \sqrt{6}}$$

$$= \frac{4\sqrt{6}}{6}$$

$$= \frac{4\sqrt{6}}{6}$$

$$= \frac{2\sqrt{6}}{3}$$

SECTION 7: CONJUGATE OF SURDS

If $a + \sqrt{b}$ is a surd, then the conjugate of $a + \sqrt{b}$ is $a - \sqrt{b}$.

The product of a surd and its conjugate is no longer a surd. Thus

$$(a + \sqrt{b})(a - \sqrt{b}) = a^2 - b$$

ILLUSTRATIVE EXAMPLES

Example 1. Simplify $(2 + \sqrt{3})(2 - \sqrt{3})$

Solution

Step 1: State the rule

$$(a + \sqrt{b})(a - \sqrt{b}) = a^2 - b$$

Step 2: Apply

$$\begin{aligned}(2 + \sqrt{3})(2 - \sqrt{3}) &= 2^2 - 3 \\ &= 4 - 3 \\ &= 1\end{aligned}$$

Example 2. Simplify $(4 + \sqrt{5})(4 - \sqrt{5})$

Solution

Step 1: State the rule

$$(a + \sqrt{b})(a - \sqrt{b}) = a^2 - b$$

Step 2: Apply

$$\begin{aligned}(4 + \sqrt{5})(4 - \sqrt{5}) &= 4^2 - 5 \\ &= 16 - 5 \\ &= 11\end{aligned}$$

SECTION 8: SKS PAPER 7

Answer all questions

Part 1

Question 1. Simplify the following:

(i). $\sqrt{54}$ (ii). $\sqrt{80}$ (iii). $\sqrt{75}$ (iv). $\sqrt{108}$ (v). $\sqrt{147}$

Answers: [(i) $3\sqrt{6}$ (ii) $4\sqrt{5}$ (iii) $5\sqrt{3}$ (iv) $6\sqrt{3}$ (v) $7\sqrt{3}$]

Question 2. Simplify: $2\sqrt{6} + 10\sqrt{6}$

Answer: $12\sqrt{6}$

Question 3. Simplify: $\frac{3}{4}\sqrt{2} + \frac{1}{4}\sqrt{2}$

Answer: $\sqrt{2}$

Question 4. Simplify: $\frac{3}{4}\sqrt{3} - \frac{1}{4}\sqrt{3}$

Answer: $\frac{\sqrt{3}}{2}$

Question 5. Simplify: $7\sqrt{5} - \frac{1}{2}\sqrt{5}$

Answer: $\frac{13\sqrt{5}}{2}$

Question 6. Simplify: $\sqrt{80} + \sqrt{125}$

Answer: $9\sqrt{5}$

Question 7. Simplify: $\sqrt{24} + \sqrt{150}$

Answer: $7\sqrt{6}$

Part 2

Question 1. Simplify: $(7 - 3\sqrt{2})(\sqrt{2} - 8)$

Answer: $31\sqrt{2} - 62$

Question 2. Simplify: $(6 - 4\sqrt{3})(\sqrt{3} - 5)$

Answer: $26\sqrt{3} - 42$

Part 3

Question 1. Simplify: $\frac{5}{\sqrt{5}}$

Answer: $\sqrt{5}$

Question 2. Simplify: $\frac{3}{\sqrt{7}}$

Answer: $\frac{3\sqrt{7}}{7}$

CHAPTER 8

FRACTIONS AND PERCENTAGES

SECTION 1: FRACTIONS

Subsection 1: Definitions and worked examples

A fraction is a part of a whole expressed as a figure placed on top of another figure. The one on top is called Numerator while the one

below is called Denominator, e. g. $\frac{2}{3}, \frac{1}{3}, \frac{7}{5}$ etc. There are essentially two types: Proper and Improper fractions.

A proper has the numerator less than the denominator, e. g. $\frac{2}{3}, \frac{1}{3}, \frac{2}{5}$ etc

An improper fraction has the numerator greater than the denominator, e. g. $\frac{4}{3}, \frac{5}{3}, \frac{7}{3}$, etc

Also we have a mixed number which consist of a whole part and fraction written together, e. g. $3\frac{4}{3}, 8\frac{5}{4}, 7\frac{7}{3}$ etc. Solving problems involving fractions does not create difficulty once a pupil remember the word "BODMAS" which stands for the order in which the operation of arithmetic is carried out. This implies.

B = Bracket

O = Of

D = Division

M = Multiplication

A = Addition

S = Subtraction

The rule states:

(a). Work out bracket

(b). Divide and multiply

(c). Add and subtract

ILLUSTRATIVE EXAMPLES

Example 1. Simplify $1\frac{2}{5} + 2\frac{1}{5}$

Solution

Step 1: Change $1\frac{2}{5}$ and $2\frac{1}{5}$ to improper fractions

$$\Rightarrow 1\frac{2}{5} = \frac{7}{5}$$

$$\Rightarrow 2\frac{1}{5} = \frac{11}{5}$$

Step 2:

$$= \frac{7}{5} + \frac{11}{5}$$

Step 3: Find the LCM

L. C. M of 5 and 5 is 5

Step 4:

$$= \frac{7 + 11}{5}$$

$$= \frac{18}{5}$$

$$= 3\frac{3}{5}$$

Example 2. Simplify $2\frac{1}{2} + 1\frac{2}{5}$

Solution

Step 1: Change $2\frac{1}{2}$ **and** $1\frac{2}{5}$ **to improper fractions**

$$2\frac{1}{2} = \frac{5}{2}$$

$$1\frac{2}{5} = \frac{7}{5}$$

Step 2:

$$= \frac{5}{2} + \frac{7}{5}$$

Step 3: Find the LCM

LCM of 2 and 5 is 10

Step 4:

$$= \frac{14 + 25}{10}$$

$$= \frac{39}{10}$$

$$= 3\frac{9}{10}$$

Example 3. Simplify: $\frac{2}{5}$ **of** $\frac{4}{5} + 1\frac{3}{10}$

Solution

Step 1: Change $1\frac{3}{10}$ **to improper fraction**

$$\frac{2}{5} \text{ of } \frac{4}{5} + \frac{13}{10}$$

Step 2: Change 'of' to multiplication

$$\frac{2}{5} \times \frac{4}{5} + \frac{13}{10}$$

According to BODMAS, we multiply before we add

$$\frac{2}{5} \times \frac{4}{5} = \frac{8}{25}$$

Step 3: Add the answer in 'step 2' with the last fraction

$$= \frac{8}{25} + \frac{13}{10}$$

LCM of 25 and 10 is 50

$$= \frac{16 + 65}{50}$$

$$= \frac{81}{50}$$

$$= 1 \frac{31}{50}$$

Example 4. Simplify: $\frac{2}{7}$ of $\frac{3}{8}$

Solution

Step 1: Change 'of' to multiplication

$$= \frac{2}{7} \times \frac{3}{8}$$

$$= \frac{2 \times 3}{7 \times 8}$$

$$= \frac{6}{56}$$

$$= \frac{3}{28}$$

Example 5. Simplify: $\frac{3}{8}$ of $\frac{7}{2}$ of $\frac{3}{2}$

Solution

Step 1: Change 'of' to multiplication

$$= \frac{3}{8} \times \frac{7}{2} \times \frac{3}{2}$$

$$= \frac{63}{32}$$

$$= 1 \frac{31}{32}$$

Example 6. Simplify $3\frac{2}{3} \div \left(\frac{2}{5} + \frac{4}{5}\right)$

Solution

Step 1: Apply BODMAS and simplify bracket first

$$\left(\frac{2}{5} + \frac{4}{5}\right)$$

LCM of 5 and 5 is 5

$$\frac{2+4}{5} = \frac{6}{5}$$

Step 2: Change $3\frac{2}{3}$ to improper fraction and divide by the answer of the bracket

$$= \frac{11}{3} \div \frac{6}{5}$$

Step 3: Change the division sign to multiplication and interchange the numerator and denominator of the fraction

after the division sign

$$= \frac{11}{3} \times \frac{5}{6}$$

$$= \frac{55}{18}$$

$$= 3\frac{1}{18}$$

Example 7. Simplify $\frac{2}{5} \times \left(\frac{2}{3} - \frac{1}{4}\right) + \frac{1}{2}$

Solution

Step 1: Apply BODMAS and simplify

$$= \left(\frac{2}{3} - \frac{1}{4}\right)$$

$$= \frac{8 - 3}{12}$$

$$= \frac{5}{12}$$

Step 2: Apply BODMAS again (solve for multiplication before addition)

$$= \frac{2}{5} \times \frac{5}{12}$$

$$= \frac{10}{60}$$

$$= \frac{1}{6}$$

Step 3: We now add

$$= \frac{1}{6} + \frac{1}{2}$$

$$= \frac{1 + 3}{6}$$

$$= \frac{4}{6}$$

$$= \frac{2}{3}$$

Example 8. Simplify: $2\frac{1}{2} \div 3\frac{3}{4}$
 $2\frac{1}{6} - 1\frac{1}{3}$

Solution

Step 1: Simplify the numerator

$$= \frac{5}{2} \div \frac{15}{4}$$

$$= \frac{5}{2} \times \frac{4}{15}$$

$$= \frac{20}{30}$$

$$= \frac{2}{3}$$

Step 2: Simplify the denominator

$$= \frac{13}{6} - \frac{4}{3}$$

$$= \frac{13 - 8}{6}$$

$$= \frac{5}{6}$$

Step 3: Divide the answer in 'step 1' by the answer in step 2

$$= \frac{2}{3} \div \frac{5}{6}$$

$$= \frac{2}{3} \times \frac{6}{5}$$

$$= \frac{12}{15}$$

$$= \frac{4}{5}$$

Subsection 2: Order of magnitude

Fractions can be arranged in either ascending or descending order of magnitude. Ascending order is the arrangement from the smallest to the biggest. Descending order is from the biggest to the smallest.

ILLUSTRATIVE EXAMPLES

Example 1. Arrange in ascending order of magnitude :

$$\frac{4}{16}, \frac{3}{30}, \frac{11}{55}, \frac{14}{28}$$

Solution

Step 1: Simplify to the lowest term

$$= \frac{1}{4}, \frac{1}{10}, \frac{1}{5}, \frac{1}{2}$$

Step 2: Find the LCM of (2, 4, 5, 10)

$$LCM = 40$$

Step 3: Apply concepts of fraction

$$\left(\frac{10}{40}, \frac{4}{40}, \frac{8}{40}, \frac{20}{40} \right)$$

Step 4: Arrange in ascending order

$$\frac{4}{40}, \frac{8}{40}, \frac{10}{40}, \frac{20}{40}$$

$$\left(\frac{3}{30}, \frac{11}{55}, \frac{4}{16}, \frac{14}{28} \right)$$

OR

Step 1: Convert to decimals

$$\frac{4}{16} = 0.25$$

$$\frac{3}{30} = 0.10$$

$$\frac{11}{55} = 0.20$$

$$\frac{14}{28} = 0.50$$

Step 2: ascending order of magnitude is 0.10, 0.20, 0.25, 0.5

Hence, the arrangement is: $\frac{3}{30}, \frac{11}{55}, \frac{4}{16}, \frac{14}{28}$

Example 2. Arrange in descending order of magnitude :

$$\frac{4}{16}, \frac{3}{30}, \frac{11}{55}, \frac{14}{28}$$

Solution

Step 1: Simplify to the lowest term

$$= \frac{1}{4}, \frac{1}{10}, \frac{1}{5}, \frac{1}{2}$$

Step 2: Find the LCM of (2, 4, 5, 10)

$$LCM = 40$$

Step 3: Apply concepts of fraction

$$\left(\frac{10}{40}, \frac{4}{40}, \frac{8}{40}, \frac{20}{40} \right)$$

Step 4: Arrange in descending order

$$\frac{20}{40}, \frac{10}{40}, \frac{8}{40}, \frac{4}{40}$$

$$\left(\frac{14}{28}, \frac{4}{16}, \frac{11}{55}, \frac{3}{30}\right)$$

OR

Step 1: Convert to decimals

$$\frac{4}{16} = 0.25$$

$$\frac{3}{30} = 0.10$$

$$\frac{11}{55} = 0.20$$

$$\frac{14}{28} = 0.50$$

Step 2: descending order of magnitude is 0.5, 0.25, 0.20, 0.10

Hence, the arrangement is: $\frac{14}{28}, \frac{4}{16}, \frac{11}{55}, \frac{3}{30}$

SECTION 2: PERCENTAGES

A percentage(%) is a fraction in which the denominator is assumed to be 100. For Example:

60% can be written as $\frac{60}{100}$

80% can be written as $\frac{80}{100}$

Subsection 1: Conversion of a fraction to a percentage

To convert a fraction to a percentage, we multiply by 100.

ILLUSTRATIVE EXAMPLES

Example 1. Convert $\frac{2}{5}$ to a percentage

Solution

Step 1: Multiply the number by 100

$$\begin{aligned} &= \frac{2}{5} \times 100 \\ &= 40\% \end{aligned}$$

Example 2. Convert $\frac{7}{50}$ to a percentage

Solution

Step 1: Multiply the number by 100

$$\frac{7}{50} \times 100 = 14\%$$

Subsection 2: Conversion of a decimal to a percentage

Example 1. Convert 0.681 to percentage

Solution

Step 1: Multiply the number by 100 and 100 can also be written as 10^2

$$= 0.681 \times 10^2$$

Step 2: Move and place the decimal point two times to the right

$$= 68.1\%$$

Example 2. Convert 0.3 to percentage

Solution

Step 1: Multiply the number by 100 and 100 can also be written as 10^2

$$= 0.3 \times 10^2$$

Step 2: Move and place the decimal point two times to the right

$$= 30.0\%$$

Subsection 3: Express a percentage as a decimal

In converting percentages to decimals, we divide the number by 100 or we multiply the number by 10^{-2} .

Example 1. Express 32% as decimal

Solution

Step 1: Multiply the number (i.e 32) by 10^{-2}

$$= 32 \times 10^{-2}$$

$$= 0.32$$

OR

$$\frac{32}{100} = 0.32$$

Example 2. Express 2.35% as decimal

Solution

Step 1: Multiply the number(i.e 2.35) by 10^{-2}

$$= 2.35 \times 10^{-2}$$

$$= 0.0235$$

OR

$$\frac{2.35}{100} = 0.0235$$

Subsection 4: Percentage of a quantity

Example 1. What is 20% of 600?

Solution

Step 1: Express 20% as a fraction and then multiply by 600

$$\frac{20}{100} \times 600 = 120$$

Example 2. What is 40% of Le 40,000.00?

Solution

Step 1: Express 40% as a fraction and then multiply by 40,000

$$\frac{40}{100} \times 40000 = \text{Le } 16,000.00$$

Example 3. What is 22.5% of Le 20,000.00?

Solution

Step 1: Express 22.5% as a fraction and then multiply by 20,000

$$\frac{22.5}{100} \times 20000.00 = \text{Le } 4,500.00$$

Example 4. What percentage is 25 of 125?

Solution

Step 1: The number coming after 'of' is the denominator

$$= \frac{25}{125}$$

Step 2: Multiply the fraction by 100

$$= \frac{25}{125} \times 100 = 20\%$$

Example 5. What percentage is 20 of 200?

Solution

Step 1:

$$= \frac{20}{200} \times 100 = 10\%$$

Note: the quantity coming after 'of' in percentages is the denominator

Subsection 5 : Increasing a quantity

Increase means :

$$100\% + x\%$$

ILLUSTRATIVE EXAMPLES

Example 1. Increase Le 240 by 20%

Solution

Step 1: Increase means

$$100\% + 20\% = 120\%$$

Step 2: Express 120% as a fraction and then multiply by 240

$$\frac{120}{100} \times 240 = \text{Le } 288$$

Example 4. Increase Le 200 by 22.5%

Solution

Step 1:

$$100\% + 22.5\% = 122.5\%$$

Step 2: Express 122.5% as a fraction and then multiply by 200

$$\frac{122.5}{100} \times 200 = \text{Le } 245$$

Subsection 6: Decreasing a quantity

Decrease means:

$$100\% - x\%$$

Example 1. Decrease Le 280 by 20%

Solution

Step 1: Decrease means

$$100\% - 20\% = 80\%$$

Step 2: Express 80% as a fraction and then multiply by 280

$$= \frac{80}{100} \times 280 = 224$$

Example 2. If Le 360,000 is decreased by 30%, what does it become?

Solution

Step 1: Decrease means

$$100\% - 30\% = 70\%$$

Step 2:

$$= \frac{70}{100} \times 360,000 = \text{Le } 252,000$$

Subsection 7: Faulty and not faulty in percentages

Example 1. A trader bought 5000 mangoes. He finds that 25% are faulty, how many mangoes are faulty and not faulty?

Solution

Step 1: Work out the number of mangoes that are faulty

$$\Rightarrow \frac{25}{100} \times 5000 = 1250$$

Step 2: Work out the number of mangoes that are not faulty

Number of mangoes that were not faulty

= Total number of mangoes – number of faulty mangoes

$$= 5000 - 1250$$

$$= 3750$$

Example 2. Mrs. Zainab Sillah Kanu bought 2000 oranges.

She found out that 20% were bruised. How many oranges were bruised and not bruised?

Solution

Step 1: Work out the number of bruised oranges

$$\frac{20}{100} \times 2000 = 400$$

Step 2: Work out the number of oranges that were not bruised

Number of oranges that were not bruised

= Total number of oranges – number of bruised oranges

$$= 2000 - 400$$

$$= 1600$$

Subsection 8: Percentage profit and loss

PROFIT: profit occurs when the selling price is higher than the cost price.

➤ $PROFIT = \text{selling price} - \text{cost price}$

LOSS: loss occurs when the cost price is higher than the selling price

➤ $LOSS = \text{cost price} - \text{selling price}$

PERCENTAGE PROFIT: is the expression of the profit in percentages.

$$\begin{aligned} \text{PERCENTAGE PROFIT} &= \frac{\text{selling price} - \text{cost price}}{\text{cost price}} \times 100 \\ &= \frac{\text{profit}}{\text{cost price}} \times 100 \end{aligned}$$

PERCENTAGE LOSS: is the expression of the loss in percentage.

$$\begin{aligned} \text{PERCENTAGE LOSS} &= \frac{\text{cost price} - \text{selling price}}{\text{cost price}} \times 100 \\ &= \frac{\text{loss}}{\text{cost price}} \times 100 \end{aligned}$$

Note: In percentage profit and loss, cost price is the denominator.

ILLUSTRATIVE EXAMPLES

Example 1. Mrs. Finda Sillah Kanu buys goods for Le 8,000 and sells it at Le 12,000. Calculate the percentage profit.

Solution

Step 1: Write the formula for percentage profit

$$\frac{\text{Selling price(s.p)} - \text{Cost price(c.p)}}{\text{Cost price(c.p)}} \times 100$$

Step 2: Substitute

$$= \frac{12000 - 8000}{8000} \times 100$$
$$= 50\%$$

Example 2. Sillah FC bought Johannes Kande (a prolific striker) for Le 14,000,000.00 from Yasmine FC and sold him to Hamoud FC for Le 7,000,000.00. What is the percentage loss?

Solution

Step 1: Write the formula for percentage loss

$$= \frac{\text{Cost price(c.p)} - \text{Selling price(s.p)}}{\text{Cost price(c.p)}} \times 100$$
$$= \frac{14000000 - 7000000}{14000000} \times 100$$
$$= \frac{7}{14} \times 100$$
$$= 50\%$$

Example 3. Leona Conteh bought a car for Le 120,000.00 and sold it to Angela Deen Swarray at a profit of 25%. Find the selling price of the car?

Solution

Step 1: Find the profit first

$$\frac{25}{100} \times 120000 = \text{Le } 30,000$$

Step 2: Selling price = Cost price + Profit

$$= \text{Le } 120,000 + \text{Le } 30,000$$
$$= \text{Le } 150,000$$

Example 4. *Kadijah Sillah Kanu bought a suit for Le 150,000 and sold it to Isha Sillah Kanu at a profit of 10%. Find the selling price of the suit?*

Solution

Step 1: *Find the profit first*

$$\frac{10}{100} \times 150000 = \text{Le } 15,000$$

Step 2: *Selling price = Cost price + Profit*

$$= \text{Le } 150000 + \text{Le } 15000$$

$$= \text{Le } 165,000$$

Example 5. *Mohamed Alpha Kamara sold his phone to Kadijah Dumbuya for Le 314,000 at a loss of $22\frac{1}{2}\%$. Find the cost price?*

Solution

Step 1: *% loss means*

$$100\% - 22\frac{1}{2}\% = 78\frac{1}{2}\%$$

Step 2: *When finding the cost price '100' is the numerator*

$$= \frac{100}{\left(\frac{157}{2}\right)} \times 314000$$

$$= 100 \div \frac{157}{2} \times 314000$$

$$= 100 \times \frac{2}{157} \times 314000$$

$$= \frac{100 \times 2 \times 314000}{157}$$

$$= \text{Le } 400,000$$

Example 6. *Alimamy Bundu Conteh sold his Sillah Kanu Series mathematics book to Marina Conteh for Le 10,000 at a loss of 20%. Find the cost price?*

Solution

Step 1: % loss means

$$100\% - 20\% = 80\%$$

Step 2: When finding the cost price, '100' is the numerator

$$\frac{100}{80} \times 10000$$

$$= \text{Le } 12,500$$

Subsection 9: Finding the original quantity in percentages

To find the original quantity make '100' the numerator.

Note: cost price is one of the original quantities.

ILLUSTRATIVE EXAMPLES

Example 1. *230 is the result of increasing a number by 15%.*

Find the number?

Solution

Step 1: Increase means

$$100\% + 15\% = 115\%$$

Step 2: Use the formula below

$$\text{Original number} = \frac{100}{\% \text{ increase}} \times \text{quantity}$$

$$= \frac{100}{115} \times 230$$

$$= 200$$

∴ The number is 200

Example 2. When an article is increase by 20% its price becomes Le 360,000. Find the marked price?

Solution

Step 1: % Increase means

$$100\% + 20\% = 120\%$$

Step 2: Use the formula below

$$\frac{100}{\% \text{ increase}} \times \text{quantity}$$

$$\frac{100}{120} \times 360000 = \text{Le } 300,000$$

SECTION 3: SKS PAPER 8

Part 1

Question 1. Simplify $3\frac{2}{5} + 4\frac{1}{5}$

Answer: $7\frac{3}{5}$

Question 2. Simplify $7\frac{1}{2} + 8\frac{2}{5}$

Answer: $15\frac{9}{10}$

Question 3. Simplify $\frac{4}{5}$ of $\frac{6}{5} + 2\frac{3}{10}$

Answer: $3\frac{13}{50}$

Question 4. Simplify $\frac{3}{7}$ of $\frac{5}{8}$

Answer: $\frac{15}{56}$

Question 5. Simplify $\frac{7}{8}$ of $\frac{7}{2}$ of $\frac{5}{2}$

Answer: $7\frac{21}{32}$

Question 6. Simplify $4\frac{2}{3} \div \left(\frac{3}{5} + \frac{4}{5}\right)$

Answer: $3\frac{1}{3}$

Question 7. Simplify $\frac{2}{5} \times \left(\frac{1}{3} - \frac{3}{4}\right) + \frac{1}{2}$

Answer: $\frac{1}{3}$

Question 8. Simplify: $\frac{4\frac{1}{2} \div 3\frac{3}{4}}{3\frac{1}{6} - 7\frac{1}{3}}$

Answer: $-\frac{36}{125}$

Part 2

Question 1. Convert $\frac{1}{5}$ to a percentage

Answer: 20%

Question 2. Convert $\frac{7}{50}$ to a percentage

Answer: 14%

Part 3

Question 1. Convert 0.8678 to percentage

Answer: 86.78%

Question 2. Convert 0.5 to percentage

Answer: 50%

Part 4

Question 1. Convert 5.123% as decimal

Answer: 0.05123

Question 2. Convert 6123% as decimal

Answer: 61.23

Part 5

Question 1. What is 50% of 1000?

Answer: 500

Question 2. What is 60% of Le 60,000.00?

Answer: Le 36,000

Question 3. What is 45.5% of Le 100,000.00?

Answer: Le 45,500

Part 6

Question 1. Increase 60%

Answer: 160%

Question 2. Increase 55.5%

Answer: 155.5%

Question 3. Increase 3600 by 30%

Answer: 4680

Question 4. Increase 200 by 25.5%

Answer: 251

Part 7

Question 1. Decrease 300 by 20%

Answer: 240

Question 2. If Le 400,000 is decreased by 50%, what does it become?

Answer: Le 200,000

Question 3. What percentage is 30 of 75?

Answer: 40

Question 4. What percentage is 80 of 200?

Answer: 40

Part 8

Question 1. A trader bought 400 mangoes. He found out that 20% are faulty, how many mangoes were faulty and not faulty?

Answer [Faulty = 80 and Not faulty = 320]

Question 2. Dalophantine bought 2500 oranges. She found out that 30% were bruised. How many oranges were bruised and not bruised?

Answer [Bruised = 750 and Not Bruised = 1750]

Part 9

Question 1. Mr. Kanu buys goods for Le 10,000 and sells it at Le 15,000. Calculate the percentage profit.

Answer = 50%

Question 2. Sillah FC bought Johannes Kande (a prolific striker) for Le 50,000,000.00 from Yasmine FC and sold him to Hamoud FC for Le 20,000,000.00. What is the percentage loss?

Answer = 60%

Question 3. Susan Sengeh bought a car for Le 50,000,000.00 and sold it to Fatmata Conteh at a profit of 25%. Find the selling price of the car?

Answer = Le 62,500,000.00

Question 4. Vidal Musa Conteh bought a suit for Le 500,000 from Alhassan Tullah at a profit of 30%. Find the selling price of the suit?

Answer = Le 650,000

Question 5. Mr. Mohamed Sellu sold his car to Mr. A. I. Turay for Le 51,000,000 at a loss of $74\frac{1}{2}\%$. Find the cost price?

Answer = Le 200,000,000

Question 6. Hassan Davies sold his Sillah – Kanu Series mathematics book to Yusuf Kamara for Le 120,000 at a loss of 40%. Find the cost price?

Answer = Le 200,000

Part 10

Question 1. 4500 is the result of increasing a number by 50%. Find the number?

Answer = 3000

Question 2. When an article is increase by 80% it price becomes Le 720,000. Find the marked price?

Answer = Le 400,000

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CHAPTER 9

ALGEBRA

SECTION 1: ADDITION AND SUBTRACTION

Subsection 1: The same sign

If the signs are the same, then factor or take out the common sign and add the numbers.

ILLUSTRATIVE EXAMPLES

Example 1. Simplify: $-2 - 2$

Solution

Step 1: Since the signs are the same, we factor out the common sign and add the numbers using the format below.

➤ *Common sign (first number + second number)*

$$= -(2 + 2)$$

$$= -4$$

Example 2. Simplify: $-4 - 5$

Solution

Step 1: Since the signs are the same, we factor out the common sign and add the numbers using the format below.

➤ *Common sign (first number + second number)*

$$= -(4 + 5)$$

$$= -9$$

Example 3. Simplify: $-7 - 3$

Solution

Step 1: Since the signs are the same, we factor out the common sign and

add the numbers using the format below.

➤ *Common sign (first number + second number)*

$$= -(7 + 3)$$

$$= -10$$

Example 4. Simplify: $-5 - 6 - 7$

Solution

Step 1: Since the signs are the same, we factor out the common sign and add the numbers using the format below.

➤ *Common sign (1^{st} number + 2^{nd} number + 3^{rd} number)*

$$= -(5 + 6 + 7)$$

$$= -18$$

Example 5. Simplify: $-5 - 5 - 9 - 8$

Solution

Step 1: Since the signs are the same, we factor out the common sign and add the numbers using the format below.

➤ *Common sign (1^{st} number + 2^{nd} number + $\dots$ + last number)*

$$= -(5 + 5 + 9 + 8)$$

$$= -27$$

Example 6. Simplify: $+4 + 5$

Solution

Step 1: Since the signs are the same, we factor out the common sign and add the numbers using the format below.

➤ *Common sign (first number + second number)*

$$= +(4 + 5)$$

$$= +9$$

Note: In mathematics, $+9$ is the same as 9.

Example 7. Simplify: $+6 + 9$

Solution

Step 1: Since the signs are the same, we factor out one sign and add the numbers using the format below.

➤ *Common sign (first number + second number)*

$$= +(6 + 9)$$

$$= +15$$

Example 8. Simplify: $+5 + 5 + 9 + 8$

Solution

Step 1: Since the signs are the same, we factor out the common sign and add the numbers using the format below.

➤ *Common sign (1^{st} number + 2^{nd} number + $\dots$ + last number)*

$$= +(5 + 5 + 9 + 8)$$

$$= +27$$

Example 9. Simplify: $-\sqrt{6} - 9\sqrt{6}$

Solution

Step 1: Factor (or take out) out the square root ($\sqrt{6}$) because it is common.

$$= (-1 - 9) \sqrt{6}$$

Step 2: Simplify the bracket using the 'same sign' principle

➤ *Common sign (first number + second number)*

$$= -(1 + 9) \sqrt{6}$$

$$= -10\sqrt{6}$$

Example 10. Simplify: $-2x^{\frac{1}{2}} - 7x^{\frac{1}{2}}$

Solution

Step 1: Factor (or take out) ' $x^{\frac{1}{2}}$ ' because it is common

$$= (-2 - 7) x^{\frac{1}{2}}$$

Step 2: Simplify the bracket using the 'same sign' principle

➤ *Common sign (first number + second number)*

$$= [-(2 + 7)] x^{\frac{1}{2}}$$

$$= -9x^{\frac{1}{2}}$$

Example 11. Simplify: $-4m - 4m$

Solution

Step 1: Factor (or take out) ' m ' because it is common.

$$= (-4 - 4) m$$

Step 2: Simplify the bracket using the 'same sign' principle

$$= -(4 + 4) m$$

$$= -8m$$

Example 12. Simplify: $-9t - 9t$

Solution

Step 1:

$$= (-9 - 9)t$$

Step 2:

$$= -(9 + 9)t$$

$$= -18t$$

Example 13. Simplify: $23\sqrt{6} + 9\sqrt{6}$

Solution

Step 1: Factor out (take out) the square root ($\sqrt{6}$) because it is common.

$$= (+23 + 9) \sqrt{6}$$

Step 2: Simplify the bracket using the 'same sign' principle

➤ *Common sign (first number + second number)*

$$= 33\sqrt{6}$$

Note: In mathematics, $+33\sqrt{6}$ is the same as $33\sqrt{6}$

Example 14. Simplify : $+2x^{\frac{1}{2}} + 7x^{\frac{1}{2}}$

Solution

Step 1: Factor or take out $x^{\frac{1}{2}}$

$$= (+2 + 7) x^{\frac{1}{2}}$$

Step 2: Simplify the bracket using the 'same sign' principle

➤ *Common sign (first number + second number)*

$$= +9x^{\frac{1}{2}}$$

Example 15. Simplify: $+4p + 46p$

Solution

Step 1: Factor out 'p' because it is common

$$= (+4 + 46)p$$

Step 2:

$$= + (4 + 46)p$$

$$= 50p$$

Example 16. Simplify: $+39y + 13y$

Solution

Step 1:

$$= +(39 + 13)y$$

Step 2:

$$= 52 y$$

Subsection 2: Different signs

❖ Case 1: Different signs involving two numbers

If the signs of two numbers are different, we take the sign of the larger number and then subtract the smaller number from the larger one.

ILLUSTRATIVE EXAMPLES

Example 1. Simplify: $-2 + 4$

Solution

Step 1: Since the signs are different, we take the sign of the larger number and then subtract the smaller number from the larger one using the format below

$$\begin{aligned} &\text{➤ Sign of larger number (larger number - smaller number)} \\ &= +(4 - 2) \\ &= +2 \end{aligned}$$

Example 2. Simplify: $-3 + 2$

Solution

Step 1: Since the signs are different, we take the sign of the larger number and then subtract the smaller number from the larger one using the format below

$$\begin{aligned} &\text{➤ Sign of larger number (larger number - smaller number)} \\ &= -(3 - 2) \\ &= -1 \end{aligned}$$

Example 3. Simplify: $4 - 6$

Solution

Step 1: Since the signs are different, we take the sign of the larger

number and then subtract the smaller number from the larger one using the format below

$$\begin{aligned}&\text{➤ Sign of larger number (larger number – smaller number)} \\&= - (6 - 4) \\&= -2\end{aligned}$$

Example 4. Simplify: $-6 + 2$

Solution

Step 1: Since the signs are different, we take the sign of the larger number and then subtract the smaller number from the larger one using the format below

$$\begin{aligned}&\text{➤ Sign of larger number (larger number – smaller number)} \\&= - (6 - 2) \\&= -4\end{aligned}$$

❖ Case 2: Different signs involving more than two numbers

Example 1. Simplify: $-2 + 1 - 7 + 4$

Solution

Step 1: Use the format below since the numbers are more than two.

$$\begin{aligned}&\text{➤ (Sum of negative numbers) + (Sum of positive numbers)} \\&= (-2 - 7) + (+1 + 4) \\&= -9 + 5 \\&= -4\end{aligned}$$

Example 2. Simplify: $-2 + 4 - 3 + 1$

Solution

Step 1: Use the format below since the numbers are more than two.

$$\begin{aligned}&\text{➤ (Sum of negative numbers) + (Sum of positive numbers)} \\&= (-2 - 3) + (+4 + 1)\end{aligned}$$

$$= -5 + 5$$

$$= 0$$

Example 3. Simplify: $-3 + 2 + 4 - 5 + 1$

Solution

Step 1: Use the format below since the numbers are more than two.

➤ **(Sum of negative numbers) + (Sum of positive numbers)**

$$= (-3 - 5) + (+2 + 4 + 1)$$

$$= -8 + 7$$

$$= -1$$

SECTION 2: MULTIPLICATION AND DIVISION

The multiplication sign is ' $\times$ ' and the division sign is ' $\div$ '

$$-(\times)-= +$$

$$-(\div)-= +$$

$$-(\times)+= -$$

$$-(\div)+= -$$

$$+(\times)-= -$$

$$+(\div)-= -$$

$$+(\times)+= +$$

$$+(\div)+= +$$

ILLUSTRATIVE EXAMPLES

Example 1. Simplify: -2×-4

Solution

Step 1: Use the format below.

$$\begin{aligned} &\text{➤ } 1^{\text{st}} \text{ sign} \times 2^{\text{nd}} \text{ sign}(1^{\text{st}} \text{ number} \times 2^{\text{nd}} \text{ number}) \\ &= - \times -(2 \times 4) \\ &= +8 \end{aligned}$$

Note: $- \times - = +$

Example 2. Simplify: -6×-4

Solution

Step 1: Use the format below.

$$\begin{aligned} &\text{➤ } 1^{\text{st}} \text{ sign} \times 2^{\text{nd}} \text{ sign}(1^{\text{st}} \text{ number} \times 2^{\text{nd}} \text{ number}) \\ &= - \times -(6 \times 4) \\ &= +24 \end{aligned}$$

Note: $- \times - = +$

Example 3. Simplify: -9×-4

Solution

Step 1: Use the format below.

$$\text{➤ } 1^{\text{st}} \text{ sign} \times 2^{\text{nd}} \text{ sign}(1^{\text{st}} \text{ number} \times 2^{\text{nd}} \text{ number})$$

$$= - \times -(9 \times 4)$$

$$= +36$$

Example 4. Simplify: -0.7×-0.2

Solution

Step 1: Use the format below.

$$\begin{aligned} &\text{➤ } 1^{\text{st}} \text{ sign} \times 2^{\text{nd}} \text{ sign}(1^{\text{st}} \text{ number} \times 2^{\text{nd}} \text{ number}) \\ &= - \times -(0.7 \times 0.2) \end{aligned}$$

$$= 0.14$$

Note: $- \times - = +$

Example 4. Simplify : -0.4×-0.6

Solution

Step 1: Use the format below.

$$\text{➤ } 1^{\text{st}} \text{ sign} \times 2^{\text{nd}} \text{ sign}(1^{\text{st}} \text{ number} \times 2^{\text{nd}} \text{ number})$$

$$= - \times -(0.4 \times 0.6)$$

$$= +0.24$$

Example 5. Simplify: $2 - (-2)$

Solution

Step 1:

$$= 2 + 2$$

$$= 4$$

Note: $-(\times)- = +$

Example 6. Simplify: $2 - (-9)$

Solution

Step 1:

$$= 2 + 9$$

$$= 11$$

Note: $(-) \times (-) = +$

Example 7. Simplify: -2×4

Solution

Step 1: Use the format below.

$$\begin{aligned} &\text{➤ } 1^{\text{st}} \text{ sign} \times 2^{\text{nd}} \text{ sign}(1^{\text{st}} \text{ number} \times 2^{\text{nd}} \text{ number}) \\ &= - \times +(2 \times 4) \\ &= -8 \end{aligned}$$

Note: $-(\times)+ = -$

Example 8. Simplify: 6×-4

Solution

Step 1: Use the format below.

$$\begin{aligned} &\text{➤ } 1^{\text{st}} \text{ sign} \times 2^{\text{nd}} \text{ sign}(1^{\text{st}} \text{ number} \times 2^{\text{nd}} \text{ number}) \\ &= + \times -(6 \times 4) \\ &= -24 \end{aligned}$$

Note: $+(\times)- = -$

Example 9. Simplify: $2 - (+12)$

Solution

Step 1:

$$\begin{aligned} &== 2 - 12 \\ &= -10 \end{aligned}$$

Note: $-(\times)+ = -$

Example 10. Simplify: $\frac{-2}{-4}$

Solution

Step 1:

$$= \frac{-2}{-4}$$

$$= \frac{1}{2}$$

Note: $-(\div)- = +$

Example 11. Simplify: $\frac{-3}{-4}$

Solution

Step 1:

$$= \frac{-3}{-4}$$

$$= \frac{3}{4}$$

Example 12. Simplify: $\frac{-3}{+4}$

Solution

Step 1:

$$= \frac{-3}{4}$$

$$= -\frac{3}{4}$$

Example 2. Simplify: $\frac{+12}{-4}$

Solution

Step 1:

$$= \frac{+12}{-4}$$

$$= -3$$

SECTION 3: WORD PROBLEMS

Subsection 1: Word problems involving sum and difference

Example 1. The sum of two numbers is 4. Write an expression for this?

Solution

Step 1: Let the 1st number be x and the 2nd be y .

Step 2: The sum of two numbers is 4 means

$x + y = 4$... This is the expression

Example 2. The sum of $2p$, $3y$ and $7z$ is 12. Write an expression for this?

Solution

Step 1: The sum of $2p$, $3y$ and $7z$ is 12 means

$2p + 3y + 7z = 12$... This is the expression

Example 3. The difference of two numbers is 8. Write an expression for this?

Solution

Step 1: Let the 1st number be x and the 2nd be y .

Step 2: The difference of two numbers is 8 means

$x - y = 8$... This is the expression

Subsection 2: Word problems involving products

Example 1. The product of $2p$ and $3q$ is 10. Write an expression for this?

Solution

Step 1: The product of $2p$ and $3q$ is 10 means

$2p \times 3q = 10$

$6pq = 10$... This is the expression

Example 2. *The product of x, y and z is 20. Write an expression for this?*

Solution

Step 1: *The product of x, y and z is 20*

$$x \times y \times z = 20$$

$$xyz = 20 \dots \text{This is the expression}$$

Example 3. *The product of three numbers is 36. Write an expression for this?*

Solution

Step 1: *Let the numbers be x, y and z*

Step 2: *The product of three numbers is 36 means*

$$x \times y \times z = 36$$

$$xyz = 36 \dots \text{This is the expression}$$

Example 4. *The product of three numbers is 0.084. If the two numbers are 0.7 and 0.2, find the third number? (BECE 1997)*

Solution

Step 1: *Let the third number be x*

Step 2: *The product of three numbers is 0.084 means*

$$0.7 \times 0.2 \times x = 0.084$$

$$0.14x = 0.084$$

Step 3: *Divide both sides by the coefficient of x*

$$\frac{0.14x}{0.14} = \frac{0.084}{0.14}$$

$$x = 0.6$$

Therefore, the third number is 0.6

SECTION 4: SUBSTITUTION

In mathematics, substitution is the replacement of a variable by a number. Variables are usually represented by letters.

Example 1. Find the value of $p + q$, if $p = 7$ and $q = 3$.

Solution

Step 1: Write the expression

$$= p + q$$

Step 2: Replace p by 7 and q by 3

$$= p + q$$

$$= 7 + 3$$

$$= 10$$

Example 2. Find the value of $x + y + z$, if $x = -1$, $y = 2$ and $z = 4$

Solution

Step 1: Write the expression

$$= x + y + z$$

Step 2: Replace x by -1 , y by 2 and z by 4

$$= -1 + 2 + 4$$

$$= 5$$

Example 3. Find the value of $p - q + r$, if $p = 2$, $q = -1$ and $r = 3$

Solution

Step 1: Write the expression

$$= p - q + r$$

Step 2: Replace p by 2, q by -1 and r by 3

$$= 2 - (-1) + 3$$

$$= 2 + 1 + 3$$

$$= 6$$

Note: $-(-1) = +1$

Example 4. Solve $4x + 6y - 2xy$ for $x = 6$ and $y = 5$

Solution

Step 1: Write the expression

$$= 4x + 6y - 2xy$$

Step 2: replace x by 6 and y by 5

$$= 4(6) + 6(5) - 2(6)(5)$$

$$= 24 + 30 - 60$$

$$= 54 - 60$$

$$= -6$$

Note: Bracket means multiplication

Example 5. Simplify: $\frac{6y + 2k}{4p + q}$, If $y = 2, k = 1, p = 3$ and $q = 4$

Solution

Step 1: Write the expression

$$\frac{6y + 2k}{4p + q}$$

Step 3: replace y by 2, k by 1, p by 3 and q by 4

$$= \frac{6(2) + 2(1)}{4(3) + (4)}$$

$$= \frac{12 + 2}{12 + 4}$$

$$= \frac{14}{16}$$
$$= \frac{7}{8}$$

Example 6. If $m = 3, n = -2$ and $p = 4$. What is the value of $2m^2n^2p^2$?

Solution

Step 1: write the expression

$$2m^2n^2p^2$$

Step 2: replace m by 3, n by -2 and p by 4

$$= 2(3)^2(-2)^2(4)^2$$
$$= 2(3 \times 3)(-2 \times -2)(4 \times 4)$$
$$= 2(9)(4)(16)$$
$$= 1152$$

Example 7. If $a = 2, b = -8$, what is the value of

$$\frac{a - 3b}{a}$$

Solution

Step 1: write the expression

$$\frac{a - 3b}{a}$$

Step 2: replace ' a ' by 2 and ' b ' by -8

$$= \frac{2 - 3(-8)}{2}$$
$$= \frac{2 + 24}{2}$$
$$= \frac{26}{2} = 13$$

Example 8. $v^2 = 2as$, find v if $a = 4$ and $s = 2$.

Solution

Step 1: write the expression

$$v^2 = 2as$$

Step 3: replace 'a' by 4 and 's' by 2

$$v^2 = 2(4)(2)$$

$$v^2 = 16$$

Step 4: Take the square root of both sides to remove v^2

$$\sqrt{v^2} = \sqrt{16}$$

$$v = 4$$

Example 9. Find v in $v = \pi r^2 h$

If $\pi = \frac{22}{7}$, $r = 4$ and $h = 14$.

Solution

Step 1:

$$v = \pi r^2 h.$$

Step 2:

$$= \frac{22}{7} \times 4^2 \times 14$$

$$= \frac{22 \times 4 \times 4 \times 14}{7}$$

$$= \frac{4928}{7}$$

$$= 704$$

Example 10. If $y = 2x - 3$. Find:

(a). y when $x = 2$

(b). x when $y = 10$

Solution

(a).

Step 1:

$$y = 2x - 3$$

Step 2:

$$y = 2(2) - 3$$

$$y = 1$$

(b).

Step 1:

$$y = 2x - 3$$

Step 2:

$$10 = 2x - 3$$

Step 3: transpose $- 3$ to the L.H.S to be $+ 3$

$$10 + 3 = 2x$$

$$\frac{13}{2} = \frac{2x}{2}$$

$$x = \frac{13}{2} = 6\frac{1}{2}$$

Example 11. If $a = 2$ and $b = 5$, find the value of

$$\frac{a^b - b^a}{ab}$$

Solution

Step 1: Substitute for a and b

$$\begin{aligned} &= \frac{2^5 - 5^2}{5 \times 2} \\ &= \frac{(2 \times 2 \times 2 \times 2 \times 2) - (5 \times 5)}{10} \\ &= \frac{32 - 25}{10} \\ &= \frac{7}{10} \end{aligned}$$

Example 12. Given $x = 2$, $y = 1$ and $z = -1$. Evaluate:

(a) $(x + y)^2$

(b) $2z^2 + x$

Solution

(a). $(x + y)^2$

$$= (2 + 1)^2$$

$$= (3)^2$$

$$= 3 \times 3$$

$$= 9$$

b) $2z^2 + x$

Solution

Step 1:

$$= 2(-1)^2 + 2$$

$$= 2(-1 \times -1) + 2$$

$$= 2(1) + 2$$

$$= 2 + 2$$

$$= 4$$

Example 13. If $y = x^2 - 10x + 24$

Find y :

(a). When $x = 0$ (b). When $x = 13$ (c). When $x = -2$

Solution

(a). When $x = 0$

$$\begin{aligned} y &= (0)^2 - 10(0) + 24 \\ &= 24 \end{aligned}$$

(b). When $x = 13$

$$\begin{aligned} y &= (13)^2 - 10(13) + 24 \\ &= 169 - 130 + 24 \\ &= 63 \end{aligned}$$

(c). When $x = -2$

$$\begin{aligned} y &= (-2)^2 - 10(-2) + 24 \\ &= 48 \end{aligned}$$

Example 14. Given that $a = -3$, $b = 4$ and $c = 5$.

find (i). $(a + b)$ (ii). $(a^2 + b^2) + c$

Solution

(i).

Step 1:

$$\begin{aligned} &= (a + b) \\ &= (-3 + 4) \\ &= 1 \end{aligned}$$

(ii)

Step 1:

$$= (a^2 + b^2) + c$$

$$[(-3)^2 + 4^2] + 5$$

$$= [(-3 \times -3) + (4 \times 4)] + 5$$

$$= 9 + 16 + 5$$

$$= 30$$

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SECTION 5: LIKE AND UNLIKE TERMS

LIKE TERMS: These are terms that have the same variable and powers.

For example: $x, 2x, 3x$, etc

UNLIKE TERMS: These are terms that do not have the same variable or powers. For example: $2x, 2y, 3p, 3m, 6z$, etc

ILLUSTRATIVE EXAMPLES

Example 1. Simplify: $12p + 3 - 2p - 2$

Solution

Step 1: Collect like terms

$$= 12p - 2p + 3 - 2$$

$$= 10p + 1$$

Note: Like terms only add and subtracts

Example 2. Simplify: $3c + 4d + 6a + 8c - 2d + a$ (BECE 1999)

Solution

Step 1: Collect like terms

$$3c + 8c + 4d - 2d + 6a + a$$

$$11c + 2d + 7a$$

Example 3. Simplify: $8a + 7b + 7a - 2b + 3c - 5a = 2c$

Solution

Step 1: Transpose $' + 2c'$ to the L.H.S of the equation to be $' - 2c'$

$$8a + 7b + 7a - 2b + 3c - 5a - 2c = 0$$

Step 2: Collect like terms

$$8a + 7a - 5a + 7b - 2b + 3c - 2c = 0$$

$$10a + 5b + c = 0$$

Example 4. Simplify: $-2a + 3b + 4a = -2b$

Solution

Step 1: Transpose $-2b'$ to the L.H.S of the equation to be $+2b'$

$$-2a + 3b + 4a + 2b = 0$$

Step 2: Collect like terms

$$-2a + 4a + 3b + 2b = 0$$

$$2a + 5b = 0$$

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SECTION 6: BRACKETS AND EXPANSION

BRACKETS are used for grouping terms together.

EXPANSION: In expansion, we multiply the variable or variables outside the bracket by each of the variables inside the bracket.

ILLUSTRATIVE EXAMPLES

Example 1. Expand

$$a(b + c)$$

Solution

Step 1: Multiply the variable outside the bracket by each of the variables inside the bracket

$$= a \times b + a \times c$$

$$= ab + ac$$

Example 2. Expand

$$3(p + q)$$

Solution

Step 1: Multiply the variable outside the bracket by each of the variables inside the bracket

$$= 3 \times p + 3 \times q$$

$$= 3p + 3q$$

Example 3. Expand

$$3a(p + q)$$

Solution

Step 1: Multiply the variable outside the bracket by each of the variables inside the bracket

$$= 3a \times p + 3a \times q$$

$$= 3ap + 3aq$$

Example 4. Expand

$$3a(2p + 3q)$$

Solution

Step 1: Multiply the variable outside the bracket by each of the variables inside the bracket

$$= 3a \times 2p + 3a \times 3q$$

$$= 6ap + 9aq$$

Example 5. Expand

$$(a + b)(c + d)$$

Solution

Step 1: Multiply the variables in the 1st bracket by each of the variables in the 2nd bracket.

$$= a(c + d) + b(c + d)$$

$$= a \times c + a \times d + b \times c + b \times d$$

$$= ac + ad + bc + bd$$

Example 6. Expand

$$(2x + 3)(x + y)$$

Solution

Step 1: Multiply the variables in the 1st bracket by each of the variables in the 2nd bracket

$$= 2x(x + y) + 3(x + y)$$

$$= 2x^2 + 2xy + 3x + 3y$$

Example 7. Expand

$$(3p - 4)(p - q)$$

Solution

Step 1: Multiply the variables in the 1st bracket by each of the variables in the 2nd bracket

$$= 3p(p - q) - 4(p - q)$$

$$= 3p^2 - 3pq - 4p + 4q$$

Example 8. Expand

$$a(b + c + d)$$

Solution

Step 1: Multiply the variable outside the bracket by each of the variables inside the bracket

$$= a \times b + a \times c + a \times d$$

$$= ab + ac + ad$$

Example 9. Expand

$$3a(2b - a + b) \dots \text{BECE 1996}$$

Solution

Step 1: Multiply the variable outside the bracket by each of the variables inside the bracket

$$= 3a \times 2b + 3a \times -a + 3a \times b$$

$$= 6ab - 3a^2 + 3ab$$

Example 10. Expand

$$(a + b)(c + d + e)$$

Solution

Step 1: Multiply the variables in the 1st bracket by each of the variables in the 2nd bracket

$$= a(c + d + e) + b(c + d + e)$$

$$= a \times c + a \times d + a \times e + b \times c + b \times d + b \times e$$

$$= ac + ad + ae + bc + bd + be$$

Example 11. Expand and Simplify

$$(5a + 4)(a - 3b + 2)$$

Solution

Step 1: Multiply the variables in the 1st bracket by each of the variables in the 2nd bracket

$$= 5a(a - 3b + 2) + 4(a - 3b + 2)$$

$$= 5a^2 - 15ab + 10a + 4a - 12b + 8$$

$$= 5a^2 - 15ab + 14a - 12b + 8$$

Example 12. Expand and Simplify

$$8(2a + b) - 3(3a - 4b)$$

Solution

Step 1: Expand each bracket

$$= 16a + 8b - 9a + 12b$$

Step 2: Collect like terms

$$= 16a - 9a + 8b + 12b$$

$$= 7a + 20b$$

Example 13. Expand and Simplify

$$2a(a - 5) - a(a - 2) + 3a$$

Solution

Step 1: Expand each bracket

$$= 2a^2 - 10a - a^2 + 2a + 3a$$

Step 2: collect like terms

$$= 2a^2 - a^2 - 10a + 2a + 3a$$

$$= a^2 - 5a$$

Example 14. Simplify

$$\frac{k+4}{4} + \frac{k-6}{5}$$

Solution

Step 1: find the LCM of 4 and 5

LCM of 4 and 5 is 20

Step 2:

$$\text{For } \frac{k+4}{4}, 4 \text{ into } 20(\text{LCM}) = 5; 5 \times (k+4) = 5k+20$$

$$\text{For } \frac{k-6}{5}, 5 \text{ into } 20(\text{LCM}) = 4; 4 \times (k-6) = 4k-24$$

Step 3: add the answers in step 2 and divide by the LCM

$$= \frac{5k+20+4k-24}{20}$$

Step 4: collect like terms

$$= \frac{9k-4}{20}$$

Example 15. Simplify $\frac{p+3}{3} + \frac{2p+1}{4}$

Solution

Step 1: find the LCM of 3 and 4

LCM of 3 and 4 is 12

Step 2:

$$\text{For } \frac{p+3}{3}, 3 \text{ into } 12(\text{LCM}) = 4; 4 \times (p+3) = 4p+12$$

$$\text{For } \frac{2p+1}{4}, 4 \text{ into } 12(\text{LCM}) = 3; 3 \times (2p+1) = 6p+3$$

Step 3: add the answers in step 2 and divide by the LCM

$$= \frac{4p + 12 + 6p + 3}{12}$$

Step 4: collect like terms

$$= \frac{10p + 15}{12}$$

Step 5: factorise

$$= \frac{5(2p + 3)}{12}$$

Example 16. Simplify

$$\frac{k + 4}{4} + \frac{k - 6}{4}$$

Solution

Step 1: find the LCM of 4 and 4

LCM of 4 and 4 is 4

Step 2:

$$\text{For } \frac{k + 4}{4}, 4 \text{ into } 4(\text{LCM}) = 1; 1 \times (k + 4) = k + 4$$

$$\text{For } \frac{k - 6}{4}, 4 \text{ into } 4(\text{LCM}) = 1; 1 \times (k - 6) = k - 6$$

Step 3: add the answers in step 2 and divide by the L.C.M

$$= \frac{k + 4 + k - 6}{4}$$

Step 4: collect like terms

$$= \frac{2k - 2}{4}$$

$$= \frac{2(k - 1)}{4}$$

$$= \frac{k - 1}{2}$$

Example 17. Simplify

$$\frac{p+3}{2} + \frac{2p+2}{4}$$

Solution

Step 1: find the LCM of 2 and 4

LCM of 2 and 4 is 4

Step 2:

For $\frac{p+3}{2}$, 2 into 4(LCM) = 2; $2 \times (p+3) = 2p+6$

For $\frac{2p+2}{4}$, 4 into 4(LCM) = 1; $1 \times (2p+2) = 2p+2$

Step 3: add the answers in step 2 and divide by the LCM

$$= \frac{2p+6+2p+2}{4}$$

Step 4: collect like terms

$$= \frac{2p+2p+6+2}{4}$$

$$= \frac{4p+8}{4}$$

Step 5: factorise

$$= \frac{4(p+2)}{4}$$

$$= p+2$$

SECTION 7: FACTORISATION

Factorisation is the process by which we factor out the common term and putting the uncommon terms inside a bracket

Subsection 1: Common factors

Example 1. Factorise

$$4p - 2$$

Solution

Step 1: find the highest common factor (H.C.F) of 4p and 2

HCF of 4p and 2 is 2.

Step 2: divide throughout by the H.C.F

$$\begin{aligned} &= \frac{4p}{2} - \frac{2}{2} \\ &= 2p - 1 \end{aligned}$$

Step 3: put the answer in 'step 2' inside a bracket and place the HCF in front of the bracket

$$= 2(2p - 1)$$

Example 2. Factorise

$$3p - 15$$

Solution

Step 1: find the HCF of 3p and 5

HCF of 3p and 15 is 3

Step 2: divide throughout by the HCF

$$\begin{aligned} &= \frac{3p}{3} - \frac{15}{3} \\ &= p - 5 \end{aligned}$$

Step 3: put the answer in step 2 inside a bracket and place the HCF in front.

$$= 3(p - 5)$$

Example 3. Factorise

$$4p - 3pk$$

Solution

Step 1: find the H.C.F of 4p and 3pk

HCF of 4p – 3pk is p

Step 2: divide throughout by the H.C.F

$$= \frac{4p}{p} - \frac{3pk}{p}$$

$$= 4 - 3k$$

Step 3: put the answer in 'step 2' inside a bracket and place the HCF in front of the bracket.

$$= p(4 - 3k)$$

Example 4. Factorise

$$3p - 15pm$$

Solution

Step 1: p is common and the HCF of 3p and 15pm is 3p

Step 2: divide throughout by the HCF

$$= \frac{3p}{3p} - \frac{15pm}{3p}$$

$$= (1 - 5m)$$

Step 3: put the answer in 'step 2' inside a bracket and place the HCF of in front of the bracket

$$= 3p(1 - 5m)$$

Example 5. Factorise

$$21x^2 - 49$$

Solution

Step 1:

HCF is 7

Step 2: divide by the HCF

$$\begin{aligned} &= \frac{21x^2}{7} - \frac{49}{7} \\ &= (3x^2 - 7) \end{aligned}$$

Step 3: put the answer in 'step 2' inside a bracket and place the HCF in front of the bracket

$$= 7(3x^2 - 7)$$

Example 6. Factorise

$$36x^2 - 24x$$

Solution

Step 1: find HCF of $36x^2$ and $24x$

HCF is $12x$

Step 2: divide throughout by the HCF

$$\begin{aligned} &= \frac{36x^2}{12x} - \frac{24x}{12x} \\ &= 3x - 2 \end{aligned}$$

Step 3: put the answer in 'step 2' inside a bracket and place the HCF in front of the bracket

$$= 12x(3x - 2)$$

Example 7. Factorise

$$20x^3 - 90x^2$$

Solution

Step 1: Find HCF of $20x^3$ and $90x^2$

HCF of $20x^3$ and $90x^2$ is $10x^2$

Step 2: Divide throughout by the HCF

$$= \frac{20x^3}{10x^2} - \frac{90x^2}{10x^2}$$

$$= 2x^1 - 9$$

Step 3: put the answer in 'step 2' inside a bracket and place the H.C.F in front of the bracket

$$= 10x^2(2x^1 - 9)$$

$$= 10x^2(2x - 9)$$

Subsection 2: Factorisation by grouping

Factorisation by grouping means that we need to group the terms with common factors before factoring.

ILLUSTRATIVE EXAMPLES

Example 1. Factorise

$$4ax + 8ay + 3bx + 6by$$

Solution

Step 1: Group common terms

$$(4ax + 8ay) + (3bx + 6by)$$

Step 2: factorise using the format below

➤ *Common term (uncommon term + uncommon term)*

$$= 4a(x + 2y) + 3b(x + 2y)$$

Step 3: factorise again using the same format

➤ *Common term (uncommon term + uncommon term)*

$$= (x + 2y)(4a + 3b)$$

Example 2. Factorise

$$4ab + 4ad + 3b + 3d$$

Solution

Step 1: Group common terms

$$(4ab + 4ad) + (3b + 3d)$$

Step 2: Factorise using the format below

$$\begin{aligned} &\text{➤ Common term (uncommon term + uncommon term)} \\ &= 4a(b + d) + 3(b + d) \end{aligned}$$

Step 3: Factorise again using the format below

$$\begin{aligned} &\text{➤ Common term (uncommon term + uncommon term)} \\ &= (b + d)(4a + 3) \end{aligned}$$

Subsection 2: factorisation by splitting the middle term

Using the general expression for a quadratic

$$ax^2 + bx + c$$

Where 'bx' is the sum and 'acx²' is the product

ILLUSTRATIVE EXAMPLES

Example 1. Factorise $x^2 - 11x + 30$

Solution

*Step 1: find two numbers which when multiplied gives ' $30x^2$ '
and when added gives ' $-11x$ '*

$$\text{Product: } -6x \times -5x = +30x^2$$

$$\text{Sum : } -6x - 5x = -11x$$

Step 2: replacing the middle term ($-11x$) by ($-6x - 5x$) gives

$$x^2 - 6x - 5x + 30$$

Step 3: put the 1st two terms inside a bracket and the last two as well.

$$(x^2 - 6x) + (-5x - 30)$$

Step 4: factorise

$$x(x - 6) - 5(x - 6)$$

Step 5: factorise again

$$(x - 5)(x - 6)$$

Example 2. Factorise $d^2 + 9d + 20$

Solution

Step 1: find two numbers which when multiplied gives $+20d^2$ and when added gives $+9d$

Product: $+5d \times +4d = +20d^2$

Sum: $+4d + 5d = +9d$

Step 2: replace the middle term $(+9d)$ by $(+4d + 5d)$

$$= d^2 + 4d + 5d + 20$$

Step 3: put the 1st two terms inside one bracket and the last two as well

$$(d^2 + 5d)(+4d + 20)$$

Step 4: factorise

$$d(d + 5) + 4(d + 5)$$

Step 5: factorise again

$$(d + 4)(d + 5)$$

Example 3. Factorise $p^2 - 9p + 14$

Solution

Step 1: find two numbers which when multiplied gives $+14p^2$ and when added gives $-9p$

Product of: $-7p \times -2p = 14p^2$

Sum of: $-7p - 2p = -9p$

Step 2: replace the middle term $(-9p)$ by $(-7p - 2p)$

$$p^2 - 7p - 2p + 14$$

Step 3: put the 1st two terms inside a bracket and the last two as well

$$(p^2 - 7p) + (-2p + 14)$$

Step 4: factorise

$$p(p - 7) - 2(p - 7)$$

Step 5: factorise again

$$(p - 7)(p - 2)$$

Example 4. Factorise $k^2 - 14k + 48$

Solution

Step 1: find two numbers which when multiplied gives $+48k^2$

and when added gives $-14k$

Product of: $-6k(\times) - 8k = +48k^2$

Sum of: $-6k - 8k = -14k$

Step 2 : replace the middle term $(-14k)$ by $(-6k - 8k)$

$$k^2 - 6k - 8k + 48$$

Step 3: put the 1st two terms inside a bracket and the last two as well

$$(k^2 - 6k) + (-8k + 48)$$

Step 4: factorise

$$k(k - 6) - 8(k - 6)$$

Step 5: factorise again

$$(k - 8)(k - 6)$$

Example 5. If $(x + 3)$ is one of the factors of $x^2 - 9x - 36$.

What is the other factor?

Solution

Step 1: find two numbers which when multiplied gives $-36x^2$ and when added gives $-9x$

Product of: $-12x (\times) + 3x = -36x^2$

Sum of: $-12x + 3x = -9x$

Step 2: replacing the middle term $(-9x)$ by $(-12x + 3x)$ gives

$$x^2 - 12x + 3x - 36$$

Step 3: put the 1st two terms inside a bracket and the last two as well

$$(x^2 - 12x) + (+3x - 36)$$

Step 4: factorise

$$x(x - 12) + 3(x - 12)$$

Step 5: factorise again

$$(x + 3)(x - 12)$$

Therefore, the other factor is $(x - 12)$

SECTION 8: DIFFERENCE OF TWO SQUARES

There are always two terms with squares that are separated by minus. For example, $a^2 - b^2$

$$\text{Where } a^2 - b^2 = (a + b)(a - b)$$

Note: try to memorize it

ILLUSTRATIVE EXAMPLES

Example 1. Factorise and simplify:

$$3^2 - 4^2$$

Solution

Step 1: write the difference of two square formula

$$a^2 - b^2 = (a + b)(a - b)$$

Step 2: apply the formula

$$3^2 - 4^2 = (3 + 4)(3 - 4)$$

$$= (7)(-1)$$

$$= -7$$

Example 2. Factorise and simplify:

$$5^2 - 4^2$$

Solution

Step 1: write the difference of two square formula

$$a^2 - b^2 = (a + b)(a - b)$$

Step 2: apply the formula

$$5^2 - 4^2 = (5 + 4)(5 - 4)$$

$$= (9)(1)$$

$$= 9$$

Example 3. Factorise and simplify:

$$6^2 - 2^2$$

Solution

Step 1: write the difference of two square formula

$$a^2 - b^2 = (a + b)(a - b)$$

Step 2: apply the formula

$$6^2 - 2^2 = (6 + 2)(6 - 2)$$

$$= (8)(4)$$

$$= 32$$

Example 4. Factorise and simplify:

$$6.25^2 - 2.34^2$$

Solution

Step 1: write the difference of two square formula

$$a^2 - b^2 = (a + b)(a - b)$$

Step 2: apply the formula

$$6.25 - 2.34^2 = (6.25 + 2.34)(6.25 - 2.34)$$

$$= (8.59)(3.91)$$

$$= 33.5869$$

Example 5. Factorise and simplify:

$$9x^2 - 4y^2$$

Solution

Step 1: write 9 and 4 in square form

$$9 = 3^2 \text{ and } 4 = 2^2$$

Step 2: replace 9 by 3^2 and 4 by 2^2

$$3^2x^2 - 2^2y^2$$

Step 3: rewrite $3^2x^2 - 2^2y^2$

$$3^2x^2 - 2^2y^2 = (3x)^2 - (2y)^2$$

Step 4 : apply the difference of two square formula

$$(3x)^2 - (2y)^2 = (3x + 2y)(3x - 2y)$$

Example 6. Factorise

$$49q^2 - 36p^2$$

Solution

Step 1: write 49 and 36 in square form

$$49 = 7^2 \text{ and } 36 = 6^2$$

Step 2: replace 49 by 7^2 and 36 by 6^2

$$7^2x^2 - 6^2y^2$$

Step 3: rewrite $7^2x^2 - 6^2y^2$

$$7^2x^2 - 6^2y^2 = (7x)^2 - (6y)^2$$

Step 4 : apply the difference of two square formula

$$(7x)^2 - (6y)^2 = (7x + 6y)(7x - 6y)$$

Example 7. Factorise $36q^2 - 25p^2$

Solution

Step 1: write 36 and 25 in square form

$$36 = 6^2 \text{ and } 25 = 5^2$$

Step 2: replace 36 by 6^2 and 25 by 5^2

$$6^2q^2 - 5^2p^2$$

Step 3: rewrite $6^2q^2 - 5^2p^2$

$$6^2q^2 - 5^2p^2 = (6q)^2 - (5p)^2$$

Step 4 : apply the difference of two square formula

$$(6q)^2 - (5p)^2 = (6q + 5p)(6q - 5p)$$

Example 8. Factorise $4m^2 - n^2$

Solution

Step 1: write 4 in square form

$$4 = 2^2$$

Step 2: replace 4 by 2^2

$$2^2m^2 - n^2$$

Step 3: rewrite $2^2m^2 - n^2$

$$2^2m^2 - n^2 = (2m)^2 - n^2$$

Step 4 : apply the difference of two square formula

$$(2m)^2 - n^2 = (2m + n)(2m - n)$$

Example 9. Factorise $10^2 - 1$

Solution

Step 1: write 1 in square form

$$1 = 1^2$$

Step 2: replace 1 by 1^2

$$10^2 - 1^2$$

Step 3 : apply the difference of two square formula

$$10^2 - 1^2 = (10 + 1)(10 - 1)$$

$$= (11)(9)$$

$$= 99$$

Example 10. Factorise $32m^2 - 50n^2$

Solution

Step 1: factorise

$$2[16m^2 - 25n^2]$$

Step 2: write 16 and 25 in square form

$$16 = 4^2 \text{ and } 25 = 5^2$$

Step 3: replace 16 by 4^2 and 25 by 5^2

$$2[4^2m^2 - 5^2n^2]$$

Step 4: rewrite $2[4^2m^2 - 5^2n^2]$

$$2[4^2m^2 - 5^2n^2] = 2[(4m)^2 - (5n)^2]$$

Step 5 : apply the difference of two square formula

$$2[(4m)^2 - (5n)^2] = 2[(4m + 5n)(4m - 5n)]$$

Example 11. Factorise $12a^2 - 27b^2$

Solution

Step 1: factorise

$$3[4a^2 - 9b^2]$$

Step 2: write 16 and 25 in square form

$$4 = 2^2 \text{ and } 9 = 3^2$$

Step 3: replace 4 by 2^2 and 9 by 3^2

$$3[2^2a^2 - 3^2b^2]$$

Step 4: rewrite $2[2^2a^2 - 3^2b^2]$

$$3[2^2a^2 - 3^2b^2] = 3[(2a)^2 - (3b)^2]$$

Step 5 : apply the difference of two square formula

$$3[(2a)^2 - (3b)^2] = 3[(2a + 3b)(2a - 3b)]$$

SECTION 9: MISCELLANEOUS EXAMPLES

Example 1. Simplify

$$(\sqrt{100} - \sqrt{81} + \sqrt{36})x^2y^2 - (-\sqrt{81} + \sqrt{100} + \sqrt{36})x^2y^2$$

Solution

Step 1: simplify the brackets

$$\begin{aligned} &= (10 - 9 + 6)x^2y^2 - (-9 + 10 + 6)x^2y^2 \\ &= 7x^2y^2 - 7x^2y^2 \\ &= 7x^2y^2 - 7x^2y^2 \\ &= 0 \end{aligned}$$

Example 2. Simplify

$$\frac{\sqrt{64}}{2}x^2 + \sqrt{49}y + \frac{\sqrt{16}}{2}x^2 - \sqrt{49}y$$

Solution

Step 1: reduce each square root to its lowest form

$$\begin{aligned} &= \frac{8}{2}x^2 + 7y + \frac{4}{2}x^2 - 7y \\ &= 4x^2 + 7y + 2x^2 - 7y \end{aligned}$$

Step 2: collect like terms

$$\begin{aligned} &= 4x^2 + 2x^2 + 7y - 7y \\ &= 6x^2 - 0 \\ &= 6x^2 \end{aligned}$$

Example 3. Simplify $2\sqrt{k} + 3\sqrt{p} + \sqrt{k} + 7\sqrt{p}$

Solution

Step 1: collect like terms

$$= 2\sqrt{k} + \sqrt{k} + 3\sqrt{p} + 7\sqrt{p}$$

$$= 3\sqrt{k} + 10\sqrt{p}$$

Example 4. Simplify:

$$k \left(\sqrt{\frac{43}{7}} \right)^0 + 2k$$

Solution

Step 1: any quantity raise to power zero is 1.

$$= k \times 1 + 2k$$

$$= k + 2k$$

$$= 3k$$

Example 5. If $c = 4, v = 2$ and $m = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$. Find the value of m .

Solution

Step 1: substitute the values of c and v into the formula

$$m = \frac{1}{\sqrt{1 - \frac{2^2}{4^2}}}$$

$$= \frac{1}{\sqrt{1 - \frac{4}{16}}}$$

Step 2: simplify the fraction inside the square root

$$= \frac{1}{\sqrt{\frac{16 - 4}{16}}}$$

$$= \frac{1}{\sqrt{\frac{12}{16}}}$$

Step 3: simplify inside the root

$$= \frac{1}{\sqrt{\frac{12}{16}}}$$

$$= \frac{1}{\sqrt{\frac{3}{4}}}$$

$$= \frac{1}{1} \div \frac{\sqrt{3}}{\sqrt{4}}$$

$$= \frac{1}{1} \times \frac{\sqrt{4}}{\sqrt{3}}$$

$$= \frac{1}{1} \times \frac{2}{\sqrt{3}}$$

$$= \frac{2}{\sqrt{3}}$$

Step 4: rationalise the denominator (multiply both the numerator and denominator by $\sqrt{3}$)

$$= \frac{2}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$$

$$= \frac{2 \times \sqrt{3}}{\sqrt{3} \times \sqrt{3}}$$

$$= \frac{2\sqrt{3}}{\sqrt{9}}$$

$$= \frac{2\sqrt{3}}{3}$$

SECTION 10: SKS PAPER 9

Part 1

Answer all questions

Question 1. Simplify: $-6000 - 6000$

Answer = -12000

Question 2. Simplify: $-8 - 9$

Answer = -17

Question 3. Simplify: $-792 - 900$

Answer = -1692

Question 4. Simplify: $+6 + 6$

Answer = $+12$

Question 5. Simplify: $+812345 + 934567$

Answer = $+1746912$

Question 6. Simplify: $+8 + 9$

Answer = $+17$

Question 7. Simplify: $-6113456 - 6346768 - 6456 - 645667$

Answer = -13112347

Question 8. Simplify: $-8 - 9 - 9 - 8$

Answer = -34

Question 9. Simplify: $+7 + 9 + 10 + 11$

Answer = $+37$

Question 10. Simplify: $-6999999999 + 6999999999$

Answer = 0

Question 11. Simplify: $-6 + 7$

Answer = 1

Question 12. Simplify: $-3 + 4 - 6 + 5 + 1 - 2$

Answer = -1

Question 13. Simplify: $-2000 + 4000 + 6000 - 8000$

Answer = 0

Part 2

Question 1. Simplify: -6000×-9000

Answer = $+54000000$

Question 2. Simplify: -9143×-4222

Answer = $+38601746$

Question 3. Simplify: $2 - (-2145678)$

Answer = $+2145680$

Question 4. Simplify: $1232 - (-256782)$

Answer = $+258014$

Question 5. Simplify: -6×9

Answer = -54

Question 6. Simplify: 9675×-4231

Answer = -40934925

Question 7. Simplify: $2000000 + (-2000000)$

Answer = 0

Question 8. Simplify: $2 - (22356772)$

Answer = -22356770

Part 3

Question 1. Simplify: $\frac{-500000000}{-25}$

Answer = $+20000000$

Question 2. Simplify: $\frac{-144}{-12}$

Answer = +12

Question 3. Simplify: $\frac{-1250000000}{-25000}$

Answer = +50000

Question 4. Simplify: $\frac{5}{-25}$

Answer = $-\frac{1}{5}$

Question 5. Simplify: $\frac{-1502500}{25}$

Answer = -60100

Part 4

Question 1. The sum of x and y is 10. Write an expression for this?

Answer [$x + y = 10$]

Question 2. The sum of x , y and z is 3. Write an expression for this?

Answer [$x + y + z = 3$]

Question 3. The difference of two numbers is 10. Write an expression for this?

Answer [$x - y = 10$]

Question 4. The product of $7p$ and q is 80. Write an expression for this?

Answer [$7pq = 80$]

Question 5. The product of a , b and z is 80. Write an expression?

Answer [$abz = 80$]

Question 6. The product of three numbers is 0.084. If two the numbers

are 0.6 and 0.2, find the third number?

Answer [Third number = 0.7]

Part 5

Question 1. Find the value of

$-q + r$, if $q = -3$ and $r = 9$

Answer = 12

Question 2. Find the value of

$4x + 6y - 2xy$, if $x = 17$ and $y = 15$

Answer = -352

Question 3. Simplify: $\frac{6y + 2k}{4p + q}$

If $y = 12$, $k = 11$, $p = 13$ and $q = 14$

Answer = $1\frac{14}{33}$

Question 4. If $m = 4$, $n = 3$ and $o = 6$.

What is the value of $2m^2n^2o^2$?

Answer = 10368

Question 5. If $a = 12$, $b = -8$, what is the value of

$$\frac{a - 3b}{a}$$

Answer = 3

Question 6. $v^2 = 2as$, find v if $a = 21$ and $s = 11$.

Leaving your answer in root form

Answer = $\sqrt{462}$

Question 7. Find the value of v in

$$v = \pi r^2 h.$$

$$\text{If } \pi = \frac{22}{7}, r = 14 \text{ and } h = 42$$

$$\text{Answer} = 25872$$

Question 8. If $y = 2x - 3$.

Find y when $x = 12$ and x when $y = 11$

$$\text{Answer } [y = 21 \text{ and } x = 7]$$

Question 9. If $a = 4$ and $b = 3$, find $\frac{a^b - b^a}{ab}$

$$\text{Answer} = -1\frac{5}{12}$$

Question 10. Given that $x = 12, y = 12$ and $z = -1$. Evaluate:

(a). $(x + y)^2$

(b). $2z^2 + x$

(a). $(x + y)^2$

$$\text{Answer} = 576$$

b) $2z^2 + x$

$$\text{Answer} = 14$$

Question 11. If $y = x^2 - 10x + 45$

Find y :

(a). When $x = 3$

(b). When $x = -8$

Answer:

(a). 24

(b). 189

Question 12. Given that $a = -6, b = 8, c = -7$.

Find: (i) $(a + b)$ (ii) $(a^2 + b^2) + c$

(i) $a + b$

Answer = +2

(ii) $(a^2 + b^2) + c$

Answer = 93

Part 6

Question 1. Simplify $8m + 7b + 7m - 2c + 3c - 5a = 2c$

Answer = $-5a + 7b - c + 15m$

Question 2. Simplify $-2h + 3b + 4a = -2b$

Answer = $4a + 5b - 2h$

Question 3. Simplify $3c + 4j + 6n + 8j - 2d + n$

Answer = $3c - 2d + 12j + 7n$

Part 7

Question 1. Expand $(3m - 4)(m - q)$

Answer = $3m^2 - 3mq - 4m + 4q$

Question 2. Expand $ab(b + c + d)$

Answer = $ab^2 + abc + abd$

Question 3. Expand $3a(2b - a + ab)$

Answer = $6ab - 3a^2 + 3a^2b$

Question 4. Expand $(a + bc)(c + d + e)$

Answer = $ac + ad + ae + bc^2 + bcd + bce$

Question 5. Expand $(5a + 6)(a - 5b + 2)$

Answer = $5a^2 - 25ab + 16a - 30b + 12$

Question 6. Simplify $8(2a + abc) - 3(3a - 4b)$

$$\text{Answer} = 7a + 8abc + 12b$$

Question 7. Expand and Simplify

$$2abc(a - 5) - a(a - 2) + 3a$$

$$\text{Answer} = 2a^2bc - 10abc - a^2 + 5a$$

Question 8. Simplify $\frac{m+4}{4} + \frac{2m-6}{5}$

$$\text{Answer} = \frac{13m-4}{20}$$

Question 9. Simplify $\frac{3h+3}{3} + \frac{4h+1}{4}$

$$\text{Answer} = \frac{8h+5}{4}$$

Part 8

Question 1. Factorise

$$2x^2 - x$$

$$\text{Answer} = x(2x - 1)$$

Question 2. Factorise $3x^2 - x$

$$\text{Answer} = x(3x - 1)$$

Question 3. Factorise $3x^2 + x$

$$\text{Answer} = x(3x + 1)$$

Question 4. Factorise $4x^3 - 16x^2$

$$\text{Answer} = 4x^2(x - 4)$$

Question 5. Factorise $k^2 + 14k + 48$

$$\text{Answer} = (k + 6)(k + 8)$$

Question 6. Factorise $p^2 + p - 2$

$$\text{Answer} = (p - 1)(p + 2)$$

Question 7. Factorise $m^2 - 2m - 8$

$$\text{Answer} = (m + 2)(m - 4)$$

Question 8. Factorise $81x^2 - 49y^2$

$$\text{Answer} = (9x + 7y)(9x - 7y)$$

Question 9. Factorise $144u^2 - 121y^2$

$$\text{Answer} = (12u - 11y)(12u + 11y)$$

Question 10. Factorise $36k^2 - 16y^2$

$$\text{Answer} = (6k - 4y)(6k + 4y)$$

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CHAPTER 10

INDICES

SECTION 1: IDENTIFICATION OF BASES AND POWERS

By inspection, the base is the number at the bottom while the power is the number at the top. For example:

1. a^n ... a is the base and n is the power

2. p^m ... p is the base and m is the power

3. p^{5+x} ... p is the base and $(5 + x)$ is the power

4. q^{x-4} ... q is the base and $(x - 4)$ is the power

SECTION 2: LAWS OF INDICES

Index notation is a method of shortening the product of numbers of equal factors or numbers that are the same. For example:

$$(i) 3 \times 3 \times 3 \times 3 \times 3 = 3^5 \quad (ii). a \times a \times a = a^3$$

Subsection 1: Multiplication law

When numbers with equal bases are multiplied their powers (indices) are added.

$$p^a \times p^b = p^{a+b}$$

ILLUSTRATIVE EXAMPLES

Example 1. Simplify: $a^m \times a^n$

Solution

Step 1: The common base is 'a'

Step 2: The powers are 'm and n'

Step 3: Take the common base and add the powers

$$= a^{m+n}$$

Example 2. Simplify: $b^p \times b^q$

Solution

Step 1: The common base is b

Step 2: The powers are p and q

Step 3: Take the common base and add the powers

$$= b^{p+q}$$

Example 3. Simplify: $k^2 \times k^3$

Solution

Step 1: The common base is k

Step 2: The powers are 2 and 3

Step 3: Take the common base and add the powers

$$= k^{2+3}$$

$$= k^5$$

Example 4. Simplify: $4^2 \times 4^3$

Solution

Step 1: The common base is 4

Step 2: The powers are 2 and 3

Step 3: Take the common base and add the powers

$$= 4^{2+3}$$

$$= 4^5$$

Step 4: Simplify

$$= 4 \times 4 \times 4 \times 4 \times 4$$

$$= 1024$$

Example 5. Simplify: $y^4 \times y^2 \times y^3$

Solution

Step 1: The common base is y

Step 2: The powers are 4, 2 and 3

Step 3: Take the common base and add the powers

$$= y^{4+2+3}$$

$$= y^9$$

Example 6. Simplify $7^4 \times 7^{-2}$

Solution

Step 1: The common base is 7

Step 2: The powers are 4 and - 2

Step 3: Take the common base and add the powers

$$= 7^{4+(-2)}$$

$$= 7^{4-2}$$

$$= 7^2$$

Step 4: Simplify

$$= 7 \times 7$$

$$= 49$$

Note: $(+) \times (-) = -$.

Example 7. Simplify: $k^{x+4} \times k^2$

Solution

Step 1: k is the common base

Step 2: $(x + 4)$ and 2 are the powers

Step 3: Take the common base and add the powers

$$= k^{x+4+2}$$

$$= k^{x+6}$$

Example 8. Simplify: $2^m k^p t^2 \times 2^n k^q t^a$

Solution

Step 1: Take the common bases and add their powers

$$= 2^m \times 2^n \times k^p \times k^q \times t^2 \times t^a$$

$$= 2^{m+n} \times k^{p+q} \times t^{2+a}$$

$$= 2^{m+n} k^{p+q} t^{2+a}$$

Example 9. Simplify: $2^a y^k p^x z^{5-v} \times 2^b y^{k-1} p^{6x} z^{4v}$

Solution

Step 1: Take the common bases and add the powers

$$= 2^{a+b} y^{k+k-1} p^{x+6x} z^{5-v+4v}$$

$$= 2^{a+b} y^{2k-1} p^{7x} z^{5+3v}$$

Example 10. Simplify: $y^k y^p y^2 \times q^2 q^4 q^w$

Solution

Step 1: Take the common bases and add their powers

$$= y^{k+p+2} \times q^{2+4+w}$$

$$= y^{k+p+2} \times q^{6+w}$$

$$= y^{k+p+2} q^{6+w}$$

Example 11. Simplify: $a^2 \times a^x \times b^2$

Solution

Step 1: Take the common base and add the powers; Leaving b^2 as it is.

$$= a^{2+x} \times b^2$$

Example 12. Simplify: $a^{k+2} \times a^{k+4} \times b^{2x+4}$

Solution

Step 1: Take the common base and add the powers; Leaving b^{2x+4} as it is.

$$= a^{k+2+k+4} \times b^{2x+4}$$

$$= a^{2k+6} \times b^{2x+4}$$

Example 13. Simplify: $k^{2x+1} \times w^{3x+4} \times w^{2x+3} \times q^{2x-1} \times q^{2x-2}$

Solution

Step 1: Take the common bases and add the power; Leaving k^{2x+1} as it is

$$= k^{2x+1} \times w^{3x+4+2x+3} \times q^{2x-1+2x-2}$$

$$= k^{2x+1} \times w^{5x+7} \times q^{4x-3}$$

Subsection 2: Division law

When dividing numbers with equal bases; the power of the divisor is subtracted from that of the dividend.

$$\frac{p^a}{p^b} = p^{a-b}$$

ILLUSTRATIVE EXAMPLES

Example 1. Simplify: $a^n \div a^m$

Solution

Step 1: Take the common base and then subtract the power of the denominator from the power of the numerator

$$= a^{n-m}$$

Example 2. Simplify: $\frac{a^p}{a^k}$

Solution

Step 1: Take the common base and then subtract the power of the denominator from the power of the numerator

$$= a^{p-k}$$

Example 3. Simplify: $\frac{2^m}{2^n}$

Solution

Step 1: Take the common base and then subtract the power of the denominator from the power of the numerator

$$= 2^{m-n}$$

Example 4. Simplify : $\frac{x^5}{x^3}$

Solution

Step 1: Take the common base and then subtract the power of the denominator from the power of the numerator

$$= x^{5-3}$$

$$= x^2$$

Example 5. Simplify: $\frac{7^4}{7^{-2}}$

Solution

Step 1: Take the common base and then subtract the power of the denominator from the power of the numerator

$$= 7^{4-(-2)}$$

$$= 7^{4+2}$$

$$= 7^6$$

$$= 7 \times 7 \times 7 \times 7 \times 7 \times 7$$

$$= 117649$$

Note: $-(-2) = +2$

Example 6. Simplify: $\frac{8^{-4}}{8^{-2}}$

Solution

Step 1: Take the common base and then subtract the power of the denominator from the power of the numerator

$$= 8^{-4-(-2)}$$

$$= 8^{-4+2}$$

$$= 8^{-2}$$

$$= \frac{1}{8^2}$$

$$= \frac{1}{8 \times 8}$$

$$= \frac{1}{64}$$

Note: $8^{-2} = \frac{1}{8^2}$

Example 4. Simplify: $k^{x+4} \div k^2$

Solution

Step 1: Take the common base and then subtract the power of the denominator from the power of the numerator

$$= k^{(x+4)-2}$$

$$= k^{x+2}$$

Example 5. Simplify: $2^m k^p t^2 \div 2^b k^k t^a$

Solution

Step 1: Take the common base and then subtract the power of the denominator from the power of the numerator

$$= 2^{m-b} k^{p-k} t^{2-a}$$

Example 6. Simplify: $2^a y^k p^x z^{5-v} \div 2^b y^{k-1} p^{-6x} z^{4v}$

Solution

Step 1: Take the common base and then subtract the power of the denominator from the power of the numerator

$$= 2^{a-b} y^{k-(k-1)} p^{x-(-6x)} z^{5-v-4v}$$

$$= 2^{a-b} y^{k-k+1} p^{x+6x} z^{5-5v}$$

$$= 2^{a-b} y^1 p^{7x} z^{5-5v}$$

$$= 2^{a-b} y p^{7x} z^{5-5v}$$

Example 7. Simplify: $y^k q^p a^2 \div y^2 q^4 a^w$

Solution

Step 1: Take the common base and then subtract the power of the denominator from the power of the numerator

$$= y^{k-2} q^{p-4} a^{2-w}$$

Example 8. Simplify: $a^2 b \div a^x$

Solution

Step 1: Take the common base and then subtract the power of the denominator from the power of the numerator; Leaving b as it is

$$= a^{2-x} \times b$$

$$= a^{2-x} b$$

Example 9. Simplify: $a^{k+2} \div a^{k+4} b^{2x+4}$

Solution

Step 1: Take the common base and then subtract the power of the denominator from the power of the numerator ; Leaving b^{2x+4}

as it is.

$$= \frac{a^{k+2}}{a^{k+4}b^{2x+4}}$$

$$= \frac{a^{k+2-(k+4)}}{b^{2x+4}}$$

$$= \frac{a^{k+2-k-4}}{b^{2x+4}}$$

$$= \frac{a^{-2}}{b^{2x+4}}$$

$$= \frac{1}{a^2b^{2x+4}}$$

Note: $a^{-2} = \frac{1}{a^2}$

Example 10. Simplify: $k^{2x+1}w^{3x+4}q^{x-1} \div w^{2x+3}q^{2x-2}$

Solution

Step 1: Take the common bases and then subtract the powers

$$= k^{2x+1} \times w^{3x+4-(2x+3)} \times q^{x-1-(2x-2)}$$

$$= k^{2x+1} \times w^{3x+4-2x-3} \times q^{x-1-2x+2}$$

$$= k^{2x+1} \times w^{x+1} \times q^{1-x}$$

$$= k^{2x+1}w^{x+1}q^{1-x}$$

Example 11. Simplify: $\frac{25a^4m^6}{5a^2m^4}$

Solution

Step 1: Separate each term and simplify

$$= \frac{25}{5} \times \frac{a^4}{a^2} \times \frac{m^6}{m^4}$$

$$= 5 \times a^{4-2} \times m^{6-4}$$

$$= 5 \times a^2 \times m^2$$

$$= 5a^2m^2$$

Note: b^{32} has no similar base, so we leave it as it is.

Subsection 3: Power law

Its states that if a base and its power raise to another power, we multiply the powers.

ILLUSTRATIVE EXAMPLES

Example 1. Simplify: $(x^y)^2$

Solution

Step 1: Multiply the powers

$$= x^{y \times 2}$$

$$= x^{2y}$$

Example 2. Simplify $(2^3)^2$

Solution

Step 1: Multiply the powers

$$= 2^{3 \times 2}$$

$$= 2^6$$

$$= 2 \times 2 \times 2 \times 2 \times 2 \times 2$$

$$= 64$$

Subsection 4: Product Power law

When the product of a different factors are raised to a certain power, perhaps in a bracket, the power is distributed over each factor in the bracket.

ILLUSTRATIVE EXAMPLES

Example 1. Simplify: $(q^2p^3)^4$

Solution

Step 1: Multiply the powers outside the bracket by each of the powers inside the bracket

$$= q^{2 \times 4} p^{3 \times 4}$$

$$= q^8 \times p^{12}$$

$$= q^8 p^{12}$$

Example 2. Simplify: $(8x^3)^2 \times (4x)^2$

Solution

Step 1: Multiply the powers outside the bracket by each of the powers inside the bracket

$$= (8^{1 \times 2} x^{3 \times 2}) \times (4^{1 \times 2} x^{1 \times 2})$$

$$= 8^2 x^6 \times 16x^2$$

$$= 8 \times 8 \times x^6 \times 16x^2$$

$$= 64x^6 \times 16x^2$$

Step 2: Multiply the numbers and the variables as well

$$= 64 \times 16 \times x^6 \times x^2$$

$$= 1024 \times x^{6+2}$$

$$= 1024x^8$$

Example 3. Simplify: $(4a^2b^3c^2)^2$

Solution

Step 1: Multiply the power outside the bracket by each of the powers inside the bracket

$$= 4^{1 \times 2} \times a^{2 \times 2} \times b^{3 \times 2} \times c^{2 \times 2}$$

$$= 4^2 a^4 b^6 c^4$$

$$= (4 \times 4) a^4 b^6 c^4$$

$$= 16a^4 b^6 c^4$$

Example 4. Simplify: $8(x^3)^2 \times (4x)^{-2}$

Solution

Step 1: Multiply the powers outside the bracket by each of the powers of

inside the brackets.

$$= 8(x^{3 \times 2}) \times (4^{1 \times -2} \times x^{1 \times -2})$$

$$= 8x^6 \times 4^{-2} \times x^{-2}$$

$$= 8x^6 \times \frac{1}{4^2} \times \frac{1}{x^2}$$

$$= \frac{8x^6}{16x^2}$$

$$= \frac{x^{6-2}}{2}$$

$$= \frac{x^4}{2}$$

Example 5. Simplify: $8(x^3)^2 - 2x^3$

Solution

Step 1: Multiply the powers outside the bracket by each of the powers of inside the brackets.

$$= 8(x^{3 \times 2}) - 2x^3$$

$$= 8x^6 - 2x^3$$

Step 2: Factorise

$$= 2x^3(4x^3 - 1)$$

Subsection 5: Zero Index

When any number or quantity is raised to power zero, the result is 1.

For example:

1. $49^0 = 1$

2. $x^0 = 1$

3. $\left(\frac{29}{8}\right)^0 = 1$

Subsection 6: Negative index

A number raised to a negative index equals the reciprocal raised to the same positive index. For example:

$$1. a^{-1} = \frac{1}{a}$$

$$2. p^{-2} = \frac{1}{p^2}$$

ILLUSTRATIVE EXAMPLES

Example 1. Simplify: 2^{-1}

Solution

Step 1: Write 1 and divide by the base and its power, ignoring the minus sign of the power

$$= \frac{1}{2^1}$$

$$= \frac{1}{2}$$

Example 2. Simplify: 4^{-2}

Solution

Step 1: Write 1 and divide by the base and its power, ignoring the minus sign of the power

$$= \frac{1}{4^2}$$

$$= \frac{1}{16}$$

Example 3. Simplify: 7^{-2}

Solution

Step 1: Write 1 and divide by the base and its power, ignoring the minus

sign of the power

$$= \frac{1}{7^2}$$

$$= \frac{1}{49}$$

Subsection 7: Fractional Index

- $a^{\frac{b}{c}} = (\sqrt[c]{a})^b$

ILLUSTRATIVE EXAMPLES

Example 1. Simplify: $a^{\frac{1}{2}}$

Solution

Step 1: State the law

$$a^{\frac{b}{c}} = (\sqrt[c]{a})^b$$

Step 2: By comparing and substituting we have

$$a^{\frac{1}{2}} = (\sqrt{a})^1 = \sqrt{a}$$

Note: Any quantity raised to power one(1) is the number itself.

Example 2. Simplify: $25^{\frac{3}{2}}$

Solution

Step 1: State the law

$$a^{\frac{b}{c}} = (\sqrt[c]{a})^b$$

Step 2: By comparing and substituting we have

$$25^{\frac{3}{2}} = (\sqrt{25})^3$$

$$= (5)^3$$

$$= 5 \times 5 \times 5$$

$$= 125$$

Note: $\sqrt{25} = 5$

Example 3. Simplify: $25^{\frac{5}{2}}$

Solution

Step 1: State the law

$$a^{\frac{b}{c}} = (\sqrt[c]{a})^b$$

Step 2: By comparing and substituting we have

$$25^{\frac{5}{2}} = (\sqrt{25})^5$$

$$= (5)^5$$

$$= 5 \times 5 \times 5 \times 5 \times 5$$

$$= 3125$$

Example 4. Simplify: $8^{\frac{2}{3}}$

Solution

Step 1: State the law

$$a^{\frac{b}{c}} = (\sqrt[c]{a})^b$$

Step 2: By comparing and substituting we have

$$8^{\frac{2}{3}} = (\sqrt[3]{8})^2$$

$$= (2)^2$$

$$= 2 \times 2$$

$$= 4$$

Note: $\sqrt[3]{8} = 2$

Example 5. Simplify: $64^{\frac{2}{3}}$

Solution

Step 1: State the law

$$a^{\frac{b}{c}} = (\sqrt[c]{a})^b$$

Step 2: By comparing and substituting we have

$$64^{\frac{2}{3}} = (\sqrt[3]{64})^2$$

$$= (4)^2$$

$$= 4 \times 4$$

$$= 16$$

Note: $\sqrt[3]{64} = 4$

Example 6. Simplify: $64^{-\frac{2}{3}}$

Solution

Step 1: Apply the negative index

$$64^{-\frac{2}{3}} = \frac{1}{64^{\frac{2}{3}}}$$

Step 2: Apply the fractional index

$$64^{-\frac{2}{3}} = \frac{1}{(\sqrt[3]{64})^2}$$

$$= \frac{1}{(4)^2}$$

$$= \frac{1}{4 \times 4}$$

$$= \frac{1}{16}$$

Example 7. Simplify: $\left(\frac{8}{64}\right)^{\frac{2}{3}}$

Solution

Step 1: Multiply the power outside the bracket by each of the numbers inside the bracket

$$= \left(\frac{8^{1 \times \frac{2}{3}}}{64^{1 \times \frac{2}{3}}} \right)$$

$$= \frac{8^{\frac{2}{3}}}{64^{\frac{2}{3}}}$$

Step 2: Apply the fractional index

$$= \frac{(\sqrt[3]{8})^2}{(\sqrt[3]{64})^2}$$

$$= \frac{(2)^2}{(4)^2}$$

$$= \frac{2 \times 2}{4 \times 4}$$

$$= \frac{4}{16}$$

$$= \frac{1}{4}$$

Example 8. Simplify: $\left[\frac{8}{64}\right]^{-\frac{2}{3}}$

Solution

Step 1: Apply the negative index

$$\left[\frac{8}{64}\right]^{-\frac{2}{3}} = \frac{1}{\left[\frac{8}{64}\right]^{\frac{2}{3}}}$$

Step 2: Apply the fractional index

$$= \frac{1}{\frac{(\sqrt[3]{8})^2}{(\sqrt[3]{64})^2}}$$

$$= \frac{1}{\frac{2^2}{4^2}}$$

$$= \frac{1}{\frac{4}{16}}$$

$$= 1 \div \frac{4}{16}$$

$$= 1 \div \frac{1}{4}$$

Changing to multiplication sign

$$= 1 \times \frac{4}{1}$$

$$= 4$$

SECTION 3: SKS PAPER 10

Answer all questions

Part 1

Question 1. Simplify $a^{-m} \times a^n$

Answer: a^{n-m}

Question 2. Simplify $b^{-6} \times b^{-q}$

Answer = b^{-6-q}

Question 3. Simplify $m^{x+3} \times m^{7x-2}$

Answer = m^{8x+1}

Question 4. Simplify $3^m k^{2p} t^2 \times 3^{3m} k^{q+2} t^{a-1}$

Answer = $3^{4m} k^{2p+q+2} t^{1+a}$

Question 5. Simplify $5^a y^{2k} p^x z^{b-4v} \times 5^{2a} y^{k-1} p^{6x-1}$

Answer: $5^{3a} y^{3k-1} p^{7x-1} z^{b-4v}$

Part 2

Question 1. $2^a y^k p^x z^{5-v} \div 2^b y^{k-1} p^{-6x} z^{4v}$

Answer: $2^{a-b} y p^{7x} z^{5-5v}$

Question 2. Simplify $y^k q^p a^2 \div y^2 q^4 a^w$

Answer: $y^{k-2} q^{p-4} a^{2-w}$

Question 3. Simplify $a^{4x+2} b \div a^x$

Answer: $a^{3x+2} b$

Question 4. Simplify $a^{k-4} \div a^{k-6} b^{7x+4}$

Answer: $\frac{a^2}{b^{7x+4}}$

Question 5. Simplify $k^{3x-1}w^{5x-4}q^{x-1} \div w^{-4x+7}q^{-3x-1}$

Answer: $k^{3x-1}w^{9x-11}q^{4x}$

Part 3

Question 1. Simplify $7(x^6)^3 - 3x^4$

Answer: $x^4(7x^{14} - 3)$

Question 2. Simplify $(4x^3)^3 \times (6x)^2$

Answer: $2304x^{11}$

Question 3. Simplify $(3a^2b^3c^2)^6$

Answer: $729a^{12}b^{18}c^{12}$

Question 4. Simplify $16(x^6)^2 \times (6x)^{-2}$

Answer: $\frac{4}{9}x^{10}$

Question 5. Simplify $(5^2)^2$

Answer: 625

Part 4

Question 1. Simplify $(2xy \times 3xy)^0$

Answer: 1

Question 2. Simplify $(2pq)^0 \times (3p^2q^3)$

Answer: $3p^2q^3$

Question 3. Simplify $(100)^1 \times (100)^0$

Answer: 100

Question 4. Simplify $(100)^1 \times (100)^1$

Answer: 10000

Question 5. Simplify $(100)^0 \times (100)^1 \times (100)^2$

Answer: 1000000

Question 6. Simplify $3x^0 \times (3x^2)$

Answer: $9x^2$

Question 7. Simplify $(2x^2) \times (3x^3)$

Answer: $6x^5$

Part 5

Question 1. Simplify 7^{-2}

Answer: $\frac{1}{49}$

Question 2. Simplify 4^{-2}

Answer: $\frac{1}{16}$

Question 3. Simplify 9^{-2}

Answer: $\frac{1}{81}$

Part 6

Question 1. Simplify $\left[\frac{125}{64}\right]^{-\frac{2}{3}}$

Answer = $\frac{16}{25}$

Question 2. Simplify $\left(\frac{125}{64}\right)^{\frac{2}{3}}$

Answer: $1\frac{9}{16}$

Question 3. Simplify $8^{\frac{2}{3}}$

Answer: 4

Question 4. Simplify $64^{-\frac{2}{3}}$

Answer: $\frac{1}{16}$

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CHAPTER 11

BINARY OPERATION

SECTION 1: DEFINITION AND WORKED EXAMPLES

The basic operation in mathematics are addition(+), subtraction (-), multiplication($\times$) and division($\div$). These operation may be defined on the set of real numbers. They are called basic operation because they combine two numbers to give another number.

ILLUSTRATIVE EXAMPLES

Example 1. *The operation $*$ is defined as:*

$$a * b = a + b + ab$$

(a) Evaluate

i). $5 * 6$

ii). $2 * 4$

Solution

(i).

Step 1: Find the values of a and b

$$a = 5 \text{ and } b = 6$$

Step 2: Substitute for a and b

$$\begin{array}{ccccccc} a * b & = & a & + & b & + & ab \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \quad \downarrow \\ 5 * 6 & = & 5 & + & 6 & + & (5)(6) \\ & & & & & & \\ & = & 5 & + & 6 & + & 30 \end{array}$$

$$= 41$$

(ii).

Step 1: Find the values of a and b

$$a = 2 \text{ and } b = 4$$

Step 2: Substitute for a and b

$$2 * 4 = 2 + 4 + (2)(4)$$

$$= 2 + 4 + 8$$

$$= 14$$

Example 2. The operation $*$ is defined by $a * b = \frac{1}{2}(a - b)$

Find :

i). $8 * 7$

ii). $12 * 4$

Solution

(i).

Step 1: Find the values of a and b

$$a = 8 \text{ and } b = 7$$

Step 2: Substitute for a and b

$$8 * 7 = \frac{1}{2}(8 - 7)$$

$$= \frac{1}{2} \times 1$$

$$= \frac{1}{2}$$

(ii).

Step 1: Find the values of a and b

$$a = 12 \text{ and } b = 4$$

Step 2: Substitute for a and b

$$12 * 4 = \frac{1}{2}(12 - 4)$$

$$= \frac{1}{2} \times 8$$

$$= 4$$

Example 3. The operation is defined by $p * q = \frac{1}{p} + \frac{1}{q}$

where $p \neq 0$ and $q \neq 0$. If $p = \frac{2}{5}$ and $q = \frac{1}{3}$.

Evaluate: $p * q$

Solution

(i).

Step 1: Substitute for p and q in

$$p * q = \frac{1}{p} + \frac{1}{q}$$

$$\text{But } p = \frac{2}{5} \text{ and } q = \frac{1}{3}$$

$$\frac{2}{5} * \frac{1}{3} = \frac{1}{\frac{2}{5}} + \frac{1}{\frac{1}{3}}$$

$$= 1 \div \frac{2}{5} + 1 \div \frac{1}{3}$$

$$= \frac{1}{1} \times \frac{5}{2} + \frac{1}{1} \times \frac{3}{1}$$

$$= \frac{5}{2} + \frac{3}{1}$$

Step 2: Find the LCM of 1 and 2

LCM of 1 and 2 is 2

$$= \frac{5 + 6}{2}$$

$$= \frac{11}{2}$$

$$= 5\frac{1}{2}$$

SECTION 2: SKS PAPER 11

Answer all questions

Question 1. *The operation $*$ is defined as:*

$$a * b = 14a + 15b + 5ab$$

(a) *Evaluate*

i). $5 * 6$

ii). $2 * 4$

Answers:

(i). 310 **and (ii).** 128

Question 2. *The operation $*$ is defined by $a * b = \frac{1}{2}(a^2 + b) + ab$*

Find :

i). $7 * 8$

ii). $12 * 4$

Answers:

(i). $84\frac{1}{2}$ **and (ii).** 122

Question 3. *The operation is defined by $p * q = 5p + 3q + 15pq$*

where $p \neq 0$ and $q \neq 0$.

(a). *If $p = \frac{2}{5}$ and $q = \frac{1}{3}$*

Evaluate:

$$p * q$$

Answer: 5

SILLAH KANU SERIES @ 2022

CHAPTER 12

CHANGE OF SUBJECT OF FORMULA

SECTION 1: DEFINITION AND WORKED EXAMPLES

The letter I is the subject of the formula $I = \frac{PRT}{100}$.

The subject stands on its own. Its value can be found directly by substituting the values of the other letters in the formula. It is often necessary to change the subject of a formula. This means to rearrange the order of the letters in the formula so that one of the other letters becomes the subject.

ILLUSTRATIVE EXAMPLES

Example 1. Make x the subject of the formula

$$y = x + 8$$

Solution

Step 1: Transpose any number or variable on the side of x to the other side of the equation

$$y - 8 = x$$

Therefore, $x = y - 8$

Note: In transposing we change the sign and position of the variable or term.

Example 2. Make x the subject of the formula

$$y = x - 3$$

Solution

Step 1: Transpose any number or variable on the side of x to the other side of the equation

$$y + 3 = x$$

Therefore, $x = y + 3$

Note: In transposing we change the sign and position of the variable or term.

Example 3. $b = a + c$

In the expression above, make:

(i) a the subject

(ii) c the subject

Solution

Solving for a

Step 1: Transpose any number or variable on the side of ' a ' to the other side of the equation

$$b - c = a$$

Therefore, $a = b - c$

Solving for c

Step 1: Transpose any number or variable on the side of ' c ' to the other side of the equation

$$b - a = c$$

Therefore, $c = b - a$

Example 4. Make x the subject of the formula

$y = 3x$

Solution

Step 1: Divide both sides by the coefficient of x

$$\frac{y}{3} = \frac{3x}{3}$$

$$x = \frac{y}{3}$$

Example 5. Make x the subject of the formula

$$y = \frac{x}{4}$$

Solution

Step 1: Cross multiply

$$\frac{y}{1} = \frac{x}{4}$$

$$y \times 4 = 1 \times x$$

$$4y = x$$

$$\text{Therefore, } x = 4y$$

Example 6. $b = ac$

In the expression above, make:

(i) a the subject

(ii) c the subject

Solution

Solving for a

Step 1: Divide both sides by the coefficient of ' a '

$$\frac{b}{c} = \frac{ac}{c}$$

$$\frac{b}{c} = a$$

$$\text{Therefore, } a = \frac{b}{c}$$

Solving for c

Step 1: Divide both sides by the coefficient of 'c'

$$\frac{b}{a} = \frac{ac}{a}$$

$$\frac{b}{a} = c$$

$$\text{Therefore, } c = \frac{b}{a}$$

Example 7. $n = 5ax$

In the expression above, make:

(i) a the subject

(ii) x the subject

Solution

Solving for a

Step 1: Divide both sides by the coefficient of 'a'

$$\frac{n}{5x} = \frac{5ax}{5x}$$

$$\frac{n}{5x} = a$$

$$\text{Therefore, } a = \frac{n}{5x}$$

Solving for x

Step 1: Divide both sides by the coefficient of x

$$\frac{n}{5a} = \frac{5ax}{5a}$$

$$\frac{n}{5a} = x$$

$$\text{Therefore, } x = \frac{n}{5a}$$

Example 8. Make x the subject of the formula

$$\frac{x}{y} = 9$$

Solution

Step 1: Cross multiply

$$\frac{x}{y} = \frac{9}{1}$$

$$x \times 1 = 9 \times y$$

$$x = 9y$$

Example 9. Make x the subject of the formula

$$\frac{y}{x} = 2$$

Solution

Step 1: Cross multiply

$$\frac{y}{x} = \frac{2}{1}$$

$$y \times 1 = 2 \times x$$

$$y = 2x$$

Step 2: Divide both sides by the coefficient of x

$$\frac{y}{2} = \frac{2x}{2}$$

$$\frac{y}{2} = x$$

$$\text{Therefore, } x = \frac{y}{2}$$

Example 10. $\frac{m}{n} = p$

In the expression above, make:

(i) m the subject

(ii) n the subject

Solution

Solving for m

Step 1: Cross multiply

$$\frac{m}{n} = \frac{p}{1}$$

$$m \times 1 = n \times p$$

$$m = np$$

Solving for n

Step 1: Divide both sides by the coefficient of n

$$\frac{m}{p} = \frac{np}{p}$$

$$\frac{m}{p} = n$$

$$\text{Therefore, } n = \frac{m}{p}$$

Example 11. Make x the subject of the formula

$$y = 6x + 11$$

Solution

Step 1: Transpose any number or variable on the side of x to the other side of the equation

$$y - 11 = 6x$$

Step 2: Divide both sides by the coefficient of x

$$\frac{y - 11}{6} = \frac{6x}{6}$$

$$\frac{y - 11}{6} = x$$

$$\text{Therefore, } x = \frac{y - 11}{6}$$

Example 12. Make x the subject of the formula

$$y = 7x - 2$$

Solution

Step 1: Transpose any number or variable on the side of x to the other side of the equation

$$y + 2 = 7x$$

Step 2: Divide both sides by the coefficient of x

$$\frac{y + 2}{7} = \frac{7x}{7}$$

$$\frac{y + 2}{7} = x$$

$$\text{Therefore, } x = \frac{y + 2}{7}$$

Example 13. Make a the subject of the formula

$$b = 5a - c$$

Solution

Step 1: Transpose any number or variable on the side of x to the other side of the equation

$$b + c = 5a$$

Step 2: Divide both sides by the coefficient of ' a '

$$\frac{b + c}{5} = \frac{5a}{5}$$

$$\frac{b + c}{5} = a$$

$$\text{Therefore, } a = \frac{b + c}{5}$$

Example 14. $x + y = 13$

In the expression above, make:

(i) x the subject

(ii) y the subject

Solution

Solving for x

Step 1: Transpose any number or variable on the side of x to the other side of the equation

$$x = 13 - y$$

Solving for y

Step 1: Transpose any number or variable on the side of y to the other side of the equation

$$x + y = 13$$

$$y = 13 - x$$

Example 15. $2p - q = 0$

In the expression above, make:

(i) p the subject

(ii) q the subject

Solution

Solving for p

Step 1: Transpose any number or variable on the side of p to the other side of the equation

$$2p = 0 + q$$

$$2p = q$$

Step 2: Divide both sides by the coefficient of p

$$\frac{2p}{2} = \frac{q}{2}$$

$$p = \frac{q}{2}$$

Solving for q

Step 1: Transpose q to the other side of the equation

$$2p = q$$

Therefore, $q = 2p$

Example 16. $2x - y = d$

In the expression above, make:

(i) x the subject

(ii) y the subject

Solution

Solving for x

Step 1: Transpose any number or variable on the side of x to the other side of the equation

$$2x = d + y$$

Step 2: Divide both sides by the coefficient of x

$$\frac{2x}{2} = \frac{d + y}{2}$$

$$x = \frac{d + y}{2}$$

Solving for y

Step 1: Transpose y to the other side of the equation since y is negative and also transpose ' d ' to the other side of the equation

$$2x - y = d$$

$$2x - d = y$$

$$\text{Therefore, } y = 2x - d$$

Example 17. Make d the subject in the formula below

$$p = 4d$$

Solution

Step 1: Divide both sides by the coefficient of d

$$\frac{p}{4} = \frac{4d}{4}$$

$$\frac{p}{4} = d$$

Therefore, $d = \frac{p}{4}$

Example 18. Make r the subject in the formula

$c = 2\pi r$

Solution

Step 1: Divide both sides by the coefficient of r

$$\frac{c}{2\pi} = \frac{2\pi r}{2\pi}$$

$$\frac{c}{2\pi} = r$$

Therefore, $r = \frac{c}{2\pi}$

Example 19. Make θ the subject in the formula

$T = \theta + 273$

Solution

Step 1: Transpose any number or variable on the side of θ to the other side of the equation

$$T - 273 = \theta$$

Therefore, $\theta = T - 273$

Example 20. $W = VI$

In the expression above, make:

(i) V the subject

(ii) I the subject

Solution

Solving for V

Step 1: Divide both sides by the coefficient of V

$$\frac{W}{I} = \frac{VI}{I}$$

$$\frac{W}{I} = V$$

$$\text{Therefore, } V = \frac{W}{I}$$

Solving for I

Step 1: Divide both sides by the coefficient of I

$$\frac{W}{V} = \frac{VI}{V}$$

$$\frac{W}{V} = I$$

$$\text{Therefore, } I = \frac{W}{V}$$

Example 21. $A = \pi rl$

In the expression above, make:

(i) r the subject

(ii) l the subject

Solution

Solving for r

Step 1: Divide both sides by the coefficients of r

$$\frac{A}{\pi l} = \frac{\pi rl}{\pi l}$$

$$\frac{A}{\pi l} = r$$

$$\text{Therefore, } r = \frac{A}{\pi l}$$

Solving for l

Step 1: Divide both sides by the coefficients of l

$$\frac{A}{\pi r} = \frac{\pi r l}{\pi r}$$

$$\frac{A}{\pi r} = l$$

$$\text{Therefore, } l = \frac{A}{\pi r}$$

Example 22. $V = lbh$

In the expression above, make:

- (i) l the subject**
- (ii) b the subject**
- (iii) h the subject**

Solution

Solving for l

Step 1: To find l, divide both sides by the coefficients of l

$$\frac{V}{bh} = \frac{lbh}{bh}$$

$$\frac{V}{bh} = l$$

$$\text{Therefore, } l = \frac{V}{bh}$$

Solving for b

Step 1: Divide both sides by the coefficient of b

$$\frac{V}{lh} = \frac{lbh}{lh}$$

$$\frac{V}{lh} = b$$

$$\text{Therefore, } b = \frac{V}{lh}$$

Solving for h

Step 1: Divide both sides by the coefficient of h

$$\frac{V}{lb} = \frac{lbh}{lb}$$

$$\frac{V}{lb} = h$$

$$\text{Therefore, } h = \frac{V}{lb}$$

Example 23. $A = 2\pi rh$

In the expression above, make:

(i) r the subject

(ii) h the subject

Solution

Solving for r

Step 1: Divide both sides by the coefficients of r

$$\frac{A}{2\pi h} = \frac{2\pi rh}{2\pi h}$$

$$\frac{A}{2\pi h} = r$$

$$\text{Therefore, } r = \frac{A}{2\pi h}$$

Solving for h

Step 1: Divide both sides by the coefficients of h

$$\frac{A}{2\pi r} = \frac{2\pi r h}{2\pi r}$$

$$\frac{A}{2\pi r} = h$$

$$\text{Therefore, } h = \frac{A}{2\pi r}$$

Example 24. $S = \frac{1}{2}vt$

In the expression above, make:

(i) v the subject

(ii) t the subject

Solution

Solving for v

Step 1: Cross multiply

$$S \times 2 = vt$$

Step 2: Divide both sides by the coefficient of v

$$\frac{2S}{t} = \frac{vt}{t}$$

$$\text{Therefore, } v = \frac{2S}{t}$$

Solving for t

Step 1: Cross multiply

$$S \times 2 = vt$$

Step 2: Divide both sides by the coefficient of t

$$S \times 2 = vt$$

$$\frac{2S}{v} = \frac{vt}{v}$$

$$\text{Therefore, } t = \frac{2S}{v}$$

Example 25. Make R the subject in the formula

$$I = \frac{PRT}{100}$$

Solution

Step 1: Cross multiply

$$100 \times I = PRT$$

Step 2: Divide both sides by the coefficient of R

$$\frac{100I}{PT} = \frac{PRT}{PT}$$

$$R = \frac{100I}{PT}$$

Example 26. $v = u + at$

In the expression above, make:

(i) u the subject

(ii) a the subject

(iii) t the subject

Solution

Step 1: Solving for u

$$u = v - at$$

Step 2: Solving for a

$$a = \frac{v - u}{t}$$

Step 3: Solving for t

$$t = \frac{v - u}{a}$$

Example 27. Make x the subject of the formula

$$\sqrt{x} = a$$

Solution

Step 1: Squaring both sides to remove the square root

$$(\sqrt{x})^2 = a^2$$

$$x = a^2$$

Example 28. Make x the subject of the formula

$$\sqrt{2x} = a$$

Solution

Step 1: Squaring both sides to remove the square root

$$(\sqrt{2x})^2 = a^2$$

$$2x = a^2$$

Step 2: Divide both sides by the coefficient of x

$$\frac{2x}{2} = \frac{a^2}{2}$$

$$x = \frac{1}{2}a^2$$

Example 29. Make x the subject of the formula

$$2\sqrt{x} = a$$

Solution

Step 1: Squaring both sides to remove the square root

$$(2\sqrt{x})^2 = a^2$$

$$2^2x = a^2$$

Step 2: Divide both sides by the coefficient of x

$$\frac{4x}{4} = \frac{a^2}{4}$$

$$x = \frac{a^2}{4}$$

Example 30. Make x the subject of the formula

$$x^2 = a$$

Solution

Step 1: Taking the square root on both sides to remove the square on x

$$\sqrt{x^2} = \sqrt{a}$$

$$x = \sqrt{a}$$

Example 31. Make x the subject of the formula

$$y = 4ax^2$$

Solution

Step 1: Divide both sides by the coefficient of x^2

$$\frac{y}{4a} = \frac{4ax^2}{4a}$$

$$x^2 = \frac{y}{4a}$$

Step 2: Taking the square root on both sides to remove the square on x

$$\sqrt{x^2} = \sqrt{\frac{y}{4a}}$$

$$x = \sqrt{\frac{y}{4a}}$$

Example 32. Make b the subject of the formula

$$\frac{a}{b} = \frac{c}{d}$$

Solution

Step 1: Cross multiply

$$a \times d = c \times b$$

$$ad = cb$$

Step 2: Divide both sides by the coefficient of b

$$\frac{ad}{c} = \frac{cb}{c}$$

$$b = \frac{ad}{c}$$

Example 33. Make b the subject of the formula

$$3(2a - b) = a$$

Solution

Step 1: Multiply the number outside the bracket by each term inside the bracket

$$3 \times 2a - 3 \times b = a$$

$$6a - 3b = a$$

Step 2: Collect like terms

$$6a - a = 3b$$

$$5a = 3b$$

Step 3: Divide both sides by the coefficient of b

$$\frac{5a}{3} = \frac{3b}{3}$$

$$b = \frac{5a}{3}$$

Example 34. Make a the subject of the formula

$$x(x + a) = ab$$

Solution

Step 1: Multiply the number outside the bracket by each term inside the bracket

$$x \times x + a \times x = ab$$

$$x^2 + ax = ab$$

Step 2: Transpose 'ax'

$$x^2 = ab - ax$$

Step 3: Factorise 'a'

$$x^2 = a(b - x)$$

Step 4: Divide both sides by the coefficient of 'a'

$$\frac{x^2}{(b - x)} = \frac{a(b - x)}{(b - x)}$$

$$a = \frac{x^2}{(b - x)}$$

Example 35. Make x the subject of the formula

$$\frac{a}{x} + b = c$$

Solution

Step 1: For the L. H. S, find the L. C. M

$$LCM = x$$

Step 2: Using the concept of fraction

$$\frac{a + bx}{x} = c$$

Step 3: Cross multiply

$$a + bx = c \times x$$

$$a + bx = cx$$

Step 4: Transpose bx

$$a = cx - bx$$

Step 5: Factorise x

$$a = x(c - b)$$

Step 6: Divide both sides by the coefficient of x

$$\frac{a}{c - b} = \frac{x(c - b)}{c - b}$$

$$x = \frac{a}{c - b}$$

Example 36. Make a the subject of the formula

$$\frac{1}{a} = 1 + \frac{1}{b}$$

Solution

Step 1: For the R. H. S, find the L. C. M

$$LCM = b$$

Step 2:

$$\frac{1}{a} = \frac{b+1}{b}$$

Step 3: Cross multiply

$$b \times 1 = a(b+1)$$

$$b = a(b+1)$$

Step 4: Divide both sides by the coefficient of 'a'

$$\frac{b}{b+1} = \frac{a(b+1)}{b+1}$$

$$a = \frac{b}{b+1}$$

Example 37. Make N the subject of the formula

$$P = \frac{N+2}{D}$$

Solution

Step 1: Cross multiply

$$P \times D = N + 2$$

$$PD = N + 2$$

Step 2: Transpose ' + 2 '

$$PD - 2 = N$$

$$\text{Therefore, } N = PD - 2$$

Example 38. Make h the subject of the formula

$$D = \sqrt{\frac{3h}{2}}$$

Solution

Step 1: Squaring both sides to remove the square root

$$D^2 = \left(\sqrt{\frac{3h}{2}} \right)^2$$

$$D^2 = \frac{3h}{2}$$

Step 2: Cross multiply

$$2 \times D^2 = 3h$$

Step 3: Divide both sides by the coefficient of h

$$\frac{2D^2}{3} = \frac{3h}{3}$$

$$h = \frac{2}{3}D^2$$

Example 39. Make I the subject of the formula

$$W = I^2 R$$

Solution

Step 1:

$$\frac{W}{R} = I^2$$

Step 2:

$$\sqrt{\frac{W}{R}} = \sqrt{I^2}$$

$$I = \sqrt{\frac{W}{R}}$$

Example 40. Make t subject of the formula

$$s = \frac{1}{2}gt^2$$

Solution

Step 1:

$$s \times 2 = gt^2$$

$$2s = gt^2$$

Step 2:

$$\frac{2s}{g} = \frac{gt^2}{g}$$

$$\frac{2s}{g} = t^2$$

Step 3:

$$\frac{2s}{g} = t^2$$

Step 4:

$$\sqrt{\frac{2s}{g}} = \sqrt{t^2}$$

$$t = \sqrt{\frac{2s}{g}}$$

Example 41. Make d the subject of the formula

$$h = \frac{2d^2}{25}$$

Solution

Step 1: Cross multiply

$$h \times 25 = 2d^2$$

Step 2: Divide both sides by the coefficient of d

$$\frac{25h}{2} = \frac{2d^2}{2}$$

$$d^2 = \frac{25h}{2}$$

Step 3: Taking the square root on both sides to remove the square

$$\sqrt{d^2} = \sqrt{\frac{25h}{2}}$$

$$d = \sqrt{\frac{25h}{2}}$$

Step 4: Separate the root

$$d = \frac{\sqrt{25} \times \sqrt{h}}{\sqrt{2}}$$

$$\text{But } \sqrt{25} = 5$$

$$d = \frac{5 \times \sqrt{h}}{\sqrt{2}}$$

$$\text{but } \frac{\sqrt{h}}{\sqrt{2}} = \sqrt{\frac{h}{2}}$$

$$\text{Therefore, } d = 5\sqrt{\frac{h}{2}}$$

Example 41. Make r the subject of the formula

$$v = \pi r^2 h$$

Solution

Step 1: Divide both sides by the coefficient of r^2

$$\frac{v}{\pi h} = \frac{\pi r^2 h}{\pi h}$$

$$r^2 = \frac{v}{\pi h}$$

Step 2: Taking the square root of both sides to remove the square

$$\sqrt{r^2} = \sqrt{\frac{v}{\pi h}}$$

$$r = \sqrt{\frac{v}{\pi h}}$$

Example 42. Make u the subject of the formula

$$s = \frac{1}{2}(u + v)t$$

Solution

Step 1: Cross multiply and expand the bracket

$$2 \times s = ut + vt$$

$$2s = ut + vt$$

Step 2: Transpose vt

$$2s - vt = ut$$

Step 3: Divide both sides by t

$$u = \frac{2s - vt}{t}$$

Step 4: Divide each numerator by the denominator

$$u = \frac{2s}{t} - \frac{vt}{t}$$

$$u = \frac{2s}{t} - v$$

Example 43. $S = 2\pi r(r + h) \dots h$

Solution

Step 1: Expand the bracket

$$S = 2\pi r^2 + 2\pi rh$$

Step 2: Transpose $2\pi r^2$

$$s - 2\pi r^2 = 2\pi rh$$

Step 3: Divide both sides by $2\pi r$

$$\frac{s - 2\pi r^2}{2\pi r} = \frac{2\pi rh}{2\pi r}$$

$$h = \frac{s - 2\pi r^2}{2\pi r}$$

Step 4: Divide each numerator by the denominator

$$h = \frac{s}{2\pi r} - \frac{2\pi r^2}{2\pi r}$$

$$h = \frac{s}{2\pi r} - r$$

Example 44. Make z the subject of the formula

$$x = \sqrt{y^2 + z^2}$$

Solution

Step 1: squaring both sides to get rid of the square root

$$(x)^2 = \left(\sqrt{y^2 + z^2}\right)^2$$

$$x^2 = y^2 + z^2$$

Step 2: transpose y^2

$$x^2 - y^2 = z^2$$

Step 3: taking the square root of both sides to get rid of the square on z

$$\sqrt{x^2 - y^2} = \sqrt{z^2}$$

$$z = \sqrt{x^2 - y^2}$$

Example 45. Make u the subject of the formula

$$\frac{1}{u} + \frac{1}{v} = \frac{2}{f}$$

Solution

Step 1: find the LCM on the LHS

LCM of u and v is uv

Step 2: using the concepts of fraction to simplify

$$\frac{v + u}{uv} = \frac{2}{f}$$

Step 3: cross multiply

$$f \times (v + u) = 2 \times uv$$

Step 4: expand the bracket and divide by the coefficient of u

$$\frac{fv + fu}{2v} = \frac{2uv}{2v}$$

$$u = \frac{fv + fu}{2v}$$

Step 5: cross multiply

$$u \times 2v = fv + fu$$

$$2uv = fv + fu$$

Step 6: transpose 'fu'

$$2uv - fu = fv$$

Step 7: factor out 'u'

$$u(2v - f) = fv$$

Step 8: divide both sides by the coefficient of 'u'

$$\frac{u(2v - f)}{(2v - f)} = \frac{fv}{(2v - f)}$$

$$u = \frac{fv}{(2v - f)}$$

SECTION 2: WORD PROBLEMS

Example 1. The mean, m , of two numbers a and b is given by

$$m = \frac{1}{2}(a + b).$$

(a). Express a in terms of m and b and

(b) Hence find a when $m = 19$ and $b = 23$

Solution

Step 1: write the relation

$$m = \frac{1}{2}(a + b)$$

Step 2: make a the subject

$m \times 2 = a + b$... we cross multiplied to obtain this result

$$2m = a + b$$

$$2m - b = a$$

Therefore, $a = 2m - b$

Step 3: substitute the given values into ' a '

$$a = 2(19) - 23$$

$$a = 38 - 23$$

$$a = 15$$

Example 2. If $x = \frac{1}{4}\sqrt{\frac{y}{a}}$, (a) express y in terms of a and x and

(b). find y when $a = 3$ and $x = \frac{1}{2}$

Solution

(a).

Step 1: write the relation

$$x = \frac{1}{4}\sqrt{\frac{y}{a}}$$

Step 2: Squaring both sides

$$(x)^2 = \left(\frac{1}{4}\sqrt{\frac{y}{a}}\right)^2$$

$$x^2 = \left(\frac{1}{4}\right)^2 \times \left(\sqrt{\frac{y}{a}}\right)^2$$

$$x^2 = \frac{1^2}{4^2} \times \frac{y}{a}$$

$$x^2 = \frac{1 \times 1}{4 \times 4} \times \frac{y}{a}$$

$$x^2 = \frac{1}{16} \times \frac{y}{a}$$

$$x^2 = \frac{1 \times y}{16 \times a}$$

$$x^2 = \frac{y}{16a}$$

Step 3: Cross multiply

$$x^2 \times 16a = y$$

$$\text{Therefore, } y = 16ax^2$$

(b).

Step 1: substitute the given values into y

$$y = 16(3) \times \left(\frac{1}{2}\right)^2$$

$$y = 48 \times \frac{1 \times 1}{2 \times 2}$$

$$y = 48 \times \frac{1}{4}$$

$$y = \frac{48 \times 1}{4}$$

$$y = \frac{48}{4}$$

$$y = 12$$

Example 3. The area A of a circle is given by the formula $A = \pi r^2$, where r is the radius of the circle. (a) Obtain a formula for r . (b) Hence find the radius of a circle of area 38.5 cm^2 , taking π to be $\frac{22}{7}$

Solution

(a)

$$A = \pi r^2$$

$$\frac{A}{\pi} = \frac{\pi r^2}{\pi}$$

$$r^2 = \frac{A}{\pi}$$

$$\sqrt{r^2} = \sqrt{\frac{A}{\pi}}$$

The square cancels the square root

$$r = \sqrt{\frac{A}{\pi}}$$

(b).

Substitute the given values into the formula

$$r = \sqrt{\frac{38.5}{\frac{22}{7}}}$$

$$r = \sqrt{\frac{38\frac{1}{2}}{\frac{22}{7}}}$$

$$r = \sqrt{\frac{77}{2} \div \frac{22}{7}}$$

$$r = \sqrt{\frac{77}{2} \times \frac{7}{22}}$$

$$r = 12\frac{1}{2}$$

Example 4. The surface of a sphere of radius r is S , where

$$S = 4\pi r^2.$$

(a). Express r in terms of S and π . (b) Find the radius of a sphere which has a surface area of 616 cm^2 . (Use the value $3\frac{1}{7}$ for π)

Solution

$$(a) S = 4\pi r^2$$

$$\frac{S}{4\pi} = \frac{4\pi r^2}{4\pi}$$

$$r^2 = \frac{S}{4\pi}$$

$$\sqrt{r^2} = \sqrt{\frac{S}{4\pi}}$$

$$r = \sqrt{\frac{S}{4\pi}}$$

(b).

$$r = \sqrt{\frac{S}{4\pi}}$$

$$r = \sqrt{\frac{616}{4\left(\frac{22}{7}\right)}} = \sqrt{\frac{616}{4 \times \frac{22}{7}}} = \sqrt{\frac{616}{\frac{88}{7}}} = \sqrt{\frac{616 \times 7}{88}} = \sqrt{\frac{4312}{88}} = \sqrt{49}$$

$$r = 7 \text{ cm}$$

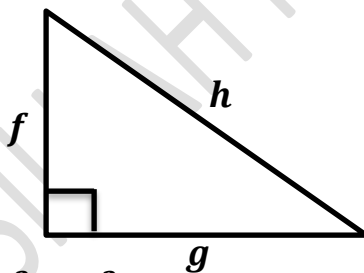
Example 5. In a right – angled triangle, the hypotenuse is h cm long and the other two sides are f cm and g cm in length.

(a) Write down a formula which connects f , g and h .

(b). Make f the subject of your formula.

(c) Find f when $g = 2$ cm and $h = 2.5$ cm

Solution



$$h^2 = f^2 + g^2$$

$$f^2 = h^2 - g^2$$

$$f = \sqrt{h^2 - g^2}$$

substitute the values

$$f = \sqrt{\left(\frac{5}{2}\right)^2 - 2^2} \longrightarrow 2.5 = \frac{25}{10} = \frac{5}{2}$$

$$f = \sqrt{\frac{25}{4} - 4}$$

$$f = \sqrt{\frac{25 - 16}{4}}$$

$$f = \sqrt{\frac{9}{4}} = \frac{3}{2} = 1\frac{1}{2}$$

Example 6. If $\frac{n^2 - 1}{m} = 4$, (a) express n in terms of m , and

(b) hence find n when $m = 12$.

Solution

(a). $\frac{n^2 - 1}{m} = 4$

Cross multiply

$$n^2 - 1 = 4m$$

$$n^2 = 4m + 1$$

$$n = \sqrt{4m + 1}$$

(b). **Substitute the values**

$$n = \sqrt{4m + 1}$$

$$n = \sqrt{4(12) + 1}$$

$$n = \sqrt{48 + 1}$$

$$n = \sqrt{49}$$

$$n = 7$$

Example 7. The volume V of a cone of height h and base radius r is $\frac{1}{3}\pi r^2 h$.

(a). Obtain a formula for r

(b). Find the base radius of a cone of height 14 cm and volume $91\frac{2}{3} \text{ cm}^3$.

(Use the value $\frac{22}{7}$ for π)

Solution

$$V = \frac{1}{3}\pi r^2 h$$

Cross multiply

$$3v = \pi r^2 h$$

Divide both sides by the coefficients of r^2

$$\frac{3v}{\pi h} = \frac{\pi r^2 h}{\pi h}$$

$$r^2 = \frac{3v}{\pi h}$$

$$\sqrt{r^2} = \sqrt{\frac{3v}{\pi h}}$$

$$r = \sqrt{\frac{3v}{\pi h}}$$

Substitute the values

$$r = \sqrt{\frac{3v}{\pi h}}$$

$$= \sqrt{\frac{3 \left(91 \frac{2}{3}\right)}{\frac{22}{7} (14)}}$$

$$= \sqrt{\frac{3 \times \frac{275}{3}}{44}}$$

$$= \sqrt{\frac{275}{44}}$$

$$= \frac{\sqrt{25} \times \sqrt{11}}{\sqrt{4} \times \sqrt{11}}$$

$$= \frac{5}{2}$$

$$= 2 \frac{1}{2}$$

Example 8. The total resistance, R ohms, of two resistances x ohms and y ohms connected in parallel is given by the formula $\frac{1}{R} = \frac{1}{x} + \frac{1}{y}$

(a) Obtain a formula for y in terms of R and x .

(b) Find y if $x = 2 \frac{2}{3}$ and $R = 2 \frac{1}{4}$

Solution

Step 1: Write the relation

$$\frac{1}{R} = \frac{y + x}{xy}$$

Step 2: Cross multiply

$$xy = R(y + x)$$

$$xy = Ry + Rx$$

Step 3: make y the subject

$$xy - Ry = Rx$$

$$y(x - R) = Rx$$

Step 4: divide both sides by the coefficient of y

$$\frac{y(x - R)}{(x - R)} = \frac{Rx}{(x - R)}$$

$$y = \frac{Rx}{(x - R)}$$

(i). *substitute the values*

$$y = \frac{Rx}{(x - R)}$$

$$y = \frac{2\frac{1}{4} \times 2\frac{2}{3}}{2\frac{2}{3} - 2\frac{1}{4}}$$

$$= \frac{\frac{9}{4} \times \frac{8}{3}}{\frac{8}{3} - \frac{9}{4}}$$

$$= \frac{6}{\frac{32 - 27}{12}}$$

$$= \frac{6}{\frac{5}{12}}$$

$$= 6 \times \frac{12}{5}$$

$$= \frac{72}{5} = 14\frac{2}{5}$$

SECTION 3: SKS PAPER 12

Answer all questions

Part 1

Question 1. Make x the subject of the formula

$$y - \frac{\sqrt{v}}{\sqrt[3]{m}} = x + 8$$

$$\text{Answer} \left[x = y - \frac{\sqrt{v}}{\sqrt[3]{m}} - 8 \right]$$

Question 2. Make N the subject

$$Nby = 3x$$

$$\text{Answer} \left[N = \frac{3x}{by} \right]$$

Question 3. Make x the subject

$$y = \frac{14\frac{1}{2}x}{4}$$

$$\text{Answer} \left[x = \frac{8y}{29} \right]$$

Question 4. Make p and q the subject

$$\frac{p \sin^2 \theta}{\cos^2 \theta} - q = 0$$

$$\text{Answer} \left[p = \frac{q \cos^2 \theta}{\sin^2 \theta} \text{ and } q = \frac{p \sin^2 \theta}{\cos^2 \theta} \right]$$

Question 5. Make r and l the subject

$$\frac{A^2 r}{\left(\frac{8k\sqrt{2}}{3y\sqrt[3]{h}} \right)} = \frac{\pi}{2} l$$

$$\text{Answer} \left[r = \frac{4\pi kl\sqrt{2}}{3A^2 y^3 \sqrt{h}} \text{ and } l = \frac{3A^2 y^3 \sqrt{h}}{4\pi k\sqrt{2}} \right]$$

Question 6. Make l, b and h the subject

$$V \sin\left(\frac{\pi}{2}\right) = lbh$$

$$\text{Answer} \left[l = \frac{V \sin\left(\frac{\pi}{2}\right)}{bh}, b = \frac{V \sin\left(\frac{\pi}{2}\right)}{lh} \text{ and } h = \frac{V \sin\left(\frac{\pi}{2}\right)}{bl} \right]$$

Question 7. Make r and h the subject

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \times A^2 = 2\pi rh\sqrt{a^2 + b^2}$$

$$\text{Answer} \left[r = \frac{-b \pm \sqrt{b^2 - 4ac}}{4\pi ah\sqrt{a^2 + b^2}} \times A^2 \text{ and } h = \frac{-b \pm \sqrt{b^2 - 4ac}}{4\pi ar\sqrt{a^2 + b^2}} \times A^2 \right]$$

Question 8. Make t the subject

$$s^2 = \frac{1}{2}v^2 t^2$$

$$\text{Answer} \left[t = \frac{s\sqrt{2}}{v} \right]$$

Question 9. Make x the subject

$$\sqrt{x} = 7ab^{\frac{2}{3}} \dots x$$

$$\text{Answer} \left[x = 49a^2 (\sqrt[3]{b})^4 \right]$$

Question 10. Make x the subject

$$\sqrt{2x} = a^{\frac{2}{3}}$$

$$\text{Answer} \left[x = \frac{a^{\frac{4}{3}}}{2} \right]$$

Question 11. Make x the subject

$$2^{\frac{6p}{7}}\sqrt{x} = a$$

$$\text{Answer} \left[x = \frac{a^2}{2^{\frac{12p}{7}}} \right]$$

Question 12. Make x the subject

$$35x^2 = a$$

$$\text{Answer} \left[x = \sqrt{\frac{a}{35}} \right]$$

Question 13. Make x the subject

$$225y = 4a^2x^2$$

$$\text{Answer} \left[x = \frac{15\sqrt{y}}{2a} \right]$$

Question 14. Make b the subject

$$\frac{3a}{4b} = \frac{6c}{7d}$$

$$\text{Answer} \left[b = \frac{7ad}{8c} \right]$$

Question 15. Make x the subject

$$312(2a - b) = 444a$$

$$\text{Answer} \left[b = \frac{15a}{26} \right]$$

Question 16. Make x the subject

$$345x(x + a) = 225ab$$

$$\text{Answer} \left[a = -\frac{23x^2}{23x - 15b} \right]$$

Question 17. Make x the subject

$$\frac{2\frac{1}{4}a}{x} + 2\sqrt[3]{b} = c$$

$$\text{Answer} \left[x = \frac{9a}{4c - 8\sqrt[3]{b}} \right]$$

Question 18. Make a the subject

$$\frac{1}{a} = 111234 + \frac{1}{b}$$

$$\text{Answer} \left[a = \frac{b}{111234b + 1} \right]$$

Question 19. Make N the subject

$$P = \frac{N + 212345}{D}$$

$$\text{Answer} [N = DP - 212345]$$

Question 20. Make h the subject

$$D = \sqrt{\frac{3h}{12}}$$

$$\text{Answer} [h = 4D^2]$$

Question 21. Make I the subject of the formula

$$1000W = 500I^2R$$

$$\text{Answer} \left[I = \sqrt{\frac{2W}{R}} \right]$$

Question 22. Make t the subject of the formula

$$23s = \frac{1}{23}gt^2$$

$$\text{Answer} \left[t = 23 \sqrt{\frac{s}{g}} \right]$$

Question 23. Make d the subject of the formula

$$312h = \frac{100d^2}{25}$$

$$\text{Answer} [d = \sqrt{78h}]$$

Question 24. Make r the subject of the formula

$$7859v = \pi r^2 h$$

$$\text{Answer} \left[r = \sqrt{\frac{7859v}{\pi h}} \right]$$

SILLAH KANU SERIES @ 2022

CHAPTER 13

EQUATIONS

Equation is a statement that the values of two mathematical expressions are equal(indicated by the sign =)

SECTION 1: LINEAR EQUATIONS

In mathematics, a linear equation is an equation that may be put in the form where the variables and the coefficients, which are often real numbers .The coefficients may be considered as parameters of the equation, and may be arbitrary expressions, provided they do not contain any of the variables.

Subsection 1: Simple linear equations

Example 1. Find the value of x in the equation below?

$$x + 6 = 8$$

Solution

Step 1. Collect like terms by transposing

$$x = 8 - 6$$

$$x = 2$$

Note. In transposing, we change the sign and position of the number.

Example 2. Find q in

$$q - 8 = 7$$

Solution

Step 1. Collect like terms by transposing

$$q = 7 + 8$$

$$q = 15$$

Example 3. Solve for x in

$$x^{\frac{1}{2}} - 3 = 6$$

Solution

Step 1. Collect like terms by transposing

$$x^{\frac{1}{2}} = 6 + 3$$

$$x^{\frac{1}{2}} = 9$$

Step 2. Convert the power of x into square root

$$\sqrt{x} = 9$$

Step 3. Squaring both sides to remove the square root

$$(\sqrt{x})^2 = 9^2$$

$$x = 9 \times 9$$

$$x = 81$$

Example 4. Solve for v in

$$v^{\frac{3}{2}} + 2 = 10$$

Solution

Step 1. Collect like terms

$$v^{\frac{3}{2}} = 10 - 2$$

$$v^{\frac{3}{2}} = 8$$

Step 2. Convert the fractional index into roots

$$(\sqrt[3]{v})^3 = 8$$

Step 3. Squaring both sides to remove the square root

$$[(\sqrt{v})^3]^2 = 8^2$$

$$v^3 = 8 \times 8$$

$$v^3 = 64$$

Step 4. Taking the cube root on both sides to get rid of the cube on "v"

$$\sqrt[3]{v^3} = \sqrt[3]{64}$$

$$v = 64^{\frac{1}{3}}$$

$$v = (4^3)^{\frac{1}{3}}$$

$$v = 4$$

Example 5. Solve for y in

$$2\sqrt{y} - 6 = 2$$

Solution

Step 1. Collect like terms

$$2\sqrt{y} = 2 + 6$$

$$2\sqrt{y} = 8$$

Step 2. Divide both sides by the coefficient of $\sqrt{y}$

$$\frac{2\sqrt{y}}{2} = \frac{8}{2}$$

$$\sqrt{y} = 4$$

Step 3. Squaring both sides to remove the square root

$$(\sqrt{y})^2 = 4^2$$

$$y = 4 \times 4$$

$$y = 16$$

Example 6. Solve for z in

$$4\sqrt[4]{z} - 8 = 8$$

Solution

Step 1. Collect like terms

$$4\sqrt[4]{z} = 8 + 8$$

$$4\sqrt[4]{z} = 16$$

Step 2. Divide both sides by the coefficient of $\sqrt[4]{z}$

$$\frac{4\sqrt[4]{z}}{4} = \frac{16}{4}$$

$$\sqrt[4]{z} = 4$$

Step 3. Raising both sides to power four(4) to remove the fourth root

$$(\sqrt[4]{z})^4 = 4^4$$

$$z = 4 \times 4 \times 4 \times 4$$

$$z = 256$$

Example 7. Find x in terms of y in

$$3xy - 6 = 9$$

Solution

Step 1. Collect like terms

$$3xy = 9 + 6$$

$$3xy = 15$$

Step 2. Divide both sides by the coefficients of x

$$\frac{3xy}{3y} = \frac{15}{3y}$$

$$x = \frac{5}{y}$$

Note: we are asked to find x in terms of y , so we make x the subject.

Example 8. Solve for e in

$$8e - 32 = 8$$

Solution

Step 1. Collect like terms

$$8e = 32 + 8$$

$$8e = 40$$

Step 2. Divide both sides by the coefficient of " e "

$$\frac{8e}{8} = \frac{40}{8}$$

$$e = 5$$

Note. In transposing we change the sign and position of the number.

Example 9. Find the value of g in

$$6g + 11 = 25 - g$$

Solution

Step 1. Collect like terms by transposing.

$$6g + g = 25 - 11$$

$$7g = 14$$

Step 2. Divide both sides by the coefficient of g

$$\frac{7g}{7} = \frac{14}{7}$$

$$g = 2$$

Example 10. Solve for p in

$$5p - 8 = 3p + 2$$

Solution

Step 1. Collect like terms by transposing

$$5p - 3p = 2 + 8$$

$$2p = 10$$

Step 2. Divide both sides by the coefficient of " p "

$$\frac{2p}{2} = \frac{10}{2}$$

$$p = 5$$

Example 11. Solve for w in

$$3(w + 1) = 10$$

Solution

Step 1. Multiply the number outside the bracket by each of the quantity inside the bracket.

$$3w + 3 = 10$$

Step 2. Collect like terms by transposing

$$3w = 10 - 3$$

$$3w = 7$$

Step 3. Divide both sides by the coefficient of " w "

$$\frac{3w}{3} = \frac{7}{3}$$

$$w = \frac{7}{3} = 2\frac{1}{3}$$

Example 12. Solve for k in the equation below

$$13k + 1 = 2(2k + 2)$$

Solution

Step 1. Multiply the number outside the bracket by each of the quantity inside the bracket.

$$3k + 1 = 4k + 4$$

Step 2. Collect like terms by transposing

$$3k - 4k = 4 - 1$$

$$-k = 3$$

Step 3. Divide both sides by the coefficient of " k "

$$\frac{-k}{-1} = \frac{3}{-1}$$

$$k = -3$$

Example 13. Solve for x in

$$4(x - 5) = 7 - 5(3 - 2x)$$

Solution

Step 1. Expand the brackets

$$4x - 20 = 7 - 15 + 10x$$

$$4x - 20 = -8 + 10x$$

Step 2. Collect like terms

$$4x - 10x = -8 + 20$$

$$-6x = 12$$

Step 3. Divide both sides by the coefficient of " x "

$$\frac{-6x}{-6} = \frac{12}{-6}$$

$$x = -2$$

Example 14. Expand and simplify

$$3(2z + 1) + 9 + 30 = 0$$

Solution

Step 1. Expand the bracket

$$6z + 3 + 9 + 30$$

$$6z + 42 = 0$$

$$6z = -42$$

Step 2. Divide both sides by the coefficient of "z"

$$\frac{6z}{6} = \frac{-42}{6}$$

$$z = -7$$

Example 15. Solve for x in

$$3\sqrt{(x-5)} = 7 - 5\sqrt{(x-5)} + 1$$

Solution

Step 1. Collect like terms

$$3\sqrt{(x-5)} + 5\sqrt{(x-5)} = 7 + 1$$

$$8\sqrt{(x-5)} = 8$$

Step 2. Divide both sides by the coefficient of $\sqrt{(x-5)}$

$$\frac{8\sqrt{(x-5)}}{8} = \frac{8}{8}$$

$$\sqrt{(x-5)} = 1$$

Step 3. Squaring both sides to remove the square root

$$(\sqrt{x-5})^2 = 1^2$$

$$x - 5 = 1 \times 1$$

$$x - 5 = 1$$

Transpose - 5

$$x = 1 + 5$$

$$x = 6$$

Subsection 2: Equations in fraction form

Example 1. Solve for x in

$$\frac{x+3}{2} + \frac{x+4}{3} = 3$$

Solution

Step 1. Find the L.C.M of 2 and 3.

L.C.M of 2 and 3 is 6

Step 2. Simplify the L.H.S using concepts of fraction

$$\frac{3(x+3) + 2(x+4)}{6} = 3$$

$$\frac{3x+9+2x+8}{6} = 3$$

$$\frac{5x+17}{6} = \frac{3}{1}$$

Step 3. Cross multiply

$$5x + 17 = 6 \times 3$$

$$5x + 17 = 18$$

Step 4. Collect like terms by transposing

$$5x = 18 - 17$$

$$5x = 1$$

Step 5. Divide both sides by the coefficient of "x"

$$\frac{5x}{5} = \frac{1}{5}$$

$$x = \frac{1}{5}$$

Example 2. Solve for x in

$$\frac{x}{5} - \frac{x}{3} = (2\sqrt{4})^2$$

Solution

Step 1. For the LHS, find the LCM of 5 and 3

LCM of 5 and 3 is 15

Step 2. Simplify the LHS using concepts of fraction

$$\frac{3x - 5x}{15} = (2\sqrt{4})^2$$

$$\frac{-2x}{15} = (2\sqrt{4})^2$$

Step 3. Simplify the RHS

$$\frac{-2x}{15} = [(2)(2)]^2$$

$$\frac{-2x}{15} = 4^2$$

$$\frac{-2x}{15} = 4 \times 4$$

$$\frac{-2x}{15} = 16$$

Step 4. Cross multiply

$$-2x = 16 \times 15$$

$$-2x = 240$$

$$\frac{-2x}{-2} = \frac{240}{-2}$$

$$x = -120$$

Example 3. Solve the equation

$$\frac{1}{3}(p + 7) - 1 = 3^2$$

Solution

Step 1: Expand the bracket

$$\frac{p + 7}{3} - 1 = 3^2$$

Step 2: For the LHS, find the LCM and simplify

$$L.C.M = 3$$

$$\frac{p + 7 - 3}{3} = 3^2$$

$$\frac{p + 4}{3} = 3^2$$

Step 3: Simplify the RHS

$$\frac{p + 4}{3} = 3 \times 3$$

$$\frac{p + 4}{3} = 9$$

Step 4: Cross multiply

$$p + 4 = 9 \times 3$$

$$p + 4 = 27$$

$$p = 27 - 4$$

$$p = 23$$

Example 4. Solve for z in the equation below

$$\frac{1}{5}(7z + 3) = \sqrt{\frac{1}{9}}$$

Solution

Step 1. For the LHS, simplify the brackets

$$\frac{7z + 3}{5} = \sqrt{\frac{1}{9}}$$

Step 2. Simplify the RHS

$$\frac{7z + 3}{5} = \frac{\sqrt{1}}{\sqrt{9}}$$

But $\sqrt{1} = 1$ and $\sqrt{9} = 3$

$$\frac{7z + 3}{5} = \frac{1}{3}$$

Step 3: Cross multiply

$$3(7z + 3) = 5(1)$$

Step 4: Expand the bracket

$$21z + 9 = 5$$

Step 5: Collect like terms

$$21z = 5 - 9$$

$$21z = -4$$

Step 6: Divide both sides by the coefficient of z

$$\frac{21z}{21} = \frac{-4}{21}$$
$$z = -\frac{4}{21}$$

Example 2. Solve for x in

$$\frac{\sqrt{x}}{4} + \frac{2\sqrt{x}}{6} = 2$$

Solution

Step 1: Find the LCM of 4 and 6

L. C. M of 4 and 6 is 12

Step 2: For the LHS, simplify using concepts of fraction

$$\frac{3\sqrt{x} + 4\sqrt{x}}{12} = 2$$

$$\frac{7\sqrt{x}}{12} = 2$$

Step 3: Cross multiply

$$7\sqrt{x} = 12(2)$$

$$7\sqrt{x} = 24$$

Step 4: Divide both sides by the coefficient of $\sqrt{x}$

$$\frac{7\sqrt{x}}{7} = \frac{24}{7}$$

$$\sqrt{x} = \frac{24}{7}$$

Step 5: Squaring both sides to remove the square root

$$(\sqrt{x})^2 = \left(\frac{24}{7}\right)^2$$

$$x = \frac{24 \times 24}{7 \times 7}$$

$$x = \frac{576}{49} = 11\frac{37}{49}$$

Example 3. Find y in terms of x in

$$\frac{4xy}{3} - \frac{5}{9} = xy$$

Solution

Step 1: For the LHS, find the LCM of 3 and 9

LCM of 3 and 9 is 9

Step 2: Simplify the fraction

$$\frac{12xy - 5}{9} = xy$$

Step 3: Cross multiply

$$12xy - 5 = 9(xy)$$

$$12xy - 5 = 9xy$$

Step 4: Collect like terms

$$12xy - 9xy = 5$$

$$3xy = 5$$

Step 5: Divide both sides by $3x$

$$\frac{3xy}{3x} = \frac{5}{3x}$$

$$y = \frac{5}{3x}$$

Example 4. Solve for x in

$$\frac{2x\sqrt{4}}{4} - \frac{x\sqrt{16}}{2} = \sqrt{25}$$

Solution

Step 1. Simplify the square roots

$$\frac{2x(2)}{4} - \frac{x(4)}{2} = 5$$

$$\frac{4x}{4} - \frac{4x}{2} = 5$$

Step 2. Find the LCM of 4 and 2

L. C. M of 4 and 2 is 4

Step 3. Simplify the fraction

$$\frac{4x - 8x}{4} = 5$$

$$\frac{-4x}{4} = 5$$

Step 4: Cross multiply

$$-4x = 4(5)$$

$$-4x = 20$$

Step 5: Divide both sides by the coefficient of x

$$\frac{-4x}{-4} = \frac{20}{-4}$$

$$x = -5$$

Example 5. Solve for x in

$$\frac{3x^{\frac{3}{2}}}{6} + \frac{7}{9} = 4x^{\frac{3}{2}}$$

[Hint: Leaving your answer in root form]

Solution

Step 1: Find the LCM of 6 and 9

LCM of 6 and 9 is 18

Step 2: Simplify the fraction

$$\frac{3(3x^{\frac{3}{2}}) + 2(7)}{18} = 4x^{\frac{3}{2}}$$

Step 3: Expand the brackets

$$\frac{9x^{\frac{3}{2}} - 14}{18} = 4x^{\frac{3}{2}}$$

Step 4: Cross multiply

$$9x^{\frac{3}{2}} + 14 = 18(4x^{\frac{3}{2}})$$

$$9x^{\frac{3}{2}} + 14 = 72x^{\frac{3}{2}}$$

Step 5: Collect like terms

$$9x^{\frac{3}{2}} - 72x^{\frac{3}{2}} = -14$$

$$-63x^{\frac{3}{2}} = -14$$

Step 6: Divide both sides by the coefficient of $x^{\frac{3}{2}}$

$$\frac{-63x^{\frac{3}{2}}}{-63} = \frac{-14}{-63}$$

$$x^{\frac{3}{2}} = \frac{14}{63}$$

$$x^{\frac{3}{2}} = \frac{2}{9}$$

Step 7: Convert the fractional index into root

$$(\sqrt{x})^3 = \frac{2}{9}$$

Step 8: Squaring both sides to remove the square root

$$(\sqrt{x^3})^2 = \left(\frac{2}{9}\right)^2$$

$$x^3 = \frac{2 \times 2}{9 \times 9}$$

$$x^3 = \frac{4}{81}$$

Step 9: Taking the cube root of both sides to get rid of the cube

$$(\sqrt[3]{x})^3 = \sqrt[3]{\frac{4}{81}}$$

$$x = \sqrt[3]{\frac{4}{81}}$$

Example 6. Solve for e in

$$\frac{5e}{8} - \frac{4e}{2} = 4$$

Solution

Step 1: LCM of 8 and 2 is 8

Step 2: Simplify

$$\frac{5e - 4(4e)}{8} = 4$$

Step 3: Expand the bracket and cross multiply

$$5e - 16e = 4(8)$$

$$-11e = 32$$

Step 4: Divide both sides by the coefficient of e

$$\frac{-11e}{-11} = \frac{32}{-11}$$

$$e = -\frac{32}{11} = -2\frac{10}{11}$$

SILLAH KANU SERIES @ 2022

SECTION 2: SIMULTANEOUS EQUATIONS

These are equations which occur at the same time and such equations contain the same unknown quantities. In this text, we are going to use two methods that are used in solving simultaneous equations which are:

- 1. Elimination method*
- 2. Substitution method*
- 3. Graphical Method*

Subsection 1: Using the elimination method

Example 1. Solve the simultaneous equations below

$$3m + 4n = 11 \dots \text{equation(1)}$$

$$m + 7n = 15 \dots \text{equation(2)}$$

Solution

To find n

Step 1. Multiply the coefficient of m in equation 2 by equation 1 to form equation 3

$$1 \times (3m + 4n = 11)$$

$$3m + 4n = 11 \dots \text{equation 3}$$

Step 2. Multiply the coefficient of m in equation 1 by equation 2 to form equation 4

$$3 \times (m + 7n = 15)$$

$$3m + 21n = 45 \dots \text{equation 4}$$

Step 3. Equation 3 – equation 4

$$3m + 4n = 11$$

$$-(3m + 21n = 45)$$

$$3m - 3m + 4n - 21n = 11 - 45$$

$$0 - 17n = -34$$

$$\frac{-17n}{-17} = \frac{-34}{-17}$$

$$n = 2$$

To find m

Step 1. Multiply the coefficient of n in equation 2 by equation 1 to form equation 3

$$7 \times (3m + 4n = 11)$$

$$21m + 28n = 77 \dots \text{equation 3}$$

Step 2. Multiply the coefficient of n in equation 1 by equation 2 to form equation 4

$$4 \times (m + 7n = 15)$$

$$4m + 28n = 60 \dots \text{equation 4}$$

Step 3. Equation 3 – equation 4

$$21m + 28n = 77$$

$$-(4m + 28n = 60)$$

$$21m - 4m + 28n - 28n = 77 - 60$$

$$17m - 0 = 17$$

$$17m = 17$$

$$\frac{17m}{17} = \frac{17}{17}$$

$$m = 1$$

Hence, $m = 1$ and $n = 2$

Example 2. Solve the simultaneous equations

$$3a - b = 5$$

$$a + b = 3 \text{ BECE JUNE 2005}$$

Solution

To find b

Step 1. Multiply the coefficient of a in equation 2 by equation 1 to form equation 3

$$1 \times (3a - b = 5)$$

$$3a - b = 5 \dots \text{equation 3}$$

Step 2. Multiply the coefficient of a in equation 1 by equation 2 to form equation 4

$$3 \times (a + b = 3)$$

$$3a + 3b = 9 \dots \text{equation 4}$$

Step 3. Equation 3 – equation 4

$$3a - b = 5$$

$$\underline{-(3a + 3b = 9)}$$

$$3a - 3a - b - 3b = 5 - 9$$

$$0 - 4b = -4$$

$$\frac{-4b}{-4} = \frac{-4}{-4}$$

$$b = 1$$

To find a

Step 1. Multiply the coefficient of b in equation(2) by equation(1) to form equation 3

$$1 \times (3a - b = 5)$$

$$3a - b = 5 \dots \text{equation 3}$$

Step 2. Multiply the coefficient of b in equation(1) by equation(2) to form equation(4)

$$-1 \times (a + b = 3)$$

$$-a - b = -3 \dots \text{equation(4)}$$

Step 3. Equation(3) – equation(4)

$$3a - b = 5$$

$$-(-a - b = -3)$$

$$3a + a - b + b = 5 + 3$$

$$4a - 0 = 8$$

$$4a = 8$$

$$\frac{4a}{4} = \frac{8}{4}$$

$$a = 2$$

$$\text{Hence, } a = 2, b = 1$$

Example 3. Solve the equations

$$b - c = 14$$

$$c = 6 - b$$

Solution

Step 1. Arrange equation 2 properly by transposing

$$b - c = 14 \dots \text{equation 1}$$

$$b + c = 6 \dots \text{equation 2}$$

Step 2. Equation 1 – equation 2

$$b - c = 14$$

$$\underline{-(b + c = 6)}$$

$$b - b - c - c = 14 - 6$$

$$-2c = 8$$

$$\frac{-2c}{-2} = \frac{8}{-2}$$

$$c = -4$$

Step 3. Equation 1 + equation 2

$$b - c = 14$$

$$\underline{+(b + c = 6)}$$

$$b + b - c + c = 14 + 6$$

$$2b = 20$$

$$\frac{2b}{2} = \frac{20}{2}$$

$$b = 10$$

Hence, $b = 10$ and $c = -4$

Example 4. Solve the simultaneous equation

$$x - y = 1 \dots \text{equation 1}$$

$$x + 2y = 7 \dots \text{equation 2} \quad \text{BECE 1997}$$

Solution

To find y

Step 1. Equation 1 – equation 2

$$x - y = 1$$

$$\underline{-(x + 2y = 7)}$$

$$x - x - y - 2y = 1 - 7$$

$$-3y = -6$$

$$\frac{-3y}{-3} = \frac{-6}{-3}$$

$$y = 2$$

To find x

Step 1:

$$2x - 2y = 2 \dots \text{equation 3}$$

$$-x - 2y = -7 \dots \text{equation 4}$$

Step 2. Equation 3 – equation 4

$$2x - 2y = 2$$

$$\underline{-(-x - 2y = -7)}$$

$$2x + x - 2y + 2y = 2 + 7$$

$$3x = 9$$

$$\frac{3x}{3} = \frac{9}{3}$$

$$x = 3$$

Hence, $x = 3$ and $y = 2$

Subsection 2: Using substitution method

Example 1. Solve the simultaneous equations

$$x - y = 1$$

$$x + 2y = 7 \quad \text{BECE 1997}$$

Solution

Step 1. Make x the subject in equation 1

$$x = 1 + y$$

Step 2. Bringing equation 2 and substitute for x

$$x + 2y = 7 \dots \text{equation 2}$$

$$(1 + y) + 2y = 7$$

$$1 + y + 2y = 7$$

$$3y = 7 - 1$$

$$3y = 6$$

$$\frac{3y}{3} = \frac{6}{3}$$

$$y = 2$$

Step 3. Substitute for y into equation 1

$$x - y = 1$$

$$x - 2 = 1$$

$$x = 3$$

Hence, $x = 3$ and $y = 2$

Example 2. Solve the simultaneous equations

$$3x + 5y = 250$$

$$5x + 3y = 270$$

Solution

Step 1: Make x the subject in equation 1

$$3x = 250 - 5y$$

$$\frac{3x}{3} = \frac{250 - 5y}{3}$$

$$x = \frac{250 - 5y}{3}$$

Step 2: Substitute for x into equation 2

$$5x + 3y = 270$$

$$5\left(\frac{250 - 5y}{3}\right) + 3y = 270$$

Step 3: Expand the bracket

$$\frac{1250 - 25y}{3} + \frac{3y}{1} = 270$$

$$\frac{1250 - 25y + 9y}{3} = 270$$

Step 4: Cross multiply

$$1250 - 16y = 810$$

$$-16y = 810 - 1250$$

$$-16y = -440$$

$$\frac{-16y}{-16} = \frac{-440}{-16}$$

$$y = 27.5$$

Step 5: Substitute for y into equation 1

$$3x + 5y = 250$$

$$3x + 5(27.5) = 250$$

$$3x + 137.5 = 250$$

$$3x = 250 - 137.5$$

$$3x = 112.5$$

$$\frac{3x}{3} = \frac{112.5}{3}$$

$$x = 37.5$$

Hence, $x = 37.5$ and $y = 27.5$

Example 3. Solve the equation

$$x + y = 9$$

$$4x - y = 1$$

Solution

Step 1: Make x the subject in equation 1

$$x = 9 - y$$

Step 2: Substitute for x into equation 2

$$4x - y = 1 \dots \text{equation 1}$$

$$4(9 - y) - y = 1$$

Step 3: Expand the bracket

$$36 - 4y - y = 1$$

Step 4: Collect like terms

$$-4y - y = 1 - 36$$

$$-5y = -35$$

Step 5: Divide both sides by the coefficient of y

$$\frac{-5y}{-5} = \frac{-35}{-5}$$

$$y = 7$$

Step 6. Substitute for y into equation 1

$$x + y = 9$$

$$x + 7 = 9$$

$$x = 9 - 7$$

$$x = 2$$

Hence, $x = 2$ and $y = 7$

Example 4. Solve the equations

$$b - c = 14$$

$$c = 6 - b$$

Solution

Step 1: Rearrange equation 2

$$b - c = 14 \dots \text{equation 1}$$

$$b + c = 6 \dots \text{equation 2}$$

Step 2: Make b the subject in equation 1

$$b = 14 + c$$

Step 3: Substitute for b into equation 2

$$b + c = 6 \dots \text{equation 2}$$

$$(14 + c) + c = 6$$

Step 4: Expand the bracket

$$14 + c + c = 6$$

$$2c = 6 - 14$$

$$2c = -8$$

Step 5: Divide both sides by the coefficient of c

$$\frac{2c}{2} = \frac{-8}{2}$$

$$c = -4$$

Step 6: Substitute for c into equation 2

$$b - c = 14 \dots \text{equation 2}$$

$$b - (-4) = 14$$

$$b + 4 = 14$$

$$b = 14 - 4$$

$$b = 10$$

Hence, $b = 10$ and $c = -4$

Example 5. Find x and y in the equation below

$$\sqrt{x} - \sqrt{y} = 1 \dots \text{equation 1}$$

$$\sqrt{x} + 2\sqrt{y} = 7 \dots \text{equation 2}$$

Solution

Step 1: Make $\sqrt{x}$ the subject in equation 2

$$\sqrt{x} = 1 + \sqrt{y}$$

Step 2: Using equation 2 and substitute for x

$$\sqrt{x} + 2\sqrt{y} = 7 \dots \text{equation 2}$$

$$(1 + \sqrt{y}) + 2\sqrt{y} = 7$$

$$1 + \sqrt{y} + 2\sqrt{y} = 7$$

$$3\sqrt{y} = 7 - 1$$

$$3\sqrt{y} = 6$$

$$\frac{3\sqrt{y}}{3} = \frac{6}{3}$$

$$\sqrt{y} = 2$$

Step 3: Squaring both sides to get y

$$(\sqrt{y})^2 = 2^2$$

$$y = 4$$

Step 4: Substitute for y into equation 1

$$\sqrt{x} - \sqrt{y} = 1$$

But $y = 4$

$$\sqrt{x} - \sqrt{4} = 1$$

$$\sqrt{x} - 2 = 1$$

$$\sqrt{x} = 1 + 2$$

$$\sqrt{x} = 3$$

Step 5: Squaring both sides to get x

$$(\sqrt{x})^2 = 3^2$$

$$x = 9$$

Hence, $x = 9$ and $y = 4$

Example 6. Solve the simultaneous equations

$$3x^{\frac{3}{2}} + 5y^{\frac{5}{3}} = 250 \dots \text{equation 1}$$

$$5x^{\frac{3}{2}} + 3y^{\frac{5}{3}} = 270 \dots \text{equation 2}$$

Find x^3 and y^5 ?

Solution

Step 1: Make $x^{\frac{3}{2}}$ the subject in equation 1

$$3x^{\frac{3}{2}} = 250 - 5y^{\frac{5}{3}}$$

$$\frac{3x^{\frac{3}{2}}}{3} = \frac{250 - 5y^{\frac{5}{3}}}{3}$$

$$x^{\frac{3}{2}} = \frac{250 - 5y^{\frac{5}{3}}}{3}$$

Step 2: Substitute for $x^{\frac{3}{2}}$ into equation 2

$$5x^{\frac{3}{2}} + 3y^{\frac{5}{3}} = 270$$

$$5\left(\frac{250 - 5y^{\frac{5}{3}}}{3}\right) + 3y^{\frac{5}{3}} = 270$$

Step 3: Expand the bracket

$$\frac{1250 - 25y^{\frac{5}{3}}}{3} + \frac{3y^{\frac{5}{3}}}{1} = 270$$

$$\frac{1250 - 25y^{\frac{5}{3}} + 9y^{\frac{5}{3}}}{3} = 270$$

Step 4: Cross multiply

$$1250 - 16y^{\frac{5}{3}} = 810$$

$$-16y^{\frac{5}{3}} = 810 - 1250$$

$$-16y^{\frac{5}{3}} = -440$$

$$\frac{-16y^{\frac{5}{3}}}{-16} = \frac{-440}{-16}$$

$$y^{\frac{5}{3}} = 27.5$$

Step 5: Substitute for $y^{\frac{5}{3}}$ into equation 1

$$3x^{\frac{3}{2}} + 5y^{\frac{5}{3}} = 250$$

$$3x^{\frac{3}{2}} + 5(27.5) = 250$$

$$3x^{\frac{3}{2}} + 137.5 = 250$$

$$3x^{\frac{3}{2}} = 250 - 137.5$$

$$3x^{\frac{3}{2}} = 112.5$$

$$\frac{3x^{\frac{3}{2}}}{3} = \frac{112.5}{3}$$

$$x^{\frac{3}{2}} = 37.5$$

$$(\sqrt{x})^3 = 37.5$$

Step 6: Squaring both sides to remove the square root

$$[(\sqrt{x})^3]^2 = (37.5)^2$$

$$x^3 = 37.5 \times 37.5$$

$$x^3 = 1406.25$$

Step 7. Find y^5

$$\text{Since } y^{\frac{5}{3}} = 27.5$$

Step 8: convert the fractional index into root

$$(\sqrt[3]{y})^5 = 27.5$$

Step 9: Raising both sides to power three(3) to remove the cube root

$$[(\sqrt[3]{y})^5]^3 = (27.5)^3$$

$$y^5 = 27.5 \times 27.5 \times 27.5$$

$$y^5 = 20796.875$$

$$\text{Hence, } x^3 = 1406.25 \text{ and } y^5 = 20796.875$$

Example 7. Solve the equation

$$\left[\frac{m}{n}\right] + \left[\frac{y}{x}\right] = 9 \dots \text{equation 1}$$

$$4\left[\frac{m}{n}\right] - \left[\frac{y}{x}\right] = 1 \dots \text{equation 2}$$

Find m in terms of n and x in terms of y .

Solution

Step 1. Make $\left[\frac{m}{n}\right]$ the subject in equation 1

$$\left[\frac{m}{n}\right] = 9 - \left[\frac{y}{x}\right]$$

Step 2: Substitute for $\left[\frac{m}{n}\right]$ into equation 2

$$4\left[\frac{m}{n}\right] - \left[\frac{y}{x}\right] = 1 \dots \text{equation 2}$$

$$4\left\{9 - \left[\frac{y}{x}\right]\right\} - \left[\frac{y}{x}\right] = 1$$

Step 3: Expand the bracket

$$36 - 4\left[\frac{y}{x}\right] - \left[\frac{y}{x}\right] = 1$$

Step 4: Collect like terms

$$-4\left[\frac{y}{x}\right] - \left[\frac{y}{x}\right] = 1 - 36$$

$$-5\left[\frac{y}{x}\right] = -35$$

Step 5: Divide both sides by the coefficient of $\left[\frac{y}{x}\right]$

$$\frac{-5\left[\frac{y}{x}\right]}{-5} = \frac{-35}{-5}$$

$$\left[\frac{y}{x}\right] = 7$$

Step 6. Substitute for $\left[\frac{y}{x}\right]$ into equation 1

$$\left[\frac{m}{n}\right] + \left[\frac{y}{x}\right] = 9$$

$$\left[\frac{m}{n}\right] + 7 = 9$$

$$\left[\frac{m}{n}\right] = 9 - 7$$

$$\left[\frac{m}{n}\right] = 2$$

Step 7. Find m in terms of n

Make m the subject

$$m = 2n$$

Step 8. Find x in terms of y

Make x the subject

$$\left[\frac{y}{x}\right] = 7$$

$$x = \frac{y}{7}$$

Hence, $x = \frac{y}{7}$ and $m = 2n$

Example 8. Solve the equation

$$xy - ms = 14 \dots \text{equation 1}$$

$$xy - 6 = -ms \dots \text{equation 2}$$

Find x in terms of y and s in terms of m

Solution

Step 1: re – arrange equation 2

$$xy - ms = 14 \dots \text{equation 1}$$

$$xy + ms = 6 \dots \text{equation 2}$$

Step 2: Make xy the subject in equation 1.

$$xy = 14 + ms$$

Step 3: Substitute for xy into equation 2

$$xy + ms = 6 \dots \text{equation 2}$$

$$(14 + ms) + ms = 6$$

Step 4: Expand the bracket

$$14 + ms + ms = 6$$

$$2ms = 6 - 14$$

$$2ms = -8$$

Step 5: Divide both sides by the coefficient of ms

$$\frac{2ms}{2} = \frac{-8}{2}$$

$$ms = -4$$

Step 6: Substitute for ms into equation 2

$$xy - ms = 14 \dots \text{equation 2}$$

$$xy - (-4) = 14$$

$$xy + 4 = 14$$

$$xy = 14 - 4$$

$$xy = 10$$

$$xy = 10$$

But x in terms of y is:

$$x = \frac{10}{y}$$

s in terms of m is:

$$s = -\frac{4}{m}$$

$$\text{Hence, } x = \frac{10}{y} \text{ and } s = -\frac{4}{m}$$

SECTION 3: EQUATION INVOLVING WORD PROBLEMS

Example 1. *The product of a certain number and 6 is equal to twice the number subtracted from 40. Find the number?*

Solution

Step 1. *Let the number be m*

The product of m and 6 means:

$$6 \times m = 6m \dots \text{part one(1)}$$

Twice m subtracted from 40 means:

$$40 - 2m \dots \text{part two(2)}$$

Step 2. *Equate part one(1) and two(2)*

$$6m = 40 - 2m$$

Step 3: *Collect like terms*

$$6m + 2m = 40$$

$$8m = 40$$

Step 4: *Divide both sides by the coefficient of m*

$$\frac{8m}{8} = \frac{40}{8}$$

$$m = 5$$

Therefore, the number is 8

Example 2. *The sum of 35 and a certain number is divided by 4. The result is equal to double the number. Find the number?*

Solution

Step 1. *Let the number be r*

The sum of 35 and r means:

$$35 + r$$

The sum divided by 4 means:

$$\frac{35 + r}{4} \dots 1^{\text{st}} \text{ sentence}$$

Double the number means:

$$2 \times r = 2r \dots 2^{\text{nd}} \text{ sentence}$$

Step 2. Equate the result of the 1st and 2nd sentence

$$\frac{35 + r}{4} = 2r$$

Step 3: Cross multiply

$$35 + r = 8r$$

Step 4: Collect like terms

$$35 = 8r - r$$

$$35 = 7r$$

$$\frac{7r}{7} = \frac{35}{7}$$

$$r = 5$$

Therefore, the number is 5

Example 3. The sum of 9 and a certain number is one and a half times the original number. Find the number?

Solution

Step 1. Let the number be p

The sum of 9 and a certain number means:

$$9 + p \dots \text{part one (1)}$$

One and a half time the original number means:

$$1\frac{1}{2} \times p = 1\frac{1}{2}p \dots \text{part two(2)}$$

Step 2. Equate part one(1) and part two(2)

$$p + 9 = 1\frac{1}{2}p$$

$$p + 9 = \frac{3}{2}p$$

Step 3: Cross multiply

$$2(p + 9) = 3p$$

$$2p + 18 = 3p$$

Step 4: Collect like terms

$$3p - 2p = 18$$

$$p = 18$$

Example 4. Four pens and six rulers cost Le 136.00. Six pens and five rulers cost Le 164.00. Find the cost of a pen and a ruler?

Solution

Step 1: Let p be the cost of pens and r be the cost of rulers.

Step 2: The 1st sentence means

$$4p + 6r = 136 \dots \text{equation 1}$$

The 2nd sentence means:

$$6p + 5r = 164 \dots \text{equation 2}$$

Step 3: Solve equation 1 and 2 simultaneously

$$24p + 36r = 816 \dots \text{equation 3}$$

$$-(24p + 20r = 656) \dots \text{equation 4}$$

$$24p - 24p + 36r - 20r = 816 - 656$$

$$16r = 160$$

Step 4: Divide both sides by the coefficient of r

$$\frac{16r}{16} = \frac{160}{16}$$

$$r = 10$$

Step 5: Substitute for r into equation 3

$$24p + 36r = 816$$

$$24p + 36(10) = 816$$

$$24p + 360 = 816$$

$$24p = 816 - 360$$

$$24p = 456$$

$$\frac{24p}{24} = \frac{456}{24}$$

$$p = 19$$

Example 5. Mohamed Sillah Kanu and Albert Memunatu Kabbasei are m and n year old respectively. The sum of their ages is 44. Sillah is eight years older than Madallo. How old is each of them?

Solution

Step 1:

Sillah = m years old

Madallo = n years old

2nd sentence means:

$$m + n = 44 \dots \text{equation 1}$$

3rd sentence means:

$$m - n = 8 \dots \text{equation 2}$$

Step 2: Solve equation 1 and 2 simultaneously

$$m + n = 44$$

$$\underline{-(m - n = 8)}$$

$$m - m + n - (-n) = 44 - 8$$

$$n + n = 36$$

$$2n = 36$$

$$\frac{2n}{2} = \frac{36}{2}$$

$$n = 18$$

Step 3. Substitute for n into equation 1

$$m + n = 44$$

$$m + 18 = 44$$

$$m = 44 - 18$$

$$m = 26$$

Hence, Sillah is 26 years old and Madallo is 18 years.

Example 6. Manager Rahdwan Hamoud gave Johannes Kande and Maya Sheik Kaikai €1,418.00. If Maya received €346.00 more than Johannes. How many did each receive?

Solution

Step 1. Let share of Johannes be j and that of Maya be m

1st sentence means:

$$m + j = 1418 \dots \text{equation(1)}$$

2nd sentence means:

$$m - j = 346 \dots \text{equation(2)}$$

Step 2. Solving simultaneously

$$m + j = 1418$$

$$-(m - j = 346)$$

$$m - m + j - (-j) = 1418 - 346$$

$$j + j = 1072$$

$$2j = 1072$$

Step 3: Divide both sides by the coefficient of j

$$\frac{2j}{2} = \frac{1072}{2}$$

$$j = 536$$

Step 4: Substitute j into equation(1)

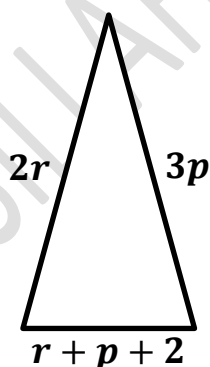
$$m + j = 1418$$

$$m = 1418 - 536$$

$$m = 882$$

Hence, Johannes received €536 and Maya received €882

Example 7. The triangle below is equilateral. Find r and p?



Solution

Step 1. In "equilateral triangle" all the sides are equal.

$$2r = 3p \dots \text{equation 1}$$

$$3p = r + p + 2 \dots \text{equation 2}$$

$$2r = r + p + 2 \dots \text{equation 3}$$

Step 2. From equation 1 make r the subject

$$r = \frac{3p}{2}$$

Step 3. Substitute for r into equation 2

$$3p = \frac{3p}{2} + p + 2$$

Step 4: Multiply throughout by the LCM

$$2(3p) = 3p + 2(p) + 2(2)$$

$$6p = 3p + 2p + 4$$

Step 5: Collect like terms

$$6p - 5p = 4$$

$$p = 4$$

Step 6: Substitute for p into equation 1

$$3p = r + p + 2 \dots \text{equation 1}$$

$$3(4) = r + 4 + 2$$

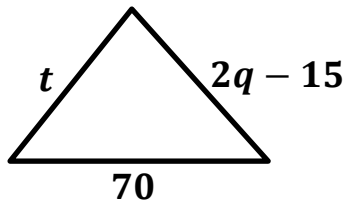
$$12 = r + 6$$

$$r = 12 - 6$$

$$r = 6$$

Hence, $r = 6$ and $p = 4$

Example 8. *The triangle below is isosceles. It has perimeter of 200cm.
Find t and q ?*



Solution

Step 1. In "isosceles triangle" two opposite sides are equal.

$$t = 2q - 15 \dots \text{equation 1}$$

Step 2. Perimeter means sum of all the sides.

$$t + 2q - 15 + 70 = 200 \dots \text{equation 2}$$

Step 3. Solving equation 1 and 2 simultaneously

$$t - 2q = -15$$

$$+(t + 2q = 145)$$

$$t + t - 2q + 2q = -15 + 145$$

$$2t = 130$$

$$\frac{2t}{2} = \frac{130}{2}$$

$$t = 65$$

Step 4. Substitute for t into equation 1

$$t - 2q = -15$$

$$65 - 2q = -15$$

$$-2q = -15 - 65$$

$$-2q = -80$$

$$\frac{-2q}{-2} = \frac{-80}{-2}$$

$$q = 40$$

Hence, $t = 65$ and $q = 40$

Example 9. The line $y = mx + c$ passes through points $(4, 10)$ and $(7, 19)$. Find the values of m and c ?

Solution

Step 1. Substitute the points $(4, 10)$ into the equation

$$y = mx + c$$

$$\text{But } x = 4, y = 10$$

$$10 = m(4) + c$$

$$10 = 4m + c \dots \text{equation 1}$$

Step 2. Substitute the points $(7, 19)$ into the equation

$$y = mx + c$$

$$\text{But } x = 7, y = 19$$

$$19 = m(7) + c$$

$$7m + c = 19 \dots \text{equation 2}$$

Step 3: Solve equation 1 and 2 simultaneously

$$4m + c = 10$$

$$-(7m + c = 19)$$

$$4m - 7m + c - c = 10 - 19$$

$$-3m = -9$$

$$\frac{-3m}{-3} = \frac{-9}{-3}$$

$$m = 3$$

Step 4: Substitute for c into equation 1

$$4m + c = 10 \dots \text{equation 1}$$

$$4(3) + c = 10$$

$$12 + c = 10$$

$$c = 10 - 12$$

$$c = -2$$

Hence, $c = -2$ and $m = 4$

SECTION 4: QUADRATIC EQUATIONS

Quadratic equations appear in the form:

$$ax^2 + bx + c = 0$$

Where bx is the sum and acx^2 is the product

ILLUSTRATIVE EXAMPLES

Example 1. *Solve the equation $x^2 - 11x + 30 = 0$*

Solution

*Step 1: find two numbers which when multiplied gives ' $30x^2$ '
and when added gives ' $-11x$ '*

Product: $6x \times -5x = +30x^2$

Sum : $-6x - 5x = -11x$

Step 2: replacing the middle term ($-11x$) by $(-6x - 5x)$ gives

$$x^2 - 6x - 5x + 30 = 0$$

Step 3: put the 1st two terms inside a bracket and the last two as well.

$$(x^2 - 6x)(-5x + 30) = 0$$

Step 4: factorise

$$x(x - 6) - 5(x - 6) = 0$$

Step 5: factorise again

$$(x - 5)(x - 6) = 0$$

Step 6: Equate each bracket to zero

(i). $x - 5 = 0$

$$x = 5$$

(ii). $x - 6 = 0$

$$x = 6$$

Hence, $x = 5$ or 6

Example 2. Solve the equation $d^2 + 9d + 20 = 0$

Solution

Step 1: find two numbers which when multiplied gives $+20d^2$ and when added gives $+9d$

Product: $+5d (\times) + 4d = +20d^2$

Sum: $+4d + 5d = +9d$

Step 2: replace the middle term $(+9d)$ by $(+4d + 5d)$

$$d^2 + 5d + 4d + 20 = 0$$

Step 3: put the 1st two terms inside one bracket and the last two as well

$$(d^2 + 5d)(+4d + 20) = 0$$

Step 4: factorise

$$d(d + 5) + 4(d + 5) = 0$$

Step 5: factorise again

$$(d + 4)(d + 5) = 0$$

Step 6: Equate each bracket to zero

$$(i). d + 5 = 0$$

$$d = -5$$

$$(ii). d - 4 = 0$$

$$d = 4$$

Hence, $d = 4$ or -5

Example 3. Solve the equation $p^2 - 9p + 14 = 0$

Solution

Step 1: find two numbers which when multiplied gives $+14p^2$ and when added gives $-9p$

Product of: $-7p \times -2p = +14p^2$

Sum of: $-7p - 2p = -9p$

Step 2: replace the middle term $(-9p)$ by $(-7p - 2p)$

$$p^2 - 7p - 2p + 14 = 0$$

Step 3: put the 1st two terms inside a bracket and the last two as well

$$(p^2 - 7p)(-2p + 14) = 0$$

Step 4: factorise

$$p(p - 7) - 2(p - 7) = 0$$

Step 5: factorise again

$$(p - 7)(p - 2) = 0$$

Step 6: Equate each bracket to zero

(i). $p - 7 = 0$

$$p = 7$$

(ii). $p - 2 = 0$

$$p = 2$$

Hence, $p = 2$ or 7

Example 4. Solve the equation $k^2 - 14k + 48 = 0$

Solution

Step 1: find two numbers which when multiplied gives $+48k^2$
and when added gives $-14k$

Product of: $-6k(\times) - 8k = +48k^2$

Sum of: $-6k - 8k = -14k$

Step 2 : replace the middle term $(-14k)$ by $(-6k - 8k)$

$$k^2 - 6k - 8k + 48 = 0$$

Step 3: put the 1st two terms inside a bracket and the last two as well

$$(k^2 - 6k)(-8k + 48) = 0$$

Step 4: factorise

$$k(k - 6) - 8(k - 6) = 0$$

Step 5: factorise again

$$(k - 8)(k - 6) = 0$$

Step 6: Equate each bracket to zero

$$(i). k - 8 = 0$$

$$k = 8$$

$$(ii). k - 6 = 0$$

$$k = 6$$

Hence, $k = 8$ or 6

SECTION 5: SKS PAPER 13

Answer all questions

Part 1

Question 1. Find the value of m in the equation below

$$m + 4 = 1$$

Answer [$m = -3$]

Question 2. Solve for k in the equation below

$$k^{\frac{1}{2}} - 10 = 6$$

Answer [$k = 256$]

Question 3. Solve for p in the equation below

$$p^{\frac{3}{2}} - 8 = 17. \text{ Leaving your answer in root form}$$

Answer [$p = \sqrt[3]{625}$]

Question 4. Solve for a in the equation below?

$$2\sqrt{a} - 8 = 4$$

Answer [$a = 36$]

Question 5. Find x in terms of y in

$$5xy - 6 = 9$$

Answer [$x = \frac{3}{y}$]

Question 6. Find the value of m in the equation below

$$7m + 16 = 25 - 2m$$

Answer [$m = 1$]

Question 7. Solve for c in the equation below

$$4(c + 1) = 10$$

$$\text{Answer} \left[c = \frac{3}{2} \right]$$

Question 8. Solve for k in the equation below

$$13k + 1 = 2(2k + 2)$$

$$\text{Answer} \left[k = \frac{1}{3} \right]$$

Question 9. Solve for j in the equation below

$$3(j - 5) = 7 - 5(3 - 2j)$$

$$\text{Answer} [j = -1]$$

Part 2

Question 1. Solve for r in the equation below?

$$\frac{r + 3}{4} + \frac{r + 4}{5} = 3$$

$$\text{Answer} \left[r = 3\frac{2}{9} \right]$$

Question 2. Solve for t in the equation below?

$$\frac{t}{3} - \frac{t}{6} = (3\sqrt{16})^2$$

$$\text{Answer} [t = 864]$$

Question 3. Find the value of e in the equation below

$$\frac{1}{4}(e + 7) - 2 = 4^2$$

$$\text{Answer} [e = 65]$$

Question 4. Find the value of z the equation below?

$$\frac{1}{6}(7z + 3) = \sqrt{\frac{4}{25}}$$

$$\text{Answer} \left[z = -\frac{3}{35} \right]$$

Question 5. find the value of g in the equation below?

$$\frac{1}{4}(g + 7) - 1 = 4^2$$

$$\text{Answer} [g = 61]$$

Question 6. Solve for a in the equation below?

$$4\sqrt{(a - 5)} = 8 - 6\sqrt{(a - 5)} + 2$$

$$\text{Answer} [a = 6]$$

Question 7. Solve for k in the equation below?

$$\frac{\sqrt{k}}{3} + \frac{2\sqrt{k}}{4} = 2$$

$$\text{Answer} \left[k = 5\frac{19}{25} \right]$$

Question 8. Solve for x in the equation below

$$\frac{4x^{\frac{3}{2}}}{3} - \frac{3}{2} = x^{\frac{3}{2}}. \text{Leaving your answer in root form}$$

$$\text{Answer} \left[x = \sqrt[3]{\frac{81}{4}} \right]$$

Question 9. Find x in terms of y in the equation below?

$$\frac{5xy}{3} - \frac{7}{4} = xy$$

Answer $\left[x = \frac{21}{8y} \right]$

Question 10. Solve for e in the equation below

$$\frac{6e}{4} - \frac{4e}{3} = 3$$

Answer $[e = 18]$

Part 3

Question 1. Solve the simultaneous equations below

$$2m + 6n = 14 \dots \text{equation(1)}$$

$$m + 7n = 11 \dots \text{equation(2)}$$

Answer $[m = 4 \text{ and } n = 1]$

Question 2. Solve the equations

$$b - c = 14 \dots \dots \text{equation(1)}$$

$$c = 6 - b \dots \dots \text{equation(2)}$$

Answer $[b = 10 \text{ and } c = -4]$

Part 4

Question 1. The product of a certain number and 6 is equal to thrice the number subtracted from 45. Find the number?

Answer $[\text{The number} = 5]$

Question 2. The sum of 42 and a certain number is divided by 4. The result is equal to double the number. Find the number?

Answer $[\text{The number is } 6]$

Question 3. The sum of 15 and a certain number is two and a half times the original number. Find the number?

Answer[$a = 10$]

Question 4. Four pens and six rulers cost Le 136.00. Six pens and five rulers cost Le 164.00. Find the cost of a pen and a ruler?

Answer[$a = 19$ and $b = 10$]

Question 5. Mohamed Sillah Kanu and Haja Fatmata Madallo Sesay are m and n years old respectively. The sum of their ages is 50. Sillah is ten years older than Madallo. How old is each of them?

Answer[$m = 30$ and $n = 20$]

Question 6. Mr. Musa Sillah Kanu gave Isha and Kadajah Le 1,118.00. If Kadajah received Le 346.00 more than Isha. How many did each receive?

Answer[$m = 732$ and $n = 386$]

Question 9. The line $y = mx + c$ passes through points (5, 11) and (8, 20). Find the values of m and c ?

Answer[$m = 3$ and $c = -4$]

Part 5

Question 1. Solve the equation $k^2 + 14k + 48 = 0$

Answer[$k = -6$ or -8]

Question 2. Solve the equation $p^2 + p - 2 = 0$

Answer [$p = 1$ or -2]

Question 3. Solve the equation $m^2 - 2m - 8 = 0$

Answer[$m = -2$ or 4]

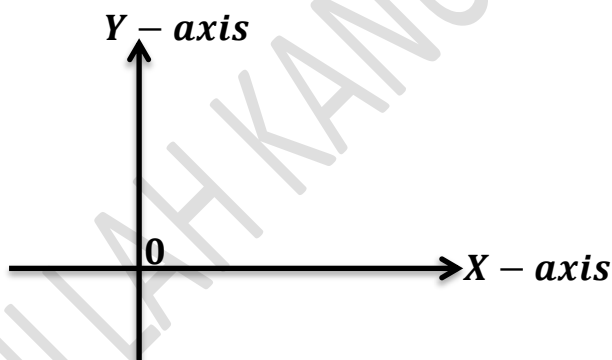
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CHAPTER 14

GRAPH

SECTION 1: INTRODUCTION

The most common forms of diagrams are the graphs. A graph of an algebraic sentence is a picture representing the meaning of that sentence. For graphs connecting two variables, two axes are drawn at right angles to each other to give a Cartesian plane. The horizontal axis is called the X – axis, and the vertical axis is called the Y – axis. The two axes crosses (intersect) at zero – point call the origin (0) of the plane.



CO – ORDINATES: *these are use to mark the points on a graph.*

LINEAR RELATION: *Any relation that can be expressed in the form $y = mx + c$, where m and c are constants and $m \neq 0$, is called a linear relation*

INTERCEPTS: the point at which the linear graph crosses the axes, are called its intercepts.

ILLUSTRATIVE EXAMPLES

Example 1a. Copy and complete the table below of the equation

$$y = 3x - 2$$

x	-3	-2	-1	0	1	2	3
y		-8					7

b). Using a scale of 2cm to 1 unit on the x – axis and 2cm to

2 units on the y – axis, draw the graph of $y = 3x - 2$

for the values between -3 and $+3$

c). find the gradient of the graph $y = 3x - 2$

Solution

i). when $x = -3$

$$y = 3x - 2$$

substitute

$$y = 3(-3) - 2$$

$$y = -9 - 2$$

$$y = -11$$

ii). when $x = -1$

$$y = 3x - 2$$

$$y = 3(-1) - 2$$

$$y = -3 - 2$$

$$y = -5$$

iii) when $x = 0$

$$y = 3x - 2$$

$$y = 3(0) - 2$$

$$y = 0 - 2$$

$$y = -2$$

iv). when $x = 1$

$$y = 3x - 2$$

$$y = 3(1) - 2$$

$$y = 3 - 2$$

$$y = 1$$

v). when $x = 2$

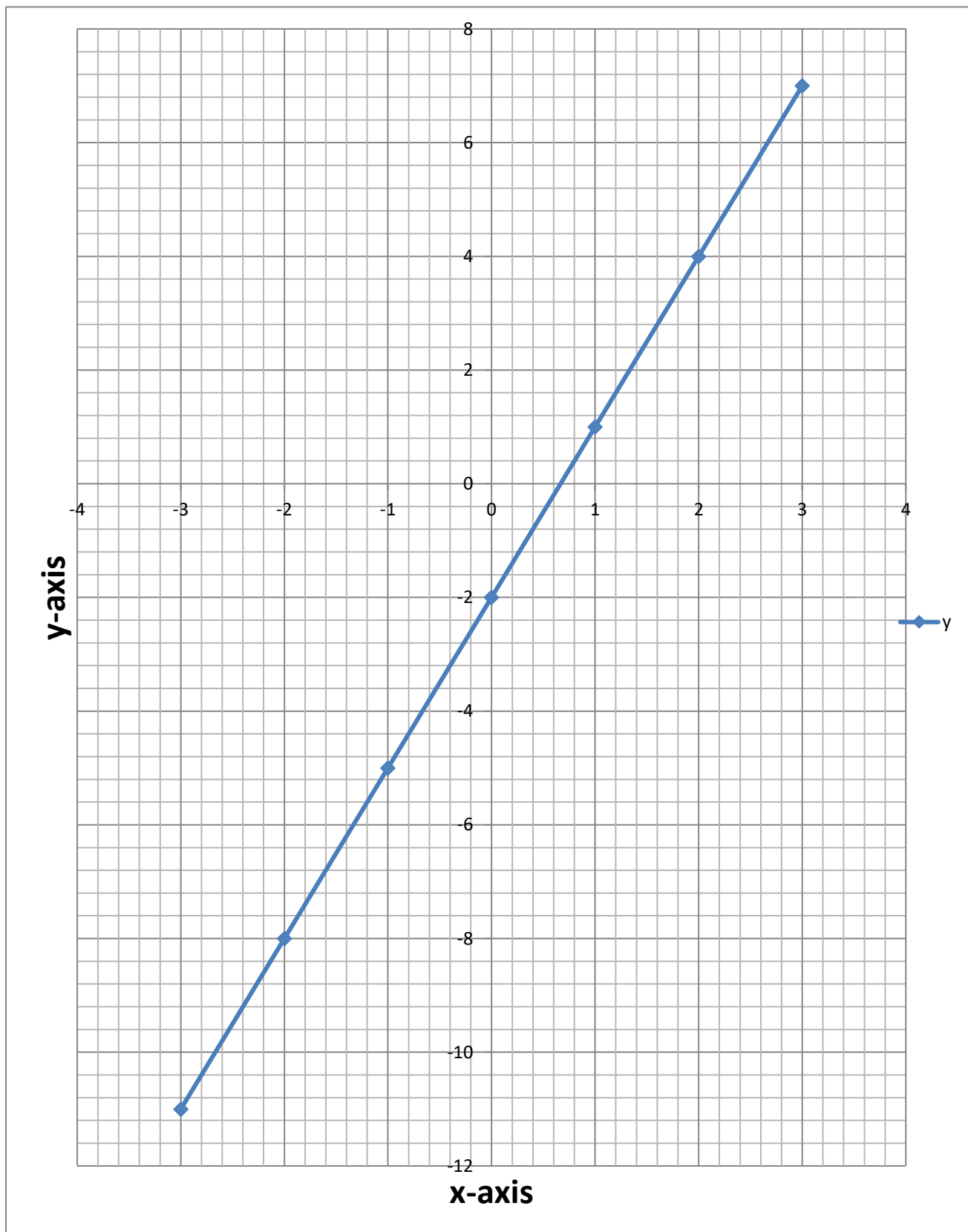
$$y = 3x - 2$$

$$y = 3(2) - 2$$

$$y = 6 - 2$$

$$y = 4$$

x	-3	-2	-1	0	1	2	3
y	-11	-8	-5	-2	1	4	7



C. To determine the gradient of the line, we select two points A(0, -2) and B(2, 4) on the line.

$$\begin{aligned}\text{Gradient} &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{4 - (-2)}{2 - 0} = \frac{4 + 2}{2} = \frac{6}{2} = 3\end{aligned}$$

From the graph, the gradient of the graph is 3.

Example 2a. Copy and complete the table below for the equation $y = 5 - 2x$

x	0	1	2	3	5
y		3			

b. Draw the graph of the equation $y = 5 - 2x$

c. Use the graph to find the

i). Gradient of the line $y = 5 - 2x$

ii). Value of y where the graph cuts the y - axis

Solution

i). when $x = 0$

$$y = 5 - 2x$$

$$y = 5 - 2(0) = 5 - 0 = 5$$

ii). when $x = 2$

$$y = 5 - 2x$$

$$y = 5 - 2(2) = 5 - 4 = 1$$

iii). when $x = 3$

$$y = 5 - 2x$$

$$y = 5 - 2(3) = 5 - 6 = -1$$

iv). when $x = 5$

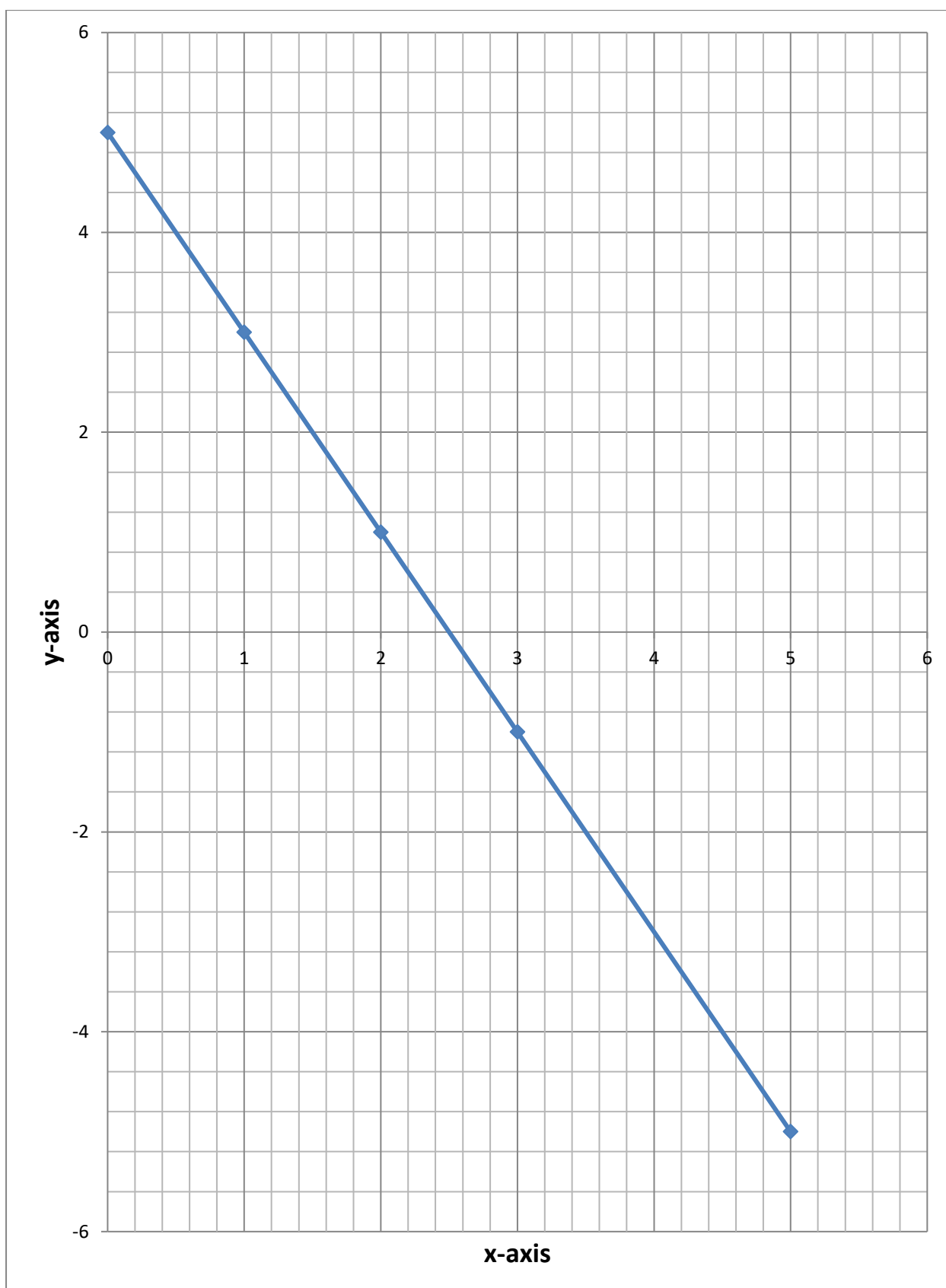
$$y = 5 - 2x$$

$$y = 5 - 2(5)$$

$$y = 5 - 10$$

$$y = -5$$

x	0	1	2	3	5
y	5	3	1	-1	-5



From the graph,

i). the gradient of the line =

$$\frac{-5 - (-1)}{5 - 3} = \frac{-5 + 1}{2} = \frac{-4}{2} = -2$$

ii). the graph cuts the y – axis at the point $(0, 5)$

Example 3a. Copy and complete the tables below of the given equations

i. $2y = x - 1$

x	-3	-2	-1	0	1	2	3
y		-1.5			0		

ii. $y = 2 - 2x$

x	-3	-2	-1	0	1	2	3
y	8				0		

b). using a scale of 2cm to 1 unit on the x – axis and 2cm to 2 units on the y – axis. Draw the graph of

$2y = x - 1$ and $y = 2 - 2x$

Solution

i). when $x = -3$

$$2y = x - 1$$

Divide both sides by the coefficient of y

$$\frac{2y}{2} = \frac{x - 1}{2}$$

$$y = \frac{x - 1}{2}$$

$$y = \frac{-3 - 1}{2} = \frac{-4}{2} = -2$$

When $x = -1$

$$y = \frac{x - 1}{2}$$

$$y = \frac{-1 - 1}{2} = \frac{-2}{2} = -1$$

When $x = 0$

$$y = \frac{x - 1}{2} = \frac{0 - 1}{2} = -\frac{1}{2} = -0.5$$

When $x = 2$

$$y = \frac{x - 1}{2} = \frac{2 - 1}{2} = \frac{1}{2} = 0.5$$

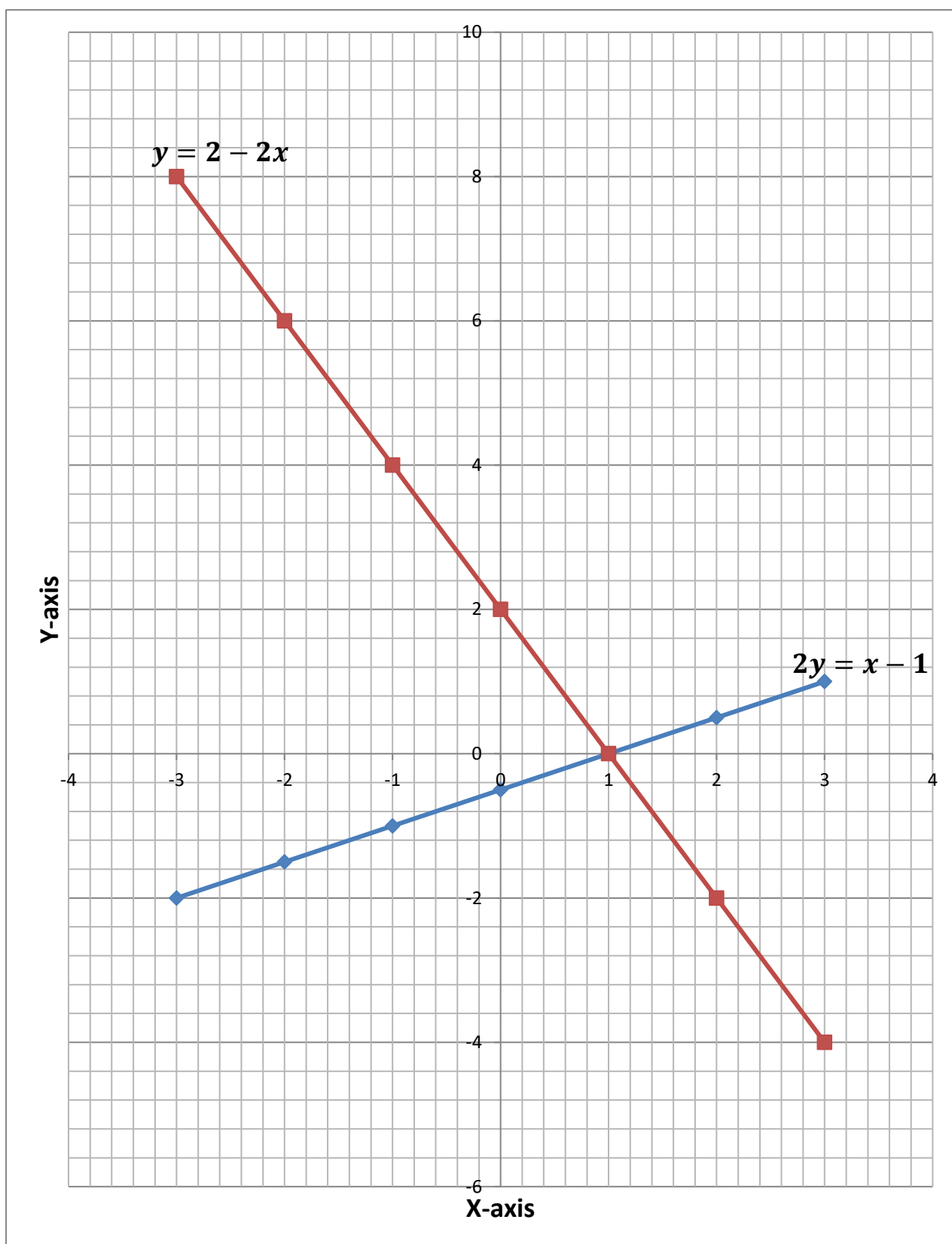
When $x = 3$

$$y = \frac{x - 1}{2} = \frac{3 - 1}{2} = \frac{2}{2} = 1$$

x	-3	-2	-1	0	1	2	3
y	-2	-1.5	-1	-0.5	0	0.5	1

ii). $y = 2 - 2x$

x	-3	-2	-1	0	1	2	3
y	8	6	4	2	0	-2	-4



SECTION 2: GENERAL EQUATIONS OF A STRAIGHT LINE

The general equation of a linear equation is given by $y = mx + c$, where m is the gradient or slope and c is the intercept of the point where the straight line cuts the y – axis.

Example 1. Find the values of m and c if the straight line $y = mx + c$, passes through the point $(-4, 6)$ and has a gradient of 8. Write the complete equation

Solution

$m = \text{Gradient} = 8, x = -4 \text{ and } y = 6$

Using the relation

$$y = mx + c$$

Substitute the values

$$6 = 8(-4) + c$$

$$6 = -32 + c$$

$$c = 6 + 32$$

$$c = 38$$

The equation is $y = 8x + 38$

SECTION 3: SKS PAPER 14

Answer all questions

Part 1

Question 1. Copy and complete the table below of the equation

$$y = 6x - 1$$

x	-3	-2	-1	0	1	2	3
y							

b). Using a scale of 2cm to 1 unit on the x – axis and 2cm to 2 units

on the y – axis, draw the graph of $y = 6x - 1$ for the values between -3 and $+3$

c). Find the gradient of the graph $y = 6x - 1$

Answer:

x	-3	-2	-1	0	1	2	3
y	-19	-13	-7	-1	5	11	17

$$\text{Gradient}(m) = 6$$

the graph cuts the y – axis at the point $(0, -1)$

Question 2. Copy and complete the table below for the equation

$$y = 8 - 4x$$

x	0	1	2	3	4
y					

b. Draw the graph of the equation $y = 8 - 4x$

c. Use the graph to find the

i). Gradient of the line $y = 8 - 4x$

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Answer:

x	0	1	2	3	4
y	8	4	0	-4	-8

Gradient(m) = -4

Question 3a. Copy and complete the tables below of the given equations

i). $4y = 4x - 1$

x	-3	-2	-1	0	1	2	3
y							

ii). $y = 2 - 4x$

x	-3	-2	-1	0	1	2	3
y							

b). using a scale of 2cm to 1 unit on both axes, draw the grap of $4y = x - 1$ and $y = 2 - 4x$

Answers:

(i)

x	-3	-2	-1	0	1	2	3
y	-3.25	-2.25	-1.25	-0.25	0.75	1.75	2.75

(ii).

x	-3	-2	-1	0	1	2	3
y	14	10	6	2	-2	-6	-10

Part 2

Question 1. Find the values of m and c if the straight line $y = mx + c$, passes through the point $(-8, 9)$ and has a gradient of 8. Write the complete equation

Answer:

Gradient(m) = 8

The equation of the straight line is $y = 8x + 73$

CHAPTER 15

EVERYDAY ARITHMETIC

SECTION 1: SIMPLE INTEREST

Interest is a payment given for saving money. It can also be the price paid for borrowing money. When interest is calculated on the basic sum of money saved (or borrowed) it is called simple interest.

Simple interest, I , can be calculated using the formula

$$SI = \frac{P \times R \times T}{100}$$

ILLUSTRATIVE EXAMPLES

Example 1. Find the simple interest on Le 9,000 invested for 4 years at 8% per annum?

Solution

Step 1: Write the formula for simple interest

$$SI = \frac{P \times R \times T}{100}$$

$$P = \text{Le } 9,000$$

$$R = 8\%$$

$$T = 4 \text{ years}$$

Step 2: Substituting into the formula

$$SI = \frac{9000 \times 8 \times 4}{100}$$

$$SI = \text{Le } 2,880$$

Example 2. Find the simple interest on Le 90,000 invested for 2 years at 40% per annum.

Solution

Step 1: Write the formula for simple interest

$$SI = \frac{P \times R \times T}{100}$$

$$P = \text{Le } 90,000$$

$$R = 40\%$$

$$T = 2 \text{ years}$$

Step 2: Substituting into the formula

$$SI = \frac{9000 \times 2 \times 40}{100}$$

$$SI = \text{Le } 72,000$$

Example 3. Calculate the simple interest on Le, 900,000.00 for 3 months at 10% per annum.

Solution

Step 1: Write the formula for simple interest

$$SI = \frac{P \times R \times T}{100}$$

$$P = \text{Le } 900,000.00$$

$$R = 10\%$$

$$T = \frac{3}{12} = \frac{1}{4} = 0.25 \text{ years}$$

Step 2: Substituting into the formula

$$SI = \frac{900,000 \times 10 \times 0.25}{100} = \text{Le } 22,500.00$$

SECTION 2: RATE

The percentage return is called the rate percent. Rate can be calculated using the formula

$$R = \frac{100 \times SI}{P \times T}$$

Example 1. Calculate the rate at which Le 200,000.00 invested for half a year gives an interest of Le 6,000.00

Solution

Step 1: Write the formula for rate

$$R = \frac{100 \times SI}{P \times T}$$

$$SI = \text{Le } 6,000.00$$

$$P = \text{Le } 200,000$$

$$T = \frac{1}{2} = 0.5 \text{ years}$$

Step 2: Substituting into the formula

$$R = \frac{100 \times SI}{P \times T}$$

$$R = \frac{100 \times 6000}{200000 \times 0.5}$$

$$R = 6\%$$

Example 2. Calculate the rate at which Le 8000.00 invested for one – quarter a year gives Le 1,200.00.

Solution

Step 1: Write the formula for rate

$$R = \frac{100 \times SI}{P \times T}$$

$$SI = Le\ 1,200.00$$

$$P = Le\ 8000$$

$$T = \frac{1}{4} = 0.25\ years$$

Step 2: Substituting into the formula

$$R = \frac{100 \times SI}{P \times T}$$

$$R = \frac{100 \times 1200}{8000 \times 0.25}$$

$$R = 3.75\%$$

SECTION 3: TIME

This is the duration of an investment. Time, T , is calculated using the formula

$$T = \frac{100 \times I}{P \times R}$$

ILLUSTRATIVE EXAMPLES

Example 1. Calculate the time on the SI Le 5,000, rate of $2\frac{1}{2}\%$ and principal of Le 40,000.00.

Solution

Step 1: Write the formula for time

$$T = \frac{100 \times I}{P \times R}$$

$$R = 2\frac{1}{2}\% = \frac{5}{2}\% = 2.5\%$$

$$P = \text{Le } 40,000$$

$$SI = \text{Le } 5,000$$

Step 2: Substituting into the formula

$$T = \frac{100 \times I}{P \times R}$$

$$T = \frac{100 \times 5000}{40000 \times 2.5}$$

$$T = 5 \text{ years}$$

Example 2. Find the time on the SI of Le 5,000, rate of 5% and principal of Le 10,000.00.

Solution

Step 1: Write the formula for time

$$T = \frac{100 \times I}{P \times R}$$

$$R = 5\%$$

$$P = \text{Le } 10,000.00$$

$$SI = \text{Le } 5,000$$

Step 2: Substituting into the formula

$$T = \frac{100 \times 5,000}{P \times R}$$

$$T = \frac{100 \times 5,000}{10,000 \times 5}$$

$$T = 10 \text{ years}$$

SECTION 4: PRINCIPAL AND AMOUNT

The money used for the investment is called the principal.

The amount is the total formed by adding the interest to the principal.

With simple interest, the principal always stays the same, no matter how many years the investment (or loan) may last.

ILLUSTRATIVE EXAMPLES

Example 1. *What sum of money must we invest to give Le 400 simple interest, if the rate is 8% per annum and the time is 3 years?*

Solution

$$P = \frac{100 \times I}{R \times T} = \frac{100 \times 400}{8 \times 3} = \frac{40000}{24} = 1666.67$$

Principal = Le 1,666.67

The sum of money (Principal) = Le 1,666.67

The Amount = Principal + Interest

= Le 1,666.67 + Le 400

= Le 2,066.67

SECTION 5: COMMISSION

The commission is usually a small percentage of the total value of goods, which they have sold.

ILLUSTRATIVE EXAMPLES

Example 1. Calculate the commission due to a car salesman if he sells a car for Le 35,000,000.00 and his commission is 10%.

Solution

Step 1:

$$\text{The commission} = \frac{10}{100} \times 35,000,000 = 350,000$$

Hence, the commission = Le 350,000

Example 2. A shopkeeper sells goods to the value of Le, 120,000. He is paid a commission of 10% on the values of these goods. he is paid a weekly salary of Le 80.00. What is his weekly earning?

Solution

Step 1:

His weekly earning = commission received + weekly salary

$$\begin{aligned}\text{The commission} &= \frac{10}{100} \times 120,000 \\ &= 12,000\end{aligned}$$

Hence, the commission = Le 12,000

His weekly earning = Le 12,000 + Le 80.00 = Le 12,080.00

SECTION 6: DISCOUNT

This is usually a percentage of the selling price, is the amount, which the retailer will take off his selling price thus reducing his profit.

ILLUSTRATIVE EXAMPLES

Example 1. *A salesman charges Le 1,000,000 for a suit of clothes but allows a discount of 20% for cash. What is the price?*

Solution

Step 1:

Discount allowed = 20% of Le 1,000,000

$$= \frac{20}{100} \times 1,000,000$$

$$= 200,000$$

Step 2:

Cash price = Le 1,000,000 – Le 200,000

$$= \text{Le } 800,000$$

Example 2. *A television marked for sale at Le 500,000 is sold for cash at a discount of 5%. What price did the customer pay?*

Solution

Step 1:

Discount allowed = 5% of Le 500,000

$$= \frac{5}{100} \times 500,000 = \text{Le } 25,000$$

Step 2:

$$\text{Cash price} = \text{Le } 500,000 - \text{Le } 25,000 = \text{Le } 475,000$$

SECTION 7: TAXES AND BILLS

Taxation is the process by which Government makes the people pay for the items of National expenditure incurred on their behalf; while rates look after the needs of the Town Council and Country Council i. e.

Taxation is a national affairs while rates are local.

ILLUSTRATIVE EXAMPLES

Example 1. A civil servant salary is Le 7,280,000. If he has Le 222,000 tax – free allowance and the remainder is taxed at 10%, how much does she pay?

Solution

Step 1: Taxable amount = salary – tax free allowance

$$= \text{Le } 7,280,000 - \text{Le } 222,000 = \text{Le } 7,058,000$$

Step 2: Tax payable = 10% of Le 7,058,000

$$= \frac{10}{100} \times \text{Le } 7,058,000$$

$$= \text{Le } 705,800$$

Example 2. A man's taxable income is Le 2,000,000. If tax is paid at 35%, find the amount paid in income tax?

Solution

Step 1: Tax payable = $x\%$ $\times$ taxable income

$$\text{Tax payable} = 35\% \times 2,000,000$$

$$= \frac{35}{100} \times 2,000,000$$

$$= 700,000$$

Hence, Taxable pay = Le 700,000

Example 3. An electricity meter reading changes from 6666 to 7777.

Calculate the cost if the first 70 units cost Le 100 each and the remaining unit cost Le 80 each.

Solution

Step 1:

Number of unit used = $7777 - 6666 = 1111$ units

70 units at Le 100 = $Le\ 7,000 \times 100 = Le\ 7,000$

Step 2: Remaining units = number of unit used – first 70 units

= $1111 - 70 = 1041$ units

1041 at Le 80 = $1041 \times 80 = Le\ 83,280$

Step 3: The total cost = 70 units at Le 100 + 1041 at Le 80

= $Le\ 7,000 + Le\ 83,280$

= $Le\ 90,280$

Therefore, the total cost is Le 90,280

Example 4. In one quarter, consumers uses 15.5 units of manufactured gas. Calculate the total cost of Le 750 per unit if the standing charge is Le 450.

Solution

Step 1:

Total unit cost = 15.5 units at Le 750

= $15.5 \times 750 = Le\ 11,625$

Step 2:

Standing charge = Le 450

Total cost = Total unit cost + Standing cost

= Le 11,625 + Le 450

= Le 12,075

Note: Standing charge is a minimum sum paid by every consumer, irrespective of the amount of gas or units of electricity used.

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SECTION 8: RATIO

A ratio is a comparison between two similar quantities. For example the ratio of 7 to 9 is written as 7:9. A ratio may be expressed as a fraction, and as ratios may be looked upon as fractions: Thus 7:9 can be written

$$\text{as } \frac{7}{9}$$

ILLUSTRATIVE EXAMPLES

Example 1. *Two length are in the ratio 10:3. If the first length is 300 meters, what is the second length?*

Solution

Step 1:

$$\text{Second length} = \frac{3}{10} \times 300 = 90 \text{ meters}$$

Example 2. *Two length are in the ratio 10:3. If the second length is 90 meters, what is the first length?*

Solution

Step 1:

$$\text{First length} = \frac{10}{3} \times 90 = 300 \text{ meters}$$

SECTION 9: PROPORTION

A proportion is a statement of equality between two ratios. It may be written with double colon or proportional sign (::) or with the sign of equality (==).

ILLUSTRATIVE EXAMPLES

Example 1. Divide Le 500 in the ratio 7:3

Solution

Step 1: Find the total number of parts

$$\text{Total number of parts} = 7 + 3 = 10$$

Step 2:

$$\text{Amount of first part} = \frac{7}{10} \times 500$$

$$= \text{Le } 350$$

$$\text{Amount of second part} = \frac{3}{10} \times 500$$

$$= \text{Le } 150$$

Example 2. A line is to be divided into three parts in the ratios 3:5:7.

If the line is 750 millimeters long. Calculate the lengths of each part.

Solution

Step 1: Find the total number of parts

$$\text{Total number of parts} = 3 + 5 + 7 = 15$$

Step 2: Find the length of each part

$$\text{Length of first part} = \frac{3}{15} \times 750 = 150 \text{ millimeters}$$

$$\text{Length of second part} = \frac{5}{15} \times 750 = 250 \text{ millimeters}$$

$$\text{Length of third part} = \frac{7}{15} \times 750 = 350 \text{ millimeters}$$

Example 3. Two villages have populations of 400 and 500 respectively. The two villages are to share a grant of Le 18,000,000 in proportion to their population. Calculate how much each village receives.

Solution

Step 1: Find the total number of parts

$$\text{Total population} = 400 + 500 = 900$$

Step 2:

$$\begin{aligned} \text{Amount of first village} &= \frac{400}{900} \times 18000000 \\ &= \text{Le } 8,000,000 \end{aligned}$$

$$\begin{aligned} \text{Amount of second village} &= \frac{500}{900} \times 18000000 \\ &= \text{Le } 1,000,000 \end{aligned}$$

Example 4. Three friends contribute sums of money to a charitable organisation in the ratio 3:2:7. If the largest amount contributed is Le 210,000, calculate the total amount contributed by the three friends

Solution

Step 1: Find the total number of parts

$$\text{Total number of parts} = 3 + 2 + 7 = 12$$

Let x = total amount contributed

Step 2:

$$\text{Largest amount} = \frac{7}{12} \times x = 210000$$

Step 3: Cross multiply

$$7x = 12 \times 240000$$

$$7x = 2520000$$

$$\frac{7x}{7} = \frac{2520000}{7}$$

$$x = 360000$$

Example 5. A car uses 120 litres of fuel for a distance of 3000 km. How many litres of fuel will be needed for a distance of 50 km?

Solution

Step 1: Find the fuel needed to cover 1 km distance

$$\Rightarrow \text{Fuel needed to cover 1 km is: } \frac{120}{3000} \text{ litres}$$

Step 2: Find the fuel needed to cover 50 km distance

$$\Rightarrow \text{Fuel needed to cover 50 km is: } \frac{120}{3000} \times 50 = 2 \text{ litres}$$

Example 6. Nine (9) men do a piece of work in 28 days. How long will it take 14 men to do it?

Solution

Step 1: Find the number of days 1 man will take to do the work

$$1 \text{ man will take: } 9 \times 28 = 252 \text{ days}$$

Step 2: Find the number of days 14 men will take to do the work

$$9 \text{ men will take: } \frac{252}{14} = 18 \text{ days}$$

SECTION 10: SPEED, DISTANCE AND TIME

$$\text{Average Speed} = \frac{\text{Total Distance covered}}{\text{Total Time Taken}}$$

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}}$$

$$\text{Distance} = \text{Time} \times \text{Speed}$$

ILLUSTRATIVE EXAMPLES

Example 1. A train travels 400km in 5 hours. What is its average speed?

Solution

Step 1: Write the formula for average speed

$$\text{Average Speed} = \frac{\text{Total Distance Covered}}{\text{Total Time Taken}}$$

$$= \frac{400 \text{ km}}{5 \text{ hr}}$$

$$= 80 \text{ km/hr}$$

Example 2. A car travels 300 km at an average speed of 60km/hr.

How long has it taken?

Solution

Step 1:

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}}$$

$$= \frac{300 \text{ km}}{60 \text{ km/hr}} = 5 \text{ hrs}$$

Example 3. *If a car travels for 5hrs at an average speed of 300km/hr. How far has it gone?*

Solution

Step 1: *Write the formula for distance*

$$\text{Distance} = \text{Time} \times \text{Speed}$$

Step 2: *Substitute*

$$= 300 \text{ km/hr} \times 5\text{hr}$$

$$= 1500 \text{ km}$$

Example 4. *A car travelling at an average speed of 70 km per hour, left Makeni at 11:30 am and arrived at 4:30 pm. How far did the car travel?*

Solution

Step 1: *Convert 4:30 pm to GMT (Greenwich Meridian Time)*

$$4:30 \text{ pm} + 12:00 \text{ pm} = 16:30 \text{ pm}$$

Step 2:

$$\text{Total time} = 16:30 - 11:30 = 5 \text{ hrs}$$

Step 3:

$$\text{Distance} = 5\text{hr} \times 70\text{km/hr} = 350 \text{ km}$$

SECTION 11: DENSITY AND POPULATION DENSITY

$$\text{Density} = \frac{\text{mass}}{\text{volumes}}$$

$$\text{Population Density} = \frac{\text{number of people}}{\text{area in square km}}$$

ILLUSTRATIVE EXAMPLES

Example 1. If 25 cm^3 of a quantity of marbels has a mass of 75 g. Calculate the density of the marble?

Solution

Step 1: Write the formula for density

$$\text{Density} = \frac{\text{mass}}{\text{volume}}$$

Step 2: Substitute the values for mass and volume

$$\text{Density} = \frac{75 \text{ g}}{25 \text{ cm}^3} = 3 \text{ g/cm}^3$$

Example 2. A town has an area of 10 km^2 and a population of 10 000 people, calculate its population density?

Solution

Step 1: Write the formula for population density

$$\text{Population Density} = \frac{\text{number of people}}{\text{area in square km}}$$

Step 2: Substitute the values for mass and volume

$$\text{Population Density} = \frac{10\,000}{10 \text{ km}} = 1\,000 \text{ people/km}$$

SECTION 12: SKS PAPER 15

Answer all questions

Part 1

Question 1. Find the simple interest on Le 50,000 invested for 5 years at 5% per annum?

Answer[$SI = \text{Le } 12,500$]

Example 2. Find the simple interest on Le 50,000 invested for 4 years at 50% per annum.

Answer[$SI = \text{Le } 100,000$]

Example 3. Calculate the simple interest on Le 600,000.00 for 3 months at 10% per annum.

Answer[$SI = \text{Le } 15,000$]

Part 2

Question 1. Calculate the rate at which Le 28,000.00 invested for six months gives an interest of Le 7,000.00

Answer[$R = 50\%$]

Question 2. Calculate the rate at which Le 900,000.00 invested for 3 months gives Le 180,000.00

Answer[$R = 80\%$]

Part 3

Question 1. Calculate the time on the SI Le 5,000 rate of 5% and principal of Le 40,000.00.

Answer $\left[T = 2\frac{1}{2} \text{ year}\right]$

Question 2. Find the time on the SI of Le 10,000, rate of 10% and principal of Le 50,000.00.

Answer $[T = 2 \text{ years}]$

Question 3. What sum of money must we invest to give Le 800 simple interest, the rate is 10% per annum and the time is 4 years?

Answer $[\text{Amount} = \text{Le } 2,800]$

Part 4

Question 1. Calculate the commission due to a car salesman if he sells a car for Le 150,000,000.00 and his commission is 15%.

Answer $[\text{Le } 22,500,000]$

Question 2. A shopkeeper sells goods to the value of Le 240,000. He is paid a commission of 8% on the values of these goods. He is paid a weekly salary of Le 60.00. What is his weekly earning?

Answer $[\text{Weekly earning} = \text{Le } 19,260]$

Question 3. A salesman charges Le 10,000,000 for a suit of clothes but allows a discount of 10% for cash. What is the price?

Answer $[\text{Discount} = \text{Le } 1,000,000]$

Question 4. A television marked for sale at Le 1,500,000 is sold for cash at a discount of 15%. What price did the customer pay?

Answer $[\text{Discount} = \text{Le } 225,000]$

Question 5. A civil servant salary is Le 7,480,000. If he has Le 540,000 tax – free allowance and the remainder is taxed at 20%,

how much does she pay?

Answer[Tax paid = Le 1,388,000]

Question 6. A man's taxable income is Le 1,000,000. If tax is paid at 15%, find the amount paid in income tax?

Answer[Tax paid = Le 150,000]

Question 7. An electricity meter reading changes from 5234 to 6124. Calculate the cost if the first 50 units cost Le 50 each and the remaining unit cost Le 30 each.

Answer[Total cost = Le 18.3]

Question 8. In a certain area, electricity is charge for a fixed rate of Le 40.00 for every unit used. If a householder receives a bill of Le 28,000, how many units has she used?

Answer[Unit used = 700 units]

Part 5

Question 1. Two length are in the ratio 2: 1. If the first length is 500 meters, what is the second length?

Answer[Second length = 250 meters]

Question 2. Two length are in the ratio 10: 3. If the second length is 60 meters, what is the first length?

Answer[Second length = 200 meters]

Question 3. Divide Le 10,000 in the ratio 7: 3

Answer[Amount of first part = Le 700 and second part = Le 300]

Question 4. A line is to be divided into three parts in the ratios 4: 6: 8.

If the line is 720 millimeters long. Calculate the lengths of each part.

Answer:

[*First length = 160 millimeters,*

Second length = 240 millimeters and

Third length = 320 millimeters]

Question 5. Two villages have populations of 800 and 600 respectively. The two villages are to share a grant of Le 28,000,000 in proportion to their population. Calculate how much each village receives.

Answer:

[*First village receive = Le 16,000,000 and*

Second village recieved = Le 12,000,000]

Question 6. Four friends contribute sums of money to a charitable organisation in the ratio 3:2:7:8. If the largest amount contributed is Le 2,000,000, calculate the total amount contributed by the four friends

Answer [*The total contributed = Le 5,000,000]*

Question 7. A car uses 120 litres of fuel for a distance of 3000 km. How many litres of fuel will be needed for a distance of 50 km?

Answer [*The fuel needed is 2 litres*]

Example 8. Five (5) men do a piece of work in 30 days. How long will it take 10 men to do it?

Answer [*10 men will take 15 days*]

Question 9. A train travels 800km in 8 hours. What is its average speed?

Answer [*Average speed = 100 km/hr*]

Question 10. *A car travels 450 km at an average speed of 90 km/h. How long has it taken?*

Answer [*Time Taken = 5 hours*]

Question 11. *If a car travels for 6hrs at an average speed of 480 km/h. How far has it gone?*

Answer [*Distance = 2880 km*]

Question 12. *A car travelling at an average speed of 90 km per hour, left Makeni at 12:30am and arrived at 6:30pm. How far did then car travels?*

Answer [*Distance = 540 km*]

Part 6

Question 1. *If 15 cm^3 of a quantity of marbels has a mass of 45 g. Calculate the density of the marble?*

Answer [*Density = 3 g/cm^3*]

Question 2. *A town has an area of 20 km^2 and a population of 20 000 people, calculate its population density?*

Answer [*1 000 people/km*]

CHAPTER 16

POLYGONS

SECTION 1: DEFINITION

A polygon is a plane figure with at least three straight sides and angles, and typically five or more.

SECTION 2: CLASSIFICATION OF POLYGONS

Polygons are classified by the number of sides they have.

<i><u>Polygons</u></i>	<i><u>Number of sides</u></i>	<i><u>Name of polygon</u></i>	<i><u>Number of sides</u></i>
<i>Triangle</i>	<i>3</i>	<i>Nonagon</i>	<i>9</i>
<i>Quadrilateral</i>	<i>4</i>	<i>Decagon</i>	<i>10</i>
<i>Pentagon</i>	<i>5</i>		
<i>Hexagon</i>	<i>6</i>		
<i>Heptagon</i>	<i>7</i>		
<i>Octagon</i>	<i>8</i>		

Polygons are either convex or concave, unless stated otherwise, "polygons", mean "convex polygon"

SECTION 3: THE SUM OF THE INTERIOR ANGLES OF A POLYGON

“In a convex polygon having n sides the sum of the interior angles is

$$(n - 2) \times 180$$

The sum of the exterior angles is 360° , no matter how many sides the polygon has”.

ILLUSTRATIVE EXAMPLES

Example 1. Find the sum of the interior angles of a convex polygons with (i) 10 sides (ii) 14 sides

Solution

(i).

Step 1: Write the formula for the sum of interior angles of a polygon

$$\text{The sum of the interior angles} = (n - 2) \times 180$$

Step 2: Substitute for $n = 10$

$$\text{The sum of the interior angles} = (10 - 2) \times 180$$

$$\text{The sum of the interior angles} = 8 \times 180$$

$$\text{The sum of the interior angles} = 1440^\circ$$

(ii)

Step 1: Write the formula for the sum of interior angles of a polygon

$$\text{The sum of the interior angles} = (n - 2) \times 180$$

Step 2: Substitute for $n = 14$

$$\text{The sum of the interior angles} = (14 - 2) \times 180$$

$$\text{The sum of the interior angles} = 12 \times 180$$

The sum of the interior angles = 2160°

Example 2. What is the sum of the interior angles of a polygon with 9 sides?

Solution

Step 1: Write the formula for the sum of interior angles of a polygon

The sum of the interior angles = $(n - 2) \times 180$

Step 2: Substitute for $n = 9$

The sum of the interior angles = $(9 - 2) \times 180$

The sum of the interior angles = 7×180

The sum of the interior angles = 1260°

**SECTION 4: FINDING THE NUMBER OF SIDES OF A
REGULAR POLYGON WHEN THE SUM OF THE INTERIOR
ANGLES IS GIVEN**

Example 1. *The sum of the interior angles of a regular polygon is 1880° .*

Find the number of sides of the regular polygon?

Solution

Step 1: Write the formula for the sum of interior angles of a polygon

The sum of the interior angles = $(n - 2) \times 180$

Step 2: Substitute the value for the sum of the interior angles

$$1880 = (n - 2) \times 180$$

$$1880 = 180n - 360$$

Step 3: Make n the subject

$$1880 + 360 = 180n$$

$$2160 = 180n$$

$$\frac{2160}{180} = \frac{180n}{180}$$

$$n = 12$$

$\therefore$ The number of sides of the polygon is 12.

SECTION 5: FINDING THE NUMBER OF SIDES OF A POLYGON

The interior angle of a regular polygon is $\frac{(n - 2) \times 180}{n}$

ILLUSTRATIVE EXAMPLES

Example 1. Find the interior angle of a regular polygon with 10 sides?

Solution

Step 1: Write the formula for the sum of interior angles of a polygon

The interior angle of a regular polygon = $\frac{(n - 2) \times 180}{10}$

Step 2: Substitute for $n = 10$

The interior angle of a regular polygon = $\frac{(10 - 2) \times 180}{10}$

The interior angle of a regular polygon = $\frac{8 \times 180}{10}$

The interior angle of a regular polygon = $\frac{1440}{10}$

The interior angle of a regular polygon = 144°

**SECTION 6: FINDING THE NUMBER OF SIDES OF A
REGULAR POLYGON WHEN THE INTERIOR ANGLE IS
GIVEN**

Example 1. *If the interior angle of a regular polygon is 140° . How many sides does it have?*

Solution

Step 1: *Write the formula for the sum of interior angles of a polygon*

$$\text{The interior angle of a regular polygon} = \frac{(n - 2) \times 180}{n}$$

Step 2: *Substitute the value for interior angle of a regular polygon*

$$140 = \frac{(n - 2) \times 180}{n}$$

Step 3: *Cross multiply*

$$140n = (n - 2) \times 180$$

$$140n = 180n - 360$$

Step 4: *Make n the subject*

$$360 = 180n - 140n$$

$$\frac{360}{40} = \frac{40n}{40}$$

$$n = 9$$

$\therefore$ The number of sides of the polygon is 9.

SECTION 7: FINDING THE EXTERIOR ANGLE OF A REGULAR POLYGON

The sum of the exterior angles of a polygon is 360° .

The exterior angle of a regular polygon = $\frac{360}{n}$

ILLUSTRATIVE EXAMPLE

Example 1. *A regular polygon has 8 sides. Find the exterior angle?*

Solution

Step 1: Write the formula for the exterior angle of a regular polygon

The exterior angle of a regular polygon = $\frac{360}{n}$

Step 2: Substitute for $n = 8$

The exterior angle of a regular polygon = $\frac{360}{8} = 45^\circ$

SECTION 8: FINDING THE NUMBER OF SIDES WHEN THE EXTERIOR ANGLE IS GIVEN

Example 1. *The exterior angle of a regular polygon is 60° . Find the number of sides?*

Solution

Step 1: Write the formula for the exterior angle of a regular polygon

$$\text{The exterior angle of a regular polygon} = \frac{360}{n}$$

Step 2: Substitute the value for the exterior angle of a regular polygon

$$60 = \frac{360}{n}$$

Step 3: Cross multiply

$$60n = 360$$

Step 4: Divide both sides by the coefficient of n

$$\frac{60n}{60} = \frac{360}{60}$$

$$n = 6$$

$\therefore$ The polygon has six (6) sides

SECTION 9: FINDING THE SUM OF THE INTERIOR ANGLES WHEN THE EXTERIOR ANGLE IS GIVEN

Example 1. *The exterior angle of a regular polygon is 60° . Find the sum of its interior angles?*

Solution

Step 1: *Write the formula for the sum of the interior angles*

The sum of the interior angles $= (n - 2) \times 180 \dots \dots$ Equation (1)

Step 2: *Write the formula for the exterior angle of a regular polygon*

The exterior angle of a regular polygon $= \frac{360}{n} \dots \dots$ Equation (2)

Step 3: *Substitute the value for the exterior angle of a regular polygon*

$$60 = \frac{360}{n}$$

Step 4: *Cross multiply*

$$60n = 360$$

Step 5: *Divide both sides by the coefficient of n*

$$\frac{60n}{60} = \frac{360}{60}$$

$$n = 6$$

Step 6: *Substitute for $n = 6$ into equation (1)*

The sum of the interior angles $= (6 - 2) \times 180$

The sum of the interior angles $= 4 \times 180 = 720^\circ$

SECTION 10: SKS PAPER 16

Answer all questions

Part 1

Question 1. Find the sum of the interior angles of a convex polygons with (i) 12 sides (ii) 8 sides

Answers:

(i). The sum of the interior angles is 1800°

(ii). The sum of the interior angles is 1080°

Part 2

Question 1. Each interior angle of a regular polygon is 140° . How many sides have it?

Answer : [Number of sides = 9]

Question 2. What is the sum of the interior angles of a polygon with 7 sides?

Answer: The sum of the interior angles is 900°

Part 3

Question 1. Find the interior angle of a regular polygon with 12 sides?

Answer: The interior angle is 150°

Question 2. Find the interior angle of a regular polygon with 15 sides?

Answer: The interior angle is 156°

Part 4

Question 1. The sum of the interior angles of a regular polygon is 1620° .

Find the number of sides of the regular polygon?

Answer: The polygon has 11 sides

Question 2. The sum of the interior angles of a regular polygon is 2520° .

Find the number of sides of the regular polygon?

Answer: The polygon has 16 sides

Part 5

Question 1. A regular polygon has 10 sides. Find the exterior angle?

Answer: The exterior angle is 36°

Question 2. A regular polygon has 9 sides. Find the exterior angle?

Answer: The exterior angle is 40°

Part 6

Question 1. The exterior angle of a regular polygon is 40° . Find the number of sides?

Answer: The polygon has nine (9) sides

Question 2. The exterior angle of a regular polygon is 30° . Find the number of sides?

Answer: The polygon has twelve (12) sides

Part 7

Question 1. The exterior angle of a regular polygon is 30° . Find the sum of its interior angles?

Answer: The sum of the interior angles = 1800°

Question 2. The exterior angle of a regular polygon is 40° . Find the sum of its interior angles?

Answer: The sum of the interior angles = 1260°

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CHAPTER 17

PLANE GEOMETRY

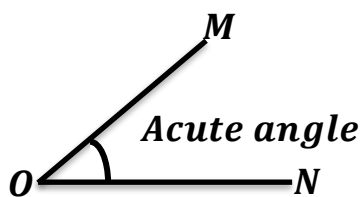
SECTION 1: INTRODUCTION

GEOMETRY: is a part of mathematics in which the properties of points, lines, surfaces and solids are investigated.

ANGLE: an angle is the amount of rotation between two straight lines.

TYPES OF ANGLE

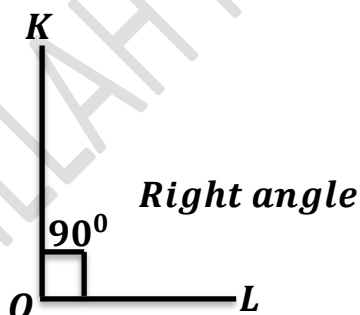
1. **ACUTE ANGLE:** an acute angle is an angle whose measure is less than 90° .



$\angle MON$ is an acute angle.

Therefore, $\angle MON = 90^\circ$.

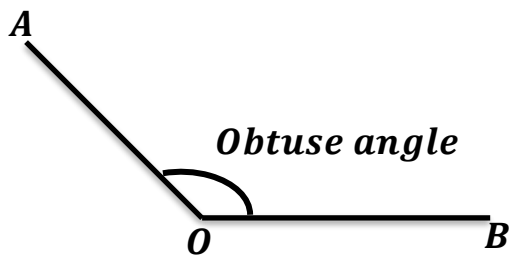
2. **RIGHT ANGLE:** a right angle is an angle whose measure is equal to 90° .



$\angle KOL$ is a right angle.

Therefore, $\angle KOL = 90^\circ$.

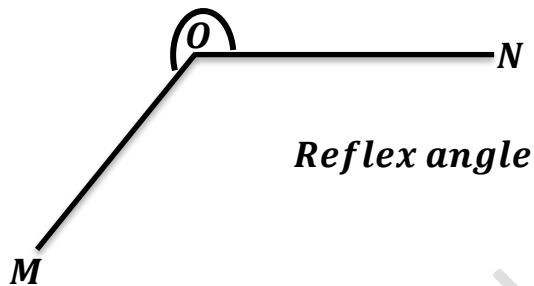
3. **OBTUSE ANGLE:** an obtuse angle is an angle whose measure is greater than 90° but less than 180° .



$\angle BOA$ is an obtuse angle.

Therefore, $\angle BOA$ is greater than 90° and less than 180° .

4. **REFLEX ANGLE**: an angle with a measure greater than 180° but less than 360° .



$\angle NOM$ is a reflex angle.

Therefore, $\angle NOM$ is greater than 180° and less than 360° .

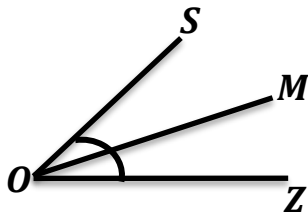
6. **STRAIGHT ANGLE**: a straight angle has a measure equal to 180° .



POQ is a straight line.

Therefore, $\angle POQ = 180^\circ$.

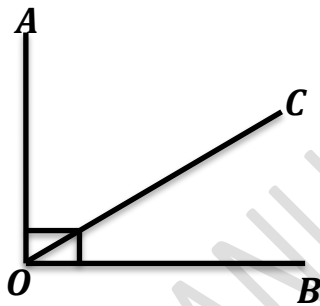
6. **ADJACENT ANGLES**: If two angles have common arm, common vertex, and the non – common arms are on the opposite sides of the common arms, they are called adjacent angles.



$\angle ZOM$ and $\angle MOS$ are the adjacent angles.

OM is the common arm, O is the common vertex and the non – common arms OZ and OS are on the opposite sides of the common arm OM .

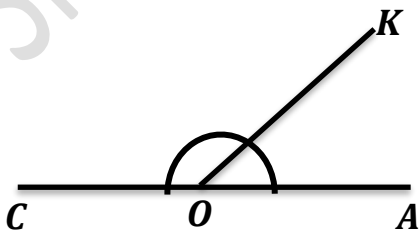
7.COMPLEMENTARY ANGLES: two angles are called complementary, if the sum of their measure is 90° .



$$\angle AOC + \angle COB = 90^\circ$$

Therefore, $\angle AOC$ and $\angle COB$ are complementary angles.

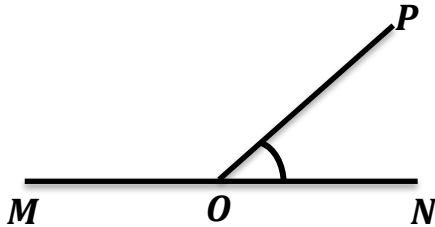
8.SUPPLEMENTARY ANGLES: two angles are called supplementary, if the sum of their measure is equal to 180° .



$$\angle AOK + \angle KOC = 180^\circ.$$

Therefore $\angle AOK$ and $\angle KOC$ are supplementary.

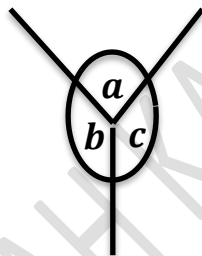
9. ADJACENT SUPPLEMENTARY ANGLES: *the sum of the adjacent angles on a straight line is equal to 180° .*



$$\angle NOP + \angle POM = 180^\circ$$

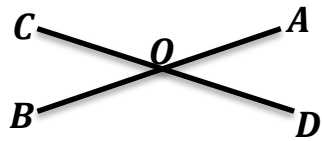
10. ANGLES AROUND A POINT: *when many lines meet at a point, they form angles around the point. The sum of all the angles formed is equal to 360 or four right angles.*

i.e. angles around a point = $360^\circ = 4$ right angles.



$$a + b + c = 360^\circ$$

11. VERTICALLY OPPOSITE ANGLES: *if two straight lines intersect at a point, the angles of the opposite sides of the point of intersection are called vertically opposite.*



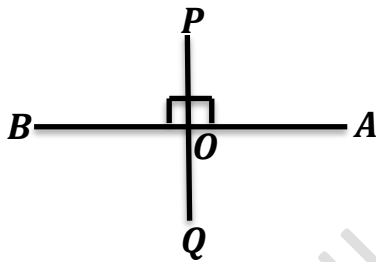
$\angle AOD$ and $\angle BOC$ are vertically opposite angles. Similarly,

$\angle AOC$ and $\angle BOD$ are vertically opposite angles.

Vertically opposite angles are equal. i. e. $\angle AOB = \angle BOC$ and

$\angle AOB = \angle BOD$.

12. PERPENDICULAR LINES: *two lines are perpendicular if they meet at right angles.*



$\angle AOP = 90^\circ$ and $\angle BOP = 90^\circ$.

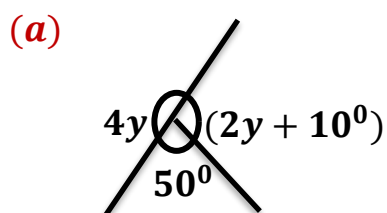
The lines AB and PQ are perpendicular to each other.

SECTION 2: ANGLES AROUND A POINT

These are angles that sum up to 360° .

ILLUSTRATIVE EXAMPLES

Example 1. Find the value of the lettered angles in the following diagrams



Solution

Step 1: add all the values and equate to 360°

$$4y + 50^\circ + 2y + 10^\circ = 360^\circ \text{ \{angles around a point sum up to } 360^\circ\}}$$

Step 2: collect like terms

$$4y + 2y + 50^\circ + 10^\circ = 360^\circ$$

$$6y + 60^\circ = 360^\circ$$

Step 3: transpose $+ 60^\circ$ (change the sign and position of $+60^\circ$)

$$6y = 360^\circ - 60^\circ$$

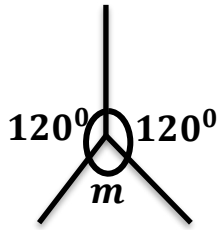
$$6y = 300^\circ$$

Step 4: divide both sides by the coefficient of y .

$$\frac{6y}{6} = \frac{300^\circ}{6}$$

$$y = 50^\circ$$

(b)



Solution

Step 1: add all and equate to 360°

$$m + 120^\circ + 120^\circ = 360^\circ \{\text{angles around a point}\}$$

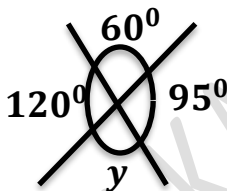
$$m + 240^\circ = 360^\circ$$

Step 2: collect like terms

$$m = 360^\circ - 240^\circ$$

$$m = 120^\circ$$

(c)



Solution

Step 1:

$$y + 95^\circ + 120^\circ + 60^\circ = 360^\circ$$

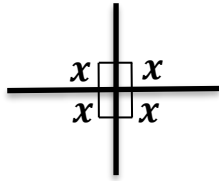
$$y + 275^\circ = 360^\circ$$

Step 2:

$$y = 360^\circ - 275^\circ$$

$$y = 85^\circ$$

(d)



Solution

Step 1:

$$x + x + x + x = 360^{\circ}$$

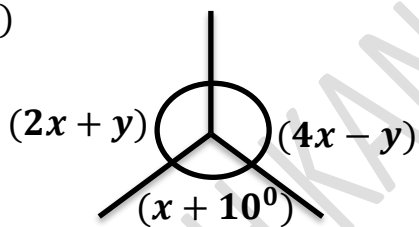
$$4x = 360^{\circ}$$

Step 2:

$$\frac{4x}{4} = \frac{360^{\circ}}{4}$$

$$x = 90^{\circ}$$

(e)



Solution

Step 1: angles around a point sum up to 360° .

$$(2x + y) + (x + 10^{\circ}) + (4x - y) = 360^{\circ}$$

$$2x + x + 4x + y - y = 360^{\circ} - 10^{\circ}$$

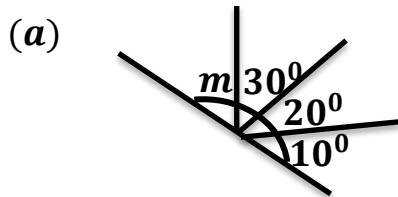
$$7x - 0 = 350^{\circ}$$

$$\frac{7x}{7} = \frac{350^{\circ}}{7} \Rightarrow x = 50^{\circ}$$

SECTION 3: ANGLES ON A STRAIGHT LINE

These are angles that sum up to 180° .

Example 1. Find the value of the lettered angles in the following diagram.



Solution

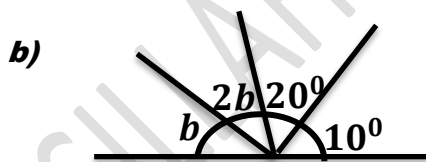
Step 1: angles on a straight line sum up to 180°

$$m + 10^\circ + 20^\circ + 30^\circ = 180^\circ$$

$$m + 60^\circ = 180^\circ$$

$$m = 180^\circ - 60^\circ$$

$$m = 120^\circ$$



Solution

Step 1: angles on a straight line sum up to 180° .

$$b + 2b + 20^\circ + 10^\circ = 180^\circ$$

$$3b + 30^\circ = 180^\circ$$

$$3b = 180^\circ - 30^\circ$$

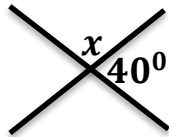
$$3b = 150^{\circ}$$

Step 2: Divide both sides by the coefficient of b

$$\frac{3b}{3} = \frac{150}{3}$$

$$b = 50^{\circ}$$

c)



Solution

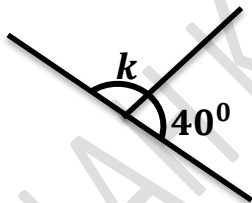
Step 1:

$$x + 40^{\circ} = 180^{\circ} \{\text{angles on a straight line}\}$$

$$x = 180^{\circ} - 40^{\circ}$$

$$x = 140^{\circ}$$

d)



Solution

Step 1:

$$k + 40^{\circ} = 180^{\circ} \{\text{angles on a straight line}\}$$

$$k = 180^{\circ} - 40^{\circ}$$

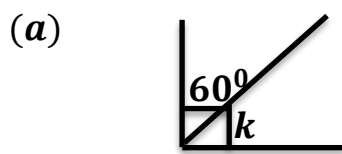
$$k = 140^{\circ}$$

SECTION 4: COMPLEMENTARY ANGLES

These are angles that sum up to 90° .

ILLUSTRATIVE EXAMPLES

Example 1. Find the value of the lettered angles in the following diagram.



Solution

Step 1: complementary angles sum up to 90°

$$60^\circ + k = 90^\circ \{\text{complementary angles}\}$$

$$k = 90^\circ - 60^\circ$$

$$k = 30^\circ$$



Solution

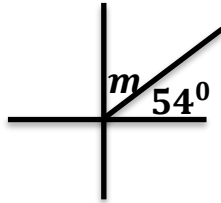
Step 1: complementary angles sum up to 90°

$$w + 38^\circ = 90^\circ \{\text{complementary angles}\}$$

$$w = 90^\circ - 38^\circ$$

$$w = 52^\circ$$

(c)



Solution

Step 1: complementary angles sum up to 90°

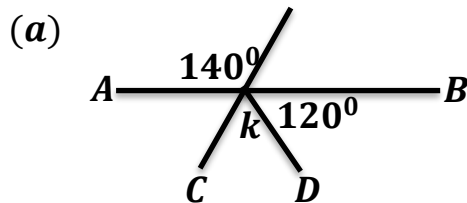
$$m + 54^\circ = 90^\circ \{\text{complementary angles sum up to } 90\}$$

$$m = 90^\circ - 54^\circ$$

$$m = 36^\circ$$

SECTION 5: VERTICALLY OPPOSITE ANGLES

Example 1. Find the value of the lettered angles in the following diagram.



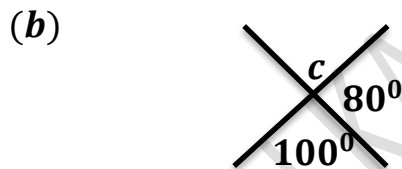
Solution

Step 1: vertically opposite angles are equal

$$140^\circ = k + 120^\circ \{\text{vertically opposite angles}\}$$

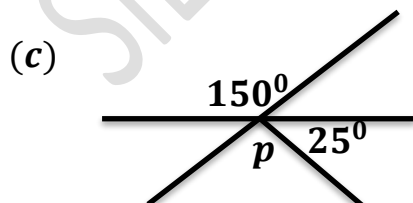
$$140^\circ - 120^\circ = k$$

$$k = 20^\circ$$



Solution

$$c = 100^\circ \{\text{vertically opposite angles are equal}\}$$



Solution

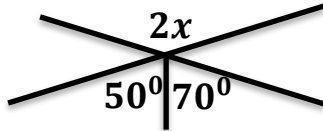
Step 1:

$$150^{\circ} = p + 25^{\circ} \{\text{vertically opposite angles are equal}\}$$

$$150^{\circ} - 75^{\circ} = p$$

$$p = 75^{\circ}$$

(d)



Solution

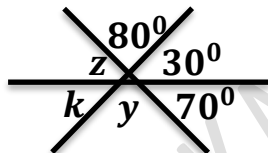
Step 1:

$$2x = 50 + 70 \{\text{vertically opposite angles are equal}\}$$

$$\frac{2x}{2} = \frac{120^{\circ}}{2}$$

$$x = 60^{\circ}$$

(e)



Solution

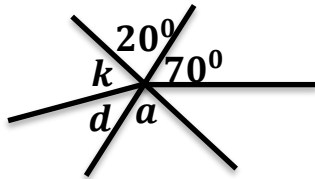
$$z = 70^{\circ} \{\text{vertically opposite angles are equal}\}$$

$$k = 30^{\circ} \{\text{vertically opposite angles are equal}\}$$

$$y = 80^{\circ} \{\text{vertically opposite angles are equal}\}$$

SECTION 7: USING TWO OR MORE ANGLE TYPES

(a)



Solution

$$a = 20^{\circ} \text{ (vertically opposite angles)}$$

$$d = 70^{\circ} \text{ (vertically opposite angles)}$$

$$a + d + k = 180^{\circ} \text{ \{angles on a straight line\}}$$

$$\text{But } d = 70^{\circ}$$

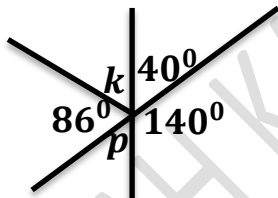
$$20^{\circ} + 70^{\circ} + k = 180^{\circ}$$

$$90^{\circ} + k = 180$$

$$k = 180^{\circ} - 90^{\circ}$$

$$k = 90^{\circ}$$

(b)



Solution

$$p = 40^{\circ} \text{ \{vertically opposite angles are equal\}}$$

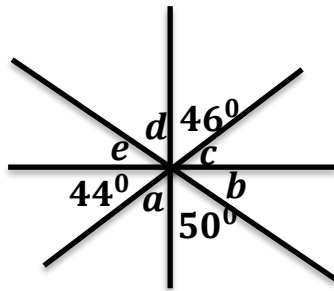
$$86^{\circ} + k + 40^{\circ} = 180^{\circ} \text{ \{angles on a straight line\}}$$

$$k + 126^{\circ} = 180^{\circ}$$

$$k = 180^{\circ} - 126^{\circ}$$

$$k = 54^{\circ}$$

(c)



Solution

$$a = 46^{\circ} \{\text{vertically opposite angles are equal}\}$$

$$d = 50^{\circ} \{\text{vertically opposite angles are equal}\}$$

$$c = 44^{\circ} \{\text{vertically opposite angles are equal}\}$$

$$e = b \{\text{vertically opposite angles are equal}\}$$

$$44^{\circ} + e + d + 46^{\circ} = 180^{\circ} \{\text{angles on a straight line}\}$$

But $d = 50^{\circ}$ and we now have:

$$44^{\circ} + e + 50^{\circ} + 46^{\circ} = 180^{\circ}$$

$$e + 140^{\circ} = 180^{\circ}$$

$$e = 180^{\circ} - 140^{\circ}$$

$$e = 40^{\circ}$$

$$\text{But } e = b$$

$$\text{Therefore, } b = 40^{\circ}$$

SECTION 8: WORD PROBLEMS INVOLVING ANGLES

Example 1. Find the complement of 29° .

Solution

Step 1: let the complement be x

$$x + 29^{\circ} = 90^{\circ} \{\text{complementary angles sum up to } 90^{\circ}\}$$

$$x = 90^{\circ} - 29^{\circ}$$

$$x = 61^{\circ}$$

Example 2. Find the complement of 45°

Solution

Step 1: let the complement be x

$$x + 45^{\circ} = 90^{\circ} \{\text{complementary angles sum up to } 90^{\circ}\}$$

$$x = 90^{\circ} - 45^{\circ}$$

$$x = 45^{\circ}$$

Example 3. Two complementary angles are in the ratio 2:3. Find the angles.

Solution

Step 1: total number of parts is:

$$2 + 3 = 5$$

Step 2:

$$\text{Angle} = 90^{\circ} \{\text{complementary angles sum up to } 90^{\circ}\}$$

Step 3: divide each number by 5 and multiply by 90°

$$\frac{2}{5} \times 90^{\circ} = 36^{\circ}$$

$$\frac{3}{5} \times 90^{\circ} = 54^{\circ}$$

Example 4. 40% of an angle is the complement of 70° . Find the angle.

Solution

Step 1: let the angle be x

40% of an angle means:

$$\frac{40}{100} \times x \dots \text{part 1}$$

Step 2: is means:

=

Step 3: complement of 70° means:

$$90^{\circ} - 70^{\circ} = 20^{\circ} \dots \text{part 2}$$

Step 4: Equate part 1 and 2

$$\frac{40}{100} \times x = 20^{\circ}$$

$$\frac{40x}{100} = \frac{20^{\circ}}{1}$$

Step 5: cross multiply

$$40x = 100 \times 20^{\circ}$$

$$40x = 2000$$

$$\frac{40x}{40} = \frac{2000^{\circ}}{40}$$

$$x = 50^{\circ}$$

Example 5. $\angle a$ and $\angle b$ are supplementary angles. If $\angle b$ is 30° more than four(4) times $\angle a$. Find angle $\angle a$ & $\angle b$.

Solution

Step 1:

1st sentence: $\angle a + \angle b = 180^\circ \dots \text{equation 1}$

2nd sentence: $\angle b = 30^\circ + 4 \times \angle a \dots \text{equation 2}$

Step 2: Using equation 1 and substitute for $\angle b$

$$\angle a + 30^\circ + 4 \angle a = 180^\circ$$

Step 3: collect like terms

$$\angle a + 4 \angle a = 180^\circ - 30^\circ$$

$$\frac{5 \angle a}{5} = \frac{150^\circ}{5}$$

$$\angle a = 30^\circ$$

Step 4: Substitute for $\angle a$ into equation 2

$$\text{But } \angle b = 30^\circ + 4 \angle a$$

$$\angle b = 30^\circ + 4 \times 30^\circ$$

$$\angle b = 30^\circ + 120^\circ$$

$$\angle b = 150^\circ$$

Example 6. Two angles are adjacent and form an angle of 160° . Their difference is 34° . Find the angles.

Solution

Step 1: let the two angles be x and y .

Step 2: 1st sentence means:

$$x + y = 160^\circ \dots \text{equation 1}$$

Step 3: 2nd sentence means:

$$x - y = 34^{\circ} \dots \text{equation 2}$$

Step 4: solving equation 1 and 2 simultaneously

$$x + y = 160^{\circ}$$

$$\pm(x - y = 34^{\circ})$$

$$x + x + y - y = 160^{\circ} + 34^{\circ}$$

$$\frac{2x}{2} = \frac{194^{\circ}}{2}$$

$$x = 97^{\circ}$$

Step 5: bringing equation 2 and substitute for x

$$x + y = 160^{\circ} \dots \text{equation 2}$$

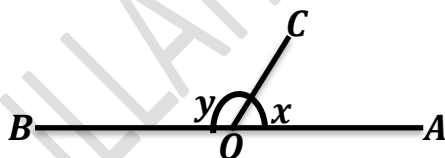
$$\text{But } x = 97^{\circ}$$

$$97^{\circ} + y = 160^{\circ}$$

$$y = 160^{\circ} - 97^{\circ}$$

$$y = 63^{\circ}$$

Example 7. In the diagram below, AOB is a straight line; x is less than y by 40° . Calculate x and y.



Solution

Step 1: angles on a straight line sum up to 180°

$$x + y = 180^{\circ} \dots \text{equation 1}$$

Step 2: x is less than y by 40° means:

$$y - x = 40^{\circ} \dots \text{equation 2}$$

Step 3: solve the equations simultaneously to find x and y

$$y + x = 180^0 \text{ ... equation 1}$$

$$+(y - x = 40^0) \text{ ... equation 2}$$

$$x - x + y + y = 180^0 + 40^0$$

$$\frac{2y}{2} = \frac{220^0}{2}$$

$$y = 110^0$$

Step 4: using equation 1 and substitute for y

$$x + y = 180^0 \text{ ... equation 1}$$

$$\text{but } y = 110^0$$

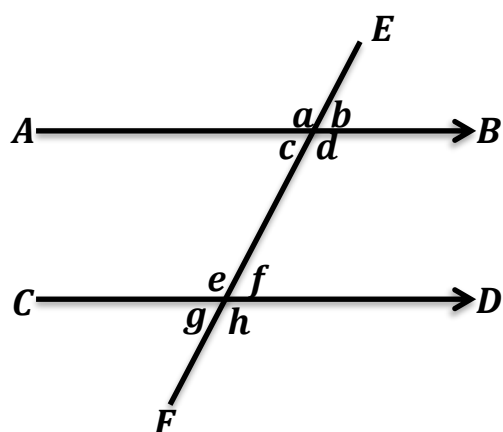
$$x + 110^0 = 180^0$$

$$x = 180^0 - 110^0$$

$$x = 70^0$$

SECTION 9: ANGLE PROPERTIES OF PARALLEL LINES

The idea of parallel lines is very important in geometry. In the figure below, AB & CD are two parallel lines. They are crossed by a slanting or oblique line EF.



This give rise to F and Z shapes.

The F shape gives two angles called corresponding angles. The Z shapes gives equal angles called alternate angles. In particular,

The straight line EF is called a transversal since it cut two parallel lines.

The pair of angles c & f; d & e; are called alternate angles

The pair of angles d & h; c & g; a & e; b & f are called corresponding angles.

The pair of angles d & f; c & e; are called co – interior angles.

From the diagram

a) Corresponding angles are equal. i. e. $b = f$, $a = e$, $d = h$ and $c = g$.

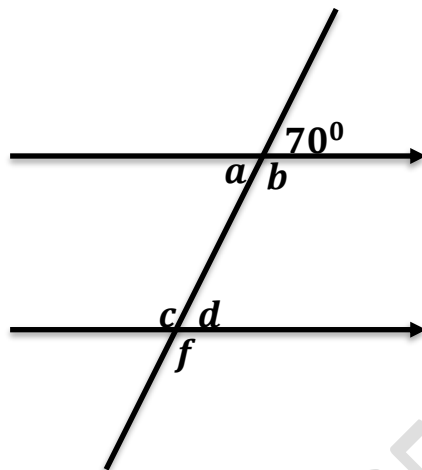
b) Alternate angles are equal I. e. $c = f$ and $d = e$.

c) Co – interior angles add up to 180° or co – interior angles

are supplementary. i. e. $d + f = 180^\circ$ and $c + e = 180^\circ$.

ILLUSTRATIVE EXAMPLES

Example 1. Find the value of the missing angles named with letters.



$$a = 70^\circ \{\text{vertically opposite angles are equal}\}$$

$$b + 70^\circ = 180^\circ \{\text{angles on a straight line sum up to } 180^\circ\}$$

$$b = 180^\circ - 70^\circ$$

$$b = 110^\circ$$

$$b + d = 180^\circ \{\text{co - interior angles sum up to } 180^\circ\}.$$

$$\text{But } b = 110^\circ$$

$$110^\circ + d = 180^\circ$$

$$d = 180^\circ - 110^\circ$$

$$d = 70^\circ$$

$$a + c = 180^\circ \{\text{co - interior angles}\}$$

But $a = 70^0$

$70^0 + c = 180^0$

$c = 180^0 - 70^0$

$c = 110^0$

$c = f$ {*vertically opposite angles are equal*}

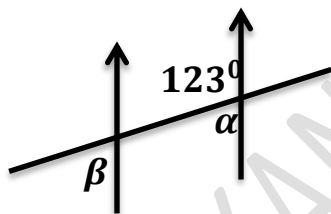
But $c = 110^0$

$f = 110^0$

Therefore, $\angle a = 70^0, \angle b = 110^0, \angle d = 70^0, \angle c = 110^0$ and

$\angle f = 110^0$.

Example 2. Find the value of α and β in the figure below



Solution

Step 1:

$\alpha + 123^0 = 180^0$ {*angles on a straight line*}

$\alpha = 180^0 - 123^0$

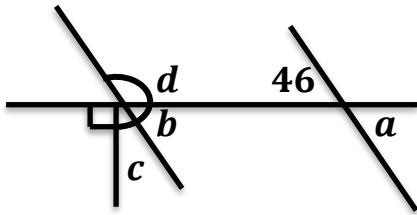
$\alpha = 57^0$

$\alpha = \beta$ {*corresponding angles*}.

But $\alpha = 57^0$

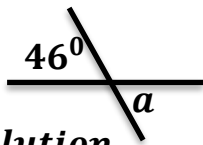
$\beta = 57^0$

Example 3. Find a, b, c & d .



Solution

Case 1;



Solution

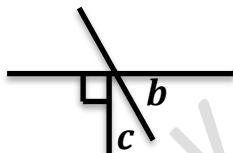
$a = 46^{\circ}$ (vertically opposite angles are equal).

$a = b$ (corresponding angles)

But $a = 46^{\circ}$

Therefore, $b = 46^{\circ}$

Case 2;



Solution

$90^{\circ} + c + b = 180^{\circ}$ (angles on a straight line)

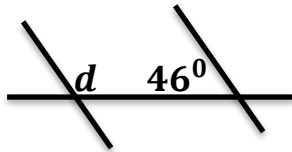
But $b = 46^{\circ}$

$90^{\circ} + c + 46^{\circ} = 180^{\circ}$

$c = 180^{\circ} - 90^{\circ} - 46^{\circ}$

$c = 44^{\circ}$

Case 3;



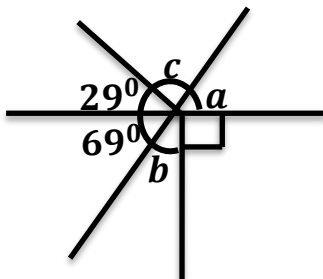
Solution

$$d + 46^{\circ} = 180^{\circ} \text{ (co - interior angle sum up to } 180^{\circ}\text{)}$$

$$d = 180^{\circ} - 46^{\circ}$$

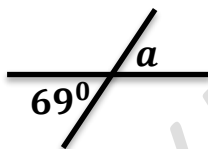
$$d = 134^{\circ}$$

Example 3. Find a , b & c .



Solution

Case 1;



Solution

$$a = 69^{\circ} \text{ (vertically opposite angles are equal)}$$

Case 2



$$29^{\circ} + c + a = 180^{\circ}$$

$$\text{But } a = 69^{\circ}$$

$$29^{\circ} + c + 69^{\circ} = 180^{\circ}$$

$$c = 180^{\circ} - 29^{\circ} - 69^{\circ}$$

$$c = 82^{\circ}$$

Case 3



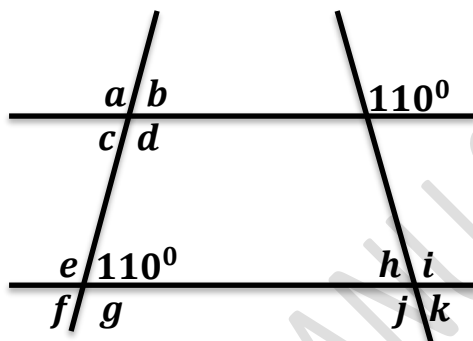
$$69^{\circ} + b + 90^{\circ} = 180^{\circ}$$

$$b = 180^{\circ} - 90^{\circ} - 69^{\circ}$$

$$b = 21^{\circ}$$

Example 4. Find the value of the lettered angles in the diagram below.

Give reasons for your answer. All angles are in degree.



Solution

$$f = 110^{\circ} \{ \text{vertically opposite angles} \}$$

$$e + 110^{\circ} = 180^{\circ} \{ \text{angles on a straight line} \}$$

$$e = 180^{\circ} - 110^{\circ}$$

$$e = 70^{\circ}$$

$$e = g, \text{ but } e = 70^{\circ}$$

$$e = 70^{\circ} \{ \text{vertically opposite angles} \}$$

$$d + 110^{\circ} = 180^{\circ} \{ \text{co - interior angles sum up to } 180^{\circ} \}$$

$$d = 180^0 - 110^0$$

$$d = 70^0$$

$$d = a \{\text{vertically opposite angles}\}$$

$$\text{But } d = 70^0$$

$$\text{Therefore, } a = 70^0$$

$$h + 110^0 = 180^0 \{\text{co - interior angles}\}$$

$$h = 180^0 - 110^0$$

$$h = 70^0$$

$$h = k \{\text{vertically opposite angles}\}$$

$$\text{But } h = 70^0$$

$$k = 70^0$$

$$h + i = 180^0 \{\text{angles on a straight line}\}$$

$$\text{But } h = 70^0$$

$$70^0 + i = 180^0$$

$$i = 180^0 - 70^0$$

$$i = 110^0$$

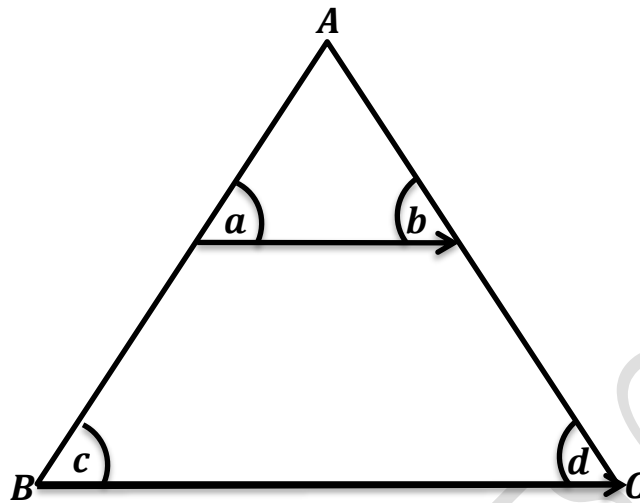
$$i = j \{\text{vertically opposite angles are equal}\}$$

$$i = 110^0$$

$$j = 110^0$$

SECTION 10: PARALLEL LINES IN A TRIANGLE

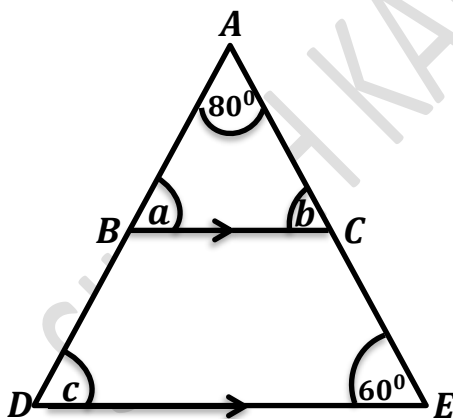
There are also some easy deductions from the idea of parallel lines.



$a = c$ {corresponding angles}

$b = d$ {corresponding angles}

Example 1. Find the value of a , b and c ?



Solution

Step 1: corresponding angles are equal

$$\angle DEC = \angle BCA$$

$$a + 60^{\circ} + 80^{\circ} = 180^{\circ} \{\text{sum of angles in a triangle}\}$$

$$a = 180^{\circ} - 60^{\circ} - 80^{\circ}$$

$$a = 40^{\circ}$$

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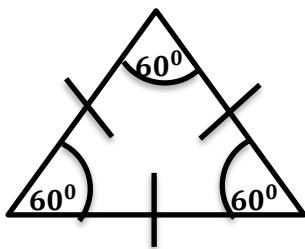
SECTION 11: TRIANGLES

There are six types of triangles:

1. EQUILATERAL TRIANGLE

An equilateral triangle has;

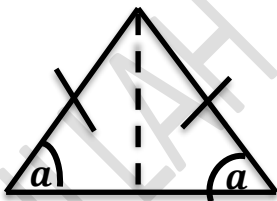
- All its sides equal*
- All interior angles are equal i. e. 60°*



2. ISOSCELES TRIANGLE

An isosceles triangle has;

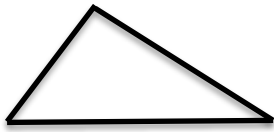
- Two of its sides equal.*
- Two of its interior angles equal.*



3. SCALENE TRIANGLE

A scalene triangle has;

- All its sides of different length.*



4. ACUTE ANGLED TRIANGLE

This is a triangle which has each of its interior angles less than 90° .

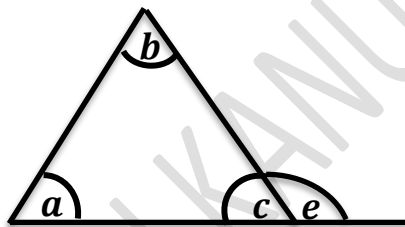
5. RIGHT ANGLE TRIANGLE

A right angle triangle is a triangle with one of its angle equal to 90° .

6. OBTUSE ANGLE TRIANGLE

This is a triangle which has one of its interior angle more than 90° .

Subsection 1: Angle properties of a triangle



a, b and c are interior angles.

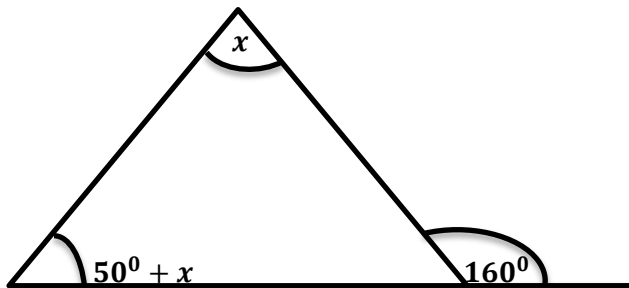
e is called an exterior angle.

The sum of two opposite interior angles equal the exterior

$$e = b + a$$

ILLUSTRATIVE EXAMPLES

Example 1. Find the value of x in the figure below



Solution

Step 1: $160^\circ = x + 50^\circ + x$

(exterior angle equals sum of the two opposite interior angles)

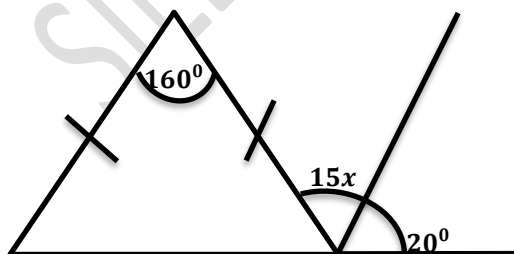
$$160^\circ = 2x + 50^\circ$$

$$160^\circ - 50^\circ = 2x$$

$$\frac{110}{2} = \frac{2x}{2}$$

Therefore, $x = 55^\circ$

Example 2. Find the value of x in the figure below



Solution

Step 1: the triangle is isosceles

Let the base angles be a .

$$a + a + 160^\circ = 180^\circ \{\text{sum of angles in a triangle}\}$$

$$2a + 160^\circ = 180^\circ$$

$$2a = 180^\circ - 160^\circ$$

$$\frac{2a}{2} = \frac{20^\circ}{2}$$

$$\underline{a = 10^\circ}$$

$$\text{Step 2: } 15x + 20^\circ = 160^\circ + 10^\circ$$

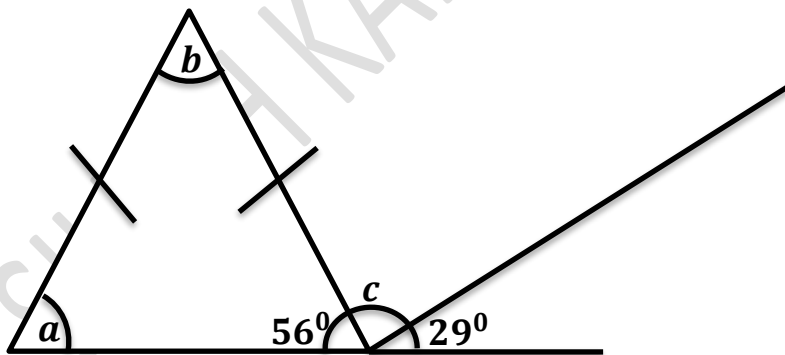
(exterior angles equal sum of two opposite interior angles)

$$15x = 170^\circ - 20^\circ$$

$$\frac{15x}{15} = \frac{150^\circ}{15}$$

$$\underline{x = 10^\circ}$$

Example 3. Find the value of a , b and c .



Solution

$$\text{Step 1: } a = 56^\circ \{\text{base angles of isosceles triangle are equal}\}$$

$$\text{Step 2: } b + a + 56^\circ = 180^\circ \{\text{sum of angle in a triangle}\}$$

$$\text{But } a = 56^\circ$$

$$b + 56^{\circ} + 56^{\circ} = 180^{\circ}$$

$$b = 180^{\circ} - 56^{\circ} - 56^{\circ}$$

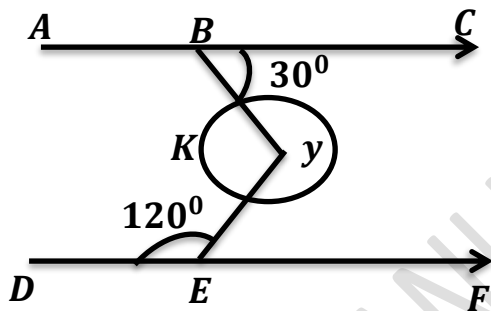
$$b = 68^{\circ}$$

Step 3: $c + 29^{\circ} + 56^{\circ} = 180^{\circ}$ {angles on a straight line}

$$c = 180^{\circ} - 29^{\circ} - 56^{\circ}$$

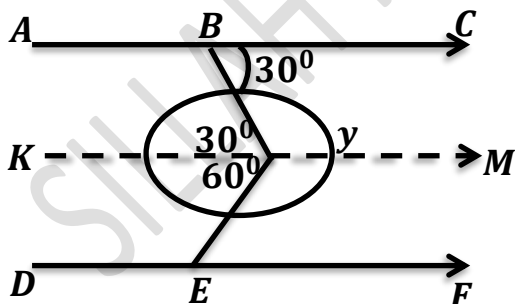
$$c = 105^{\circ}$$

Example 4. In the diagram below, $AC \parallel DF$. Angle CBT is 30° and angle DEK is 120° . Find the value of the reflex angle y .



Solution

Draw a line parallel to AC and DF at the point K as shown below



Step 1: Alternate angles are equal

$$\angle CBT = \angle BTS = 30^{\circ}$$

Step 2: Co – interior angles add up to 180°

$$\angle STE + \angle TED = 180^{\circ}$$

But $\angle TED = 120^\circ$

$\angle STE = 180^\circ - 120^\circ$

$\angle STE = 60^\circ$

Step 3: Sum of angles at a point is 360°

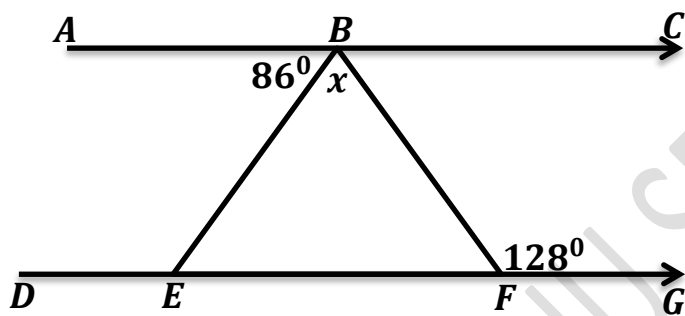
$y + 60^\circ + 30^\circ = 360^\circ$

$y = 360^\circ - 30^\circ - 60^\circ$

$y = 360^\circ - 90^\circ$

$y = 270^\circ$

Example 5.



In the diagram above, AC is parallel to DG , angle $BFG = 128^\circ$ and angle $ABE = 86^\circ$.

Find the value of:

i). Angle CBF

ii). x

Solution

i) $\angle BFE + \angle BFG = 180^\circ$ {sum of angles on a straight line}

$\angle BFE + 128^\circ = 180^\circ$

$\angle BFE = 180^\circ - 128^\circ$

$\angle BFE = 52^\circ$

But $\angle CBF = \angle BFE$ {alternate angles are equal}

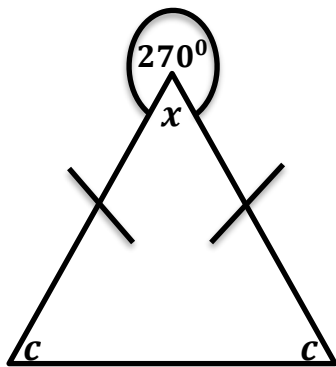
Therefore, $\angle CBF = 52^\circ$

ii) $52^\circ + 86^\circ + x = 180^\circ$ (sum of angles on a straight line)

$$x = 180^\circ - 86^\circ - 52^\circ$$

$$\underline{x = 42^\circ}$$

Example 6. Find the value of c in the figure below



Solution

Step 1: $x + 270^\circ = 360^\circ$ {Angles around a point sum up to 360° }

$$x = 360^\circ - 270^\circ$$

$$x = 90^\circ$$

Step 2: $c + c + x = 180^\circ$ {sum of angles in a triangle}

$$\text{But } x = 90^\circ$$

$$c + c + 90^\circ = 180^\circ$$

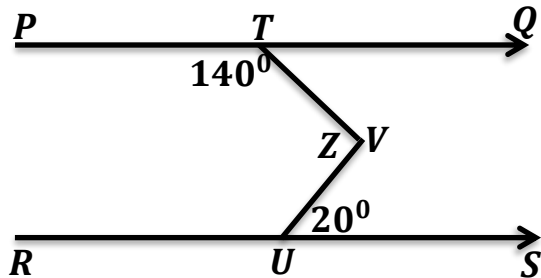
$$2c + 90^\circ = 180^\circ$$

$$2c = 180^\circ - 90^\circ$$

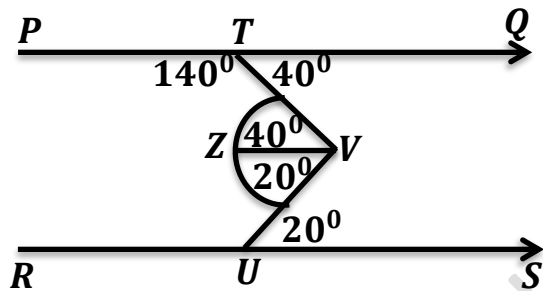
$$\frac{2c}{2} = \frac{90^\circ}{2}$$

$$\underline{c = 45^\circ}$$

Example 7. Find the value of Z .



Solution



Step 1: $140^\circ + \angle QTV = 180^\circ$ {angles on a straight line}

$$\angle QTV = 180^\circ - 140^\circ$$

$$\angle QTV = 40^\circ$$

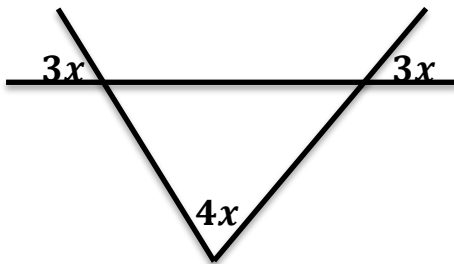
Step 2: $\angle TVZ = \angle QTV = 40^\circ$ {alternate angles are equal}

Step 3: $\angle UVZ = 20^\circ$

Step 4: $Z = 20^\circ + 40^\circ$

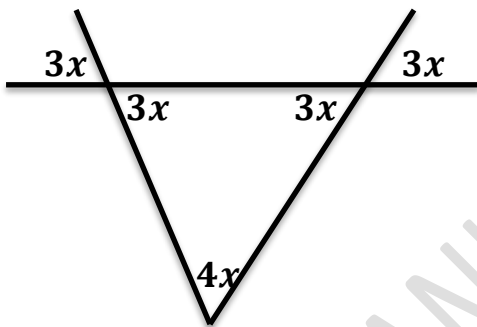
$$Z = 60^\circ$$

Example 8. Find the value of x .



Solution

Step 1: Vertically opposite angles are equal.



$$3x + 4x + 3x = 180^0 \text{ \{sum of angles in a triangle\}}$$

$$\frac{10x}{10} = \frac{180^0}{10}$$

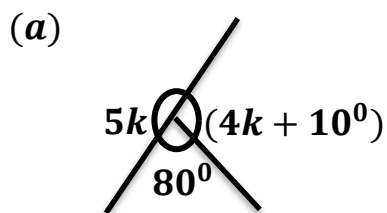
$$\underline{\underline{x = 18^0}}$$

SECTION 12: SKS PAPER 17

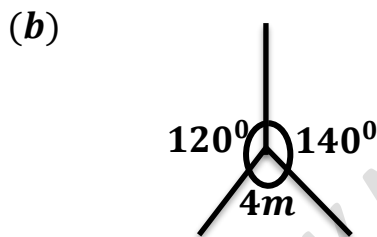
Part 1

Answer all questions

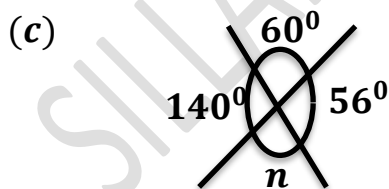
Question 1. Find the value of the lettered angles in the following diagrams



Answer [$k = 30^\circ$]

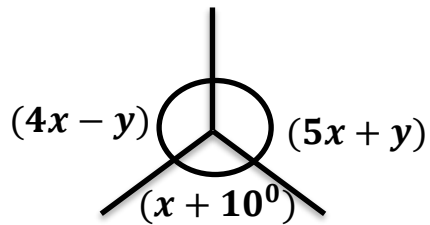


Answer [$k = 25^\circ$]



Answer [$n = 104^\circ$]

(d)

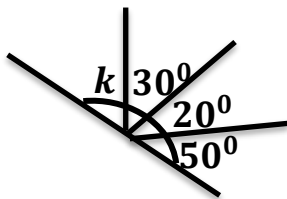


Answer [$x = 35^0$]

Part 2

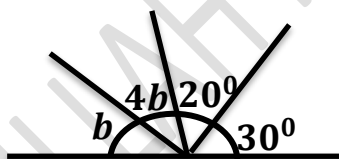
Question 1. Find the value of the lettered angles in the following diagram?

(a)



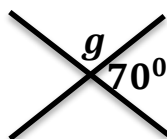
Answer [$k = 80^0$]

b)



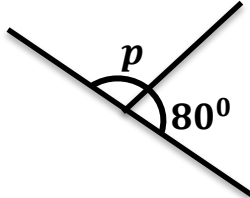
Answer [$b = 26^0$]

c)



Answer $[g = 110^{\circ}]$

d)

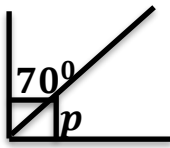


Answer $[p = 100^{\circ}]$

Part 3

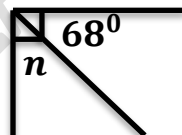
Question 1. Find the value of the lettered angles in the following diagram.

(a)



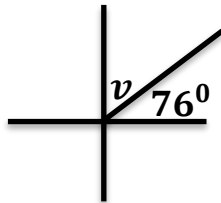
Answer $[p = 20^{\circ}]$

(b)



Answer $[n = 22^{\circ}]$

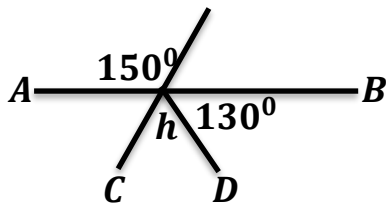
(c)



Answer $[v = 14^\circ]$

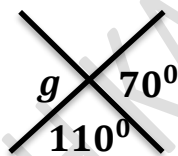
Part 4

(a)



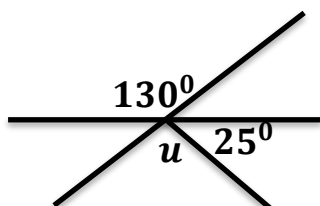
Answer $[h = 20^\circ]$

(b)



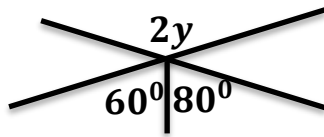
Answer $[g = 70^\circ]$

(c)



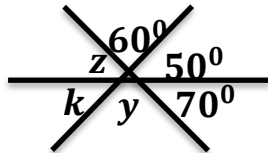
Answer $[u = 105^\circ]$

(d)



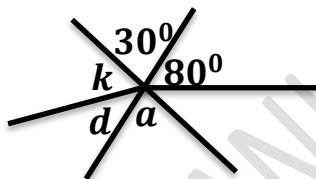
Answer [$y = 70^\circ$]

(e)



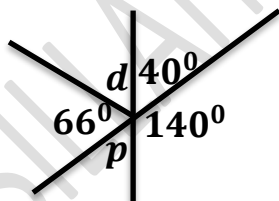
Answer [$y = 60^\circ, k = 50^\circ$ and $z = 70^\circ$]

(a)



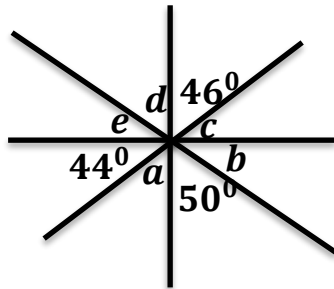
Answer [$a = 30^\circ, d = 80^\circ$ and $k = 70^\circ$]

(b)



Answer [$p = 40^\circ$ and $d = 74^\circ$]

(c)



Answers: [$a = 46^\circ$, $b = 40^\circ$, $c = 44^\circ$, $d = 50^\circ$ and $e = 40^\circ$]

Part 5

a) Find the complement of 58° .

Answer = 32°

b) Find the complement of 32° .

Answer = 58°

c) Two complementary angles are in the ratio 4: 5. Find the angles.

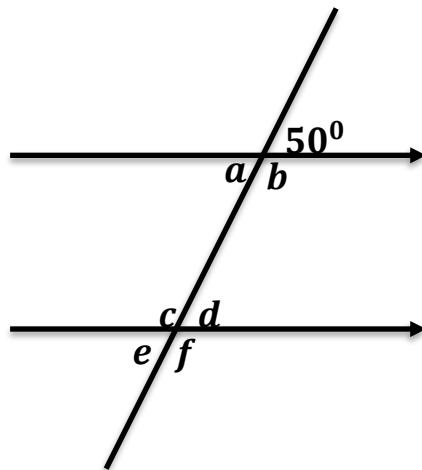
Answer [40° and 50°]

d) 50% of an angle is the complement of 40° . Find the angle.

Answer [The angle is 100°]

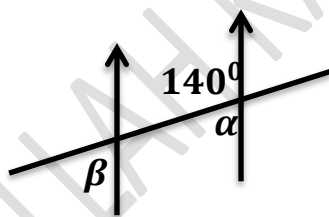
Part 6

Question 1. Find the value of the missing angles named with letters?



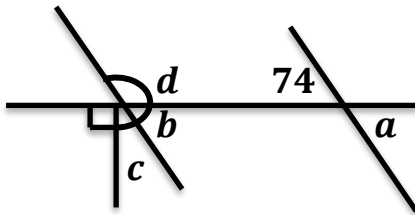
Answer [$a = 50^\circ$, $b = 130^\circ$, $c = 130^\circ$, $d = 50^\circ$, $e = 50^\circ$ and $f = 130^\circ$]

Question 2. Find the value of α and β in the figure below



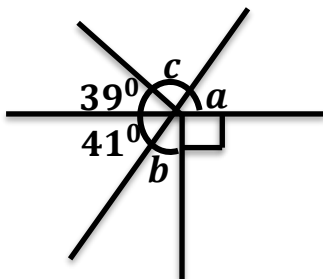
Answer [$\alpha = 40^\circ$ and $\beta = 40^\circ$]

3. Find a, b, c & d .



Answer [$a = 74^\circ, b = 74^\circ, c = 16^\circ$ and $d = 106^\circ$]

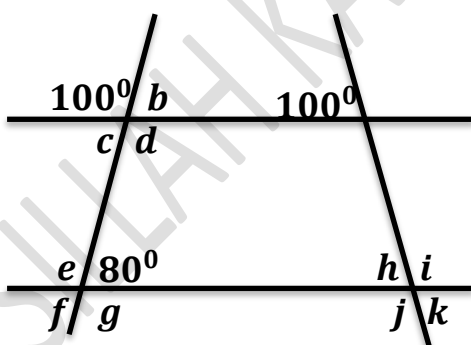
Question 3. Find a, b & c .



Answer [$a = 41^\circ, b = 49^\circ$ and $c = 100^\circ$]

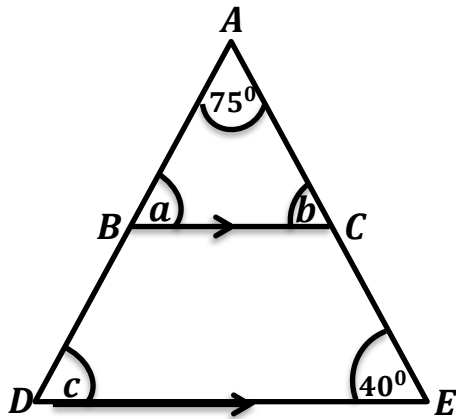
Question 4. Find the value of the lettered angles in the diagram below.

All angles are in degree.



Answer [$b = 80^\circ, c = 80^\circ, d = 100^\circ, e = 100^\circ, f = 80^\circ, g = 100^\circ,$
 $h = 100^\circ, i = 80^\circ, j = 80^\circ$ and $k = 100^\circ$]

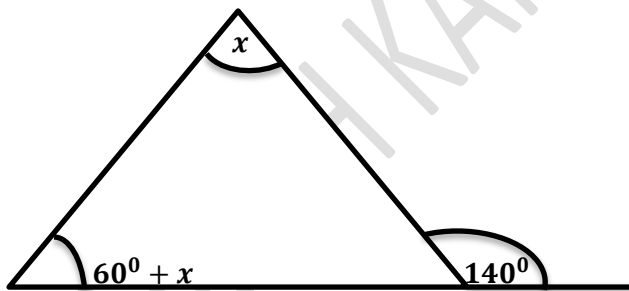
Question 5. Find the value of a , b and c ?



Answer [$a = 65^\circ$, $b = 40^\circ$, $c = 65^\circ$]

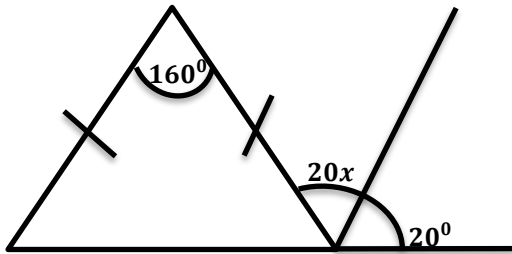
Part 7

Question 1. Find the value of x in the figure below



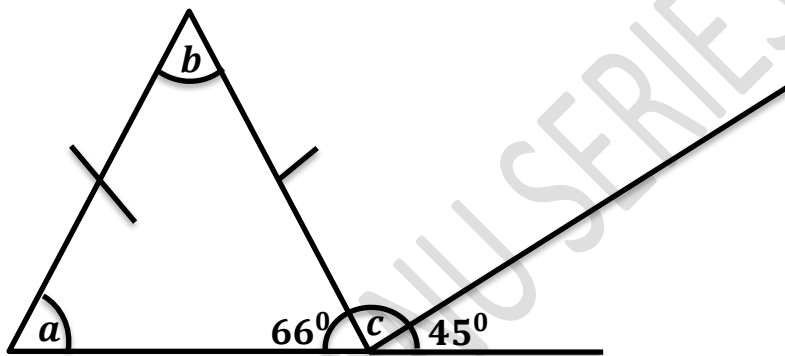
Answer [$x = 40^\circ$]

Question 2. Find the value of x in the figure below?



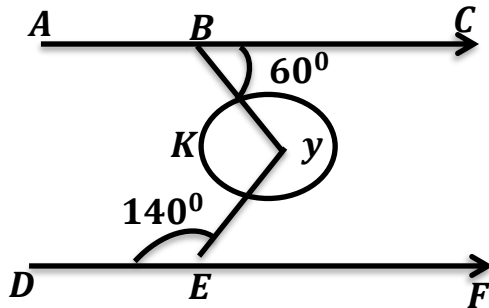
Answer [$x = 7.5^\circ$]

Question 3. Find the value of a , b and c .



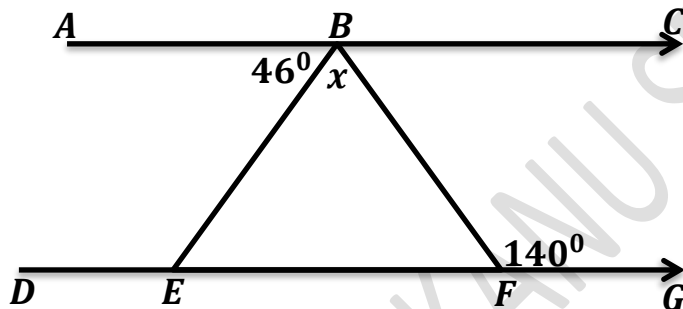
Answer [$a = 66^\circ$, $b = 48^\circ$ and $c = 69^\circ$]

Question 4. In the diagram below, $AC \parallel DF$. Angle CBT is 30° and angle DEK is 120° . Find the value of the reflex angle y .



Answer [$y = 260^\circ$]

Question 5.



In the diagram above, AC is parallel to DG , angle $BFG = 140^\circ$ and angle $ABE = 46^\circ$.

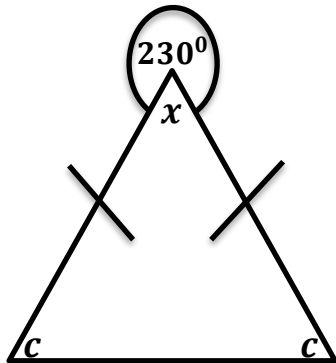
Find the value of:

i) Angle CBF

ii) x

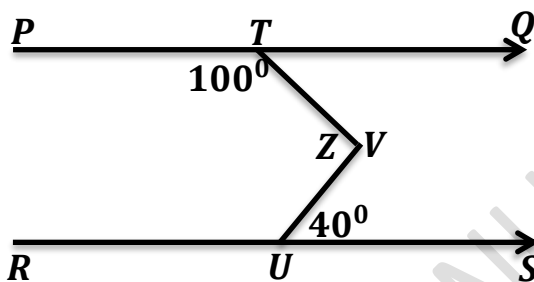
Answers [$\angle CBF = 40^\circ$ and $x = 94^\circ$]

Question 6. Find the value of c in the figure below



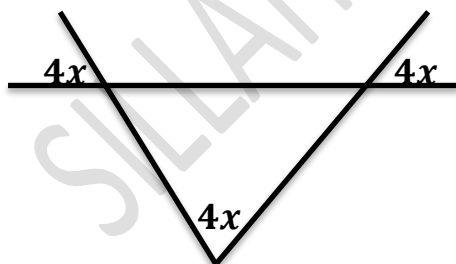
Answer [$c = 25^\circ$]

Question 7. Find the value of Z .



Answer [$Z = 120^\circ$]

Question 8. Find the value of x .



Answer [$x = 15$]

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CHAPTER 18

LOGARITHM

SECTION 1: DEFINITION

The most common logarithms used are logarithms to base 10, therefore if the base is omitted it means it is in base 10. The table of logarithms gives only the decimal part of each logarithm called the mantisa.

The whole number called the characteristic can easily be found by expressing the number in the standard form.

The logarithm table has three main columns:

- 1. column one: this gives the first two digits of the number*
- 2. column two: this gives the third digit*
- 3. column three: (the difference column) gives the fourth digit*

SECTION 2: FINDING LOGARITHMS

Example:

1. $\log 2847 = 3.4544$ (from tables)

2. $\log 2.324 = 0.36622$ (from tables)

3. $\log 147.26 = 2.1681$ (from tables)

4. $\log 0.000246 = \bar{4} + \log 2.46 = \bar{4} + 0.3909$

5. $\log 0.024 = \bar{2} + \log 2.4 = \bar{2} + 0.3802$

SECTION 3: ADDITION IN LOGARITHM

Example 1. Evaluate:

$$\begin{array}{r} 1.2406 \\ +2.1246 \\ \hline 3.3652 \end{array}$$

Example 2. Evaluate:

$$\begin{array}{r} 2.2410 \\ +\bar{3}.1246 \\ \hline \bar{1}.3656 \end{array}$$

Note: $2 + \bar{3} = 2 - 3 = -1 = \bar{1}$

Example 3. Evaluate:

$$\begin{array}{r} 7.4612 \\ +\bar{4}.2213 \\ \hline 3.6825 \end{array}$$

SECTION 4: SUBTRACTION IN LOGARITHM

Example 1. Evaluate:

$$\begin{array}{r} 1.2468 \\ -2.1102 \\ \hline \bar{1}.1366 \end{array}$$

Example 2. Evaluate:

$$\begin{array}{r} 2.4621 \\ -1.2468 \\ \hline \bar{1}.2153 \end{array}$$

Example 3. Evaluate:

$$\begin{array}{r} \bar{1}.777 \\ -2.078 \\ \hline \bar{3}.699 \end{array}$$

SECTION 5: MULTIPLICATION IN LOGARITHM

Example 1. Evaluate:

$$\bar{1}.2468$$

$$\times \quad \underline{2}$$

$$\underline{\bar{2}.4936}$$

Example 2. Evaluate:

$$\bar{2}.2684$$

$$\times \quad \underline{3}$$

$$\underline{\bar{6}.8052}$$

SECTION 6: USING LOGARITHM TO EVALUATE EXPRESSIONS

Subsection 1: Products

Step 1: Find the logarithm of the number

Step 2: Add the logarithm

Step 3: Find the antilogarithm of the result

Note: for products, add the logarithms

ILLUSTRATIVE EXAMPLE

Example 1. Evaluate 2.426×4.024 using logarithms

Solution

Steps 1 and 2

NUMBER	LOGARITHM
2.426	0.3849
4.024	0.6047
Add the logs for multiplication	0.9896

Step 3: Find the antilog of 0.9896

Antilog of 0.9896 = 9.7634

Therefore, $2.426 \times 4.024 = 9.7634$

Example 2. Evaluate 4.0246×2.102 using logarithms

Solution

Steps 1 and 2

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<i>NUMBER</i>	<i>LOGARITHM</i>
4.0246	0.6047
2.102	0.3226
<i>Add the logs for multiplication</i>	0.9273

Step 3: Find the antilog of 0.9273

Antilog of 0.9273 = 8.4586

Therefore, $4.0246 \times 2.102 = 8.4586$

Subsection 2: Division

Step 1: Find the logarithms of the numbers

Step 2: Subtract the logarithm of the denominator from the numerator

Note: If the denominator is a product treat it as in the example above

Step 3: Find the antilogarithm of the result

Note: For division, subtract the logarithms

ILLUSTRATIVE EXAMPLES

Example 1. Use logarithms to evaluate

24.67

21.42

Solution

Steps 1 and 2

<i>NUMBER</i>	<i>LOGARITHM</i>
24.67	1.3922
21.42	1.3308
<i>Subtract the logs for division</i>	0.0614

Step 3: Find the antilog of 0.0614 = 1.1519

Therefore, $\frac{24.67}{21.42} = 1.1519$

Example 2. Use logarithms to evaluate

6.024

4.222

Solution

Steps 1 and 2

NUMBER	LOGARITHM
6.024	0.7799
4.222	0.6255
Subtract the logs for division	0.1544

Step 3: find the antilog of 0.1544 = 1.4269

Therefore, $\frac{6.024}{4.222} = 1.4269$

Subsection 3: Powers

POWERS: we can find any power of a number very easily.

Step 1: find the logarithm of the number

Step 2: multiply the logarithm by the power

Step 3: find the antilogarithm of the result

ILLUSTRATIVE EXAMPLES

Example 1. Evaluate $(23)^2$

Solution

Step 1: $\log 23 = 1.3617$

Step 2: $1.3617 \times 2 = 2.7234$

Step 3: *antilog* $2.7234 = 528.9322$

Therefore, $(23)^2 = 528.9322$

Example 2. Evaluate $(2.12)^3$

Solution

Step 1: *log* $2.12 = 0.3263$

Step 2: $0.3263 \times 3 = 0.9789$

Step 3: *antilog* $0.9789 = 9.5258$

Therefore, $(2.12)^3 = 9.5289$

Subsection 4: Roots

It is just as easy to find squares, cube, or other roots.

Step 1: *find the logarithm of the number*

Step 2: *divide the logarithm by the root*

Step 3: *find the antilogarithm of the result*

ILLUSTRATIVE EXAMPLES

Example 1. Use logarithms to evaluate the following

(i) $\sqrt{11}$

(ii) $\sqrt[3]{17}$

Solution

(i) **Step 1:** *log* $11 = 1.0414$

Step 2: $1.0414 \div 2 = 0.5207$

Note: we divide by two(2) because of the square root

Step 3: *antilog* $0.5207 = 3.3167$

Therefore, $\sqrt{11} = 3.3167$

(ii). **Step 1:** *log* $17 = 1.2304$

Step 2: $1.2304 \div 3 = 0.4101$

Step 3: *antilog* $0.4101 = 2.5710$

Therefore, $\sqrt[3]{17} = 2.5710$

Note: *we divide by three(3) because of the cube root*

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SECTION 7: SKS PAPER 18

Part 1

Question 1. Evaluate 2.433×4.0222 using logarithms

Answer = 9.7859

Question 2. Evaluate $4.0345 \times 2.222 \times 2.122$ using logarithms

Answer = 19.0195

Question 3. Use logarithms to evaluate

$$\frac{44.67}{33.42}$$

Answer = 1.3366

Question 4. Use logarithms to evaluate

$$\frac{7.134}{3.456}$$

Answer = 2.0640

Part 2

Question 1. Evaluate $(43)^2$

Answer = 1849.2686

Question 2. Evaluate $(1.11)^3$

Answer = 1.3674

Question 3. Use logarithms to evaluate the following

(i) $\sqrt{19}$

Answer = 4.3591

(ii) $\sqrt[3]{11}$

Answer = 2.2238

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CHAPTER 19

STATISTICS

SECTION 1: INTRODUCTION

Statistics is the discipline that concerns the collection, organization, analysis, interpretation and presentation of data.

SECTION 2: MEASURES OF CENTRAL TENDENCY

The measures of central tendency are the averages of the frequency distribution. These averages are the mean, mode and median.

Subsection 1: Mode

The mode of a set of data values is the value that appears most often.

ILLUSTRATIVE EXAMPLES

Example 1. Find the mode of the following?

1, 1, 2, 1, 2, 3, 4 and 5.

Solution

Mode = 1 [because it appears the highest]

Example 2. Find the mode of the following?

4, 5, 2, 3, 2, 6, 2, 4 and 2.

Solution

Mode = 2 [because it appears the highest]

Example 3. Find the mode of the following?

2, 1, 3, 4, 1, 2 and 2.

Solution

Mode = 2 [*because it appears the highest*]

Example 4. Find the mode of the following?

1, 1, 2, 1, 2, 3, 4 and 2.

Solution

Mode = 1 and 2 [*because they appear more than the others*]

Example 5. Find the mode of the following?

3, 3, 3, 3, 3 and 3.

Solution

There is no mode because the numbers are the same

Example 6. Find the mode of the table below?

Marks	0	1	2	3	4
Frequency	8	6	9	7	4

Solution

Mode = 2 [*because it has the highest frequency*]

Example 7. Find the mode of the table below?

Age	0	1	2	3	4	5	6
Frequency	9	5	7	8	4	3	2

Solution

Mode = 0 [*because it has the highest frequency*]

Example 8. Find the mode of the table below?

Age	0	1	2	3
Frequency	5	3	4	9

Solution

Mode = 3 [because it has the highest frequency]

Example 9. The total number of pupils in a class is 20. Find the modal age of the class?

Age	10	11	12	13
Frequency	4	x	5	4

Solution

Step 1: Find the value of x

Total number of pupils = sum of all the frequencies

$$\Rightarrow 20 = 4 + x + 5 + 4$$

$$\Rightarrow 20 = 13 + x$$

$$\Rightarrow 20 - 13 = x$$

$$\Rightarrow x = 7$$

Step 2: Fill in the value of x in the table

Age	10	11	12	13
Frequency	4	7	5	4

Step 3: Find the modal age

$\Rightarrow$ The modal age is 11 [because it has the highest frequency]

Example 8. The total number of pupils in a class is 30. Find the modal age of the table below ?

Age	12	14	16	18
Frequency	$9 - x$	$x + 4$	x	8

Solution

Step 1: Find the value of x

$\Rightarrow$ Total number of pupils = sum of all the frequencies

$$\Rightarrow 30 = 9 - x + x + 4 + x + 8$$

$$\Rightarrow 30 = x + 21$$

$$\Rightarrow x = 30 - 21$$

$$\Rightarrow x = 9$$

Step 2: Fill in the value of x in the table

Age	2	4	6	8
Frequency	0	13	9	8

Step 3: Find the modal age

$\Rightarrow$ The modal age is 4 [because it has the highest frequency]

Example 9. If the total number of boys in a class is 30. Find the modal age of the boys?

Age	3	6	9
Frequency	x	$x + 1$	$x - 1$

Solution

Step 1: Find the value of x

$\Rightarrow$ Total number of boys = sum of all the frequencies

$$30 = x + x + 1 + x - 1$$

$$30 = 3x + 0$$

$$\frac{30}{3} = \frac{3x}{3}$$

$$x = 10$$

Step 2: Fill in the value of x in the table

Age	3	6	9
Frequency	10	11	9

Step 3: Find the modal age

$\Rightarrow$ The modal age is 6 [because it has the highest frequency]

Subsection 2: Mean

➤ Case 1: The mean of a set of numbers or values

The mean is the average of the numbers. It is easy to calculate: add up all the numbers, then divide by how many numbers there are. In other words it is the sum divided by the count. The mean is denoted by: $\bar{x}$ i. e.

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n} = \frac{\sum x}{n}$$

ILLUSTRATIVE EXAMPLES

Example 1. Find the mean of the following numbers?

1, 2, 3, 4 and 5.

Solution

Step 1: Write the formula for mean of numbers

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \cdots + x_n}{n} = \frac{\sum x}{n}$$

Step 2: Substitute and simplify

$$\bar{x} = \frac{1 + 2 + 3 + 4 + 5}{5}$$

$$\bar{x} = \frac{15}{5}$$

$$\bar{x} = 3$$

Example 2. Find the mean of 2, 4, 6, 8, and 10?

Solution

Step 1: Write the formula for mean of numbers

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \cdots + x_n}{n} = \frac{\sum x}{n}$$

Step 2: Substitute and simplify

$$\bar{x} = \frac{2 + 4 + 6 + 8 + 10}{5}$$

$$\bar{x} = \frac{30}{5}$$

$$\bar{x} = 6$$

Example 3. Find the mean of 3, 6, 9, and 12?

Solution

Step 1: Write the formula for mean of numbers

$$\bar{x} = \frac{\sum x}{n}$$

Step 2: Substitute and simplify

$$\bar{x} = \frac{3 + 6 + 9 + 12}{4}$$

$$\bar{x} = \frac{30}{4}$$

$$\bar{x} = 7.5$$

Example 4. If the mean is 3. Find the value of x in the following?

1, 2, x , 4 and 5

Solution

Step 1: Write the formula for mean of numbers and simplify

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \cdots + x_n}{n} = \frac{\sum x}{n}$$

Step 2: Substitute and simplify

$$3 = \frac{1 + 2 + x + 4 + 5}{5}$$

$$\frac{3}{1} = \frac{12 + x}{5}$$

Step 3: Cross multiply

$$3 \times 5 = 12 + x$$

$$15 = 12 + x$$

Step 4: Transpose

$$15 - 12 = x$$

$$x = 3$$

Example 5. If the mean is 6. Find the value of x in the following

2, 4, 6, $x + 1$ and 10

Solution

Step 1: Write the formula for mean of numbers

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \cdots + x_n}{n} = \frac{\sum x}{n}$$

Step 2: Substitute and simplify

$$6 = \frac{23 + x}{5}$$

$$\frac{6}{1} = \frac{23 + x}{5}$$

Step 3: Cross multiply

$$6 \times 5 = 23 + x$$

$$30 = 23 + x$$

Step 4: Transpose

$$30 - 23 = x$$

$$x = 7$$

➤ **Case 2: Calculating the mean from a frequency distribution table**

If the values $x_1, x_2, x_3, \dots, x_n$ occur with frequencies $f_1, f_2, f_3, \dots, f_n$ respectively, then the mean is given by:

$$\bar{x} = \frac{\sum fx}{\sum f}$$

Where f is the frequency, x is the class marks and $\sum$ means the "sum of"

ILLUSTRATIVE EXAMPLES

Example 1. Find the mean mark of the table below?

Marks	1	2	3	4
Frequency	2	4	6	8

Solution

Step 1: Represent it in this format

Marks(x)	f	fx
1	2	2

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2	4	8
3	6	18
4	8	32
	$\Sigma f = 20$	$\Sigma fx = 60$

Step 2: Write the formula for mean and substitute

$$\bar{x} = \frac{\Sigma fx}{\Sigma f}$$

$$\bar{x} = \frac{60}{20}$$

$$\bar{x} = 3$$

Note: fx means $f \times x$

Example 2. If the mean is 10, find the value of x in the table below?

Marks	1	2	3	4
Frequency	3	6	9	12

Solution

Step 1: Represent it in this format

Marks(x)	f	fx
1	3	3
2	6	12
3	9	27
4	12	48
	$\Sigma f = 30$	$\Sigma fx = 90$

Step 2: Write the formula for mean and substitute

$$\bar{x} = \frac{\Sigma fx}{\Sigma f}$$

$$\bar{x} = \frac{90}{30} = 3$$

Subsection 3: Median**➤ Case 1: The median of a set of numbers or values**

The median is the value separating the higher half from the lower half of a data sample. For a data set, it may be thought of as the “middle” value. For example, in the data set [1, 3, 3, 6, 7, 8, 9], the middle number is 6, the fourth largest and also the fourth smallest, number in the sample. To find the median of a set of ‘n’ numbers: $x_1, x_2, x_3, \dots, x_n$

1. Arrange the numbers in order of magnitude

2. If n is odd then the median is the middle item i.e. the

$\frac{1}{2}(n + 1)$ th item or observation

3. If n is even then the median is the arithmetic mean of the two middle

items i.e. the $\frac{1}{2}(n)$ th item and $\frac{1}{2}(n + 1)$ th item or $\frac{1}{2}(n)$ th item

and the item after it

ILLUSTRATIVE EXAMPLES

Example 1. Find the median of the following numbers?

- (i) 3, 1, 2, 4 and 5.
- (ii) 2, 6, 4, 8, 10, 12 and 14.
- (iii) 3, 9, 6, 12, 15 and 18
- (iv) 8, 12, 4, 16, 20 and 24

Solution

(i).

Step 1: Rearrange the numbers in ascending order

1, 2, 3, 4, 5

Step 2: Find the middle number

Middle number is 3

Therefore, the median is 3

(ii).

Step 1: Rearrange in ascending order

2, 4, 6, 8, 10, 12, 14

Step 2: Find the middle number

The middle number is 8.

Therefore, the median is 8

(iii).

Step 1: Rearrange in ascending order

3, 6, 9, 12, 15, 18

Step 2: Find the middle number

The middle numbers are 9 and 12

Step 3: Add the middle numbers and divide by two

$$\text{Median} = \frac{9 + 12}{2}$$

$$\text{Median} = \frac{21}{2}$$

$$\text{Median} = 10.5$$

Therefore, the median is 10.5

(iv).

Step 1: rearrange in ascending order

4, 8, 12, 16, 20, 24

Step 2: Find the middle number

The middle numbers are 12 and 16

Step 3: Add the middle numbers and divide by two

$$\text{Median} = \frac{12 + 16}{2}$$

$$\text{Median} = \frac{28}{2}$$

$$\text{Median} = 14$$

Therefore, the median is 14

➤ *Case 2: Median from an ungrouped frequency distribution*

Example 1. Find the median of the table below?

<i>Marks</i>	1	2	3	4
<i>Frequency</i>	2	4	6	8

Solution

Method 1

Step 1: Write the number of times each mark appears in ascending order

1, 1, 2, 2, 2, 2, 3, 3, 3, 3, 3, 3, 4, 4, 4, 4, 4, 4, 4

Step 2: Find the middle numbers

The middle numbers are 3 and 3

Step 3: Add the middle numbers and divide by two

$$\text{Median} = \frac{3 + 3}{2}$$

$$\text{Median} = \frac{6}{2}$$

$$\text{Median} = 3$$

$\therefore$ The median mark is 3

Method 2

Step 1: Add all the frequencies

$$\text{Total frequency} = 2 + 4 + 6 + 8 = 20$$

Step 2: Since the total frequency is even (i. e 20) and

$$\left(\frac{20}{2}\right)^{\text{th}} = 10^{\text{th}} \text{ observation,}$$

The median will be the average of the 10th and 11th items.

From the table the 10th item is 3 and the 11th item is also 3

Hence, the median mark

$$\frac{3 + 3}{2} = 3 \text{ marks}$$

$\therefore$ The median is 3

Example 2. Find the median of the table below?

Marks	1	2	3	4
Frequency	2	4	6	9

Solution

Method 1

Step 1: Write the number of times each mark appears in ascending order

1, 1, 2, 2, 2, 2, 3, 3, 3, 3, 3, 3, 4, 4, 4, 4, 4, 4, 4, 4

Step 2: find the middle number

The middle number is 3

∴ The median mark is 3

Method 2

Step 1: add all the frequencies

Total frequency = 2 + 4 + 6 + 9 = 21

Step 2: Since the total frequency is odd (i.e 21) and $\left(\frac{21 + 1}{2}\right)^{th}$
 $= \left(\frac{22}{2}\right)^{th} = 11^{th}$

the median will be the average of the 11th and 11th items.

From the table the 11th item is 3 and the 11th item is also 3

Hence, the median mark is:

$$\frac{3 + 3}{2} = 3 \text{ marks}$$

SECTION 3: THE USE OF AT LEAST

Example 1. *These are the marks obtained by some students*

Marks	1	2	3	4	5
Frequency	2	4	6	8	10

Find the number of students that score:

- (i) **At least one mark**
- (ii) **At least two marks**
- (iii) **At least three marks**
- (iv) **At least four marks**
- (v) **At least five marks**

Solution

- (i) **At least 1 mark means: the sum of the frequencies of one mark and above.**

$$= 2 + 4 + 6 + 8 + 10$$

$$= 30 \text{ students}$$

- (ii) **At least 2 marks means: the sum of the frequencies of two marks and above.**

$$4 + 6 + 8 + 10 = 28 \text{ students}$$

- (iii) **At least 3 marks means: the sum of the frequencies of three marks and above.**

$$6 + 8 + 10 = 24 \text{ students}$$

- (iv) **At least 4 marks means: the sum of the frequencies of four marks and above.**

$$8 + 10 = 18$$

- (v) **At least 5 marks means: the sum of the frequencies of five marks and above.**

$$= 10 \text{ students}$$

SECTION 4: THE USE OF AT MOST

Example 1. *These are the marks obtained by some students.*

Marks	1	2	3	4	5
Frequency	2	4	6	8	10

Find the number of student that score:

- (i) **At most 1 mark**
- (ii) **At most 2 marks**
- (iii) **At most 3 marks**
- (iv) **At most 4 marks**
- (v) **At most 5 marks**

Solution

(i). **At most 1 mark means: the frequency of one mark and below.**

$\Rightarrow 2$ students

(ii). **At most 2 marks means: the sum of the frequencies of two marks and below.**

$\Rightarrow 2 + 4 = 6$ students

(iii). **At most 3 marks means: the sum of the frequencies of three marks and below.**

$\Rightarrow 2 + 4 + 6 = 12$ students

(iv). **At most 4 marks means: the sum of the frequencies of four marks and below.**

$\Rightarrow 2 + 4 + 6 + 8 = 20$ students

(v). **At most 5 marks means: the sum of the frequencies of five marks and below.**

$$\Rightarrow 10 + 8 + 6 + 4 + 2 = 30 \text{ students}$$

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SECTION 5: THE USE OF ABOVE

Example 1. These are the marks obtained by some students.

Marks	1	2	3	4	5
Frequency	2	4	6	8	10

Find the number of student that score:

- (i) **Above 1 mark**
- (ii) **Above 2 marks**
- (iii) **Above 3 marks**
- (iv) **Above 4 marks**
- (v) **Above 5 marks**

Solution

- (i) **Above 1 mark means: the sum of all the frequencies which marks are greater than one.**
 $4 + 6 + 8 + 10 = 28$ students
- (ii) **Above 2 marks means: the sum of all the frequencies which marks are greater than two.**
 $6 + 8 + 10 = 24$ students
- (iii) **Above 3 marks means: the sum of all the frequencies which marks are greater than three.**
 $8 + 10 = 18$ students
- (iv) **Above 4 marks means: the sum of all the frequencies which marks are greater than four.**
 $= 10$ students
- (v) **Above 5 marks means: the sum of all the frequencies which marks are greater than five.**
 $=$ No student

Example 2. In a certain college, these are the marks obtained by the students.

Marks	1	2	3	4	5
Frequency	2	4	6	8	10

Find the number of students that score:

- (i) ***Below 1 mark***
- (ii) ***Below 2 marks***
- (iii) ***Below 3 marks***
- (iv) ***Below 4 marks***
- (v) ***Below 5 marks***

Solution

- (i) ***Below 1 mark = no student***
- (ii) ***Below 2 marks = 2 students***
- (iii) ***Below 3 marks = $2 + 4 = 6$ students***
- (iv) ***Below 4 marks = $2 + 4 + 6 = 12$ students***
- (v) ***Below 5 marks = $2 + 4 + 6 + 8 = 20$ students***

SECTION 6: FREQUENCY TABLE

A frequency table is a method of organizing raw data in a compact form by displaying a series of scores in ascending or descending order, together with their frequencies – the number of times each score occurs in the respective data.

ILLUSTRATIVE EXAMPLES

Example 1. Construct a frequency table for the following marks.

23, 42, 42, 51

51, 45, 25, 50

64, 37, 51, 42

45, 71, 71, 51

Solution

Step 1: construct a table in ascending order of the marks

MARKS	TALLY	FREQUENCY
23	/	1
25	/	1
37	/	1
42	///	3
45	//	2
50	/	1
51	////	4
64	/	1
71	//	2

Example 2. *The following were the mathematics scores for 30 candidates at a certain school.*

4, 8, 9, 4, 6, 6

6, 6, 7, 2, 8, 5

2, 5, 8, 3, 5, 4

1, 3, 7, 2, 7, 8

3, 8, 9, 5, 9, 7

Construct a frequency table for this information.

Solution

MARKS	TALLY	FREQUENCY
1	/	1
2	///	3
3	///	3
4	///	3
5	////	4
6	////	4
7	////	4
8	////-	5
9	///	3

SECTION 7: GRAPHICAL REPRESENTATION OF DATA

In Statistics, data can be represented graphically in many forms depending on the data and purpose for which it is intended. We shall consider the following graphs:

1. Pie chart

2. Bar chart

Subsection 1: Pie chart

A pie chart is used to represent a given information or data pictorially, using the sector of a circle.

The following steps show how a given data is represented or illustrated on a pie chart.

Step 1: Find the total of the category values

Step 2: Calculate the angle of each category.

$$\text{Angle of category} = \frac{\text{Category value}}{\text{Sum of category values}} \times 360^\circ$$

Step 3: Draw the sectors on a circle diagram using the angles. To draw the diagram

i). Draw a circle using a pair of compasses

ii). Draw a radius as a straight line

iii). Use a protractor to draw the angles of each sector. Make sure the protractor is on the center of the circle. Measure the angle very carefully.

iv). Label each sector carefully and the size of the angle also.

ILLUSTRATIVE EXAMPLES

Example 1. The table below show the various items sold in a city shop in freetown.

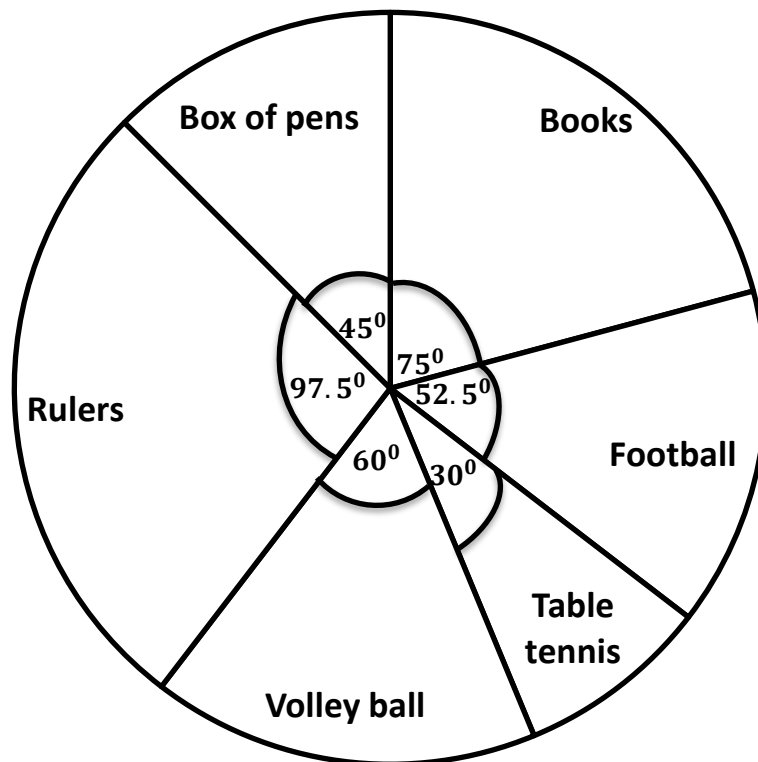
ITEMS	NUMBER
Books	50
Football	35
Table tennis	20
Volley ball	40
Rulers	65
Box of pens	30

a). Draw a pie chart to represent the information.

b). What percentage of rulers are sold

Solution

ITEMS	ANGLE
Books	$\frac{50}{240} \times 360^{\circ} = 75$
Football	$\frac{35}{240} \times 360^{\circ} = 52.5$
Table tennis	$\frac{20}{240} \times 360^{\circ} = 30$
Volley ball	$\frac{40}{240} \times 360^{\circ} = 60$
Rulers	$\frac{65}{240} \times 360^{\circ} = 97.5$
Box of pens	$\frac{30}{240} \times 360^{\circ} = 45$



Example 2. A group of 300 mathematics teacher were classified as follows:

University Graduates **120**

Diplomats **90**

Specialists **50**

Others **y**

a). Calculate the value of y .

b). Draw a pie chart to illustrate the above information

c). What fraction of teachers are Specialists

Solution

a).

Step 1: To find y we equate all and equate it to the total as shown below

$$120 + 90 + 50 + y = 300$$

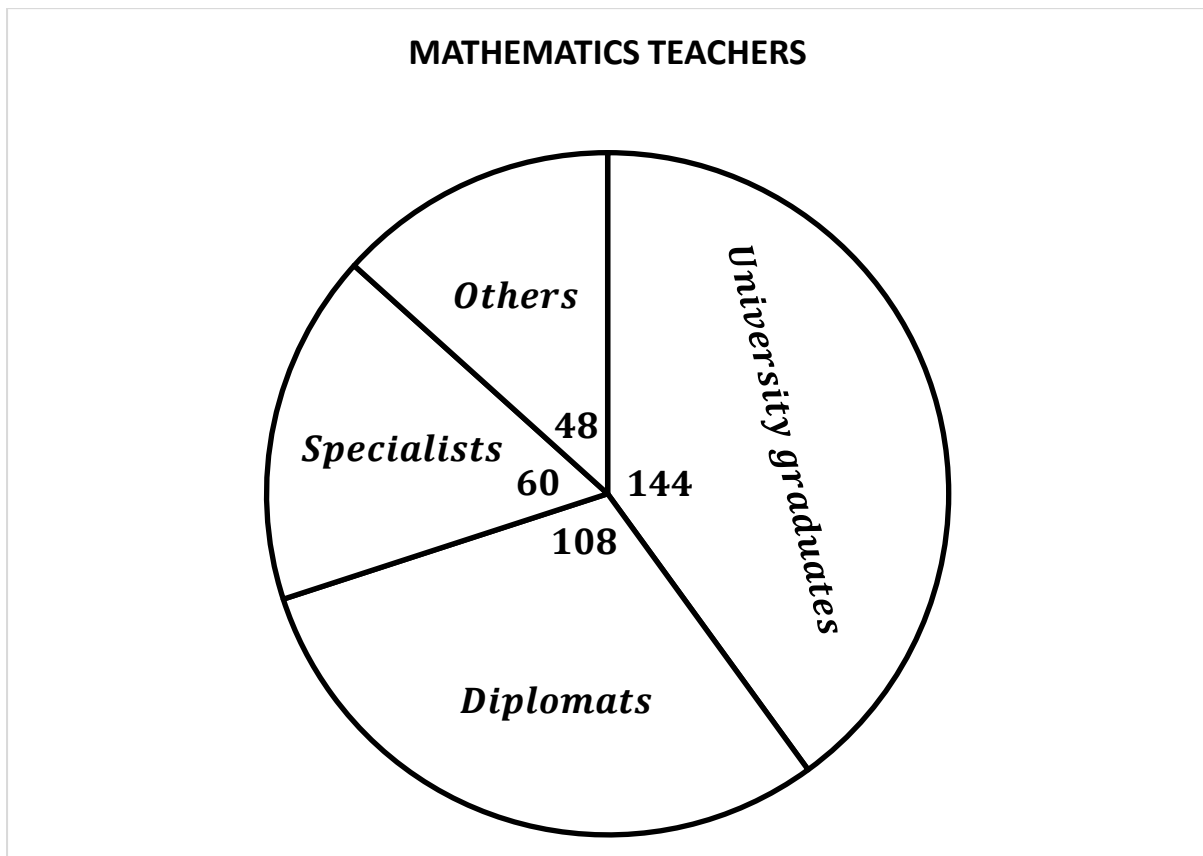
$$260 + y = 300$$

$$y = 300 - 260$$

$$y = 40$$

b).

<i>MATHEMATICS TEACHERS</i>	<i>ANGLE</i>
<i>University graduates</i>	$\frac{120}{300} \times 360^\circ = 144$
<i>Diplomats</i>	$\frac{90}{300} \times 360^\circ = 108$
<i>Specialists</i>	$\frac{50}{300} \times 360^\circ = 60$
<i>Others</i>	$\frac{40}{300} \times 360^\circ = 48$



c). Fraction of teachers that are specialist = $\frac{40}{300} = \frac{2}{15}$

Subsection 2: Bar chart or graph

A bar chart or graph is a chart or graph that presents categorical data with rectangular bars with heights or lengths proportional to the values that they represent. The bars can be plotted vertically or horizontally. A vertical bar chart is sometimes called a column chart.

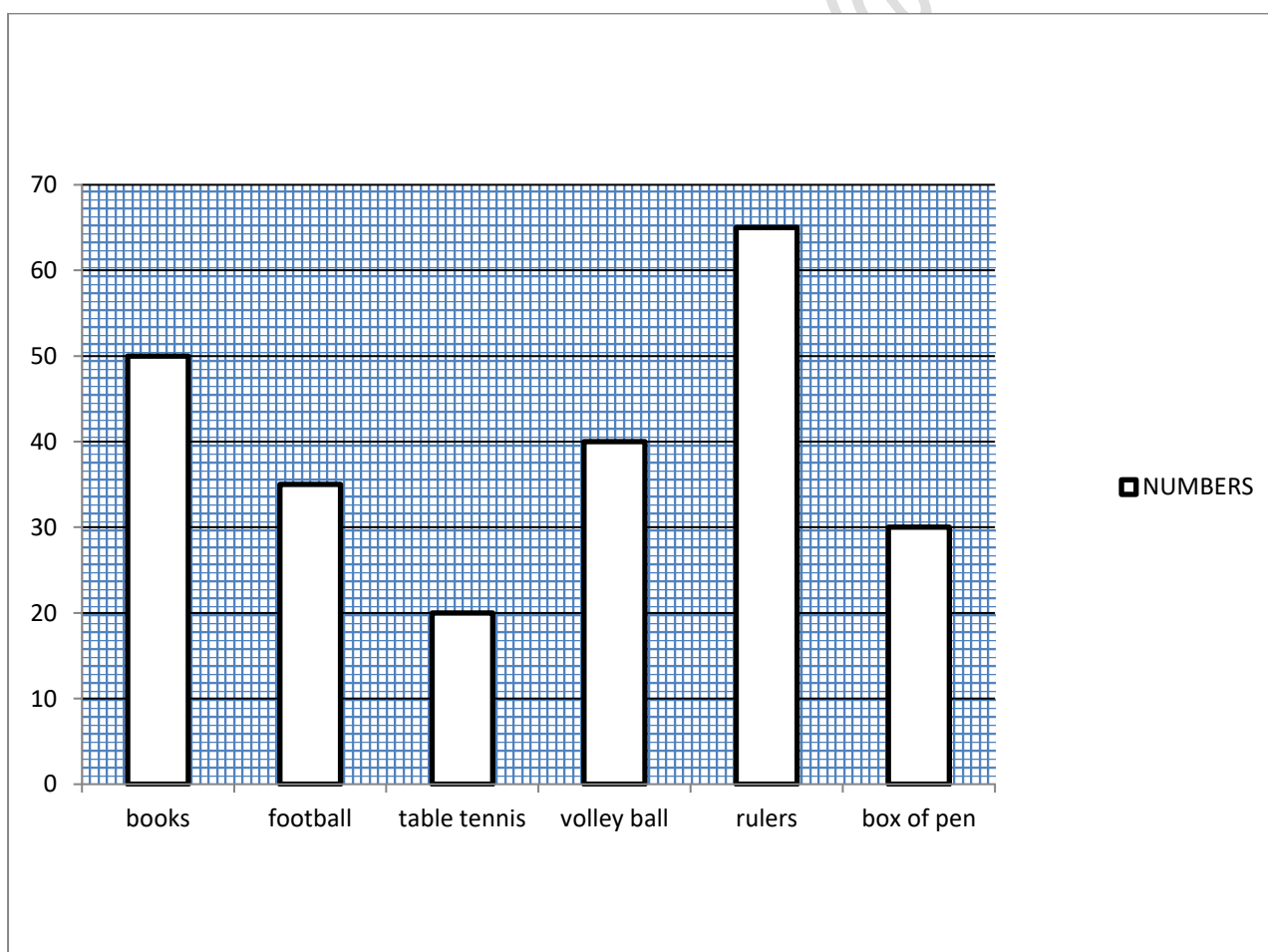
ILLUSTRATIVE EXAMPLES

Example 1. The table below show the various items sold in a city shop in Freetown.

ITEMS	NUMBER
Books	50
Football	35
Table tennis	20
Volley ball	40
Rulers	65
Box of pen	30

a). Draw a bar chart to represent the information?

Solution



Example 2. Five hundred students in a certain university were surveyed regarding which courses they are following. The responses were

as follows:

Travel and Tourism 125

Health and social care 83

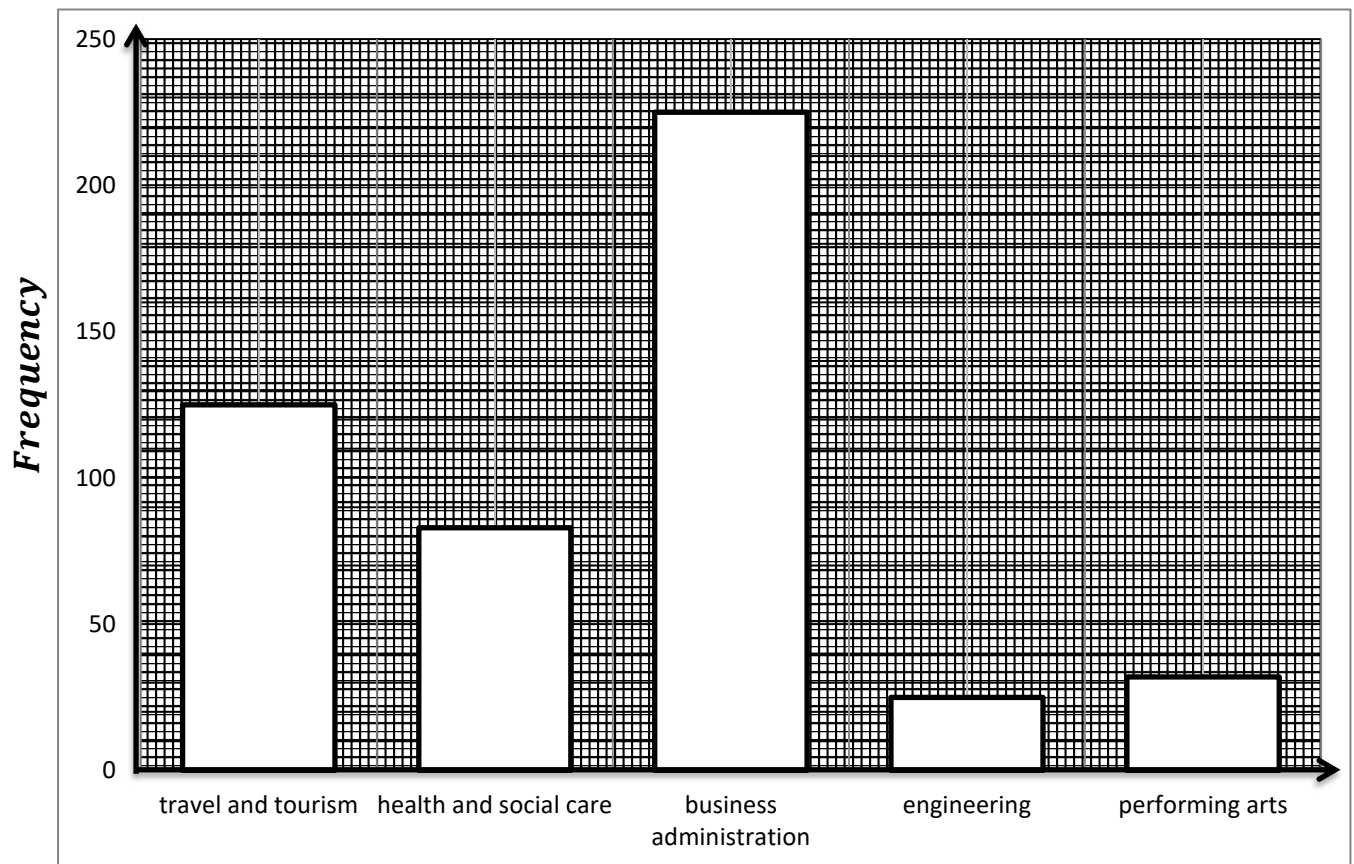
Business administration 225

Engineering 25

Performing Arts 32

a). Draw a bar chart to represent this information

Solution



SECTION 8: SKS PAPER 19

Answer all questions

Part 1

Question 1. Find the mode of the following?

1, 1, 2, 1, 2, 3, 4, 2, 2, 2 and 5.

Answer[mode = 2]

Question 2. Find the mode of the following?

4, 5, 2, 3, 2, 6, 2, 4, 3, 3, 3 and 2.

Answer[mode = 3]

Question 3. Find the mode of the following?

2, 1, 3, 4, 1, 1, 1, 2 and 2.

Answer[mode = 1]]

Example 4. Find the mode of the table below?

Marks	0	1	2	3	4
Frequency	2	6	13	7	4

Answer[mode = 2]

Question 5. Find the mode of the table below?

Age	0	1	2	3	4	5	6
Frequency	9	5	7	8	14	3	2

Answer[mode = 4]

Question 6. Find the mode of the table below?

Age	0	1	2	3
Frequency	5	3	4	9

Answer[mode = 3]

Question 7. If the total age of boys in a school is 30. Find the mode of the table below?

Ages	0	1	2	3
Frequency	7	x	5	4

Answer[mode = 1]

Question 8. If the total number of girls in a school is 35, find the mode?

Ages	2	4	6	8
Frequency	$14 - x$	$x + 4$	x	8

Answer[mode = 4]

Question 9. If the total age in a class is 36, find the mode of the table below?

Ages	3	6	9
Frequency	x	$2x + 1$	$x - 1$

Answer[mode = 6]

Part 2

Question 1. Find the mean of the following numbers?

1, 2, 3, 4, 5, 6 and 7

Answer [mean = 4]

Question 2. Find the mean of 2, 3, 4, 6, 8, and 10?

Answer [mean = 5.5]

Question 3. Find the mean of 3, 4, 6, 9, and 12?

Answer[mean = 6.8]

Question 4. Find the mean of the table below?

Marks	1	2	3	4
Frequency	2	6	8	8

Answer [mean = 2.917 correct to 3 decimal places]

Question 5. Find the mean of the table below?

Marks	1	2	3	4
Frequency	3	5	9	12

Answer[mean = 3.034]

Part 3

Question 1. If the mean is 4. Find the value of x in the following?

1, 2, x , 4, 5, 6

Answer [$x = 6$]

Question 2. If the mean is 8. Find the value of x in the following

2, 4, 6, $2x + 1$, 10

Answer[$x = 8.5$]

Part 4

Question 1. These are the marks obtained by some students

Marks	1	2	3	4	5
Frequency	3	6	9	12	14

Find the number of students that score:

i). At least one mark

ii). At least two marks

iii). At least three marks

iv). At least four marks

v). At least five marks

Answers:

i). At least one mark = 44

ii). At least two marks = 41

iii). At least three marks = 35

iv). At least four marks = 26

v). At least five marks = 14]

Part 5

Question 1. These are the marks obtained by some students.

Marks	1	2	3	4	5
Frequency	3	6	9	12	15

Find the number of student that score:

i). At most 1 mark

ii). At most 2 marks

iii). At most 3 marks

iv). At most 4 marks

v). At most 5 marks

Answers:

i). At most one mark = 3

ii). At least two marks = 9

iii). At least 3 marks = 18

iv). At least four marks = 30

iv). At least five marks = 45]

Part 6

Question 1. These are the marks obtained by some students.

Marks	1	2	3	4	5
Frequency	3	6	9	12	15

Find the number of student that score:

i). Above 1 mark

ii). Above 2 marks

iii). Above 3 marks

iv). Above 4 marks

v). Above 5 marks

Answers:

i). Above one mark = 42

ii). Above two marks = 36

iii). Above three marks = 27

iv). Above four marks = 15

v). Above five marks = 0

Question 2. *In a certain school, these are the marks obtained by the pupils.*

Marks	1	2	3	4	5
Frequency	3	6	12	15	18

Find the number of students that score:

i). Below 1 mark

ii). Below 2 marks

iii). Below 3 marks

iv). Below 4 marks

v). Below 5 marks

Answers:

i). Below one mark = 0

ii). Below two marks = 3

iii). Below three marks = 9

iv). Below four marks = 21

v). Below five marks = 36

Part 7

Question 1. *Find the median of the following numbers?*

i). 7, 1, 2, 4 and 5.

ii). 2, 6, 9, 8, 10, 12 and 14.

iii). 3, 9, 6, 12, 15 and 19

iv). 8, 12, 4, 11, 20 and 24

Answers:

(i). *median* = 4

(ii). *median* = 9

(iii). *median* = 10.5

(iv). *median* = 11.5

Question 2. Find the median of the table below?

Marks	1	2	3	4
Frequency	3	6	9	12

Answer[*median* = 3]

Question 3. Find the median of the table below?

Marks	1	2	3	4
Frequency	2	4	8	9

Answer[*median* = 3]

Part 8

Question 1. Construct a frequency table for the following marks.

24 , 42, 42, 51

51, 45, 25, 50

64, 37, 51, 42

45, 71, 71, 24

Question 2. The following were the mathematics scores for 30 candidates at a certain school.

4, 8, 9, 4, 6, 6, 4

6, 6, 7, 2, 8, 5, 5

2, 5, 8, 3, 5, 4, 7

1, 3, 7, 2, 7, 8, 8

3, 8, 9, 5, 9, 7, 9

Construct a frequency table for this information.

Part 9

Question 1. The table below show the various items sold in a city shop in Freetown.

<i>ITEMS</i>	<i>NUMBER</i>
<i>Books</i>	<i>50</i>
<i>Football</i>	<i>35</i>
<i>Table tennis</i>	<i>35</i>
<i>Volley ball</i>	<i>40</i>
<i>Rulers</i>	<i>55</i>
<i>Box of pen</i>	<i>30</i>

Draw a bar chart to represent the information?

Question 2. 500 students in a certain university were surveyed regarding which courses they are following. The responses were as follows:

Travel and Tourism 125

Health and social care 83

Business administration 230

Engineering 30

Performing Arts 32

Draw a bar chart to represent the information

Part 10

Question 1. *The table below show the various items sold in a city shop in Freetown.*

ITEMS	NUMBER
Books	50
Football	35
Table tennis	20
Volley ball	21
Rulers	65
Box of pen	34

Draw a pie chart to represent the information.

Question 2. *Five hundred students at a certain college surveyed regarding which courses they are following. The responses were as follows:*

Travel and Tourism 125

Health and social care 83

Business administration 230

Engineering 30

Performing Arts 32

Draw a pie chart to represent this information

CHAPTER 20

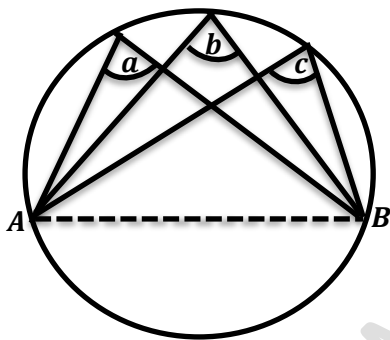
CIRCLE THEOREMS

SECTION 1: PROPERTIES OF ANGLES

There are eight (8) angle properties in a circle.

Subsection 1: Property one (1)

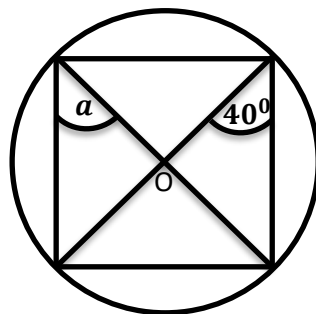
The angle a chord or arc subtends at the circumference in the same segment of a circle are equal.



$a = b = c$ {The angle a chord or arc subtends at the circumference in the same segment of a circle are equal}

ILLUSTRATIVE EXAMPLES

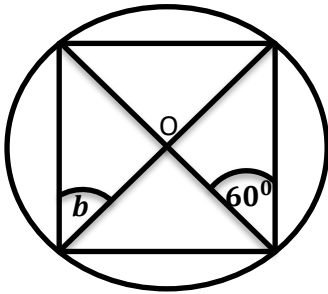
Example 1. Find the value of a in the figure below?



Solution

Step 1: $a = 40^\circ$ {The angle a chord or arc subtends at the circumference in the same segment of a circle are equal}

Example 2. Find the value of b in the figure below?

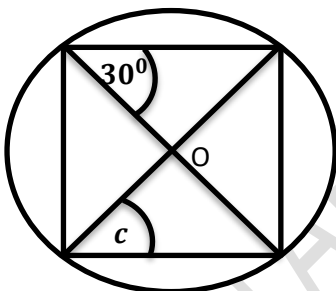


Solution

Step 1: $b = 60^\circ$ {The angle a chord or arc subtends at the circumference in the same segment of a circle are equal}

Example 3. Find the value of c in the figure below?

Solution



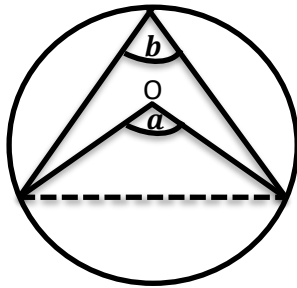
Solution

Step 1: $c = 30^\circ$ {The angle a chord or arc subtends at the circumference in the same segment of a circle are equal}

Subsection 2: Property two (2)

The angle a chord subtends at the center is twice the angle it subtends

at the circumference of a circle.



$a = 2b$ {The angle a chord subtends at the center is twice the angle it subtends at the circumference of a circle. }

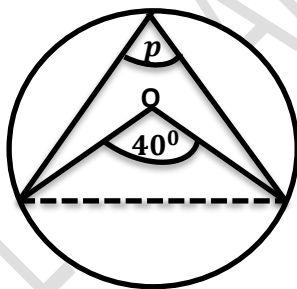
a = angle subtends at the center

b = angle subtends at the circumference

Therefore, $b = \frac{a}{2}$

ILLUSTRATIVE EXAMPLES

Example 1. Find the value of p in the figure below?



Solution

Step 1: $40^\circ = 2 \times p$ {The angle a chord subtends at the center is twice the angle it subtends at the circumference of a circle}

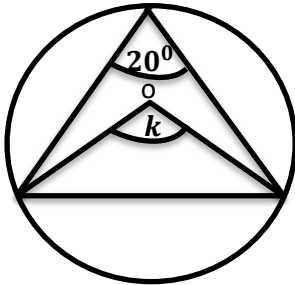
$$40^\circ = 2p$$

Step 2: Divide both sides by the coefficient of p .

$$\frac{40^{\circ}}{2} = \frac{2p}{2}$$

$$p = 20^{\circ}.$$

Example 2. Find the value of k in the figure below.

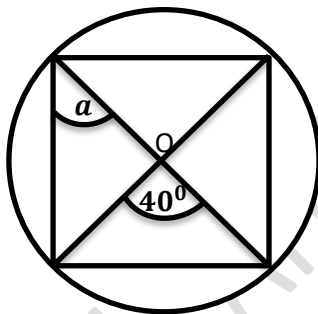


Solution

Step 1: $k = 2 \times 20^{\circ}$ {The angle a chord subtends at the center is twice the angle it subtends at the circumference of a circle}

$$k = 40^{\circ}$$

Example 3. Find the value of a in the figure below?



Solution

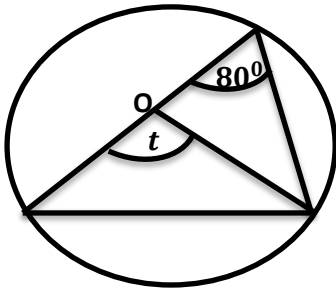
Step 1: $40^{\circ} = 2 \times a$ {the angle a chord subtends at the center is twice the angle it subtends at the circumference of a circle}

Step 2: divide both sides by the coefficient of a

$$\frac{40^{\circ}}{2} = \frac{2a}{2}$$

$$a = 20^{\circ}$$

Example 4. Find the value of t in the figure below?

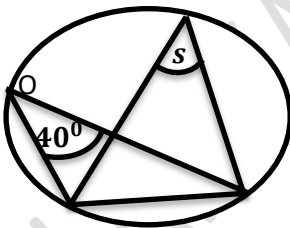


Solution

Step 1: $t = 2 \times 80^{\circ}$ {the angle a chord subtends at the center is twice the angle it subtends at the circumference of a circle}

$$t = 160^{\circ}$$

Example 5. Find the value of s in the figure below



Solution

Step 1: $40^{\circ} = 2s$ {The angle a chord subtends at the center is twice the angle it subtends at the circumference of a circle}

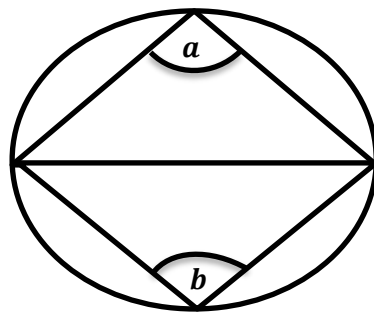
Step 2: divide both sides by the coefficient of s

$$\frac{40^{\circ}}{2} = \frac{2s}{2}$$

$$s = 20^{\circ}$$

Subsection 3: Property three (3)

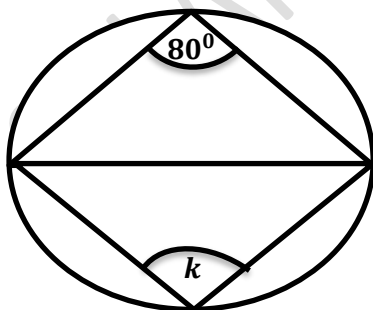
The sum of angles a chord or an arc subtends at the circumference of opposite segment in a circle is equal to 180.



$a + b = 180^{\circ}$ { the sum of angles a chord or an arc subtends at the circumference of opposite segment in a circle is equal to 180° }

ILLUSTRATIVE EXAMPLES

Example 1. Find the value of k in the figure below?



Solution

Step 1: $k + 80^{\circ} = 180^{\circ}$ {the sum of angles a chord or an arc subtends at the circumference of opposite segment in a circle is

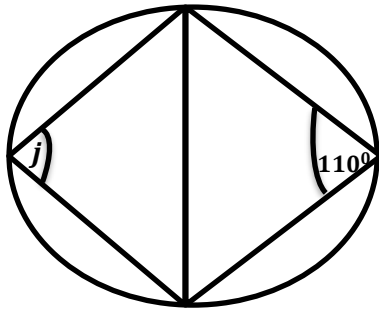
equal to 180^0 }

Step 2: collect like terms

$$k = 180^0 - 80^0$$

$$k = 100^0$$

Example 2. Find the value of j in the figure below?



Solution

Step 1: $j + 110^0 = 180^0$ {the sum of angles a chord or an arc subtends at the circumference of opposite segment in a circle is equal to 180^0 }

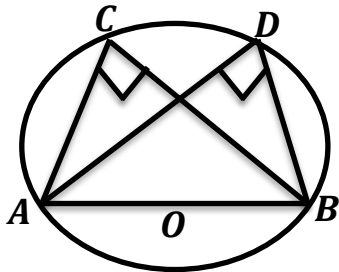
Step 2: collect like terms

$$j = 180^0 - 110^0$$

$$j = 70^0$$

Subsection 4: Property four (4)

The angle the diameter of a circle subtends at the circumference is 90° .

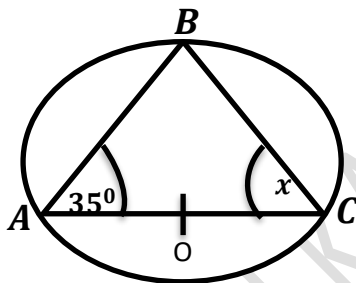


$$\angle ADB = 90^\circ \text{ and } \angle ACB = 90^\circ$$

Note: the letter in the middle represents angle.

ILLUSTRATIVE EXAMPLES

Example 1. Find the value of x in the figure below?



Solution

Step 1: $\angle ABC = 90^\circ$ {The angle the diameter of a circle subtends at the circumference is 90° }

Step 2: $\angle ABC + 35^\circ + x = 180^\circ$ {sum of angles in a triangle}

But $\angle ABC = 90^\circ$

$$90^\circ + 35^\circ + x = 180^\circ$$

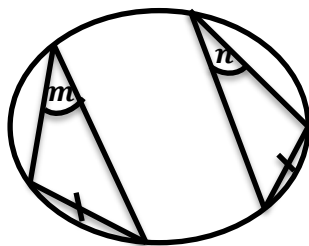
$$125^\circ + x = 180^\circ$$

$$x = 180^{\circ} - 125^{\circ}$$

$$x = 55^{\circ}$$

Subsection 5: Property five (5)

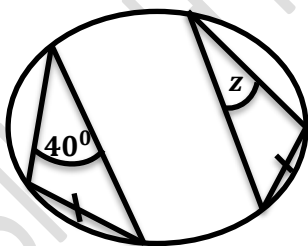
Equal chords or arcs subtends the same angles at circumference of a circle.



$m = n$ {Equal chords or arcs subtends the same angles at circumference of a circle}

ILLUSTRATIVE EXAMPLES

Example 1. Find the value of z in the figure below.



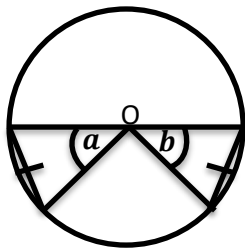
Solution

$z = 40^{\circ}$ {Equal chords or arcs subtends the same angles at circumference of a circle}

Subsection 6: Property six (6)

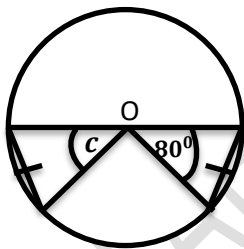
Equal chords or arcs subtends at the center of a circle.

$$a = b$$



ILLUSTRATIVE EXAMPLES

Example 1. Find the value of c in the figure below?

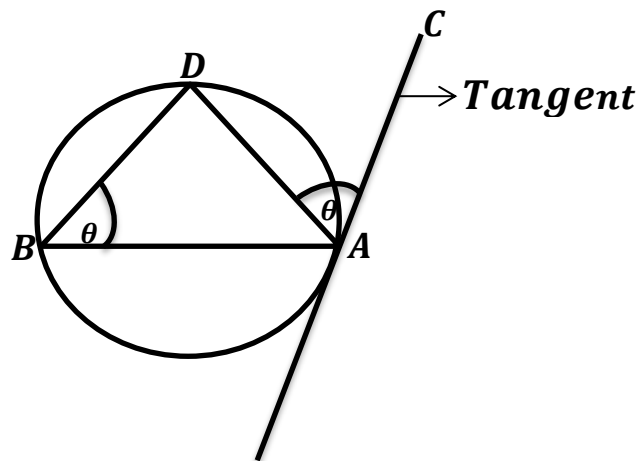


Solution

Step 1: $c = 80^\circ$ {Equal chords or arcs subtends at the center of a circle}

Subsection 7: Property seven (7)

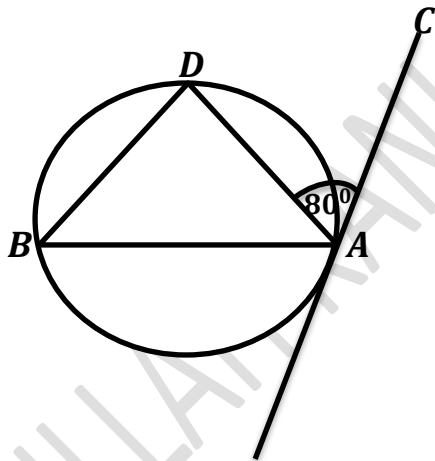
The angle between a chord and a tangent at the end of the chord equal angle in alternate segment.



$\angle DAC = \angle DBA$ {The angle between a chord and a tangent at the end of the chord equal angle in alternate segment}

ILLUSTRATIVE EXAMPLES

Example 1. Find the value of $\angle DBA$ in the figure below?



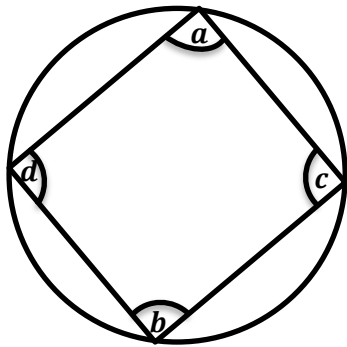
Solution

Step 1: $\angle DAC = \angle DBA$ {The angle between a chord and a tangent at the end of the chord equal angle in alternate segment}

Therefore, $\angle DBA = 80^\circ$

Subsection 8: Property 8

The opposite angles of a cyclic quadrilateral add up to 180° .

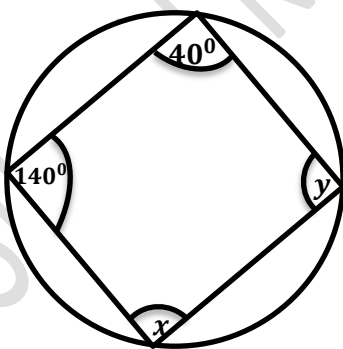


$a + b = 180^\circ$ {The opposite angles of a cyclic quadrilateral add up to 180° }

$c + d = 180^\circ$ {The opposite angles of a cyclic quadrilateral add up to 180° }

ILLUSTRATIVE EXAMPLES

Example 1. Find the value of x and y in the figure below?



Solution

Step 1: $x + 40^\circ = 180^\circ$ {The opposite angles of a cyclic quadrilateral add up to 180}

$$x = 180^{\circ} - 40^{\circ}$$

$$x = 140^{\circ}$$

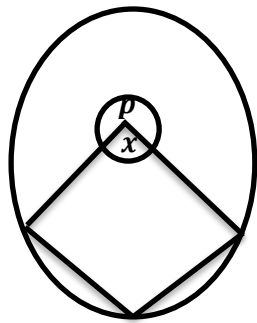
Step 2: $y + 140^{\circ} = 180^{\circ}$ {The opposite angles of a cyclic quadrilateral add up to 180° }

$$y = 180^{\circ} - 140^{\circ}$$

$$y = 40^{\circ}$$

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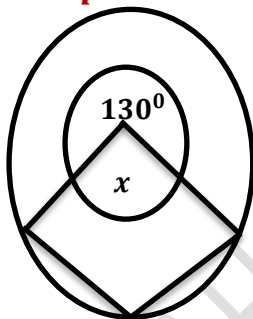
SECTION 2: ANGLES AROUND A POINT ON A CYCLIC QUADRILATERAL



$$x + p = 360^0 \text{ \{angles around a point sum up to } 360^0\}}$$

ILLUSTRATIVE EXAMPLES

Example 1. Find the value of x in the figure below?



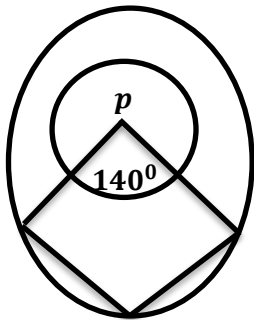
Solution

Step 1: $x + 130^0 = 360^0 \text{ \{angles around a point sum up to } 360\}}$

$$x = 360^0 - 130^0$$

$$x = 230^0$$

Example 2. Find the value of p in the figure below?



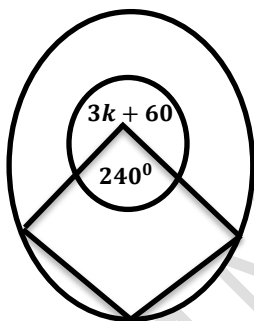
Solution

Step 1: $p + 140^\circ = 360^\circ$ {angle around a point sum up to 360° }

$$x = 360^\circ - 140^\circ$$

$$x = 120^\circ$$

Example 3. Find the value of k in the figure below?



Solution

Step 1: $3k + 60^\circ + 240^\circ = 360^\circ$ {angle around a point sum up to 360° }

$$3k + 300^\circ = 360^\circ$$

$$3k = 360^\circ - 300^\circ$$

$$3k = 60^\circ$$

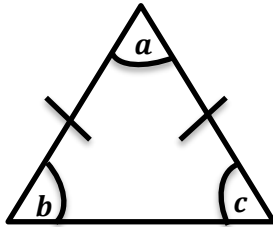
Step 2: divide both sides by the coefficient of k

$$\frac{3k}{3} = \frac{60^0}{3} \Rightarrow k = 20^0$$

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SECTION 3: ISOSCELES TRIANGLE

The base angles of an isosceles triangle are equal.

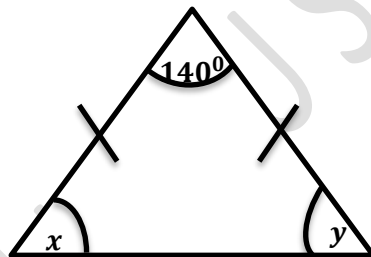


$$a = c \text{ \{base angles of isosceles triangle are equal\}}$$

$$a + b + c = 180^0 \text{ \{sum of angles in a triangle is } 180^0 \}}$$

ILLUSTRATIVE EXAMPLES

Example 1. Find the values of x and y in the figure below?



Solution

Step 1: $x = y$ \{base angles of isosceles triangle are equal\}

$$x + y + 140^0 = 180^0 \text{ \{sum of angles in a triangle\}}$$

$$x + x + 140^0 = 180^0$$

$$2x + 140^0 = 180^0$$

$$2x = 180^0 - 140^0$$

$$2x = 40^0$$

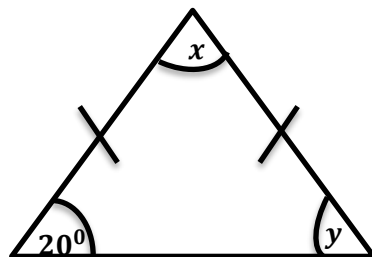
Step 2: divide both sides by the coefficient of 'x'.

$$\frac{2x}{2} = \frac{40^0}{2}$$

$$x = 20^0$$

$$\therefore x = 20^0 \text{ and } y = 20^0$$

Example 2. Find the values of x and y in the figure below?



Solution

Step 1: $y = 20^0$ {base angles of isosceles triangle are equal}

Step 2: $y + 20 + x = 180$ {sum of angles in a triangle}

but $y = 20^0$

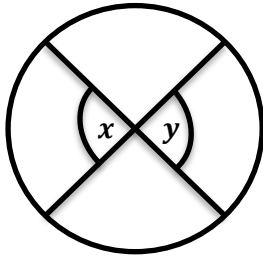
$$20^0 + 20^0 + x = 180^0$$

$$40^0 + x = 180^0$$

$$x = 180^0 - 40^0$$

$$x = 140^0$$

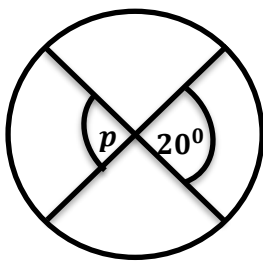
SECTION 4: OPPOSITE ANGLES



$x = y$ {opposite angles are equal}

ILLUSTRATIVE EXAMPLES

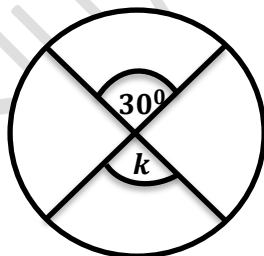
Example 1. Find the value of p in the figure below?



Solution

$p = 20^\circ$ {opposite angles are equal}

Example 2. Find k in the figure below?

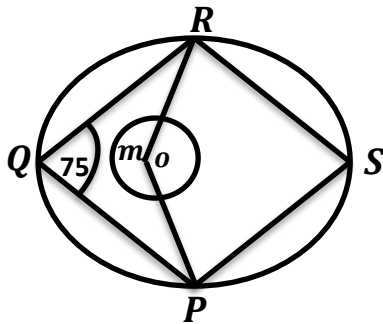


Solution

$k = 30^\circ$ {opposite angles are equal}

SECTION 5: EXAMPLES INVOLVING MORE THAN ONE THEOREM

Example 1. Find the value of m in the figure below?



Solution

Step 1: $\angle POR = 2 \times 75^\circ$ {the angle a chord subtends at the center is twice the angle it subtends at the circumference of a circle}

$$\angle POR = 150^\circ$$

Step 2: $\angle POR + m = 360^\circ$ {angles around a point sum up to 360° }

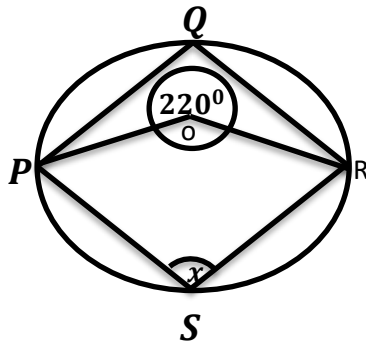
$$\text{But } \angle POR = 150^\circ$$

$$\text{Therefore, } 150^\circ + m = 360^\circ$$

$$m = 360^\circ - 150^\circ$$

$$m = 210^\circ$$

Example 2. Find the value of x in the figure below?



Solution

Step 1: $220^\circ = 2 \times x$ {the angle a chord subtends at the center is twice the angle it subtends at the circumference of a circle}

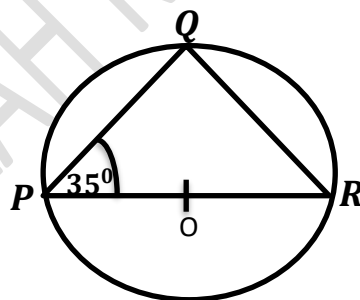
$$220^\circ = 2x$$

Step 2: divide both sides by the coefficient of x .

$$\frac{220^\circ}{2} = \frac{2x}{2}$$

$$x = 110^\circ$$

Example 3. In the figure below, find $\angle PRQ$?



Solution

Step 1: $35^\circ + \angle PQR + \angle PRQ = 180^\circ$ {sum of angles in a triangle sum up to 180° }

But $\angle PQR = 90^\circ$ {angle a diameter subtend equal 90° }

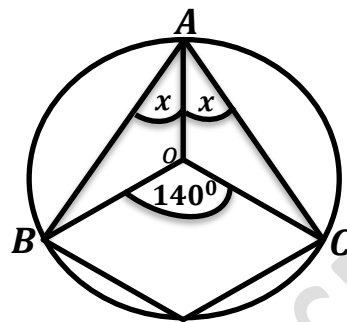
$$35^\circ + 90^\circ + \angle PRQ = 180^\circ$$

$$\angle PRQ + 125^\circ = 180^\circ$$

$$\angle PRQ = 180^\circ - 125^\circ$$

$$\angle PRQ = 55^\circ$$

Example 4. In the figure below, find the value of x ?



Solution

Step 1: $\angle BOC = 2 \times \angle BAC$ {The angle a chord subtends at the center is twice the angle it subtends at the circumference of a circle}

But, $\angle BAC = x + x = 2x$

$$2 \times 2x = 140^\circ$$

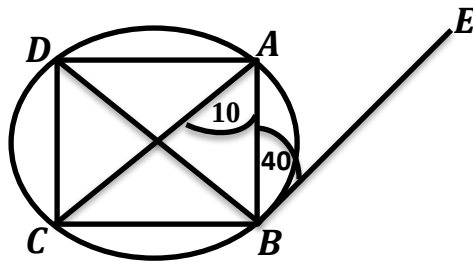
$$4x = 140^\circ$$

Step 2: divide both sides by the coefficient of x

$$\frac{4x}{4} = \frac{140^\circ}{4}$$

$$x = 35^\circ$$

Example 5. In the figure below, find $\angle ADB$ and $\angle CDB$.

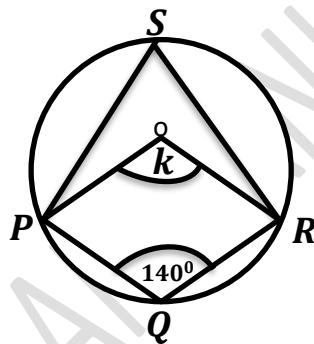


Solution

Step 1: $\angle ADB = 40^\circ$ {the angle between a chord and a tangent at the end of the chord equal angle in alternate segment}

Step 2: $\angle BAC = \angle CDB = 10^\circ$ {The angle a chord or arc subtends at the circumference in the same segment of a circle are equal}

Example 6. In the figure below, find the value of k and $\angle PSR$.



Solution

Step 1: $\angle PQR + \angle PSR = 180$ {sum of angles in a triangle}

But $\angle PQR = 140$

Therefore, $140 + \angle PSR = 180$

$\angle PSR = 180 - 140$

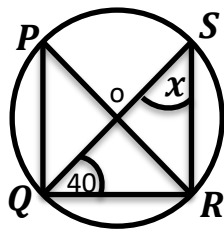
$\angle PSR = 40^\circ$

Step 2: $k = 2 \times \angle PSR$ {The angle a chord subtends at the center is twice the angle it subtends at the circumference of a circle}

$$k = 2 \times 40$$

$$k = 80^\circ$$

Example 7. Find the value of x in the figure below?



Solution

Step 1: $\angle SQR = \angle PRQ = 40^\circ$ {base angles of isosceles triangle are equal}

Step 2: $\angle QOR + \angle SQR + \angle PRQ = 180^\circ$ {sum of angles in a triangle sum up to 180}

$$\angle QOR + 40 + 40 = 180$$

$$\angle QOR + 80 = 180$$

$$\angle QOR = 180 - 80$$

$$\angle QOR = 100$$

Step 3: $\angle QOR = 2 \times x$ {The angle a chord subtends at the center is twice the angle it subtends at the circumference of a circle}

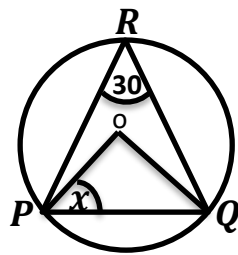
$$100 = 2x$$

Step 4: divide both sides by the coefficient of x

$$\frac{100}{2} = \frac{2x}{2}$$

$$x = 50^0$$

Example 8. Find the value of x in the figure below?



Solution

Step 1: $\angle POQ = 2 \times \angle PRQ$ {The angle a chord subtends at the center is twice the angle it subtends at the circumference of a circle}

But, $\angle PRQ = 30$

$$\angle POQ = 2 \times 30$$

$$\angle POQ = 60$$

Step 2: $\angle PQO = x$ {base angles of isosceles triangle are equal}

Step 3: $x + \angle POQ + \angle PQO = 180$ {The angle a chord subtends at the center is twice the angle it subtends at the circumference of a circle}

$$x + 60 + x = 180 \text{ {sum of angles in a triangle is 180}}$$

$$2x + 60 = 180$$

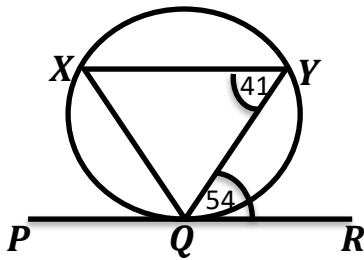
$$2x = 180 - 60$$

$$2x = 120$$

$$\frac{2x}{2} = \frac{120}{2}$$

$$x = 60^{\circ}$$

Example 9. Find $\angle YXQ$ and $\angle YQX$ in the figure below?



Solution

Step 1: $\angle YQR = \angle YXQ = 54^{\circ}$ {The angle between a chord and a tangent at the end of the chord equal angle in alternate segment}

Therefore, $\angle YXQ = 54^{\circ}$

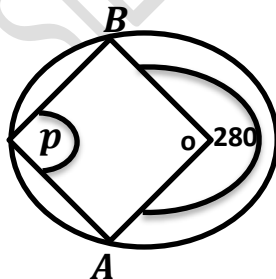
Step 2: $\angle YQX + 54 + 41 = 180$ {sum of angles in a triangle}

$$\angle YQX + 95 = 180$$

$$\angle YQX = 180 - 95$$

$$\angle YQX = 85^{\circ}$$

Example 10. Find the value of p in the diagram below?



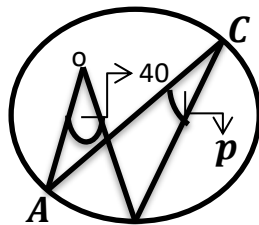
Solution

Step 1: $280 = 2 \times p$ {The angle a chord subtends at the center is twice the angle it subtends at the circumference of a circle}

$$\frac{280}{2} = \frac{2p}{2}$$

$$p = 140^{\circ}$$

Example 11. Find the value of p in the figure below?



Solution

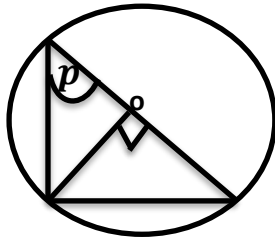
Step 1: $2 \times p = 40$ {The angle a chord subtends at the center is twice the angle it subtends at the circumference of a circle}

$$2p = 40$$

$$\frac{2p}{2} = \frac{40}{2}$$

$$p = 20^{\circ}$$

Example 13. Find the value of p in the figure below?



Solution

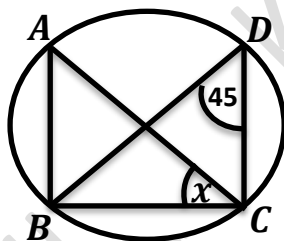
Step 1: $2 \times p = 90$ {The angle a chord subtends at the center is twice the angle it subtends at the circumference of a circle}

$$2p = 90$$

$$\frac{2p}{2} = \frac{90}{2}$$

$$p = 45^\circ$$

Example 14. If $\angle ABC = 70^\circ$, find the value of x in the figure below?



Solution

Step 1: $x + \angle ABC + \angle CAB = 180$ {sum of angles in a triangle}

But $\angle CAB = 45$ {The angle a chord or arc subtends at the circumference in the same segment of a circle are equal}

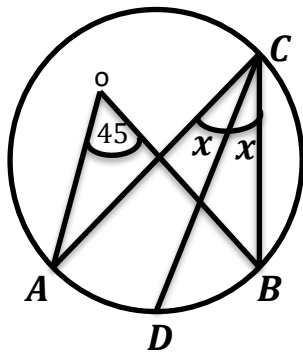
$$x + 70 + 45 = 180$$

$$x + 115 = 180$$

$$x = 180 - 115$$

$$x = 65^{\circ}$$

Question 15. Find the value of x in the figure below?



Solution

Step 1: $\angle AOB = 2 \times \angle ACB$ {The angle a chord subtends at the center is twice the angle it subtends at the circumference of a circle}

But $\angle ACB = x + x = 2x$

$$45 = 2 \times 2x$$

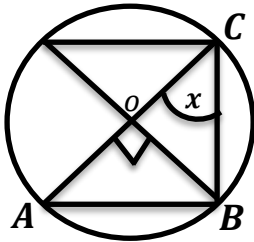
$$45 = 4x$$

Step 2 : divide both sides by the coefficient of x

$$\frac{45}{4} = \frac{4x}{4}$$

$$x = 11.25^{\circ}$$

Example 16. Find the value of x in the figure below?



Solution

Step 1: $\angle AOB = 2 \times \angle ACB$ {The angle a chord subtends at the center is twice the angle it subtends at the circumference of a circle}

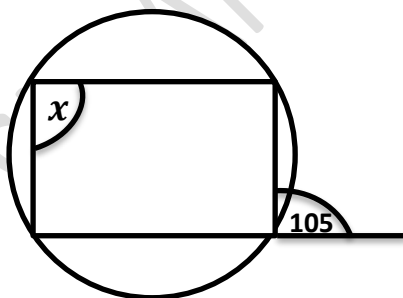
But, $\angle AOB = 90^\circ$

$$90 = 2x$$

$$\frac{90}{2} = \frac{2x}{2}$$

$$x = 45^\circ$$

Example 17. Find the value of x in the figure below?

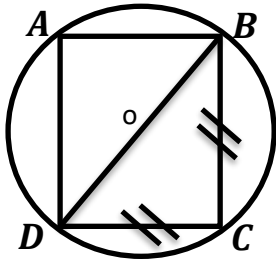


Solution

Step 1: $x = 105^\circ$ {The angle between a chord and a tangent

at the end of the chord equal angle in alternate segment}

Example 18. Find the value of $\angle DBC$ in the figure below?



Solution

Step 1: $\angle DCB = 90^\circ$ {The angle the diameter of a circle subtends at the circumference is 90° }

Step 2: $\angle DBC = \angle CDB$ {base angles of isosceles triangle are equal }

Step 3: let $\angle DBC = x$

Step 4: $\angle DCB + \angle DBC + \angle CDB = 180^\circ$ {sum of angles in a triangle}

$$x + x + 90 = 180$$

$$2x + 90 = 180$$

$$2x = 180 - 90$$

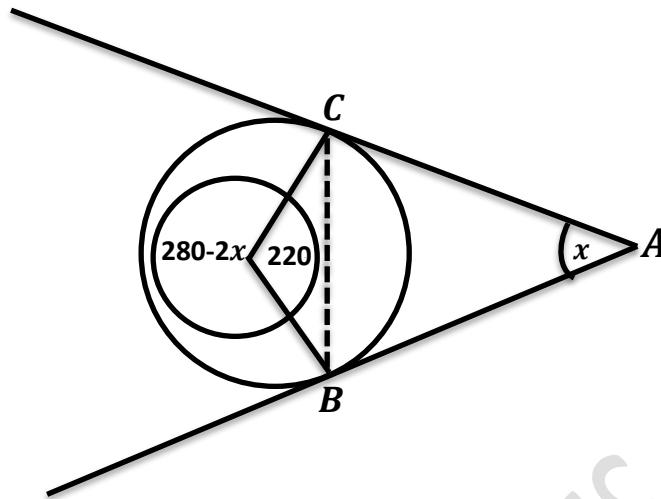
$$2x = 90$$

$$\frac{2x}{2} = \frac{90}{2}$$

$$x = 45^\circ$$

Therefore, $\angle DBC = 45^\circ$

Example 19. Find the value of x in the figure below?



Solution

Step 1: $280 - 2x = 2 \times x$ {The angle a chord subtends at the center is twice the angle it subtends at the circumference of a circle}

Step 2: collect like terms

$$280 - 2x = 2x$$

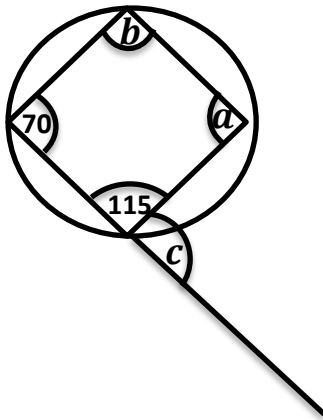
$$280 = 2x + 2x$$

$$280 = 4x$$

$$\frac{280}{4} = \frac{4x}{4}$$

$$x = 70^0$$

Example 20. Find the values of a , b and c ?



Solution

Step 1: $c + 115 = 180$ {angles on a straight line}

$$c = 180 - 115$$

$$c = 65^{\circ}$$

Step 2: $115 + b = 180$ {opposite angles in a cyclic quadrilateral sum up to 180}

$$b = 180 - 115$$

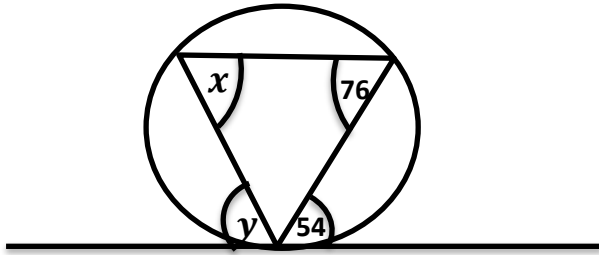
$$b = 65^{\circ}$$

Step 3: $a + 70 = 180$ {opposite angles in a cyclic quadrilateral sum up to 180}

$$a = 180 - 70$$

$$a = 110^{\circ}$$

Example 21 . Find the values of x and y in the figure below?

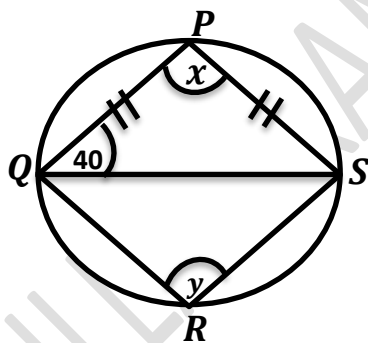


Solution

Step 1: $y = 76^\circ$ {The angle between a chord and a tangent at the end of the chord equal angle in alternate segment}

Step 2: $x = 54^\circ$ {The angle between a chord and a tangent at the end of the chord equal angle in alternate segment}

Example 22 . Find the values of x and y in the figure below?



Solution

Step 1: $\angle PSQ = 40^\circ$ {base angle of isosceles triangle are equal}
 $x + 40 + 40 = 180$ {sum of angles in a triangle}

$$x + 80 = 180$$

$$x = 180 - 80$$

$$x = 80^\circ$$

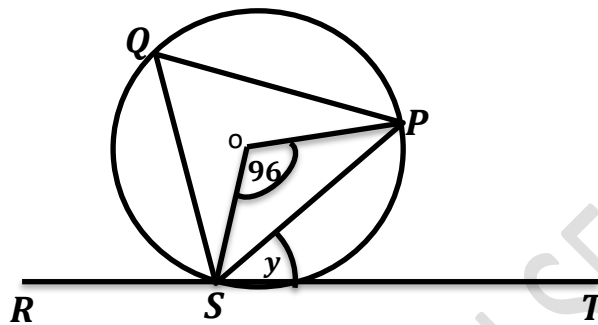
Step 2: $x + y = 180$ {The opposite angles in a cyclic quadrilateral sum up to 180}

$$100 + y = 180$$

$$y = 180 - 100$$

$$y = 80^\circ$$

Example 23 . Find the value of y in the figure below?



Solution

Step 1: $\angle PQS = \angle PST = y$ {The angle between a chord and a tangent at the end of the chord equal angle in alternate segment}

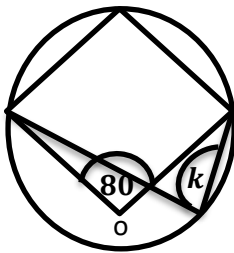
Step 2: $2 \times \angle PQS = 96$ {The angle a chord subtends at the center is twice the angle it subtends at the circumference of a circle}

$$\frac{2 \times \angle PQS}{2} = \frac{96}{2}$$

$$\angle PQS = 48^\circ$$

Therefore, $y = 48^\circ$

Example 24. Find the value of k in the figure below?



Solution

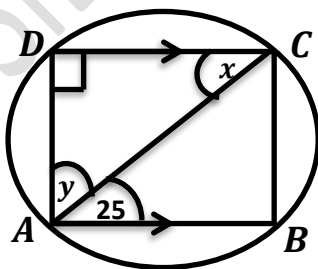
Step 1: $2 \times k = 80$ {The angle a chord subtends at the center is twice the angle it subtends at the circumference of a circle}

$$2k = 80$$

$$\frac{2k}{2} = \frac{80}{2}$$

$$k = 40^\circ$$

Example 25. Find x and y in the figure below?



Solution

Step 1: $x = 25$ {*alternate angles are equal*}

Step 2: $x + y + 90 = 180$ {*sum of angles in a triangle is 180*}

$$25 + y + 90 = 180$$

$$y + 115 = 180$$

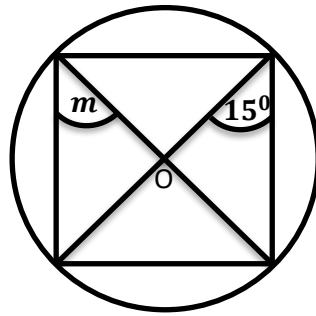
$$y = 180 - 115$$

$$y = 65^{\circ}$$

SECTION 6: SKS PAPER 20

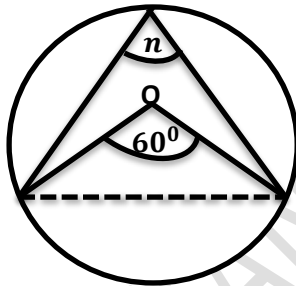
Answer all questions

Question 1. Find the value of m in the figure below?



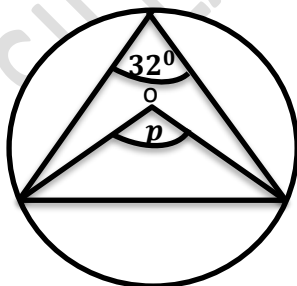
Answer [$m = 15^\circ$]

Question 2. Find the value of n in the figure below?



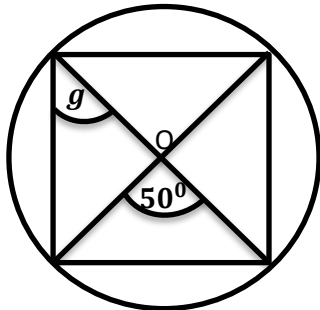
Answer [$n = 30^\circ$]

Question 3. Find the value of p in the figure below?



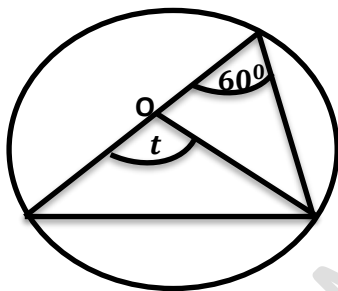
Answer [$p = 64^\circ$]

Question 4. Find the value of g in the figure below?



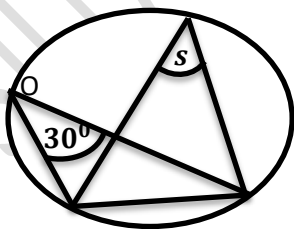
Answer [$g = 25^\circ$]

Question 5. Find the value of t in the figure below?



Answer [$t = 120^\circ$]

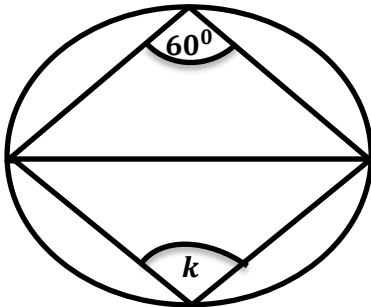
Question 6. Find the value of s in the figure below



Answer [$s = 15^\circ$]

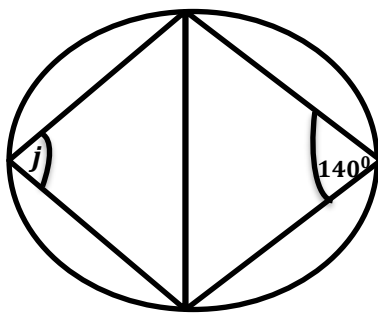
Part 2

Question 1. Find the value of k in the figure below?



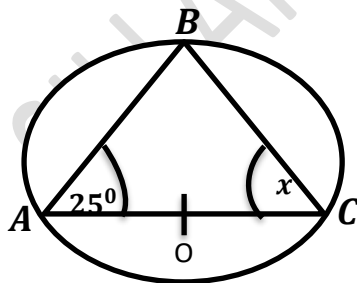
Answer [$n = 120^\circ$]

Question 2. Find the value of j in the figure below?



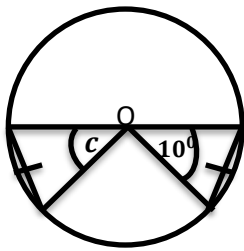
Answer [$j = 40^\circ$]

Question 3. Find the value of x in the figure below?



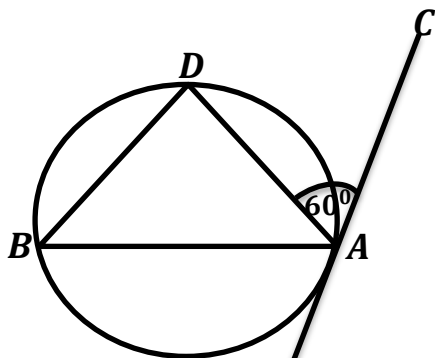
Answer [$x = 65^\circ$]

Question 4. Find the value of c in the figure below?



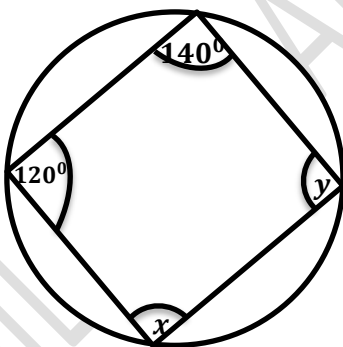
Answer [$c = 10^\circ$]

Question 5. Find the value of $\angle DBA$ in the figure below?



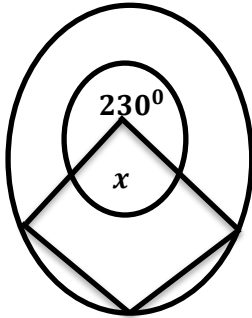
Answer [$\angle DBA = 60^\circ$]

Question 6. Find the value of x and y in the figure below?



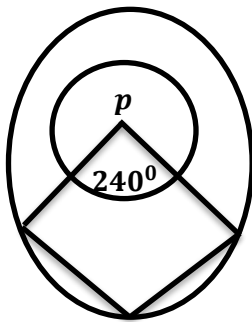
Answer [$x = 40^\circ$ and $y = 60^\circ$]

Question 7. Find the value of x in the figure below?



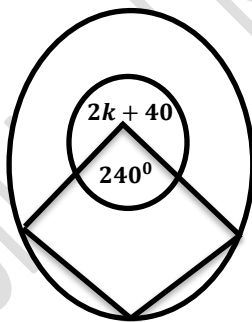
Answer [$x = 40^\circ$ and $y = 60^\circ$]

Question 8. Find the value of p in the figure below?



Answer [$x = 40^\circ$ and $y = 60^\circ$]

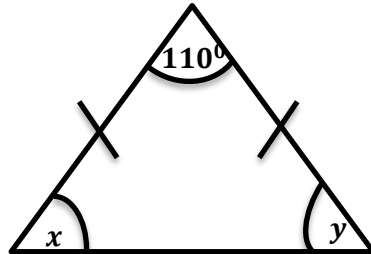
Question 9. Find the value of k in the figure below?



Answer [$k = 40^\circ$]

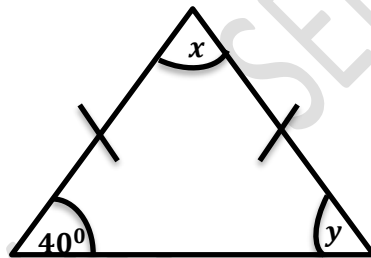
Part 3

Question 1. Find the values of x and y in the figure below?



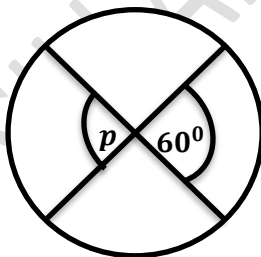
Answer [$x = 35^\circ$ and $y = 35^\circ$]

Question 2. Find the values of x and y in the figure below?



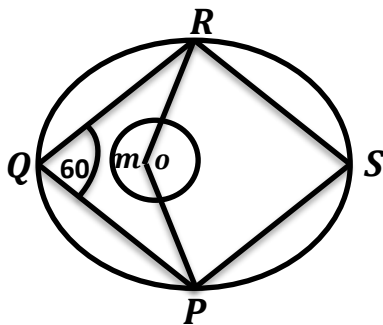
Answer [$x = 100^\circ$ and $y = 40^\circ$]

Question 3. Find the value of p in the figure below?



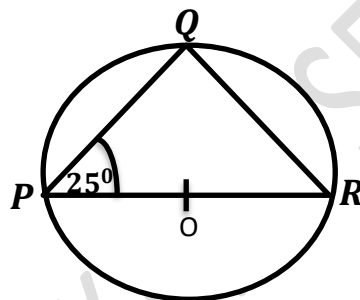
Answer [$p = 60^\circ$]

Question 4. Find the value of m ?



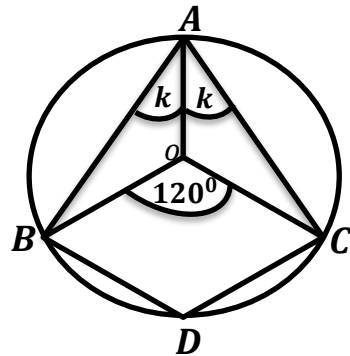
Answer [$x = 40^\circ$ and $y = 60^\circ$]

Question 5. In the figure below, find $\angle PQR$?



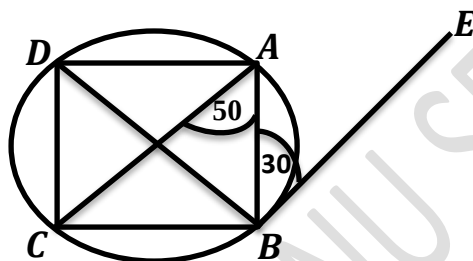
Answer [$\angle PQR = 65^\circ$]

Question 6. In the figure below, find the value of k ?



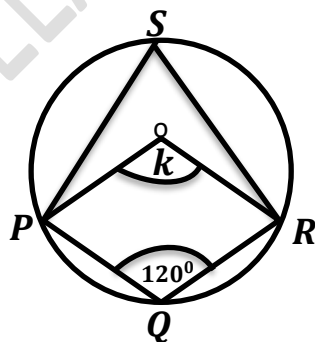
Answer [$k = 30^\circ$]

Question 7. In the figure below, find $\angle ADB$ and $\angle BDC$.



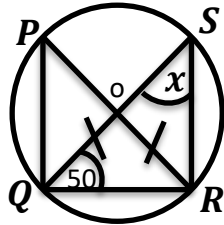
Answer [$\angle ADB = 30^\circ$ and $\angle BDC = 50^\circ$]

Question 8. In the figure below, find the value of k and $\angle PSR$.



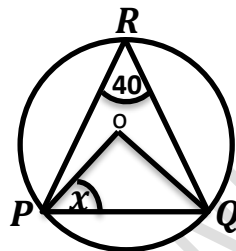
Answer [$\angle PSR = 60^\circ$ and $k = 120^\circ$]

Question 9. Find the value of x in the figure below?



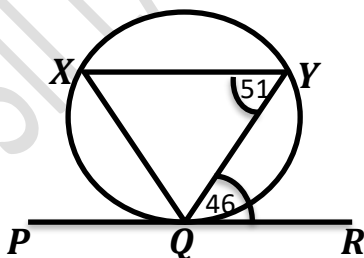
Answer [$x = 40^\circ$]

Question 10. Find the value of x in the figure below?



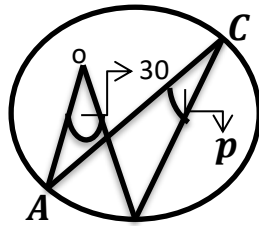
Answer [$x = 50^\circ$]

Question 11. Find $\angle YXQ$ and $\angle YQX$ in the figure below?



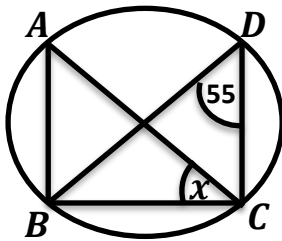
Answer [$\angle YXQ = 46^\circ$ and $\angle YQX = 83^\circ$]

Question 12. Find the value of p in the figure below?



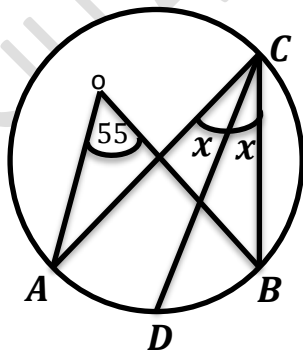
Answer [$p = 15^\circ$]

Question 13. If $\angle ABC = 100^\circ$, find the value of x in the figure below?



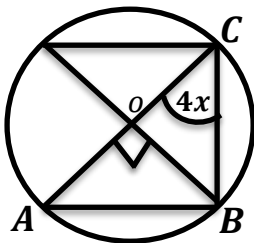
Answer [$x = 25^\circ$]

Question 14. Find the value of x in the figure below?



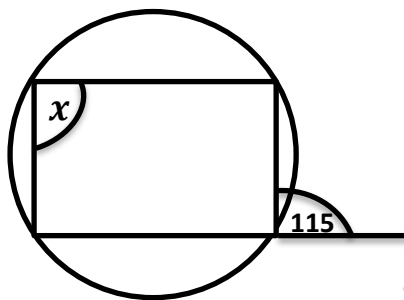
Answer [$x = 13.75^\circ$]

Question 13. Find the value of x in the figure below?



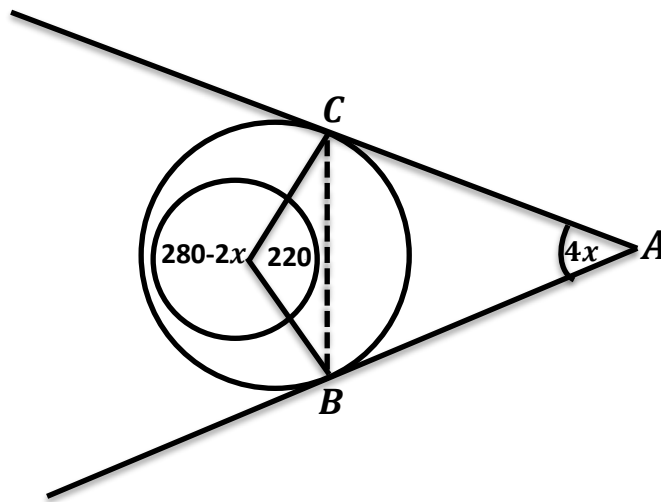
Answer [$x = 11.25^\circ$]

Question 14. Find the value of x in the figure below?



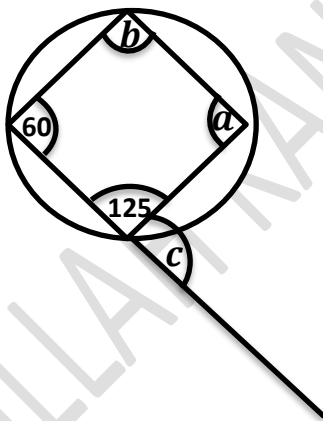
Answer [$x = 115^\circ$]

Question 15. Find the value of x in the figure below?



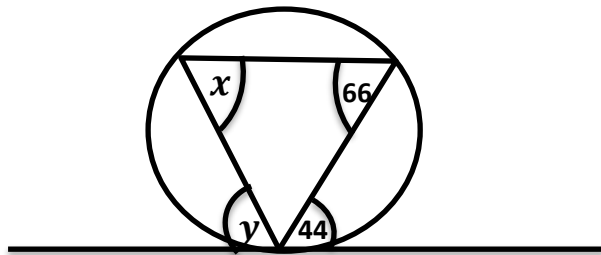
Answer [$x = 28^{\circ}$]

Question 16. Find the values of a , b and c ?



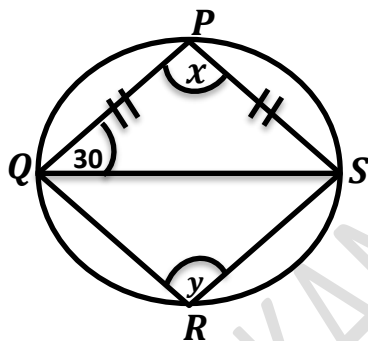
Answer [$a = 120^{\circ}$, $b = 55^{\circ}$ and $c = 55^{\circ}$]

Question 17 . Find the values of x and y in the figure below?



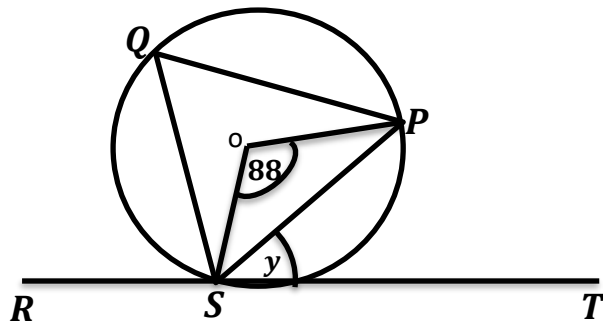
Answer [$x = 44^\circ$ and $b = 66^\circ$]

Question 18. Find the values of x and y in the figure below?



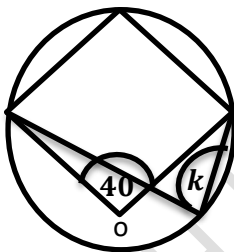
Answer [$x = 120^\circ$ and $y = 60^\circ$]

Question 19 .Find the value of y in the figure below?



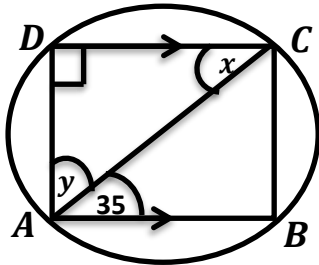
Answer $[y = 44^\circ]$

Question 20. Find the value of k in the figure below?



Answer $[k = 20^\circ]$

Question 21. Find x and y in the figure below?

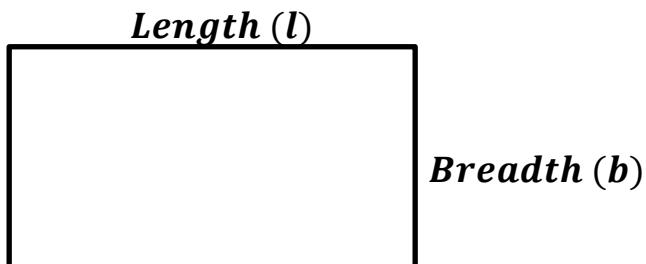


Answer [$x = 35^\circ$ and $y = 55^\circ$]

CHAPTER 21

MEASURATION

SECTION 1: RECTANGLE

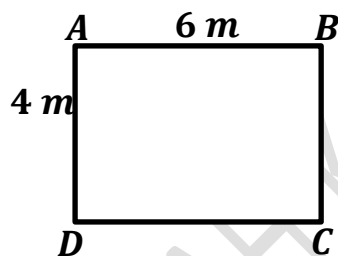


$$\text{Area} = \text{length}(l) \times \text{breadth}(b)$$

$$\text{Perimeter} = 2(\text{length} + \text{breadth})$$

ILLUSTRATIVE EXAMPLES

Example 1. Find the area and perimeter of the figure below?



Solution

Solving for perimeter

Step 1: Write the formula for the perimeter of a rectangle

$$\text{Perimeter of a rectangle} = 2(l + b)$$

Step 2: Substitute for l and b

$$= 2(6 + 4) \text{ m}$$

$$= 2 \times 10$$

$$= 20 \text{ m}$$

Solving for Area

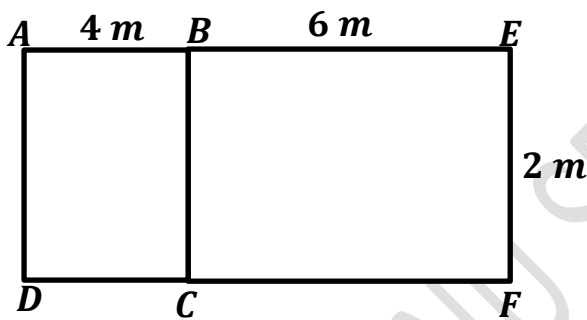
Step 1: Write the formula for the area of a rectangle

$$\text{Area} = l \times b$$

Step 2: Substitute for l and b

$$= 6 \text{ m} \times 4 \text{ m} = 24 \text{ m}^2$$

Example 2. Find the perimeter and area of rectangle ABCD and BEFC



Solution

Solving for perimeter

Step 1:

$$\text{Perimeter of Rectangle ABCD} = 2(l + b)$$

$$= 2(2 + 4)m$$

$$= 12m$$

Step 2:

$$\text{Perimeter of Rectangle BEFC} = 2(l + b)$$

$$= 2(6 + 2) \text{ m}$$

$$= 16 \text{ m}$$

Solving for Area

Step 1:

Area of Rectangle ABCD = $l \times b$

$$= 4 \text{ m} \times 2 \text{ m}$$

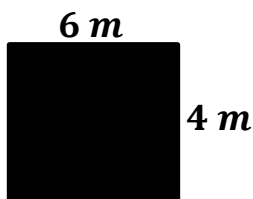
$$= 8 \text{ m}^2$$

Step 2:

Area of Rectangle BEFC = $6 \text{ m} \times 2 \text{ m}$

$$= 12 \text{ m}^2$$

Example 3. Find the area of the shaded portion?

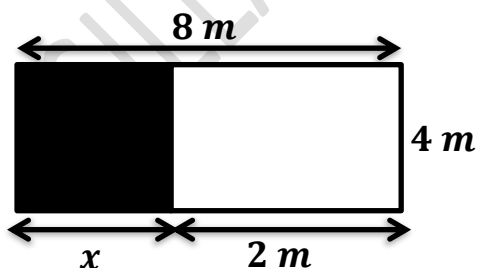


Solution

$$\Rightarrow \text{Perimeter} = 2(6 + 4) \text{ m} = 20 \text{ m}$$

$$\Rightarrow \text{Area} = 6 \text{ m} \times 4 \text{ m} = 24 \text{ m}^2$$

Example 4. Find the perimeter and area of the shaded portion?



Solution

Step 1: considering length and breadth of the shaded portion

$$\begin{aligned}\text{Perimeter} &= 2(6 + 4) \text{ m} \\ &= 12 \text{ m}\end{aligned}$$

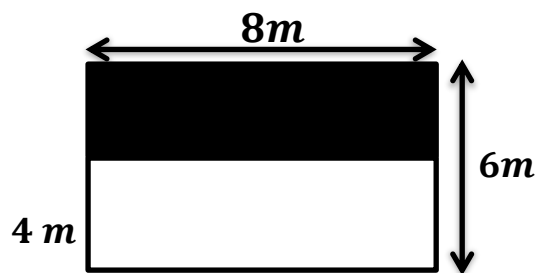
$$\text{Area} = 4 \text{ m} \times 6 \text{ m} = 24 \text{ m}^2$$

Note:

$$1. x = (8 - 2) \text{ m} = 6 \text{ m}$$

2. In rectangles, two opposite sides are equal.

Example 5. Find the perimeter and area of the shaded portion.



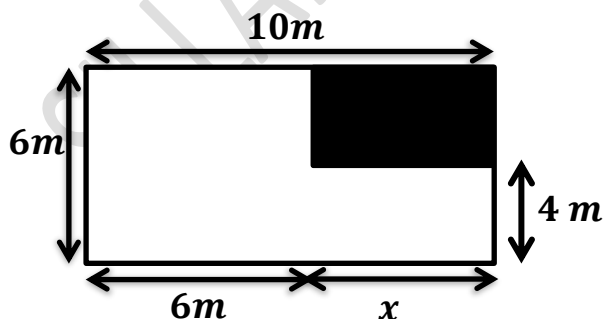
Solution

Step 1: considering length and breadth of the shaded portion.

$$\text{Perimeter} = 2(8 + 2) \text{ m} = 20 \text{ m}$$

$$\text{Area} = 8 \text{ m} \times 2 \text{ m} = 16 \text{ m}^2$$

Example 6. Find the perimeter and area of the shaded portion.



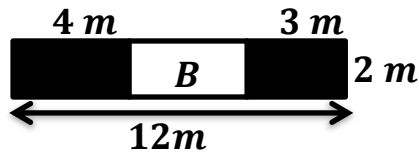
Solution

Step 1: considering length and breadth of the shaded portion.

$$\text{Perimeter} = 2(2 + 4) \text{ m} = 12 \text{ m}$$

$$\text{Area} = 4 \text{ m} \times 2 \text{ m} = 8 \text{ m}^2$$

Example 7. Find the perimeter and area of rectangle B?



Solution

Step 1: considering length and breadth of rectangle B

$$\text{Perimeter of rectangle B} = 2(5 + 2) \text{ m}$$

$$= 14 \text{ m}$$

Step 2:

$$\text{Area of rectangle B} = l \times b$$

$$= 5 \text{ m} \times 2 \text{ m}$$

$$= 10 \text{ m}^2$$

Example 8. The area of a rectangle is 24 cm^2 . If its length is 6 cm, find its breadth.

Solution

Step 1: write the formula for area of a rectangle

$$\text{Area of a rectangle} = \text{length} \times \text{breadth}$$

Step 2: substitute

$$24 \text{ cm}^2 = 6 \text{ cm} \times b$$

Step 3: divide both sides by the coefficient of b

$$\frac{24 \text{ cm}^2}{6 \text{ cm}} = \frac{6 \text{ cm} \times b}{6 \text{ cm}}$$

$$b = 4 \text{ cm}$$

Example 9. The area of a rectangle is $64^{\frac{1}{2}} \text{ cm}^2$. Find its breadth, if its length is 8 cm.

Solution

Step 1: write the formula for area of a rectangle

Area of a rectangle = length $\times$ breadth

$$\text{But Area} = 64^{\frac{1}{2}} \text{ cm}^2 = \sqrt{64} = 8 \text{ cm}^2$$

Step 2: substituting into the formula

$$8 \text{ cm}^2 = 4 \text{ cm} \times b$$

Step 3: divide both sides by the coefficient of b

$$\frac{8}{4} = \frac{4b}{4}$$

$$b = 2 \text{ cm}$$

Example 10. If the length of a rectangle is $(2\sqrt{4} + \sqrt{9}) \text{ cm}$ and its area is 21 cm^2 . Find its breadth?

Solution

Step 1: write the formula for area of a rectangle and substitute

Area of a rectangle = length $\times$ breadth

$$21 = (2\sqrt{4} + \sqrt{9}) \times b$$

$$\text{But } \sqrt{4} = 2 \text{ and } \sqrt{9} = 3$$

$$21 = [2(2) + 3] \times b$$

$$21 = [4 + 3] b$$

$$21 = 7b$$

Step 2: Divide both sides by the coefficient of b

$$\frac{21}{7} = \frac{7b}{7}$$

$$b = 3 \text{ cm}$$

Example 11. The area of a rectangle is $(\sqrt{100} + 3x) \text{ cm}^2$.

The length and breadth are $(16^{\frac{1}{2}} - x) \text{ cm}$ and 2^3 cm respectively.

Show that the area and length of the rectangle are 16 cm^2 and 2 cm respectively.

Solution

Step 1: write the formula for area of a rectangle and simplify

Area of a rectangle = length $\times$ breadth

Step 2: But

$$\text{Area} = (\sqrt{100} + 3x) = (10 + 3x) \text{ cm}^2$$

$$\text{Length} = (16^{\frac{1}{2}} - x) \text{ cm} = (4 - x) \text{ cm}$$

$$\text{Breadth} = 2^3 = 2 \times 2 \times 2 = 8$$

Step 3: substituting into the area of a rectangle

$$10 + 3x = (4 - x) \times 8$$

Step 3: Expand the bracket

$$10 + 3x = 32 - 8x$$

Step 4: Collect like terms by transposing

$$3x + 8x = 32 - 10$$

$$11x = 22$$

Step 5: Divide both sides by the coefficient of x

$$\frac{11x}{11} = \frac{22}{11}$$

$$x = 2$$

Step 6: Substitute to find the area and length of the rectangle

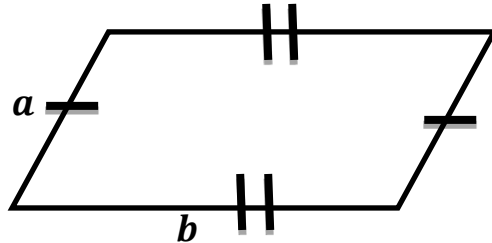
$$\Rightarrow \text{Area} = 10 + 3 \times 2 = 16 \text{ cm}^2$$

$$\Rightarrow \text{Length} = (4 - 2) \text{ cm} = 2 \text{ cm}$$

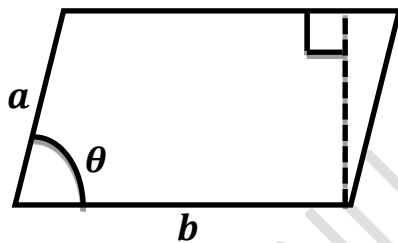
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SECTION 2: PARALLELOGRAM

A parallelogram has opposite sides equal, opposite sides equal and opposite sides parallel.



The area of a parallelogram can be got in two ways as a result of the dimension given.



Perimeter of a parallelogram = $2(a + b)$

Area = $b \times h$ i.e. base $\times$ perpendicular height

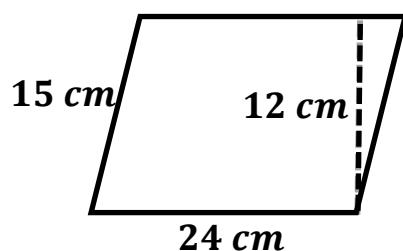
OR

Area = $a \times b \times \sin \theta$

= $ab \sin \theta$ i.e. product of two sides and the sine of the included angle

ILLUSTRATIVE EXAMPLES

Example 1. Calculate the perimeter and area of the figure below



Solution

Solving for perimeter

Step 1: write the formula for perimeter of a parallelogram

$$\text{Perimeter of a parallelogram} = 2(a + b)$$

Step 2: Substitute the given values into the formula

$$= 2(15 + 24)$$

$$= 2 \times 39$$

$$= 78 \text{ cm}$$

Solving for area

Step 1: write the formula for area of a parallelogram

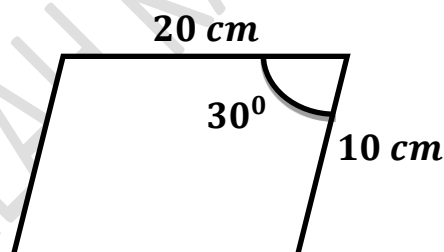
$$\text{Area of a parallelogram} = \text{base} \times \text{height}$$

Step 2: Substitute the given values into the formula

$$= 21 \times 12$$

$$= 252 \text{ cm}^2$$

Example 2. Calculate the perimeter and area of the figure below



Solution

Solving for perimeter

Step 1: write the formula for perimeter of a parallelogram

$$\text{Perimeter of a parallelogram} = 2(a + b)$$

Step 2: Substitute the given values into the formula

$$= 2(20 + 10)$$

$$= 2 \times 30$$

$$= 60 \text{ cm}$$

Solving for area

Step 1: Write the formula for area of a parallelogram

$$\text{Area} = ab \sin \theta$$

Step 2: Substitute the given values into the formula

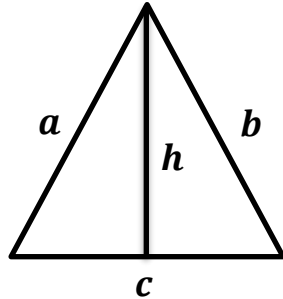
$$= 20 \times 10 \times \sin 30^\circ$$

$$= 200 \times 0.5$$

$$= 100 \text{ cm}^2$$

SECTION 3: TRIANGLES

A triangle has three sides and a height.

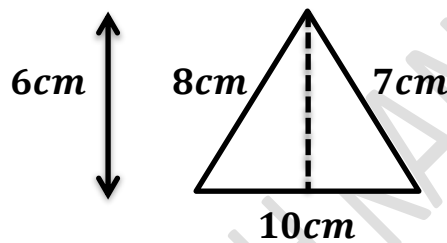


Perimeter of a triangle = $a + b + c$

Area of a triangle = $\frac{1}{2} \times \text{base} \times \text{height}$

ILLUSTRATIVE EXAMPLES

Example 1. Find the perimeter and area of the triangle.



Solution

Solving for perimeter

Step 1: Write the formula for perimeter of a triangle

$$\text{Perimeter} = a + b + c$$

Step 2: Substitute for a, b and c

$$\text{Perimeter} = (8 + 7 + 10) \text{ cm} = 25 \text{ cm}$$

-

Solving for area

Step 1: Write the formula and substitute

$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$$

$$\text{Area} = \frac{1}{2} \times 10 \times 6 = 30 \text{ cm}^2$$

Example 2. If the area of a triangle is 16cm^2 and its width is 4cm , Find its height?

Solution

Step 1: Write the formula and substitute

$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$$

$$16 = \frac{1}{2} \times 4 \times \text{height}$$

Step 2: Cross multiply

$$\frac{16 \times 2}{4} = \frac{4h}{4}$$

$$\text{Height, } h = 8 \text{ cm}$$

Example 4. If the area of the triangle is $\frac{\sqrt{16}}{2} \text{ m}^2$ and the height is 4 m . Find the width?

Solution

Step 1: Write the formula for area of a triangle

$$\text{Area of a triangle} = \frac{1}{2} \times \text{base} \times \text{height}$$

Step 2: Substitute

$$\frac{\sqrt{16}}{2} = \frac{1}{2} \times b \times 4$$

$$\frac{4}{2} = \frac{4b}{2}$$

$$2 = \frac{4b}{2}$$

Step 2: Cross multiply

$$2 \times 2 = 4b$$

$$4 = 4b$$

Step 3: Divide both by the coefficient of b

$$b = 1 \text{ m.}$$

Therefore, the width is 1 m.

Example 5. *The perimeter of a triangle is 52 cm and its base is $(x\sqrt{36} + x\sqrt{49})$ cm. If the two sides of the triangle are $(2x + x\sqrt{9})$ cm and $(4x + x\sqrt{16})$ cm. Show that the base of the triangle is 26 cm.*

Solution

Step 1: Write the formula for perimeter of a triangle

Perimeter of a triangle = sum of all the sides (i.e. $a + b + c$)

Step 2: Substituting into the formula

$$52 = (x\sqrt{36} + x\sqrt{49}) + (2x + x\sqrt{9}) + (4x + x\sqrt{16})$$

$$\text{But } \sqrt{36} = 6, \sqrt{49} = 7 \text{ and } \sqrt{16} = 4.$$

$$52 = x \times 6 + x \times 7 + x \times 3 + 4x + x \times 4$$

$$52 = 6x + 7x + 2x + 3x + 4x + 4x$$

$$52 = 26x$$

Step 3: Divide both sides by the coefficient of x

$$\frac{52}{26} = \frac{26x}{26}$$

$$x = 2$$

Step 4: Find the value of the base

$$\text{Base} = (x\sqrt{36} + x\sqrt{49})$$

$$x = 2$$

$$\text{Base} = 2 \times 6 + 2 \times 7$$

$$\text{Base} = 12 + 14$$

$$\text{Base} = 26 \text{ cm ... as required}$$

Example 6. Show that the area of a triangle is 50 cm^2 , if the base and height of the triangle are $10^{\frac{1}{2}} \text{ cm}$ and $10^{\frac{3}{2}} \text{ cm}$ respectively.

Solution

Step 1: Write the formula for area of a triangle

$$\text{Area of a triangle} = \frac{1}{2} \times \text{base} \times \text{height}$$

$$\text{Area of a triangle} = \frac{1}{2} \times 10^{\frac{1}{2}} \times 10^{\frac{3}{2}}$$

Step 2: using the multiplication law of indices

$$\text{Area of a triangle} = \frac{1}{2} \times 10^{\frac{3+1}{2}}$$

$$\text{Area of a triangle} = \frac{1}{2} \times 10^{\frac{4}{2}}$$

$$\text{Area of a triangle} = \frac{1}{2} \times 10^2$$

$$\text{Area of a triangle} = \frac{1}{2} \times 10 \times 10$$

$$\text{Area of a triangle} = \frac{100}{2}$$

$$\text{Area of a triangle} = 50 \text{ cm}^2$$

Therefore, the area of the triangle is 50 cm^2

Example 5. Prove that the base of a triangle is $10^{\frac{1}{2}} \text{ cm}$, if the area and height of the triangle are 50 cm^2 and $10^{\frac{3}{2}} \text{ cm}$ respectively.

Solution

Step 1: Write the formula for area of a triangle

$$\text{Area of a triangle} = \frac{1}{2} \times \text{base} \times \text{height}$$

$$50 = \frac{1}{2} \times b \times 10^{\frac{3}{2}}$$

Step 2: Cross multiply

$$50 \times 2 = b \times 10^{\frac{3}{2}}$$

Step 3: Divide both sides by the coefficient of b

$$\frac{100}{10^{\frac{3}{2}}} = \frac{b \times 10^{\frac{3}{2}}}{10^{\frac{3}{2}}}$$

$$\text{But } 100 = 10^2$$

$$\frac{10^2}{10^{\frac{3}{2}}} = b$$

Step 4: Using the division law of indices

$$b = 10^{2-\frac{3}{2}}$$

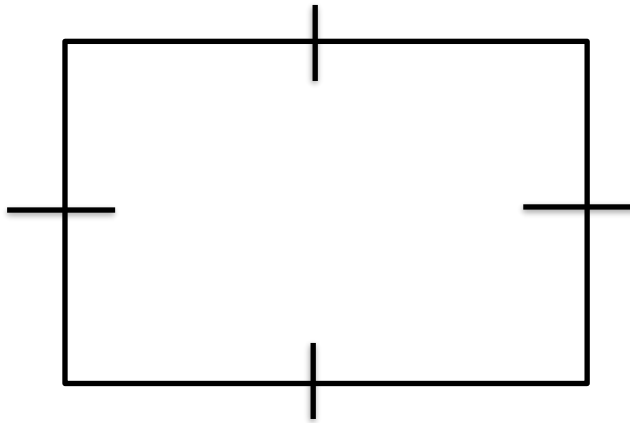
$$b = 10^{\frac{1}{2}}$$

Therefore ,the base = $10^{\frac{1}{2}}\text{cm}$... As required

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SECTION 4 : SQUARE

All its sides are equal.



Perimeter of a square = $4l$

Area of a square = $l \times l$

ILLUSTRATIVE EXAMPLES

Example 1. Find the area and perimeter of a square of length 4 cm.

Solution

Solving for perimeter

Step 1: Write the formula for perimeter of a square

Perimeter = $4l$

Perimeter = 4×4

Perimeter = 16 cm

Solving for area

Step 1: Write the formula for area of a square

Area = $l \times l$

Area = 4×4

Area = 16 cm^2

Example 2. *If the area of a square is 64 m^2 . Find its length?*

Solution

Step 1: *Write the formula for area of a square*

$$\text{Area} = l \times l \Rightarrow \text{Area} = l^2$$

Step 2: *Substitute*

$$64 = l^2$$

Step 3: *Taking the square root of both sides to remove the square on l*

$$\sqrt{64} = \sqrt{l^2}$$

$$l = 8 \text{ m}$$

Example 3. *If the perimeter of a square is 40 m . Find its length?*

Solution

Step 1: *Write the formula for perimeter of a square*

$$\text{perimeter} = 4l$$

Step 2: *Substitute*

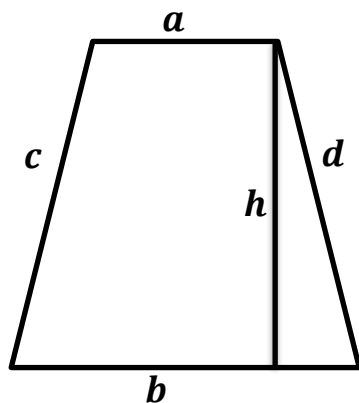
$$40 = 4l$$

Step 3: *Divide both sides by the coefficient of l*

$$\frac{40}{4} = \frac{4l}{4}$$

$$l = 10 \text{ m}$$

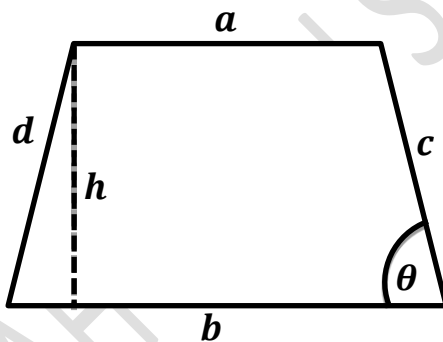
SECTION 5: TRAPEZIUM



Perimeter of a trapezium $= a + b + c + d$

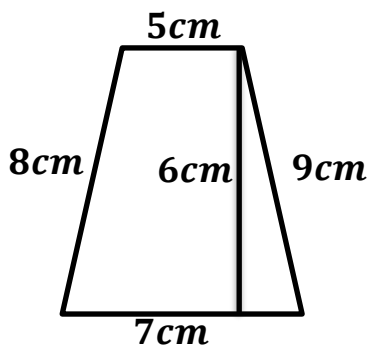
(i). *Area of a trapezium* $= \frac{1}{2}(a + b) \times h$

(ii). *Area of a trapezium* $= \frac{1}{2}(a + b) \times c \sin \theta$



ILLUSTRATIVE EXAMPLES

Example 1. Find the perimeter and area of the trapezium.



Solution

Solving for perimeter

Step 1: Write the formula for perimeter of a trapezium

$$\text{Perimeter of a trapezium} = a + b + c + d$$

Step 2: Substitute the given values

$$\text{Perimeter} = (5 + 7 + 9 + 8) \text{ cm} = 29 \text{ cm}$$

Solving for area

Step 1: Write the formula for area of a trapezium

$$\text{Area of a trapezium} = \frac{1}{2}(a + b) \times h$$

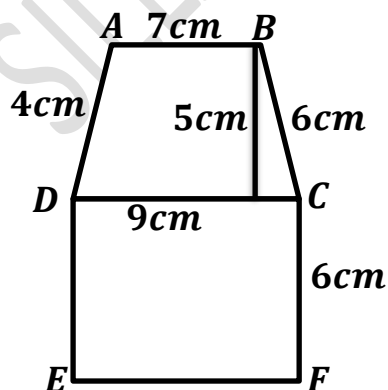
Step 2: Substitute the given value

$$\text{Area} = \frac{1}{2}(5 + 7) \times 6$$

$$\text{Area} = \frac{12}{2} \times 6$$

$$\text{Area} = 36 \text{ cm}^2$$

Example 2. Find the perimeter and area of the figure below



Solution

Solving for perimeter

Step 1: Write the formula

**Perimeter of the figure = perimeter of the trapezium +
perimeter of the rectangle**

Step 2: Substitute

$$= 6 + 4 + 7 + 9 + 2(6 + 9)$$

$$= 3 + 4 + 7 + 9 + 12 + 18$$

$$= 53 \text{ cm}$$

Solving for area

Step 1: Write the formula

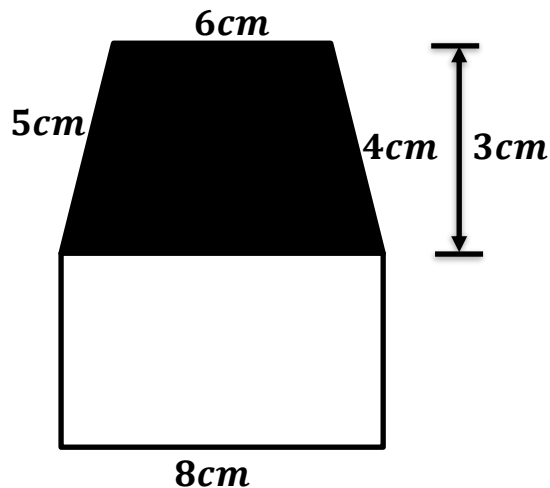
Area of the figure = area of trapezium + area of rectangle

Step 2: Substitute

$$\text{Area of the figure} = \frac{1}{2}(7 + 9) \times 5 + (6 \times 9)$$

$$\text{Area of the figure} = (40 + 54) \text{ cm}^2 = 94 \text{ cm}^2$$

Example 4. Find the perimeter of the figure and the area of the shaded portion.



Solution

Solving for perimeter

Step 1: Write the formula

**Perimeter of the figure = perimeter of trapezium +
perimeter of rectangle**

Step 2: Substitute

$$\begin{aligned} &= 6 + 3 + 4 + 8 + 2(6 + 8) \\ &= (6 + 3 + 4 + 8 + 12 + 16) \text{ cm} \\ &= 49 \text{ cm} \end{aligned}$$

Solving for area

Step 1: Write the formula for area of a trapezium (i.e Shaded portion)

$$\text{Area of trapezium} = \frac{1}{2}(a + b) \times h$$

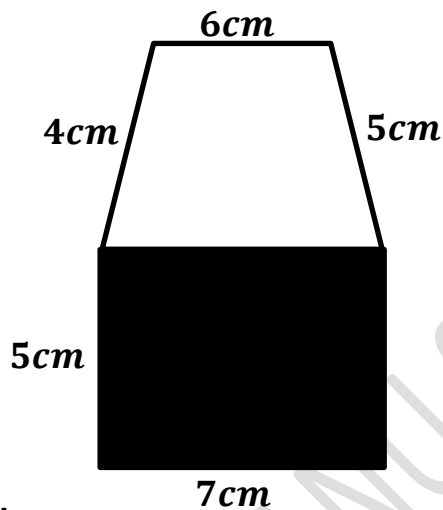
Step 2: Substitute and simplify

$$\text{Area of shaded portion} = \frac{1}{2} (6 + 8) \times 5$$

$$\text{Area of shaded portion} = \frac{14}{2} \times 5$$

$$\text{Area of shaded portion} = 35 \text{ cm}^2$$

Example 6. Find the perimeter of the structure and the area of the shaded portion.



Solution

Step 1:

Perimeter = Sum of all the sides

$$\text{Perimeter} = (6 + 4 + 5 + 7 + 5 + 7 + 5) \text{ cm} = 39 \text{ cm}$$

Step 2: Find the area of the shaded portion

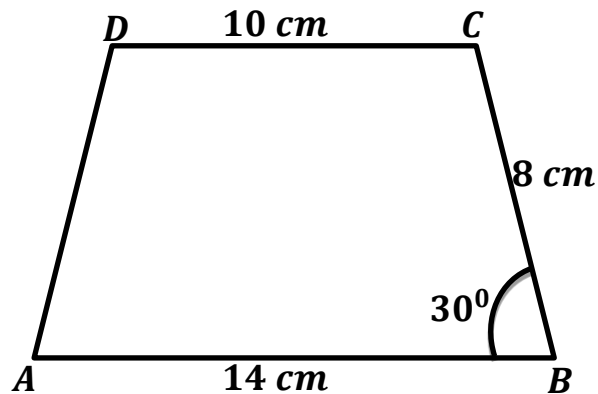
Area of the shaded portion (i.e rectangle) = $l \times b$

$$\text{Area of the shaded portion} = (5 \times 7) \text{ cm}^2 = 35 \text{ cm}^2$$

Example 7. ABCD is a trapezium in which $\overline{AB} \parallel \overline{DC}$. $|\overline{AB}| = 14\text{cm}$, $|\overline{BC}| = 8\text{cm}$, $|\overline{CD}| = 10\text{cm}$ and $\widehat{ABC} = 30^\circ$.

Solution

Step 1: Draw the diagram of the information given



Step 2: Write the formula for area of a trapezium

$$\text{Area of a trapezium} = \frac{1}{2}(a + b) \times c \sin \theta$$

Step 3: Substitute the given values

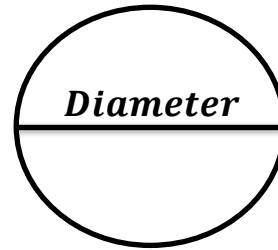
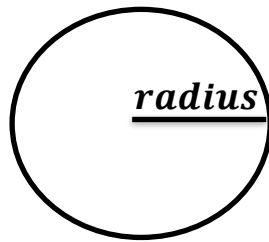
$$\text{Area of a trapezium} = \frac{1}{2}(10 + 14) \times 8 \times \sin 30$$

$$\text{Area of a trapezium} = \frac{1 \times 24 \times 8 \times 0.5}{2}$$

$$\text{Area of a trapezium} = \frac{96}{2}$$

$$\text{Area of a trapezium} = 48 \text{ cm}^2$$

SECTION 6: CIRCLE



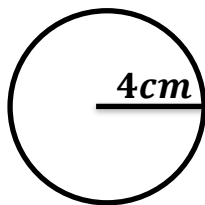
Perimeter or circumference of a circle = $2\pi r$ or πd

Area of a circle = πr^2

Where $\pi = \frac{22}{7}$, r = radius and d = diameter

ILLUSTRATIVE EXAMPLES

Example 1. Find the perimeter and area of the circle?



Solution

Solving for perimeter

Step 1: Write the formula for perimeter of a circle

Perimeter of a circle = $2\pi r$

Step 2: Substitute and simplify

$$\text{Perimeter of a circle} = 2 \times \frac{22}{7} \times 4$$

$$\text{Perimeter of a circle} = \frac{176}{4}$$

$$\text{Perimeter of a circle} = 44\text{cm}$$

Solving for area

Step 1: Write the formula for area of a circle

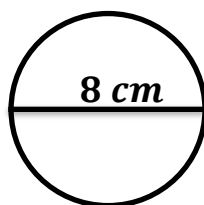
$$\text{Area of a circle} = \pi r^2$$

$$\text{Area of a circle} = \frac{22}{7} \times 4^2$$

$$\text{Area of a circle} = \frac{352}{7}$$

$$\text{Area of a circle} = 50.285 \text{ cm}^2$$

Example 2. Find the perimeter and area of the circle?



Solution

Solving for perimeter

Step 1: Write the formula for perimeter of a circle

$$\text{Perimeter of a circle} = \pi d$$

Step 2: Substitute and simplify

$$\text{Perimeter of a circle} = \frac{22}{7} \times 8$$

$$\text{Perimeter of a circle} = \frac{22 \times 8}{7}$$

$$\text{Perimeter of a circle} = \frac{176}{4}$$

$$\text{Perimeter of a circle} = 44 \text{ cm}$$

Solving for area

Step 1: Write the formula for area of a circle

$$\text{Area of a circle} = \pi r^2$$

Step 2: Substitute

$$\text{But } d = 2r$$

$$r = \frac{d}{2} = \frac{8}{2} = 4 \text{ cm}$$

$$\text{Area of a circle} = \frac{22}{7} \times 4^2$$

$$\text{Area of a circle} = \frac{352}{7}$$

$$\text{Area of a circle} = 50.285 \text{ cm}^2$$

Example 3. If the perimeter of a circle is 100 cm, find the radius of the circle?

Solution

Step 1: Write the formula for perimeter of a circle

$$\text{Perimeter of a circle} = 2\pi r \text{ or } \pi d$$

Step 2: Substitute and simplify

$$44 = \frac{44 \times r}{7}$$

Step 3: Cross multiply

$$44 \times 7 = 44r$$

Step 4: Divide both sides by the coefficient of r

$$\frac{308}{44} = \frac{44r}{44}$$

$$r = 7 \text{ cm}$$

Example 4. *If the area of a circle is 154 cm^2 , find the radius of the circle?*

Solution

Solving for perimeter

Step 1: *Write the formula for perimeter of a circle*

Perimeter of a circle $= 2\pi r$

Step 2: *Substitute*

$$= 2 \times \frac{22}{7} \times 6$$

$$= \frac{264}{7}$$

$$= 37.714 \text{ cm}$$

Solving for perimeter

Step 1: *Write the formula for area of a circle*

Area of a circle $= \pi r^2$

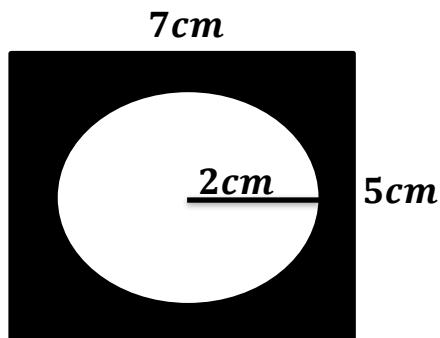
$$\text{Area of a circle} = \frac{22}{7} \times 6 \times 6$$

$$\text{Area of a circle} = \frac{22 \times 6 \times 6}{7}$$

$$\text{Area of a circle} = \frac{792}{7}$$

$$\text{Area of a circle} = 113.142 \text{ cm}^2$$

Example 6. Find the area of the shaded portion



Solution

Step 1:

Area of the shaded portion = area of rectangle – area of circle

Area of the shaded portion = $l \times b - \pi r^2$

Step 2: Substitute

Area of the shaded portion = $7 \times 5 - \frac{22}{7} \times 2 \times 2$

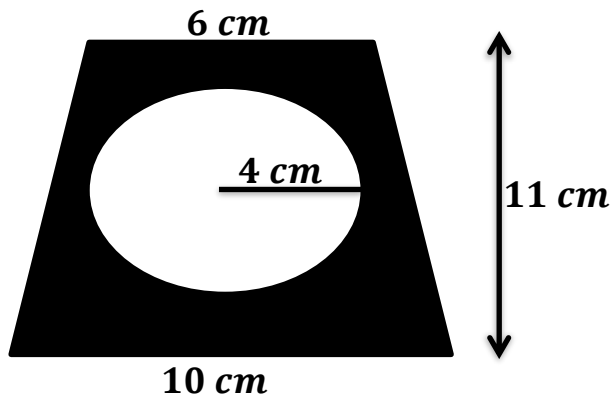
Area of the shaded portion = $35 - \frac{22 \times 2 \times 2}{7}$

Area of the shaded portion = $35 - \frac{88}{7}$

Area of the shaded portion = $(35 - 12.57) \text{ cm}^2$

Area of the shaded portion = 22.43 cm^2

Example 7. Find the area of the shaded portion?



Solution

Step 1:

Area of shaded portion = area of trapezium – area of circle

$$\text{Area of the shaded portion} = \frac{1}{2}(a + b) \times h - \pi \times r^2$$

Step 2: Substitute

$$\text{Area of the shaded portion} = \frac{1}{2}(6 + 10) \times 11 - \frac{22}{7} \times 4 \times 4$$

$$\text{Area of the shaded portion} = \frac{16 \times 11}{2} - \frac{22 \times 4 \times 4}{7}$$

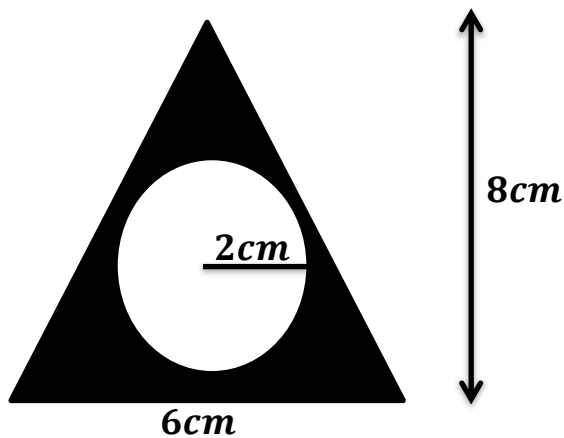
$$\text{Area of the shaded portion} = \frac{16 \times 11}{2} - \frac{22 \times 4 \times 4}{7}$$

$$\text{Area of the shaded portion} = \frac{176}{2} - \frac{352}{7}$$

$$\text{Area of the shaded portion} = (88 - 50.28) \text{ cm}^2$$

$$\text{Area of the shaded portion} = 37.72 \text{ cm}^2$$

Example 8. Find the area of the shaded portion?



Solution

Step 1: Write the formula

Area of shaded portion = area of triangle - area of circle

$$\text{Area of the shaded portion} = \frac{1}{2} \times b \times h - \pi r^2$$

Step 2: Substitute and simplify

$$\text{Area of the shaded portion} = \frac{1}{2} \times 6 \times 8 - \frac{22}{7} \times 2 \times 2$$

$$\text{Area of the shaded portion} = \frac{6 \times 8}{2} - \frac{22 \times 2 \times 2}{7}$$

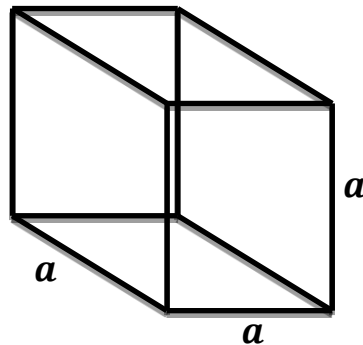
$$\text{Area of the shaded portion} = \frac{48}{2} - \frac{88}{7}$$

$$\text{Area of the shaded portion} = (24 - 12.57) \text{ cm}^2$$

$$\text{Area of the shaded portion} = 11.43 \text{ cm}^2$$

SECTION 7: CUBE

A cube has six faces of equal dimensions. This means that it has length, breadth(width) and height all of which are equal to each other.



Surface area of a cube = $6a^2$ (square units)

Volume of a cube = a^3 (square units)

ILLUSTRATIVE EXAMPLES

Example 1. *A cube has a length of 7 cm, calculate its surface area and volume*

Solution

Solving for surface area

Step 1: *Write the formula for surface area of a cube*

Surface area of a cube = $6a^2$

Step 2: *Substitute the given value*

Surface area of a cube = $6(7)^2$

= $6 \times 7 \times 7$

= 294 cm^2

Solving for volume

Step 1: Write the formula for volume of a cube

$$\text{Volume of a cube} = a^3$$

Step 2: Substitute the given value

$$\text{Volume of a cube} = 7^3 \text{ cm}^3$$

$$\text{Volume of a cube} = (7 \times 7 \times 7) \text{ cm}^3$$

$$\text{Volume of a cube} = 343 \text{ cm}^3$$

Example 2. The surface area of a cube is 294 cm^2 , find its length?

Solution

Step 1: Write the formula for surface area of a cube

$$\text{Surface area of a cube} = 6a^2$$

Step 2: Substitute the given value

$$294 = 6a^2$$

Step 3: Divide both sides by the coefficient of a^2

$$\frac{294}{6} = \frac{6a^2}{6}$$

$$49 = a^2$$

Step 4: Taking the square root of both sides

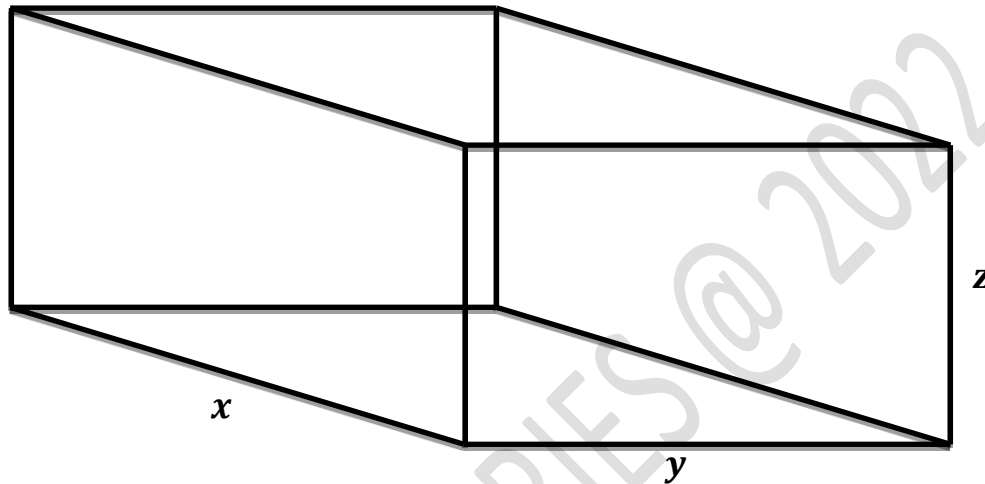
$$\sqrt{a^2} = \sqrt{49}$$

$$a = 7 \text{ cm}$$

Therefore, the length of the cube is 7 cm.

SECTION 8: CUBOID

A cuboid is a solid with a rectangular base and sides. It has six rectangular faces if all sides are closed.



Surface Area of a cuboid = $2(xy + xz + yz)$

ILLUSTRATIVE EXAMPLES

Example 1. *A cuboid has length 15cm, width 11cm and height 10cm. Calculate its surface area*

Solution

Step 1: *Write the formula for surface area of a cuboid*

Surface Area of a cuboid = $2(xy + xz + yz)$

Step 2: *Substitute the given values*

Surface Area of a cuboid = $2(15 \times 11 + 15 \times 10 + 11 \times 10)$

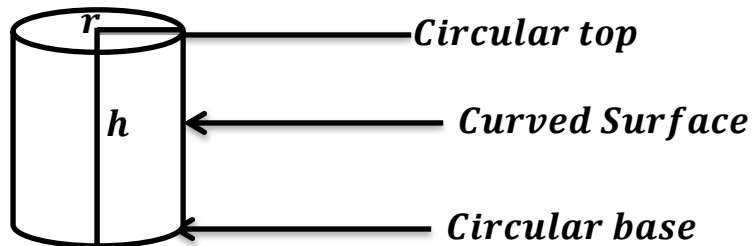
Surface Area of a cuboid = $2(165 + 150 + 110)$

Surface Area of a cuboid = 2×425

Surface Area of a cuboid = 850 cm^2

SECTION 9: CYLINDER

A cylinder is a prism whose cross – section is a circle.



Curved surface area of a cylinder = $2\pi rh$

Total surface area = Areas of all faces

(i). When both tops are closed:

= Area of base + Area of top + curved surface area

= $\pi r^2 + \pi r^2 + 2\pi rh$

= $2\pi r^2 + 2\pi rh$

= $2\pi r(r + h)$ square units

(ii). When one top is opened

= Area of base + curved surface area

= $\pi r^2 + 2\pi rh$

= $\pi r(r + 2h)$ square units

Volume of a cylinder = $\pi r^2 h$ cubic units

ILLUSTRATIVE EXAMPLES

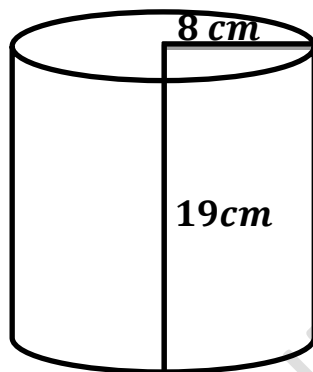
Example 1. *A cylindrical container has a base radius of 8 cm and its height is 19 cm. Calculate its:*

(a). *Total surface area*

(b). Curved surface area

(c). Volume

$\left[\text{Take } \pi = \frac{22}{7} \right]$



Solution

(a).

Step 1: Write the formula for total surface area of a cylinder

(i). Total surface area when both tops are closed:

$$= 2\pi r(r + h) \text{ square units}$$

Step 2: Substitute the given values

$$= 2 \times 3.142 \times 8(8 + 19)$$

$$= 2 \times 3.142 \times 8(8 + 19)$$

$$= 50.272(27)$$

$$= 1357.3 \text{ cm}^2$$

(ii). Total surface area when one top is opened

$$= \pi r(r + 2h) \text{ square units}$$

Step 3: Substitute the given values

$$= 3.142 \times 8(8 + 2 \times 19)$$

$$= 25.136(46)$$

$$= 1156.3 \text{ cm}^2$$

(b).

Step 1: Write the formula for curved surface area of a cylinder

$$\text{Curved surface area of a cylinder} = 2\pi rh$$

Step 2: Substitute the given values

$$\text{Curved surface area of a cylinder} = 2 \times 3.142 \times 8 \times 19$$

$$\text{Curved surface area of a cylinder} = 955.2 \text{ cm}^2$$

(c).

Step 1: Write the formula for volume of a cylinder

$$\text{Volume of a cylinder} = \pi r^2 h$$

Step 2: Substitute the given values

$$\text{Volume of a cylinder} = \frac{22}{7} \times 8^2 \times 19$$

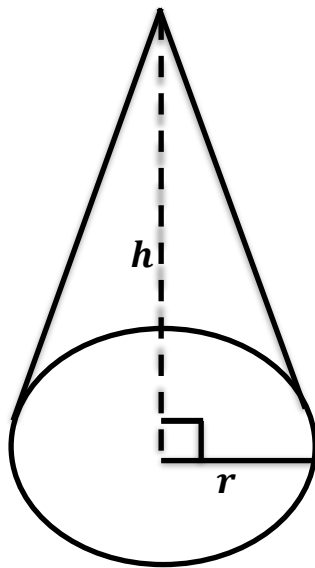
$$\text{Volume of a cylinder} = \frac{22}{7} \times 8 \times 8 \times 19$$

$$\text{Volume of a cylinder} = \frac{22 \times 8 \times 8 \times 19}{7}$$

$$\text{Volume of a cylinder} = \frac{26752}{7}$$

$$\text{Volume of a cylinder} = 3821.7 \text{ cm}^3$$

SECTION 10: CONE



Curve Surface Area of a Cone = πrl

Total Surface Area of a cone = $\pi r(l + r)$

Volume of a cone = $\frac{1}{3}\pi r^2 h$

Where, l = slant height, r = radius and h = height

ILLUSTRATIVE EXAMPLES

Example 1. Find the curved surface area of a cone, if its radius and slant height are 3 cm and 5 cm respectively $\left[\text{Take } \pi = \frac{22}{7} \right]$

Solution

Step 1: Write the formula for curved surface area of a cone

Curved surface area of a cone = πrl

Curved surface area of a cone = $\frac{22}{7} \times 3 \times 5$

$$\text{Curved surface area of a cone} = \frac{330}{7}$$

$$\text{Curved surface area of a cone} = 47.14 \text{ cm}^2$$

Example 2. Find the total surface area of a cone, if its radius and slant height are 3 cm and 5 cm respectively. [Take $\pi = \frac{22}{7}$]

Solution

Step 1: Write the formula for total surface area of a cone

$$\text{Total surface area of a cone} = \pi r(l + r)$$

$$\text{Total surface area of a cone} = \frac{22}{7} \times 3(5 + 3)$$

$$\text{Total surface area of a cone} = \frac{22}{7} \times 3(8)$$

$$\text{Total surface area of a cone} = \frac{22}{7} \times 24$$

$$\text{Total surface area of a cone} = \frac{528}{7}$$

$$\text{Total surface area of a cone} = 57.43 \text{ cm}^2$$

Example 3. Find the volume of a cone, if its radius and height are 3 cm and 5 cm respectively. [Take $\pi = \frac{22}{7}$]

Solution

Step 1: Write the formula for volume of a cone

$$\text{Volume of a cone} = \frac{1}{3} \pi r^2 h$$

$$= \frac{1}{3} \times \frac{22}{7} \times 3^2 \times 5$$

$$= \frac{1}{3} \times \frac{22}{7} \times 3 \times 3 \times 5$$

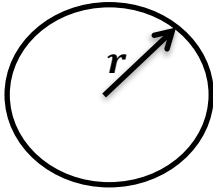
$$= \frac{1 \times 22 \times 3 \times 3 \times 5}{3 \times 7}$$

$$= \frac{990}{21}$$

$$= 47.14 \text{ cm}^3$$

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SECTION 11: SPHERE



Surface area of a sphere = $4\pi r^2$

Volume of a sphere = $\frac{4}{3}\pi r^3$

ILLUSTRATIVE EXAMPLES

Example 1. Find the surface area and volume of the sphere of radius

4 cm. [Take $\pi = \frac{22}{7}$]

Solution

Solving for surface area

Step 1: Write the formula for surface area of a sphere

Surface area of a sphere = $4\pi r^2$

Step 2: Substitute the given values

Surface area of a sphere = $4 \times \frac{22}{7} \times 4^2$

Surface area of a sphere = $4 \times \frac{22}{7} \times 4 \times 4$

Surface area of a sphere = $\frac{4 \times 22 \times 4 \times 4}{7}$

Surface area of a sphere = $\frac{1408}{7}$

Surface area of a sphere = 201.14 cm^2

Solving for volume

Step 1: Write the formula for volume of a sphere

$$\text{Volume of a sphere} = \frac{4}{3}\pi r^3$$

Step 2: Substitute the given values

$$\text{Volume of a sphere} = \frac{4}{3} \times \frac{22}{7} \times 4^3$$

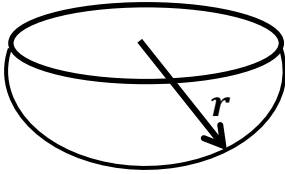
$$\text{Volume of a sphere} = \frac{4}{3} \times \frac{22}{7} \times 4 \times 4 \times 4$$

$$\text{Volume of a sphere} = \frac{4 \times 22 \times 4 \times 4 \times 4}{3 \times 7}$$

$$\text{Volume of a sphere} = \frac{5632}{21}$$

$$\text{Volume of a sphere} = 268.19 \text{ cm}^3$$

SECTION 12: HEMISPHERE



Curve surface area of a hemisphere = $2\pi r^2$

Volume of a hemisphere = $\frac{2}{3}\pi r^3$

ILLUSTRATIVE EXAMPLES

Example 1. Find the surface area and volume of the hemisphere of radius 4 cm. [Take $\pi = \frac{22}{7}$]

Solution

Solving for surface area

Step 1: Write the formula for surface area of a hemisphere

Surface area of a hemisphere = $2\pi r^2$

Step 2: Substitute the given values

Surface area of a hemisphere = $2 \times \frac{22}{7} \times 4^2$

Surface area of a hemisphere = $2 \times \frac{22}{7} \times 4 \times 4$

Surface area of a hemisphere = $\frac{2 \times 22 \times 4 \times 4}{7}$

Surface area of a hemisphere = $\frac{704}{7}$

Surface area of a hemisphere = 100.57 cm^2

Solving for volume

Step 1: Write the formula for volume of a sphere

$$\text{Volume of a sphere} = \frac{2}{3}\pi r^3$$

Step 2: Substitute the given values

$$\text{Volume of a sphere} = \frac{2}{3} \times \frac{22}{7} \times 4^3$$

$$\text{Volume of a sphere} = \frac{2}{3} \times \frac{22}{7} \times 4 \times 4 \times 4$$

$$\text{Volume of a sphere} = \frac{2 \times 22 \times 4 \times 4 \times 4}{3 \times 7}$$

$$\text{Volume of a sphere} = \frac{2816}{21}$$

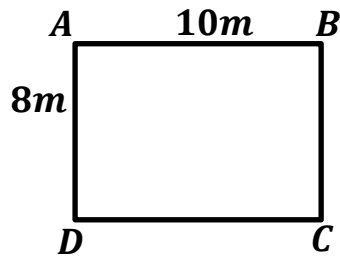
$$\text{Volume of a sphere} = 134.10 \text{ cm}^3$$

SECTION 13: SKS PAPER 21

Answer all questions

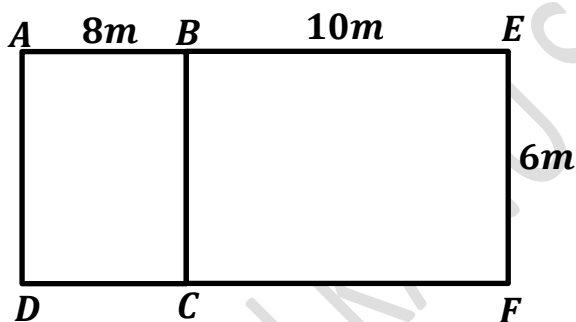
Part 1

Question 1. Find the area and perimeter of the figure below?



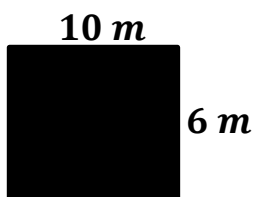
Answer [Perimeter = 36m and Area = 80m^2]

Question 2. Find the perimeter and area of rectangle ABCD and BEFC



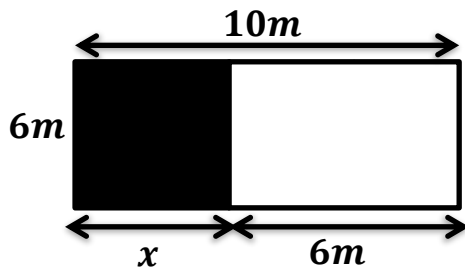
Answer [Perimeter of ABCD = 28m and Area = 48m^2
Perimeter of BEFC = 32m and Area = 60m^2]

Question 3. Find the area of the shaded portion?



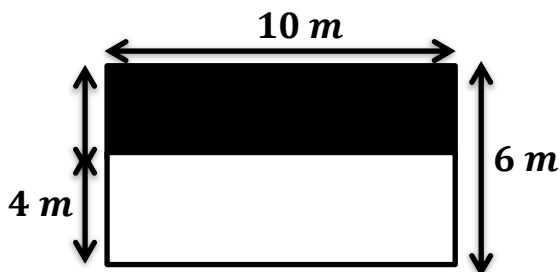
Answer [Area = 60m^2]

Question 4. Find the perimeter and area of the shaded portion?



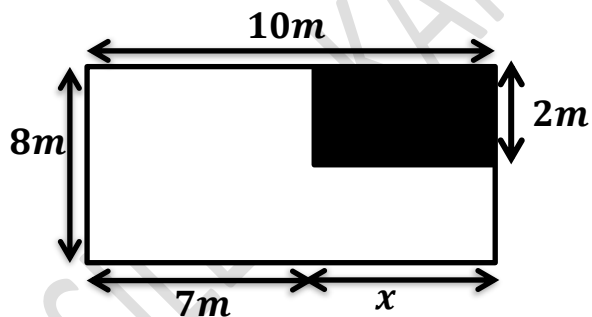
Answer[Perimeter = 20m and Area = $24m^2$]

Question 5. Find the perimeter and area of the shaded portion.



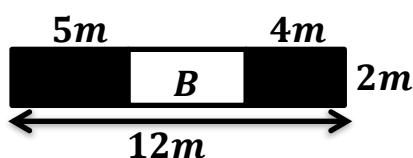
Answer[Perimeter = 24m and Area = $20m^2$]

Question 6. Find the perimeter and area of the shaded portion.



Answer[Perimeter = 10 m and Area = $6 m^2$]

Question 7. Find the perimeter and area of rectangle B?



Answer [Perimeter = 10 m and Area = 6 m²]

Part 2

Question 1. The area of a rectangle is 24cm². If its length is 4 cm, find its breadth.

Answer [Breadth = 6 cm]

Question 2. The area of a rectangle is 100¹/₂ cm². Find its breadth, if its length is 5 cm.

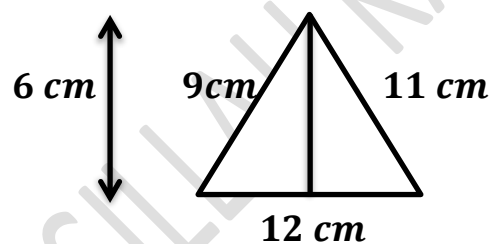
Answer [Breadth = 2 cm]

Question 3. If the length of a rectangle is $(2\sqrt{16} + \sqrt{4})$ cm and its area is $\sqrt{100}$ cm². Find its breadth?

Answer [Breadth = 1 cm]

Part 3

Question 1. Find the perimeter and area of the triangle.



Answer [Perimeter = 32m and Area = 36m²]

Question 2. If the area of a triangle is 20cm² and its width is 4cm, Find its height?

Answer [Height = 10cm]

Question 3. If the area of the triangle is $\frac{\sqrt{100}}{2} \text{ cm}^2$ and the height is 2 cm.

Find the width?

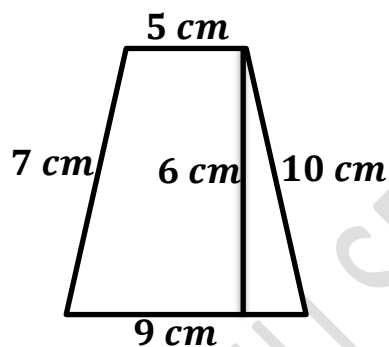
Answer [width = 5 cm]

Part 4

Question 1. Find the area and perimeter of a square of length 8cm.

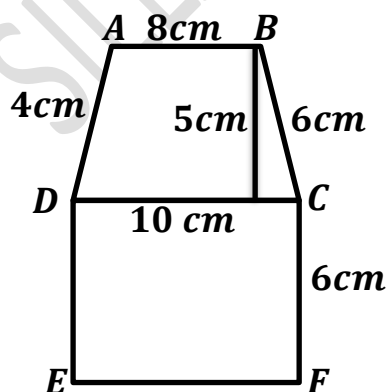
Answer [Perimeter = 32 cm and Area = 64 cm²]

Question 2. Find the perimeter and area of the trapezium.



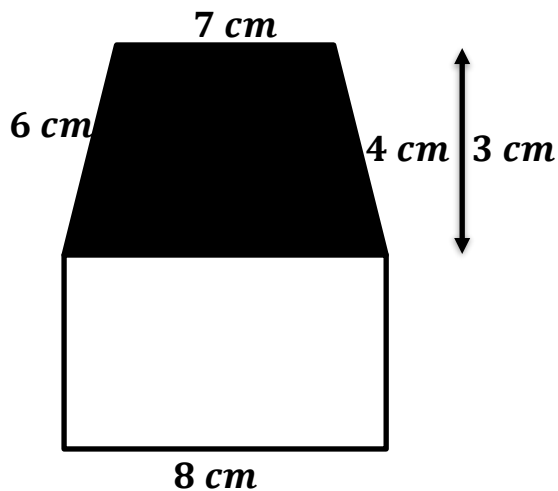
Answer [Perimeter = 31cm and Area = 42 cm²]

Question 3. Find the perimeter and area of the figure below



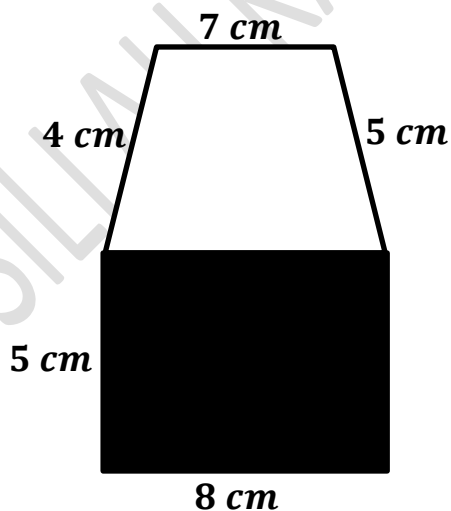
Answer[*Perimeter = 60 cm and Area = 105 cm²*]

Question 4. Find the perimeter of the figure and the area of the shaded portion.



Answer [*Perimeter = 25 cm and Area = $22\frac{1}{2}$ cm²*]

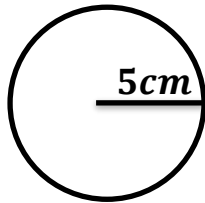
Question 6. Find the perimeter of the structure and the area of the shaded portion



Answer[*Perimeter = 50 cm and Area = 40 cm²*]

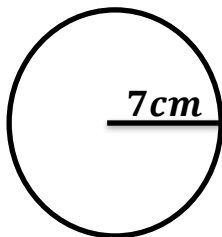
Part 5

Question 1. Find the perimeter and area of the circle?



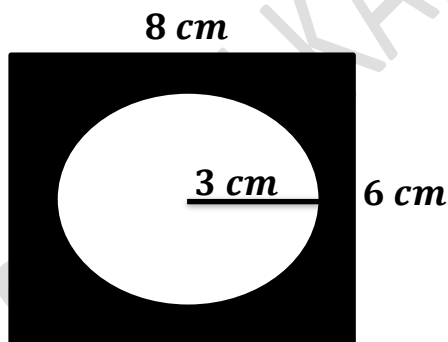
Answer $\left[\text{Area} = 78\frac{4}{7} \text{ cm}^2 \right]$

Question 2. Find the perimeter and area of the circle?



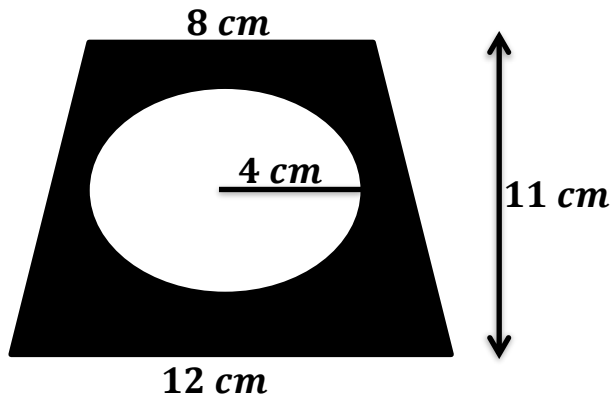
Answer $\left[\text{Area} = 154 \text{ cm}^2 \right]$

Question 3. Find the area of the shaded portion



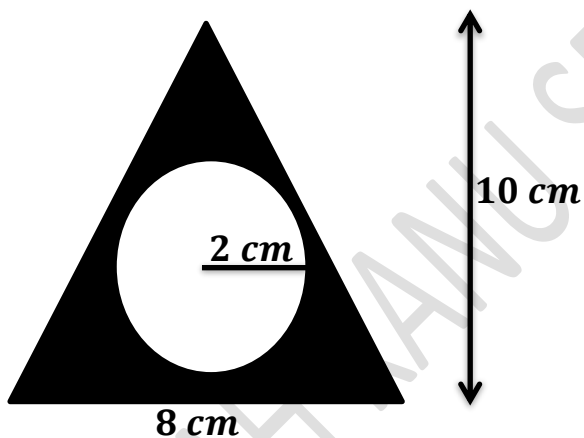
Answer $\left[\text{Area} = 19\frac{5}{7} \text{ cm}^2 \right]$

Question 4. Find the area of the shaded portion?



Answer $\left[\text{Area} = 59\frac{5}{7} \text{ cm}^2 \right]$

Question 5. Find the area of the shaded portion?



Answer $\left[\text{Area} = 27\frac{3}{7} \text{ cm}^2 \right]$

Part 6

Question 1. Find the surface area and volume of the cylinder of radius 6m and height 10m?

Answer $\left[\text{Surface Area} = 603\frac{3}{7} \text{ cm}^2 \right]$

Question 2. Find the curved surface area of a cone, if its radius and slant height are 5 cm and 7 cm respectively $\left[\text{Take } \pi = \frac{22}{7} \right]$

Answer $[\text{Curve Surface Area} = 110 \text{ cm}^2]$

Question 3. Find the total surface area of a cone, if its radius and slant height are 6 cm and 10 cm respectively. $\left[\text{Take } \pi = \frac{22}{7} \right]$

Answer $\left[\text{Total Surface Area} = 301\frac{5}{7} \text{ cm}^2 \right]$

Question 3. Find the volume of a cone, if its radius and height are 5 cm and 8 cm respectively. $\left[\text{Take } \pi = \frac{22}{7} \right]$

Answer $\left[\text{Volume} = 209\frac{11}{21} \text{ cm}^3 \right]$

Part 7

Question 1. Find the surface area and volume of the sphere of radius 8 cm. $\left[\text{Take } \pi = \frac{22}{7} \right]$

Answer $\left[\text{Surface Area} = 804\frac{4}{7} \text{ cm}^2 \text{ and Volume} = 2145\frac{11}{21} \text{ cm}^3 \right]$

Question 2. Find the surface area and volume of the sphere of radius 4 cm. $\left[\text{Take } \pi = \frac{22}{7} \right]$

Answer $\left[\text{Surface area} = 201\frac{1}{7} \text{ cm}^2 \text{ and Volume} = 268\frac{4}{21} \text{ cm}^3 \right]$

Part 8

Question 1. Find the surface area and volume of the hemisphere of radius 8 cm. [Take $\pi = \frac{22}{7}$]

Answer [Surface Area = $402\frac{2}{7} \text{ cm}^2$ and Volume = $1072\frac{16}{21} \text{ cm}^3$]

Question 2. Find the surface area and volume of the hemisphere of radius 4 cm. [Take $\pi = \frac{22}{7}$]

Answer [Surface area = $100\frac{4}{7} \text{ cm}^2$ and Volume = $134\frac{2}{21} \text{ cm}^3$]

Part 9

Question 1. Find the area and volume of a cube of length 4 cm

Answer [Area = 96 cm^2 and Volume = 64 cm^3]

Question 2. Find the area and volume of a cube of length 8 cm

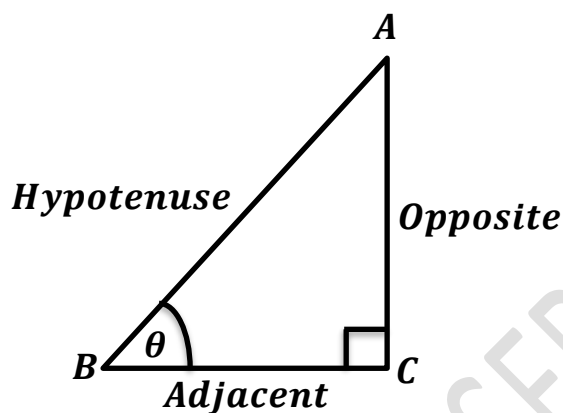
Answer [Area = 384 cm^2 and Volume = 512 cm^3]

CHAPTER 22

TRIGONOMETRY

SECTION 1: TRIGONOMETRIC RATIOS

For a right – angle triangle the following can be defined:



$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{|AC|}{|AB|}$$

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{|BC|}{|AB|}$$

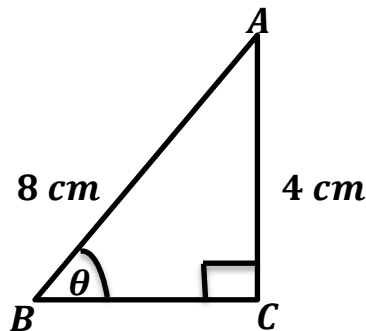
$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{|AC|}{|BC|}$$

- *The side that forms angle with the hypotenuse is called the Adjacent*
- *The opposite is the remaining side of the right – angle triangle after the hypotenuse and adjacent are known.*

ILLUSTRATIVE EXAMPLES

Example 1. Find the value of θ in the diagram below?

[Hint: $\sin^{-1}(0.5) = 30^\circ$]



Solution

Step 1: list the trigonometric ratios

$$\sin \theta = \frac{\text{Opposite}}{\text{Hypotenuse}}$$

$$\cos \theta = \frac{\text{Adjacent}}{\text{Hypotenuse}}$$

$$\tan \theta = \frac{\text{Opposite}}{\text{Adjacent}}$$

Step 2: list the given sides

$$\text{Hypotenuse} = 8 \text{ cm}$$

$$\text{Opposite} = 4 \text{ cm}$$

Step 3: use the trigonometric ratio that have opposite and hypotenuse.

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\sin \theta = \frac{4}{8}$$

$$\sin \theta = 0.5$$

Step 4: Make θ the subject

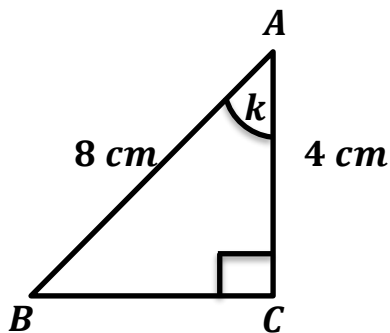
$$\theta = \sin^{-1}(0.5)$$

$$\text{But } \sin^{-1}(0.5) = 30^\circ$$

$$\therefore \theta = 30^\circ$$

Example 2. Find the value of k in the figure below?

[Hint: $\cos^{-1}(0.5) = 60^\circ$]



Solution

Step 1: list the trigonometric ratios

$$\sin \theta = \frac{\text{Opposite}}{\text{Hypotenuse}}$$

$$\cos \theta = \frac{\text{Adjacent}}{\text{Hypotenuse}}$$

$$\tan \theta = \frac{\text{Opposite}}{\text{Adjacent}}$$

Step 2: list the given sides

Hypotenuse = 8 cm

Adjacent = 4 cm (because it forms angle with the hypotenuse)

Step 3: use the trigonometric ratio that have opposite and hypotenuse

$$\cos k = \frac{\text{Adjacent}}{\text{Hypotenuse}}$$

$$\cos k = \frac{4}{8}$$

$$\cos k = 0.5$$

Step 4: Make k the subject

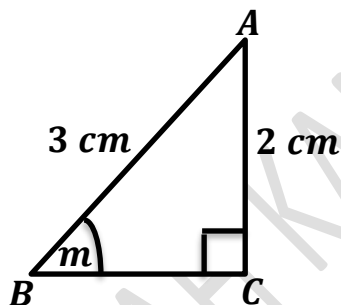
$$k = \cos^{-1}(0.5)$$

$$\text{But } \cos^{-1}(0.5) = 60^{\circ}$$

$$\therefore k = 60^{\circ}$$

Example 3. Find the value of m in the figure below?

[Hint: $\sin^{-1}(0.667) = 41.83^{\circ}$]



Solution

Step 1: using the trigonometric ratios

$$\sin \theta = \frac{\text{Opposite}}{\text{Hypotenuse}}$$

$$\cos \theta = \frac{\text{Adjacent}}{\text{Hypotenuse}}$$

$$\tan \theta = \frac{\text{Opposite}}{\text{Adjacent}}$$

Step 2: list the given sides

$$\text{Hypotenuse} = 3\text{cm}$$

$$\text{Opposite} = 2\text{cm}$$

Step 3: use the trig. ratio that have opposite and hypotenuse

$$\sin m = \frac{\text{Opposite}}{\text{Hypotenuse}}$$

$$\sin m = \frac{2}{3}$$

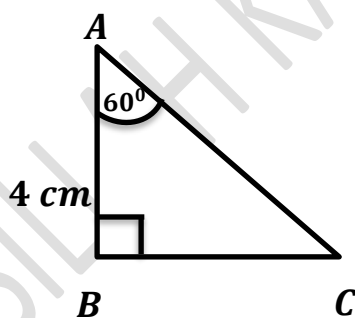
$$\sin m = 0.667$$

Step 4: Make m the subject

$$m = \sin^{-1}(0.667)$$

$$m = 41.83^{\circ}$$

Example 4. Find the value of / BC/?



Solution

Step 1: list the trigonometric ratios

$$\sin \theta = \frac{\text{Opposite}}{\text{Hypotenuse}}$$

$$\cos \theta = \frac{\text{Adjacent}}{\text{Hypotenuse}}$$

$$\tan \theta = \frac{\text{Opposite}}{\text{Adjacent}}$$

Step 2: list the given sides

$$\text{Adjacent} = 4 \text{ cm}$$

$$\text{Opposite} = /BC/$$

Step 3: Substitute

$$\tan 60^\circ = \frac{/BC/}{4}$$

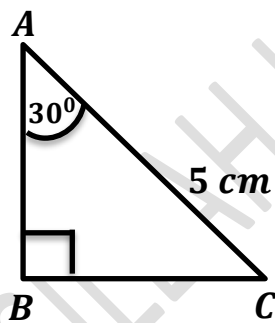
Step 4: cross multiply

$$\tan 60^\circ \times 4 = /BC/$$

$$1.732 \times 4 = /BC/$$

$$/BC/ = 6.928 \text{ cm}$$

Example 5. Find the value of /AB/?



Solution

Step 1: list the trigonometric ratios

$$\sin \theta = \frac{\text{Opposite}}{\text{Hypotenuse}}$$

$$\cos \theta = \frac{\text{Adjacent}}{\text{Hypotenuse}}$$

$$\tan \theta = \frac{\text{Opposite}}{\text{Adjacent}}$$

Step 2: list the given sides

$$\text{Adjacent} = /AB/$$

$$\text{Hypotenuse} = 5 \text{ cm}$$

Step 3: Substitute

$$\cos 30^\circ = \frac{/AB/}{5}$$

Step 4: cross multiply

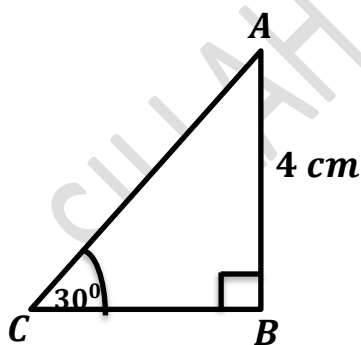
$$\cos 30^\circ \times 5 = /AB/$$

$$0.8660 \times 5 = /AB/$$

$$/AB/ = 4.33 \text{ cm}$$

Example 6. Find the value of /AC/?

[Hint: $\sin 30^\circ = 0.5$]



Solution

Step 1: list the trigonometric ratios

$$\sin \theta = \frac{\text{Opposite}}{\text{Hypotenuse}}$$

$$\cos \theta = \frac{\text{Adjacent}}{\text{Hypotenuse}}$$

$$\tan \theta = \frac{\text{Opposite}}{\text{Adjacent}}$$

Step 2: list the given sides

$$\text{Opposite} = 4 \text{ cm}$$

$$\text{Hypotenuse} = /AC/$$

Step 3: substitute

$$\sin 45^\circ = \frac{4}{/AC/}$$

Step 4: cross multiply

$$\sin 30^\circ \times /AC/ = 4$$

$$\text{But } \sin 30^\circ = 0.5$$

$$0.5 \times /AC/ = 4$$

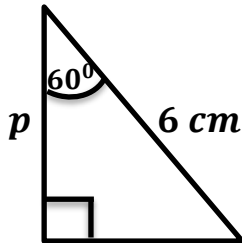
Step 5: divide both sides by the coefficient of /AC/

$$\frac{\cancel{0.5} \times /AC/}{\cancel{0.5}} = \frac{4}{\cancel{0.5}}$$

$$/AC/ = 8 \text{ cm}$$

Example 7. Find the value of p ?

[Hint: $\cos 60 = 0.5$]



Solution

Step 1: list the trigonometric ratios

$$\sin \theta = \frac{\text{Opposite}}{\text{Hypotenuse}}$$

$$\cos \theta = \frac{\text{Adjacent}}{\text{Hypotenuse}}$$

$$\tan \theta = \frac{\text{Opposite}}{\text{Adjacent}}$$

Step 2: list the given sides

$$\text{Adjacent} = p$$

$$\text{Hypotenuse} = 6\text{ cm}$$

Step 3: Substitute

$$\cos 60 = \frac{p}{6}$$

Step 3: cross multiply

$$\cos 60 \times 6 = p$$

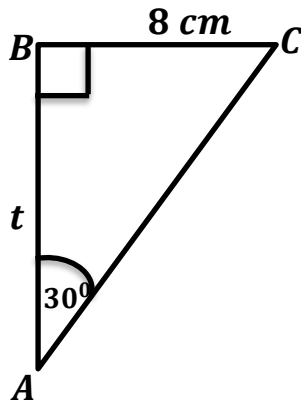
$$\text{But } \cos 60 = 0.5$$

$$p = 0.5 \times 6$$

$$p = 3 \text{ cm}$$

Example 8. Find the value of t ?

[Hint: $\tan 30 = 0.577$]



Solution

Step 1:

$$\tan 30 = \frac{8}{t}$$

$$\tan 30 \times t = 8$$

$$\text{But } \tan 30 = 0.577$$

$$0.577 \times t = 8$$

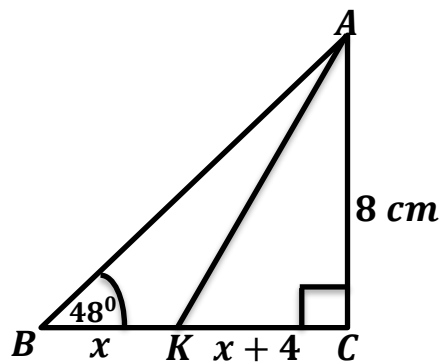
Step 2: Divide both sides by the coefficient of t

$$\frac{0.577 \times t}{0.577} = \frac{8}{0.577}$$

$$t = 13 \text{ cm}$$

Example 9. Find the value of x in the figure below?

[Hint: $\tan 48 = 1.1106$]



Solution

Considering Triangle ABC

Step 1: $\tan \theta = \frac{\text{Opposite}}{\text{Adjacent}} = \frac{|AC|}{|BC|}$

$$\tan 48 = \frac{8}{x + (x + 4)}$$

Step 2: Cross multiply

$$\tan 48 \times (2x + 4) = 8$$

Step 3: Expand the bracket

$$\tan 48 \times 2x + \tan 48 \times 4 = 8$$

$$2x \times (1.1106) + 1.1106 \times 4 = 8$$

$$2.2212x + 4.4424 = 8$$

Step 4: Collect like terms

$$2.2212x = 8 - 4.4424$$

$$2.2212x = 3.5576$$

Step 5: Divide both sides by the coefficient of x

$$\frac{2.2212x}{2.2212} = \frac{3.5576}{2.2212} = 1.6 \text{ (correct to one decimal place)}$$

OR

Step 1: $\tan 48 = \frac{8}{|BC|}$

Cross multiply

$$|BC| = \frac{8}{\tan 48} = \frac{8}{1.1106} = 7.2 \text{ (1dp)}$$

Step 2: *Substitute for |BC|*

$$|BC| = x + x + 4 = 7.2$$

$$2x = 7.2 - 4$$

$$2x = 3.2$$

Divide both sides by the coefficient of x

$$\frac{2x}{2} = \frac{3.2}{2}$$

$$x = 1.6 \text{ (Correct to 1 decimal place)}$$

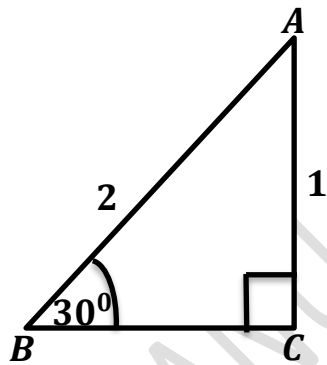
Example 10. If $\sin 30^\circ = \frac{1}{2}$, prove that $\cos 30^\circ = \frac{\sqrt{3}}{2}$ and

$$\tan 30^\circ = \frac{1}{\sqrt{3}}$$

Solution

Step 1: draw the right – angle triangle and indicate the lengths of the given sides

$$\text{But } \sin \theta = \frac{\text{Opposite}}{\text{Hypotenuse}} = \frac{1}{2} \text{ (given in the question)}$$



Step 2: using Pythagoras theorem to find the remaining side

$$/AB/^2 = /BC/^2 + /AC/^2$$

$$2^2 = /BC/^2 + 1^2$$

$$4 = /BC/^2 + 1$$

$$4 - 1 = /BC/^2$$

$$3 = /BC/^2$$

Step 3: Taking the square root of both sides to get rid of the square

$$\sqrt{/BC/^2} = \sqrt{3}$$

$$/BC/ = \sqrt{3}$$

Step 4: $\cos 30^\circ = \frac{\text{Adjacent}}{\text{Hypotenuse}}$

But adjacent = $/BC/ = \sqrt{3}$ and hypotenuse = $/AB/ = 2$

Therefore, $\cos 30^\circ = \frac{\sqrt{3}}{2}$

Step 5: $\tan \theta = \frac{\text{Opposite}}{\text{Adjacent}}$

Therefore, $\tan 30^\circ = \frac{1}{\sqrt{3}}$

Example 12. If $\tan 30^\circ = \frac{1}{\sqrt{3}}$, find $\sin 30^\circ$ and $\cos 30^\circ$.

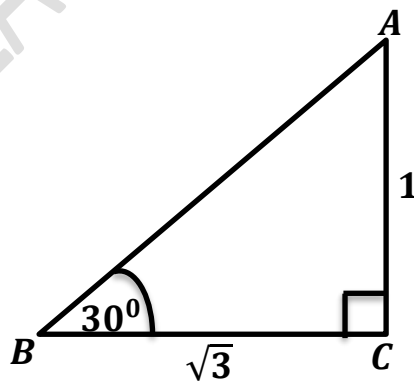
Solution

Step 1: draw the right – angle triangle and indicate the lengths of the given sides

But $\tan \theta = \frac{\text{Opposite}}{\text{Adjacent}}$

Opposite = 1

Adjacent = $\sqrt{3}$



Step 2: using Pythagoras theorem to find the remaining side

$$/AB/^2 = (\sqrt{3})^2 + 1^2$$

$$/AB/^2 = \sqrt{3} \times \sqrt{3} + 1 \times 1$$

$$/AB/^2 = \sqrt{9} + 1$$

$$\text{But } \sqrt{9} = 3$$

$$/AB/^2 = 3 + 1$$

$$/AB/^2 = 4$$

Step 3: Taking the square root of both sides to get rid of the square

$$\sqrt{/AB/^2} = \sqrt{4}$$

$$\text{but } \sqrt{4} = 2$$

$$/AB/ = 2$$

Step 4: $\cos \theta = \frac{\text{Adjacent}}{\text{Hypotenuse}}$

But adjacent = $/BC/ = \sqrt{3}$ and hypotenuse = $/AB/ = 2$

$$\text{Therefore, } \cos 30^\circ = \frac{\sqrt{3}}{2}$$

Step 5: $\sin 30^\circ = \frac{\text{Opposite}}{\text{Hypotenuse}}$

But opposite = 1 and hypotenuse = 2

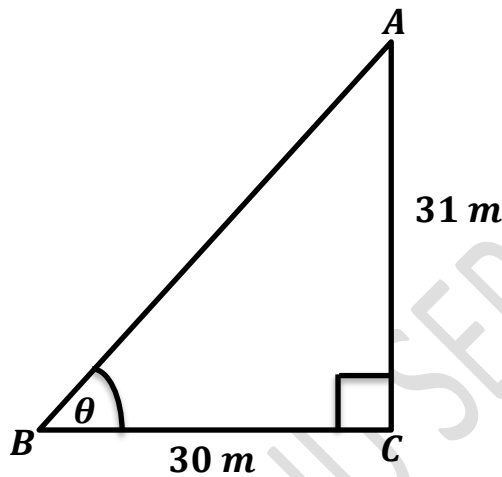
$$\text{Therefore, } \sin 30^\circ = \frac{1}{2}$$

SECTION 2: WORD PROBLEMS

Example 1. A path is 30 m away from a vertical pole. The height of the pole is 31 m. Calculate the angle which the path makes the horizontal. [Hint: $\tan^{-1}(1.03) = 45.85$]

Solution

Step 1: draw the right – angle triangle



Step 2: find the angle

$$\tan \theta = \frac{31}{30}$$

$$\tan \theta = 1.03$$

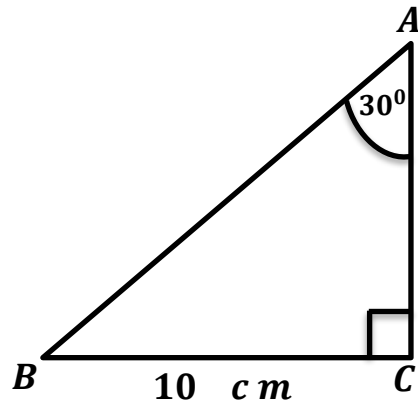
$$\theta = \tan^{-1}(1.03)$$

$$\theta = 45.85^{\circ}$$

Example 2. A ladder leans against a vertical pole at an angle of 30° to the wall. If the foot of the ladder is 10 m away from the wall, calculate the length of the ladder.

Solution

Step 1: Draw the diagram



Step 2:

$$\sin \theta = \frac{\text{Opposite}}{\text{Hypotenuse}}$$

$$\sin 30 = \frac{|BC|}{|AB|}$$

$$\sin 30 = \frac{10}{|AB|}$$

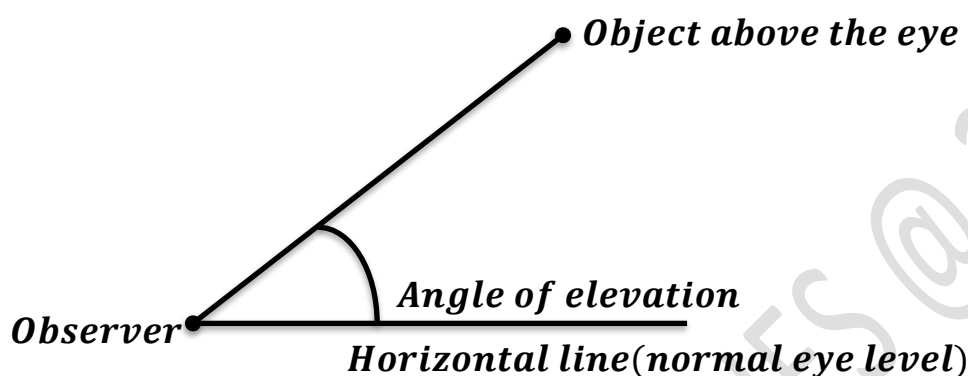
$$|AB| = \frac{10}{\sin 30} = \frac{10}{0.5} = 20$$

Therefore, the length of the ladder is 20 m.

SECTION 3: ANGLE OF ELEVATION AND DEPRESSION

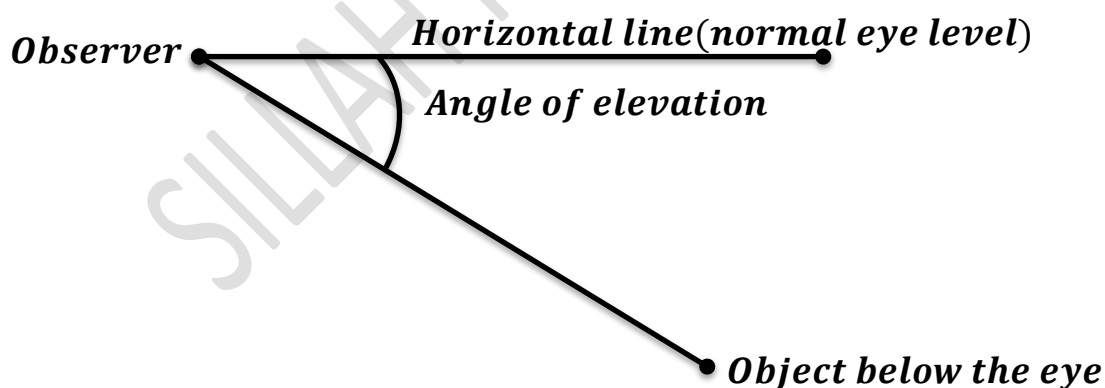
- **Angle of Elevation**

The angle of elevation is the angle between normal eye level (horizontal line) and the line through which an object is above the observer's.



- **Angle of Depression**

On the other hand, the angle of depression is the angle between the horizontal line and the line through which the object which is below the observer's eye can be seen.



ILLUSTRATIVE EXAMPLES

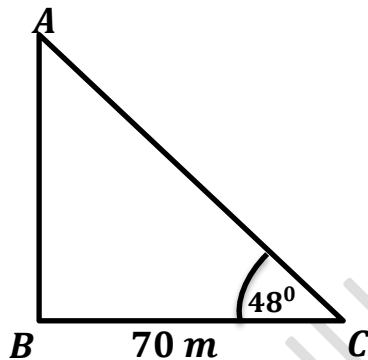
Example 1. From a point 70 m from the foot of a vertical pole, along level ground, the angle of elevation of the top of the pole is 48° .

What is the length of the pole?

[Hint: $\tan 48 = 1.111$]

Solution

Step 1: Draw a right – angle triangle and indicates all the given data



Step 2: use the appropriate trigonometric ratio to find /AB/

$$\tan 48^\circ = \frac{AB}{70}$$

Step 3: cross multiply

$$\tan 48^\circ \times 70 = AB$$

$$AB = 1.111 \times 70$$

$$AB = 77.77\text{m}$$

Therefore, the length is 77.77m

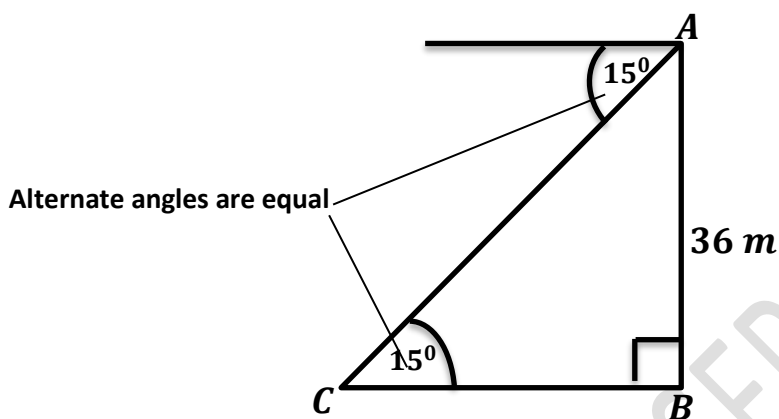
Example 2. A house is 36m above sea level. If the angle of depression of a boat from the top of the tower is 15° , how far is the boat from

the house?

[Hint: $\tan 15 = 0.268$]

Solution

Step 1: Draw the right – angle triangle indicating all the given data



Step 2: use the appropriate trigonometric ratio to find /BC/

$$\tan 15^{\circ} = \frac{36}{/BC/}$$

Step 3: Cross multiply

$$\tan 15^{\circ} \times /BC/ = 36$$

Step 4: Divide both sides by the coefficient of /BC/

$$\frac{\tan 15^{\circ} \times /BC/}{\tan 15^{\circ}} = \frac{36}{\tan 15^{\circ}}$$

$$/BC/ = \frac{36}{0.268}$$

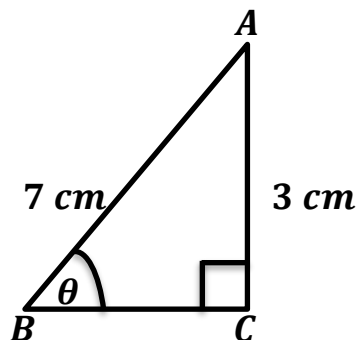
$$/BC/ = 134.33 \text{ m}$$

SECTION 4: SKS PAPER 22

Part 1

Question 1. Find the value of θ in the diagram below?

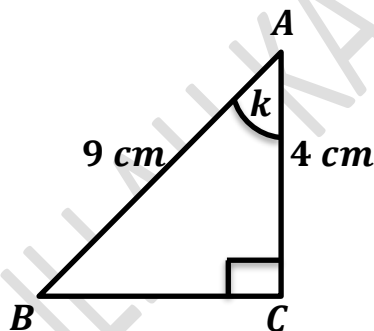
[Hint: $\sin^{-1} 0.429 = 25.404$]



Answer [$\theta = 25.404^{\circ}$ correct to 3 decimal places]

Question 2. Find the value of k in the figure below?

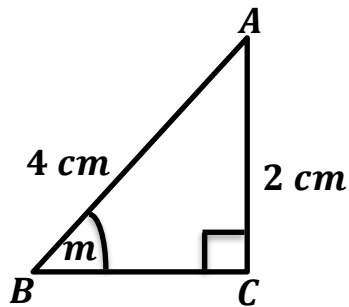
[Hint: $\cos^{-1} 0.444 = 63.640$]



Answer [$k = 63.640^{\circ}$ correct to 3 decimal places]

Question 3. Find the value of m in the figure below?

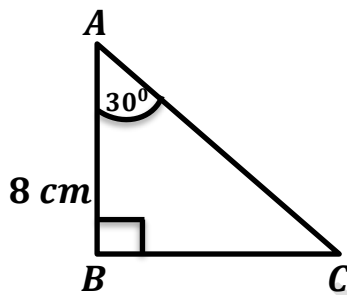
[Hint: $\sin^{-1} 0.5 = 30$]



Answer [$m = 30^\circ$]

Question 4. Find the value of $\angle BC$ /?

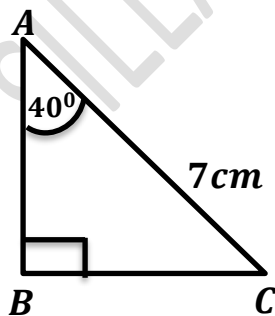
[Hint: $\tan 30 = 0.577$]



Answer [$\angle BC = 4.616 \text{ cm}$ correct to 3 decimal places]

Question 5. Find the value of $\angle AB$ /?

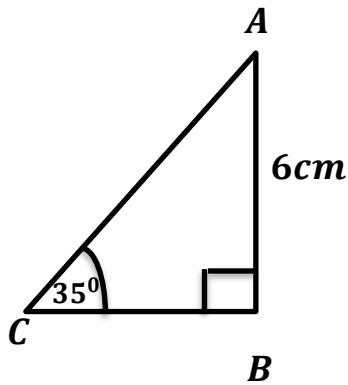
[Hint: $\cos 40 = 0.766$]



Answer [$\angle AB = 5.362 \text{ cm}$ correct to 3 decimal places]

Question 6. Find the value of $\angle AC$ /?

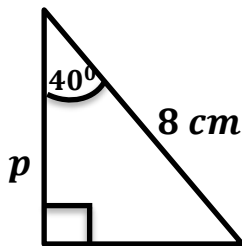
[Hint: $\sin 35 = 0.574$ and $\frac{6}{0.574} = 10.453$]



Answer [$\angle AC$ = 10.453 cm correct to 3 decimal places]

Question 7. Find the value of p ?

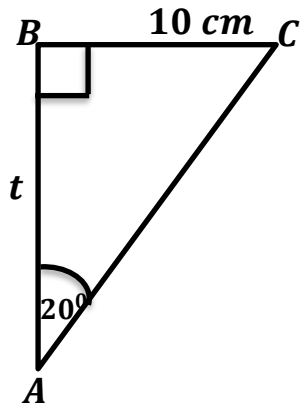
[Hint: $\cos 40 = 0.766$]



Answer [$p = 6.128$ cm correct to three decimal places]

Question 8. Find the value of t ?

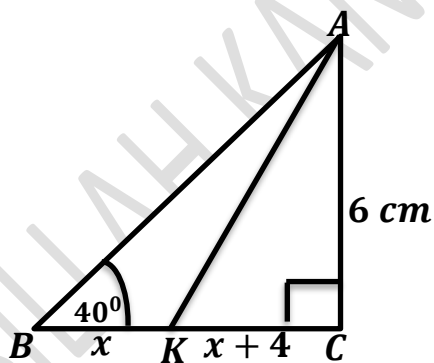
[Hint: $\tan 20 = 0.364$]



Answer [$t = 27.473$ cm correct to 3 decimal places]

Question 9. Find the value of x in the figure below?

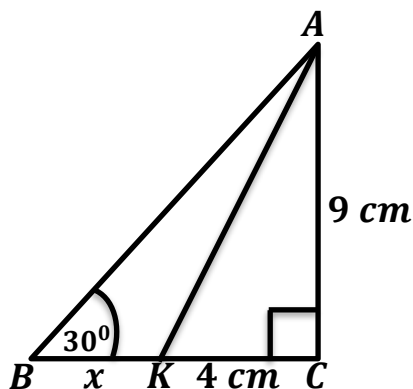
[Hint: $\tan 40 = 0.839$ and $\frac{2.644}{1.678} = 1.576$]



Answer [$x = 1.576$ cm correct to 3 decimal places]

Question 10. Find the value of x in the figure below?

[Hint: $\tan 30^\circ = 0.577$ and $\frac{6.692}{0.577} = 11.598$]



Answer [$x = 11.598 \text{ cm}$]

Part 2

Question 1. A path is 350 m away from the foot of a vertical pole. If the height of the vertical pole is 36 m. Calculate the angle which the path makes the horizontal.

Answer [$\theta = 5.873^\circ$ correct to 3 decimal places]

Question 2. A path is inclined at 35° to the horizontal of a vertical pole. If the hill is 36 m high. Calculate the length of the path?

Answer [Length of the path is 51.413 m correct to 3 decimal places]

Part 3

Question 1. From a point 90 m from the foot of a vertical pole, along level ground, the angle of elevation of the top of the pole is 30° .

What is the length of the pole?

Answer [*Length of the pole is 51.962m correct to 3 decimal places*]

Question 2. *A house is 40 m above sea level. If the angle of depression of a boat from the top of the tower is 45° , how far is the boat from the house?*

Answer [*Distance of the boat from the house is 40 m*]

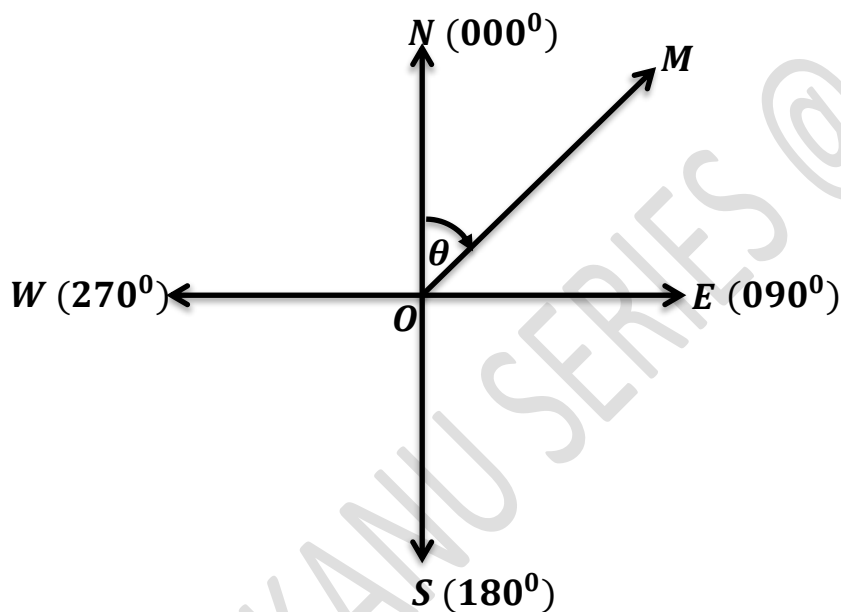
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CHAPTER 23

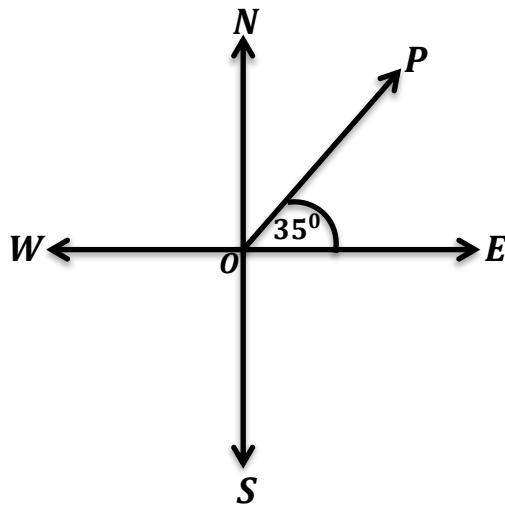
BEARING AND DISTANCE

SECTION 1: BEARING OF A POINT FROM ANOTHER.

When the direction of a point M from another point O is measured in the clockwise direction from the geographic north, it is called bearing.



Example 1. Find the bearing of P from O?

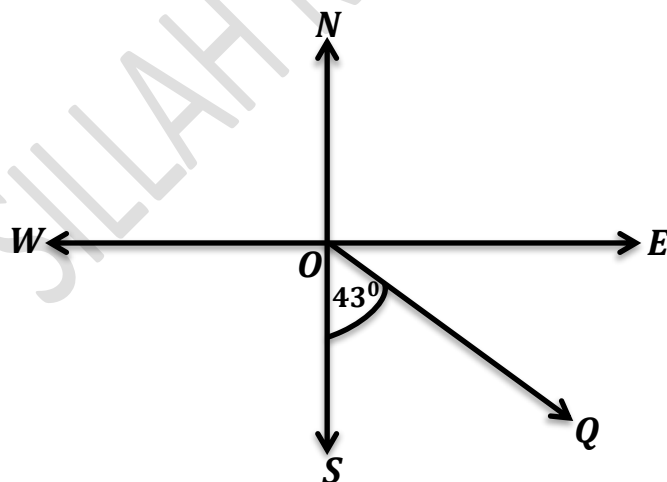


Solution

Step 1: Since the angle lies in North – East or 1st quadrant, we then subtract the angle from 90° .

Therefore, the bearing of P from O = $90^\circ - 35^\circ = 55^\circ$

Example 2. Find the bearing of Q from O?



Solution

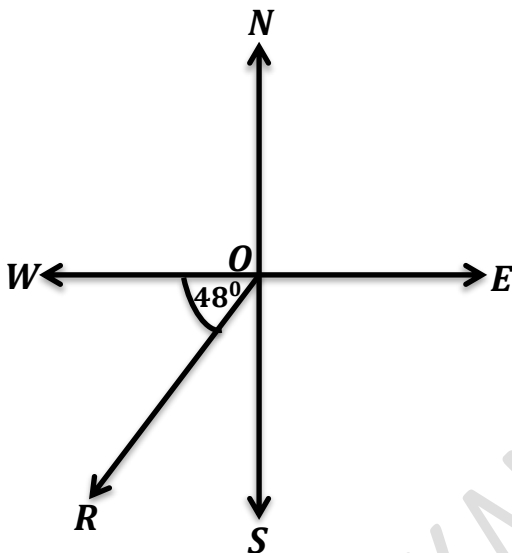
Step 1: Since the angle lies in the South – East or 2nd quadrant, we then subtract the angle from 180° .

Therefore, the bearing of Q from O = $180^\circ - 043^\circ = 137^\circ$

Example 3. Find the bearing of R from O?

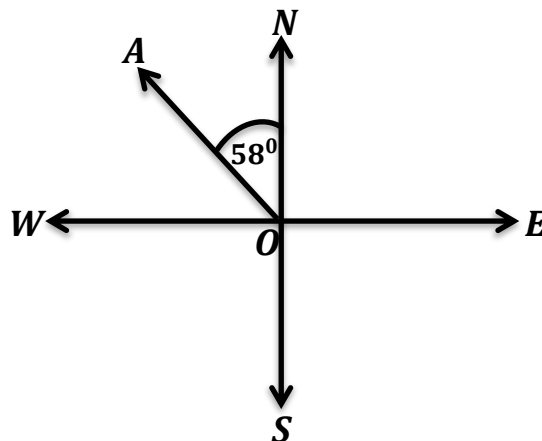
Solution

Step 1: Since the angle lies in the South – West or 3rd quadrant, we then subtract the angle from 270° .



Therefore, the bearing of R from O = $270^\circ - 048^\circ = 222^\circ$

Example 4. Find the bearing of A from O?



Solution

Step 1: Since the angle lies in the North – West or 4th quadrant, we then subtract the angle from 360° .

Therefore, the bearing of A from O = $360^\circ - 058^\circ = 302^\circ$.

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SECTION 2: REVERSAL BEARING

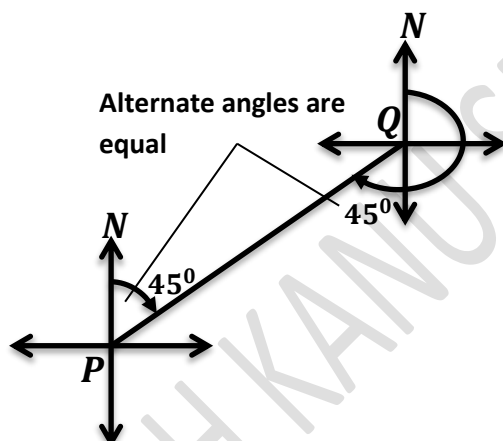
❖ Case 1:

If the bearing of Q from P is θ , for $000^\circ \leq \theta \leq 180^\circ$, then the bearing of P from Q is $(180^\circ + \theta)$.

Example 1. If the bearing of Q from P is 045° , find the bearing of P from Q .

Solution

Step 1: Represent the information on a diagram



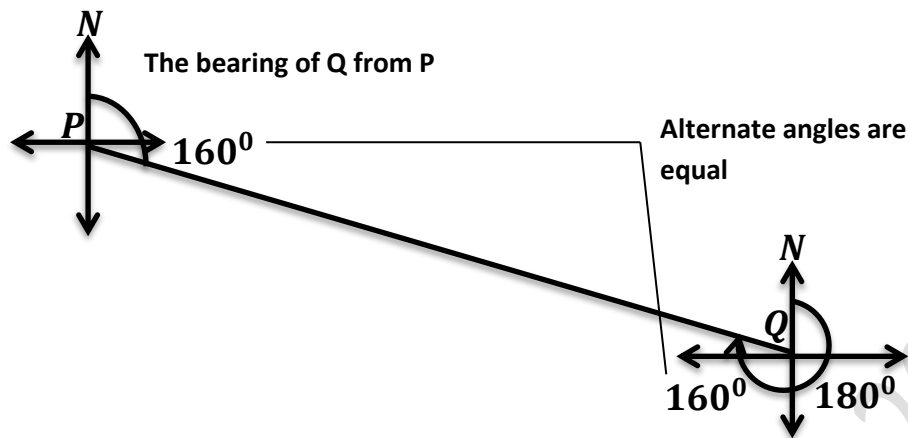
Step 2: Since the angle is less than 180° , we find the sum of 180° and the angle.

Therefore, the bearing of P from $Q = 180^\circ + 45^\circ = 225^\circ$

Example 2. If the bearing of Q from P is 160° , find the bearing of P from Q ?

Solution

Step 1: Represent the information on a diagram



Step 2: Since the angle is less than 180° , we find the sum of 180° and the angle.

Therefore, the bearing of P from Q = $180^\circ + 150^\circ = 340^\circ$

Example 3 .If the bearing of B from A is 020° , find the bearing of A from B?

Solution

Step 1: Since the angle is less than 180° , we find the sum of 180° and the angle.

Therefore, the bearing of A from B = $180^\circ + 20^\circ = 200^\circ$

Example 4.If the bearing of B from A is 140° , find the bearing of A from B?

Solution

Step 1: Since the angle is less than 180° , we find the sum of 180° and the angle.

Therefore, the bearing of A from B = $180^0 + 140^0 = 320^0$

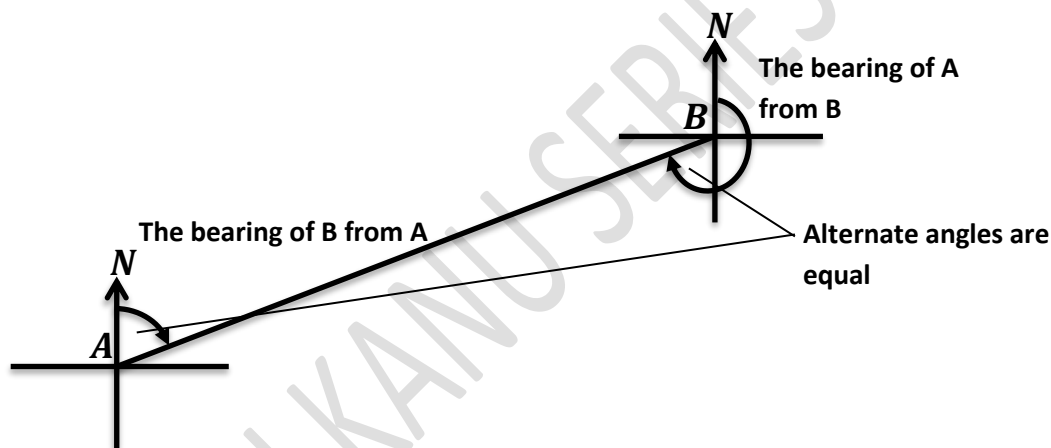
❖ **Case 2:**

If the bearing of B from A is θ , where $180^0 \leq \theta \leq 360^0$, then the bearing of A from B is $(\theta - 180^0)$.

Example 1. If the bearing of A from B is 240^0 , find the bearing of B from A?

Solution

Step 1: Draw a diagram to represent the information



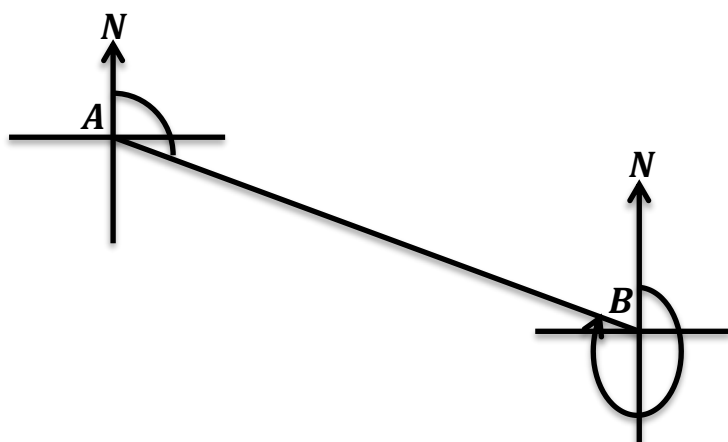
Step 2: Since the angle is greater than 180^0 , we subtract 180^0 from the angle.

Therefore, the bearing of B from A = $240^0 - 180^0 = 60^0$

Example 2. If the bearing of A from B is 340^0 , find the bearing of B from A?

Solution

Step 1: Draw the diagram to represent the information



Step 2: Since the angle is greater than 180° , we subtract 180° from the angle.

Therefore, the bearing of B from A = $340^\circ - 180^\circ = 160^\circ$

Example 3 .If the bearing of Q from P is 210° , find the bearing of P from Q?

Solution

Step 1: Since the angle is greater than 180° , we subtract 180° from the angle.

Therefore, the bearing of P from Q = $210^\circ - 180^\circ = 30^\circ$.

Example 4.If the bearing of Q from P is 310° . Find the bearing of P from Q?

Solution

Step 1: Since the angle is greater than 180° , we subtract 180° from the angle.

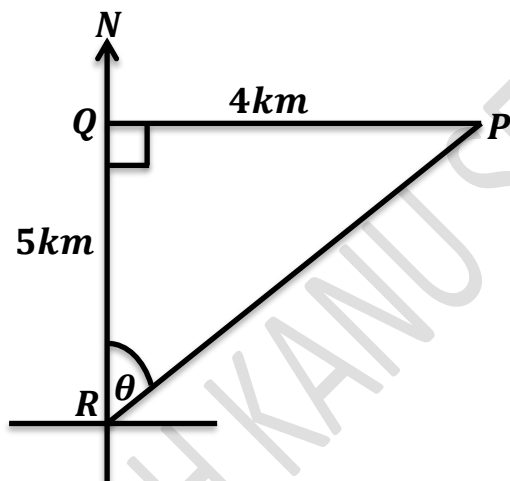
Therefore, the bearing of P from Q = $310^\circ - 180^\circ = 130^\circ$.

SECTION 3: DISTANCE – BEARING FORM

The position of the point B from A is denoted by $\overrightarrow{AB}$, with the arrow over the top representing the direction of movement.

ILLUSTRATIVE EXAMPLES

Example 1. In the diagram below, find the bearing and the distance of P from R?



Solution

Step 1: By Pythagoras' theorem

$$/PR/^2 = /PQ/^2 + /QR/^2$$

$$/PR/^2 = 4^2 + 5^2$$

$$/PR/^2 = 16 + 25$$

$$/PR/^2 = 41$$

Step 3: Taking the square root on both sides

$$(\sqrt{PR})^2 = \sqrt{41}$$

$$PR = 6.4031$$

Step 4: Find the angle

$$\tan \theta = \frac{4}{5}$$

$$\tan \theta = 0.8$$

$$\theta = \tan^{-1}(0.8)$$

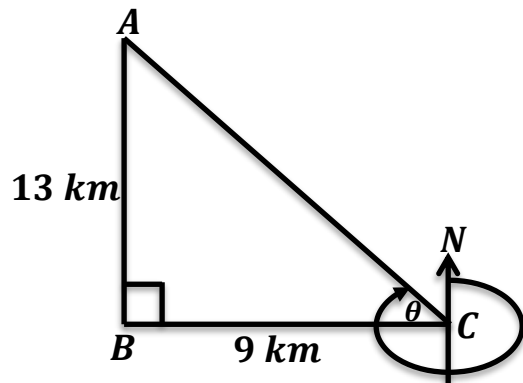
$$\theta = 38.659$$

*Therefore, the distance of P from R = 6,4031km and
the bearing of P from R = 38.659*

*Example 2. A ship sails 13km from port A due south to a point B. It then
sails due east from point B to a point C, a distance of 9km. Find :
position of the port A from the point C.*

Solution

Step 1: Draw the diagram to represent the information



Step 2: By Pythagoras' theorem

$$AC^2 = AB^2 + BC^2$$

$$AC^2 = 13^2 + 9^2$$

$$AC^2 = 169 + 81$$

$$AC^2 = 250$$

Step 3: Taking the square root of both sides

$$\sqrt{AC^2} = \sqrt{250}$$

$$AC = 15.8113 \text{ km}$$

Step 4: Find the angle

$$\tan \theta = \frac{13}{9}$$

$$\tan \theta = 1.4$$

$$\theta = \tan^{-1}(1.4)$$

$$\theta = 54.46^\circ$$

Therefore, the distance of P from R = 6,4031km and

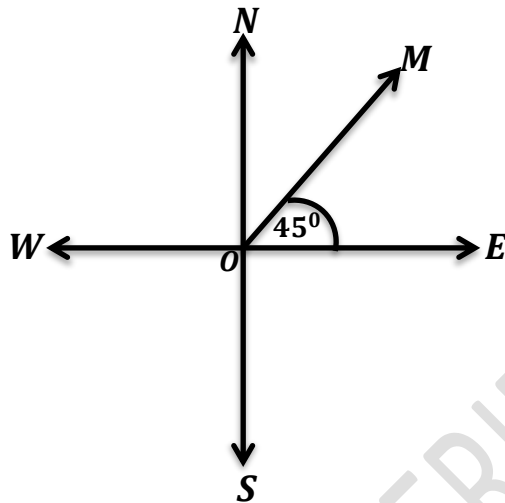
the bearing of P from R = $270^\circ + 54.46^\circ = 324.46^\circ$

SECTION 4: SKS PAPER 5

Answer all questions

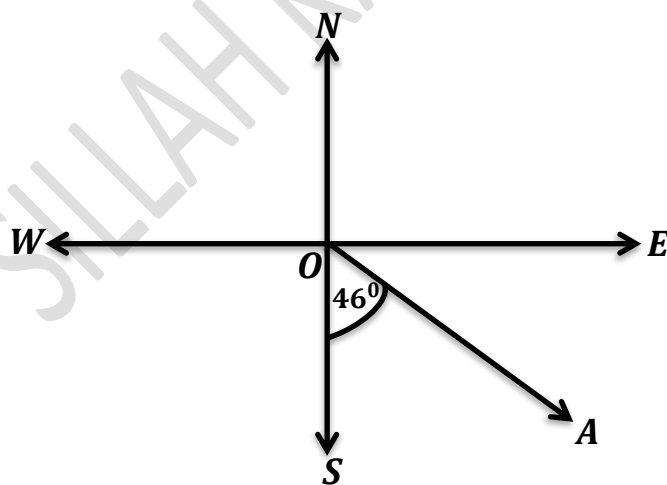
Part 1

Question 1. Find the bearing of M from O?



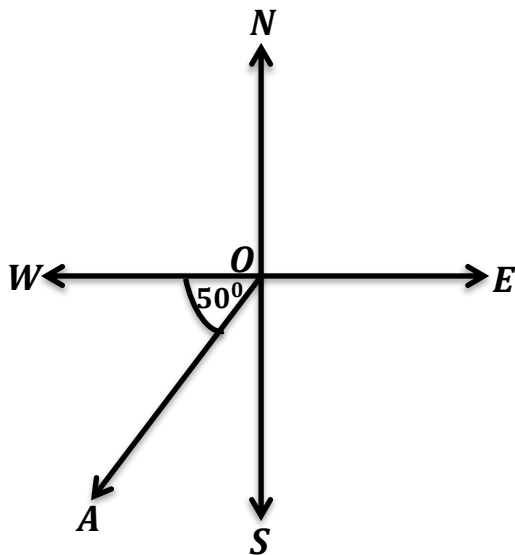
Answer [The bearing of M from O is 45°]

Question 2. Find the bearing of Q from O?



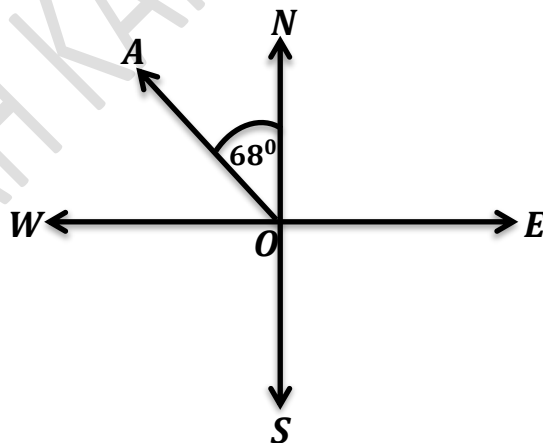
Answer [The bearing of A from O is 134°]

Question 3. Find the bearing of A from O?



Answer [The bearing of A from O is 220°]

Question 4. Find the bearing of A from O?



Answer [The bearing of A from O is 292°]

Part 2

Question 1. *If the bearing of Q from P is 050^0 , find the bearing of P from Q?*

Answer *[The bearing of P from Q is 230^0]*

Question 2. *If the bearing of Q from P is 120^0 , find the bearing of P from Q?*

Answer *[The bearing of P from Q is 300^0]*

Question 3 . *If the bearing of B from A is 040^0 , find the bearing of A from B?*

Answer *[The bearing of A from B is 220^0]*

Question 4. *If the bearing of B from A is 110^0 , find the bearing of A from B?*

Answer *[The bearing of A from B is 290^0]*

Question 5. *If the bearing of A from B is 250^0 , find the bearing of B from A?*

Answer *[The bearing of B from A is 70^0]*

Question 6. *If the bearing of B from A is 290^0 , find the bearing of A from B?*

Answer *[The bearing of A from B is 110^0]*

Question 7 . *If the bearing of Q from P is 220^0 , find the bearing of P from Q?*

Answer *[The bearing of P from Q is 40^0]*

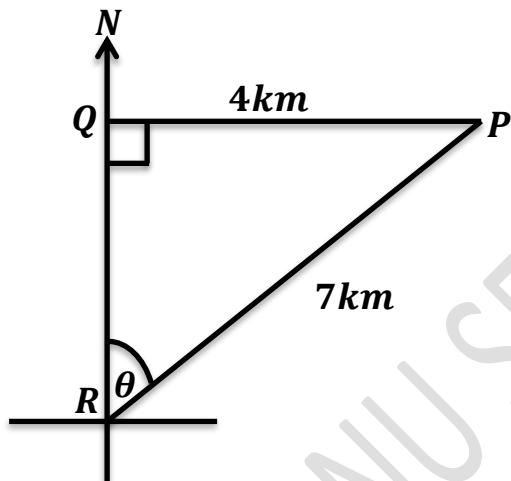
Question 8. *If the bearing of Q from P is 320^0 , find the bearing of P from*

Q?

Answer [The bearing of P from Q is 140^0]

Part 3

Question 1. In the diagram below, the point P is 5km due east of the point Q, find the bearing and the distance of P from R.



Answer [The bearing of R from Q is 38.659^0]

Question 2. A ship sails 13 km from port A due south to a point B. It then sails due east from point B to a port C, a distance of 9km. find :
position of the port A from the port C.

Answer [The bearing of P from R is 324.46^0]

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CHAPTER 24

CONSTRUCTION

SECTION 1: TO BISECT A STRAIGHT LINE

A line segment is the part of the line between P and Q, including the part P and Q.

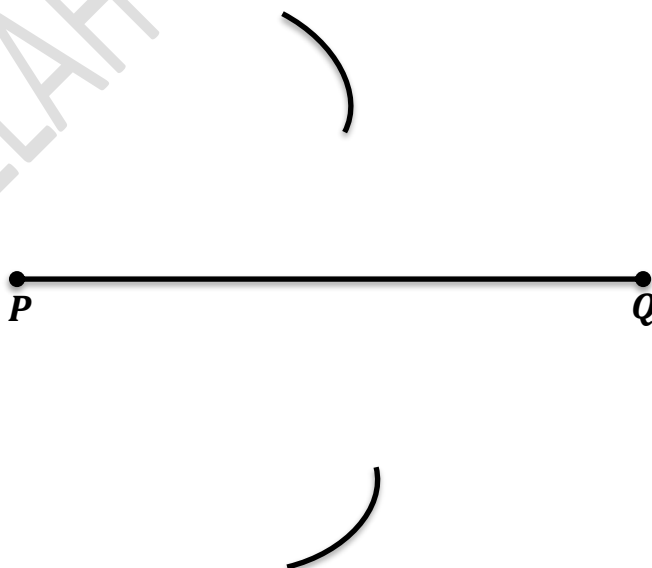


To bisect the line segment PQ means to divide it into two equal parts.

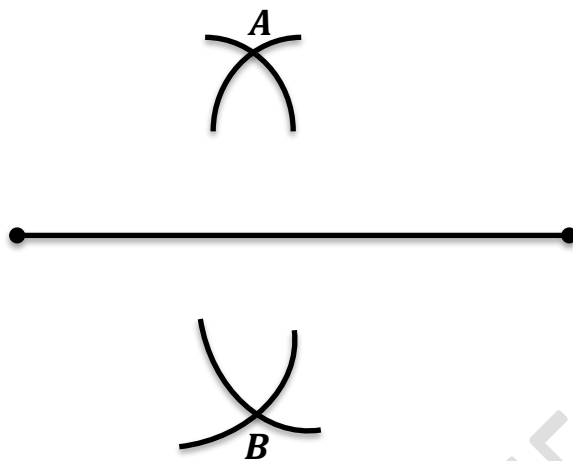
➤ *Steps involved in bisecting a line segment*

Step 1 : *Open a pair of compasses so that the radius is more than half the length of PQ.*

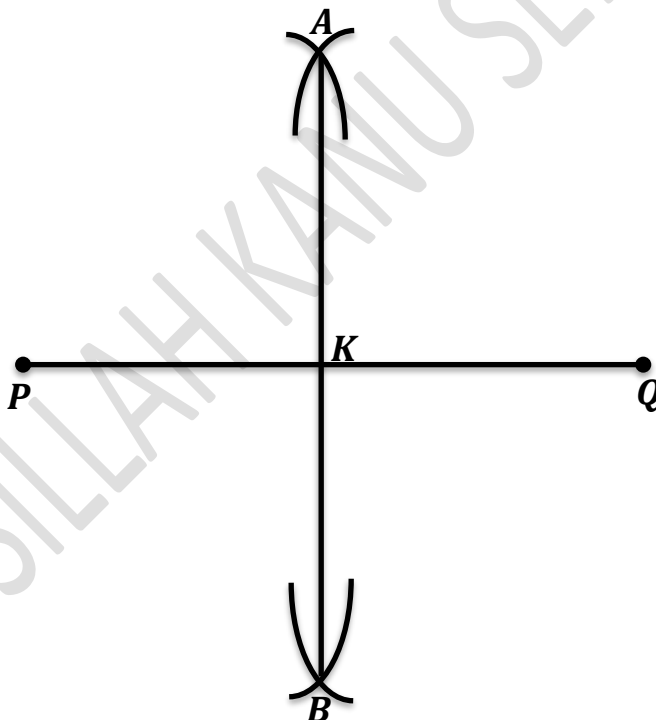
Step 2: *Place the sharp point of the compasses on P. Draw two arcs, one above, the other below the middle of PQ, as shown below.*



Step 3: Keep the same radius and place the sharp point of the compasses on Q. Draw two arcs so that they cut the first arcs at A and B as shown below.



Step 4: Draw a straight line through A and B so that it cuts PQ at K.



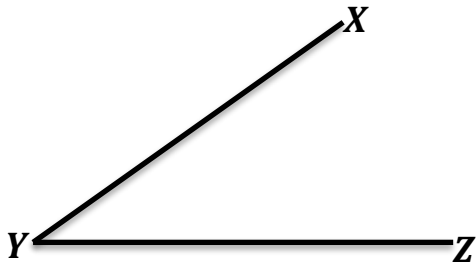
Note: *K is the mid point of PQ. AB meets PQ perpendicularly. AB is the perpendicular bisector of PQ. Use a protractor to check that*

$PK = KQ$ and $\angle PKA = \angle QKB = 90^\circ$ as show in the figure above.

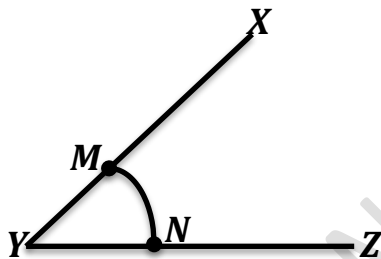
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SECTION 2: TO BISECT A GIVEN LINE.

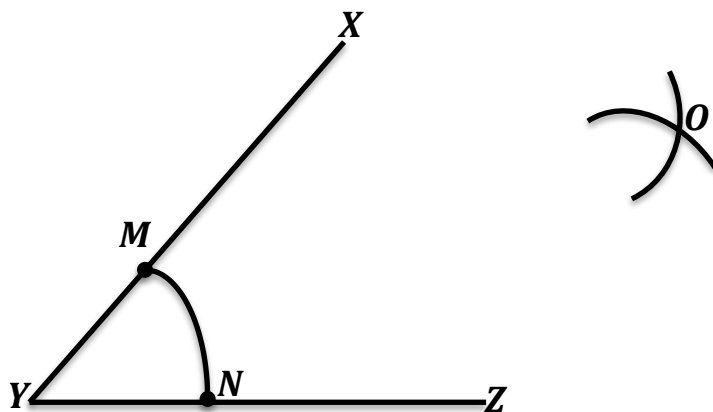
Given any angle XYZ.



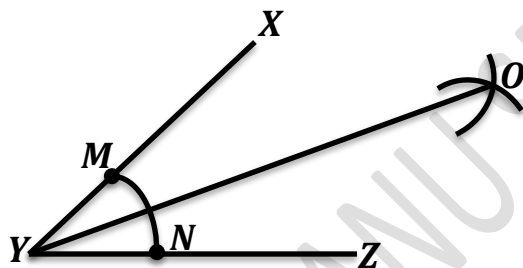
Step 1: with center Y any radius (i. e. open a pair of compasses to any radius and place the sharp point at Y) draw an arc to cut the arms YX, YZ, at M, N as shown below.



Step 2: with centers M, N and equal radii, draw arcs to cut each other at O.



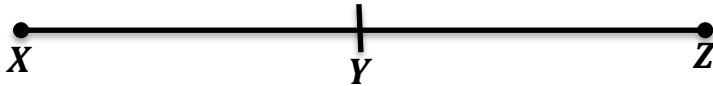
Step 3: join YO .



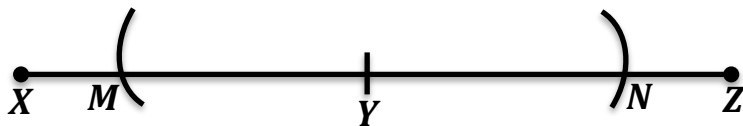
YO bisects $\angle XYZ$. YO is the bisector of ABC . Use a protractor to check that $\angle XYO = \angle ZYO$ in the figure above.

SECTION 3: TO CONSTRUCT AN ANGLE OF 90° .

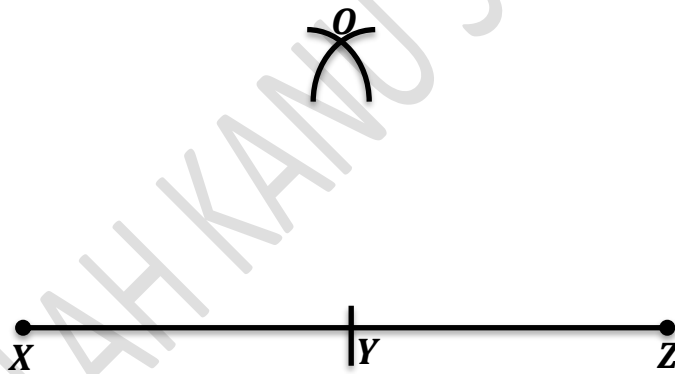
Given a point Y on a straight line XZ .



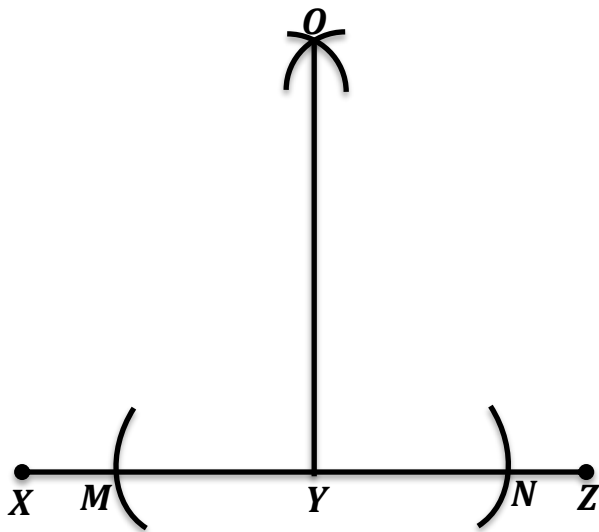
Step 1: with center Y and any radius draw arcs to cut XZ at M and N , as shown below.



Step 2: with centers M, N and equal radii, draw arcs to cut each other at O .



Step 3: join YO



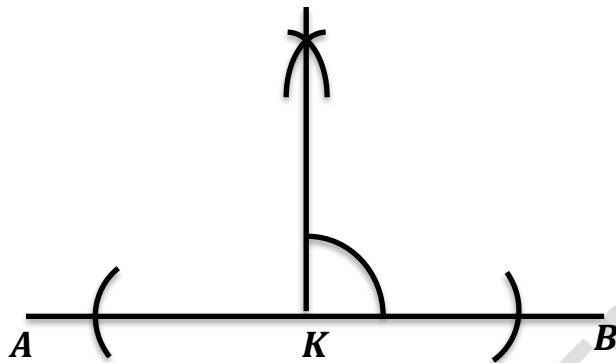
YO is perpendicular to XZ. Thus $\angle OYX = \angle OYZ = 90^\circ$.

Use a protractor to check this result.

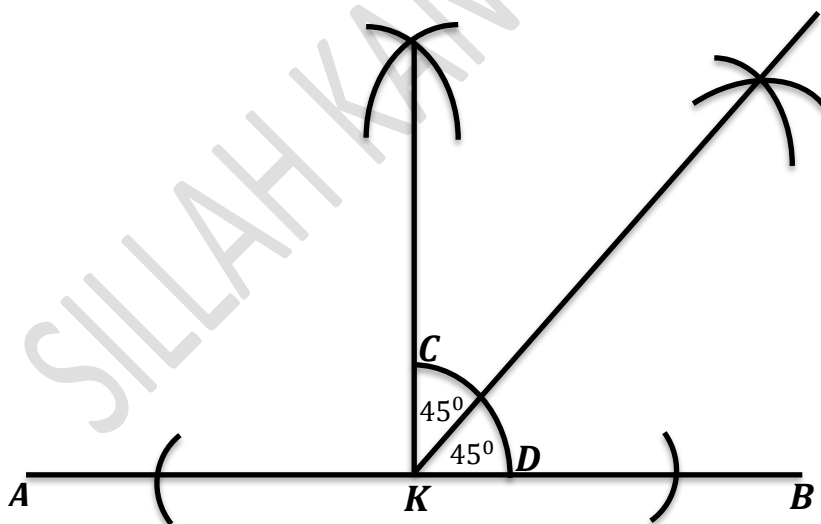
SECTION 4: TO CONSTRUCT AN ANGLE OF 45° .

Step 1: $45^\circ = \frac{1}{2}$ of 90° .

Step 2: To construct an angle of 45° , first construct an angle of 90° .



Step 3: bisect the above



Use a protractor to check your angle.

SECTION 5: TO CONSTRUCT AN ANGLE OF 60°

Given a straight line AB:

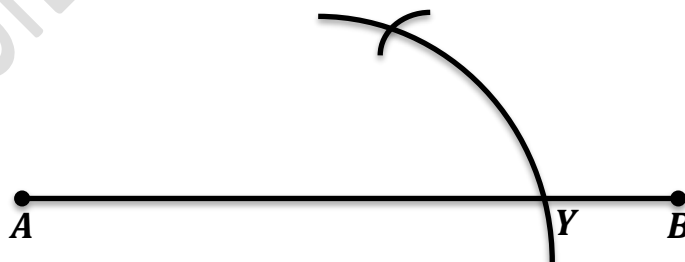


To construct a point C such that $\angle CAB = 60^\circ$.

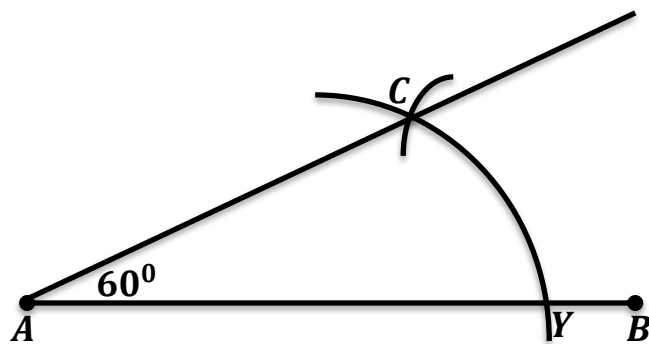
Step 1: with center A and any radius, draw an arc to cut CA at Y.



Step 2: with center and the same radius, drawn an arc to cut then first arc at C.



Step 3: join CA

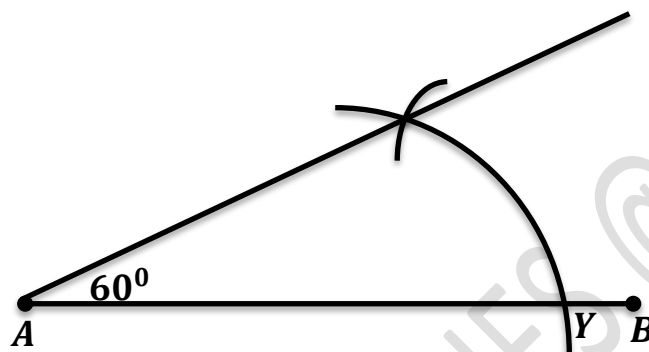


$\angle CAB = 60^\circ$. Use a protractor to check your result.

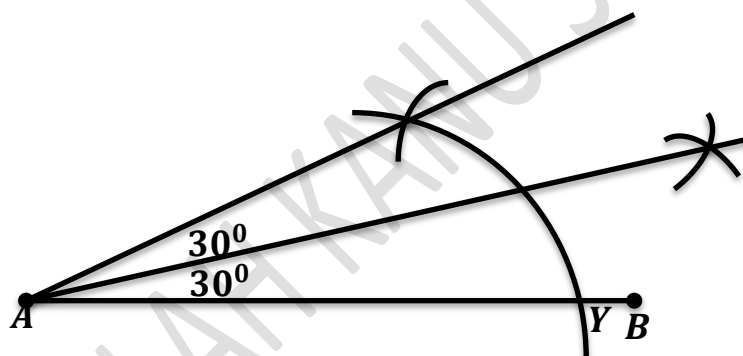
SECTION 6: TO CONSTRUCT AN ANGLE OF 30°

Step 1: $\frac{1}{2}$ of $60^\circ = 30^\circ$.

Step 2: to construct an angle of 30° , first construct an angle of 60° .



Step 3: bisect the above



NOT DRAWN TO SCALE.

Use a protractor to check the data above.

SECTION 7: CONSTRUCTION OF TRIANGLES.

To be able to draw a triangle we must be given:

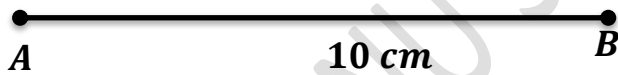
- 1. The length of all the the three sides.*
- 2. The length of two sides and an angle between anyn two sides or*
- 3. The sides of any two angles and the length of one side.*

Subsecton 1: Construction of triangles given three sides

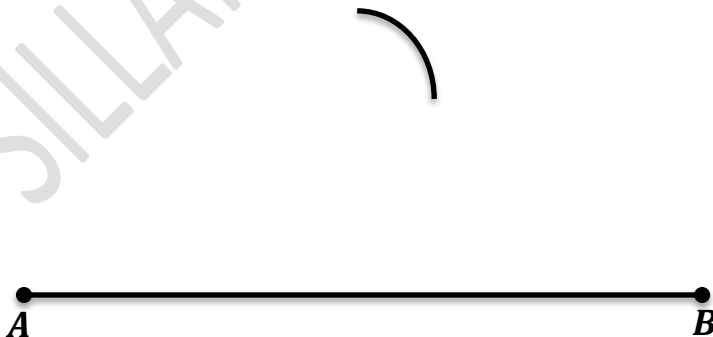
**Example 1. Construct a triangle ABC, where $AB = 10\text{ cm}$,
 $AC = 11\text{ cm}$ and $BC = 9\text{ cm}$.**

Solution

Step 1: draw a line segment of length 10cm.



Step 2: open your compasses to a radius of 11cm and with center at A, draw an arc above the line segment.

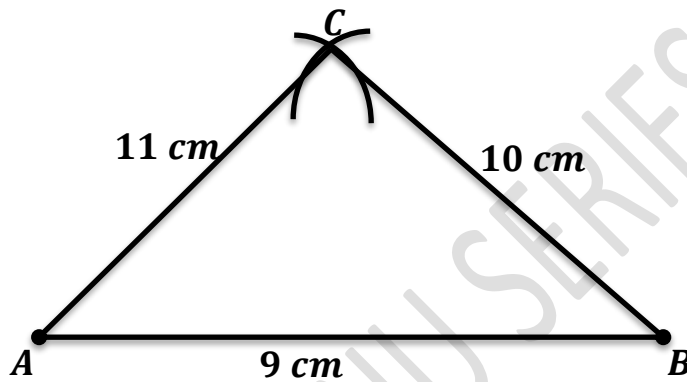


Step 3: again open your compasses to a radius of 9cm and

with B as center describe another arc to intersect the first one at C.



Step 4: join A to C and B to C.



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SECTION 8: MORE WORKED EXAMPLES ON CONSTRUCTION

Example 1(a). Draw $\triangle ABC$ in which $\overline{AB} = 7\text{ cm}$, $\angle CAB = 72^\circ$
and $\angle ABC = 54^\circ$.

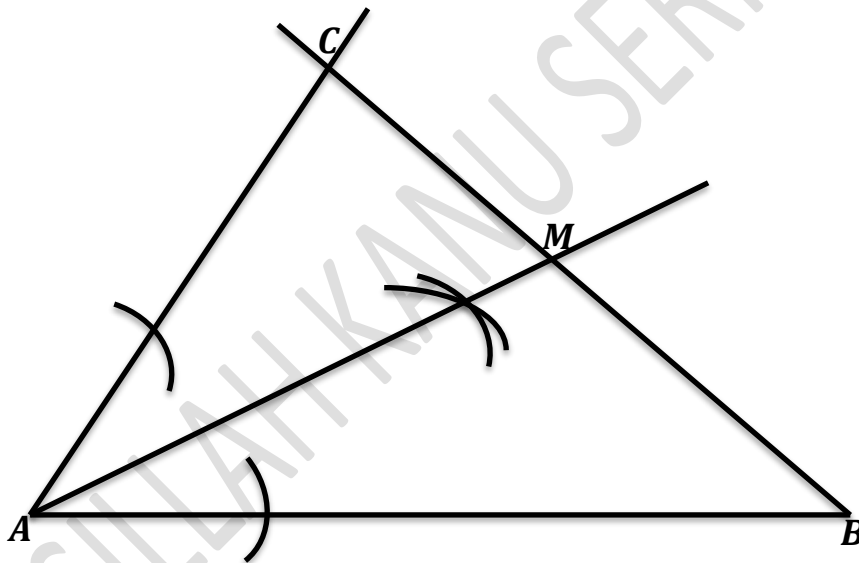
(b). Measure $\overline{AC}$ and $\angle ACB$

(c). Name the type of triangle you have drawn

(d). Bisect $\angle BAC$ and let the bisector meet $\overline{BC}$ at M

(e). Measure $\overline{BM}$ and $\overline{MC}$

Solution



$AC = 7\text{ cm}$

$\text{Angle } ACB = 54^\circ$

The triangle is isosceles because $\angle ACB = \angle ABC$

$BM = 4\text{ cm}$

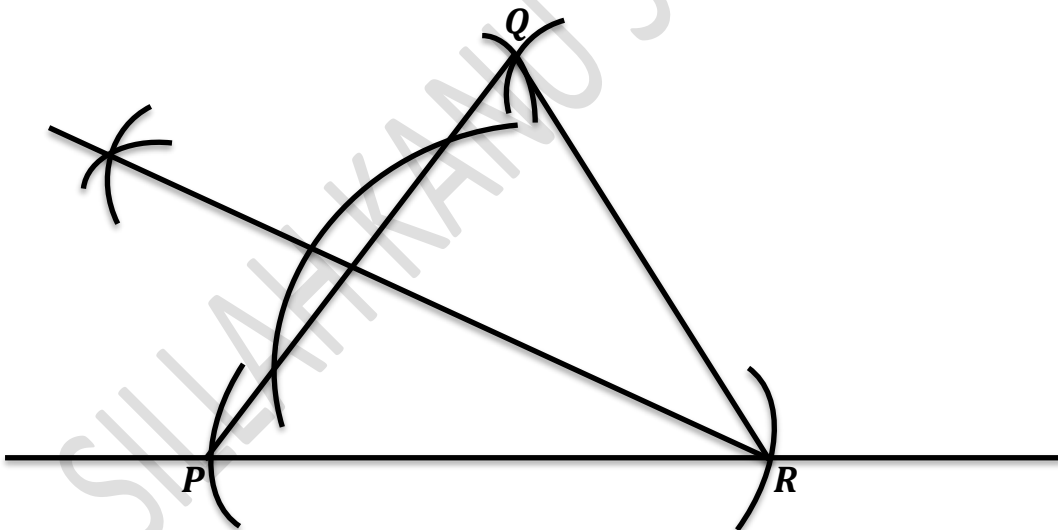
$$MC = 4.2\text{cm}$$

Example 2(a). use a ruler and a pair of compasses only, construct a triangle PQR with $PR = 6\text{cm}$, $PQ = 7\text{cm}$ and $QR = 6\text{cm}$

(b). Construct the perpendicular from R to PQ

(c). Measure the length of the perpendicular and then find the area of the triangle PQR .

Solution



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c. the length of the perpendicular is 5.2cm , which is the height of the triangle.

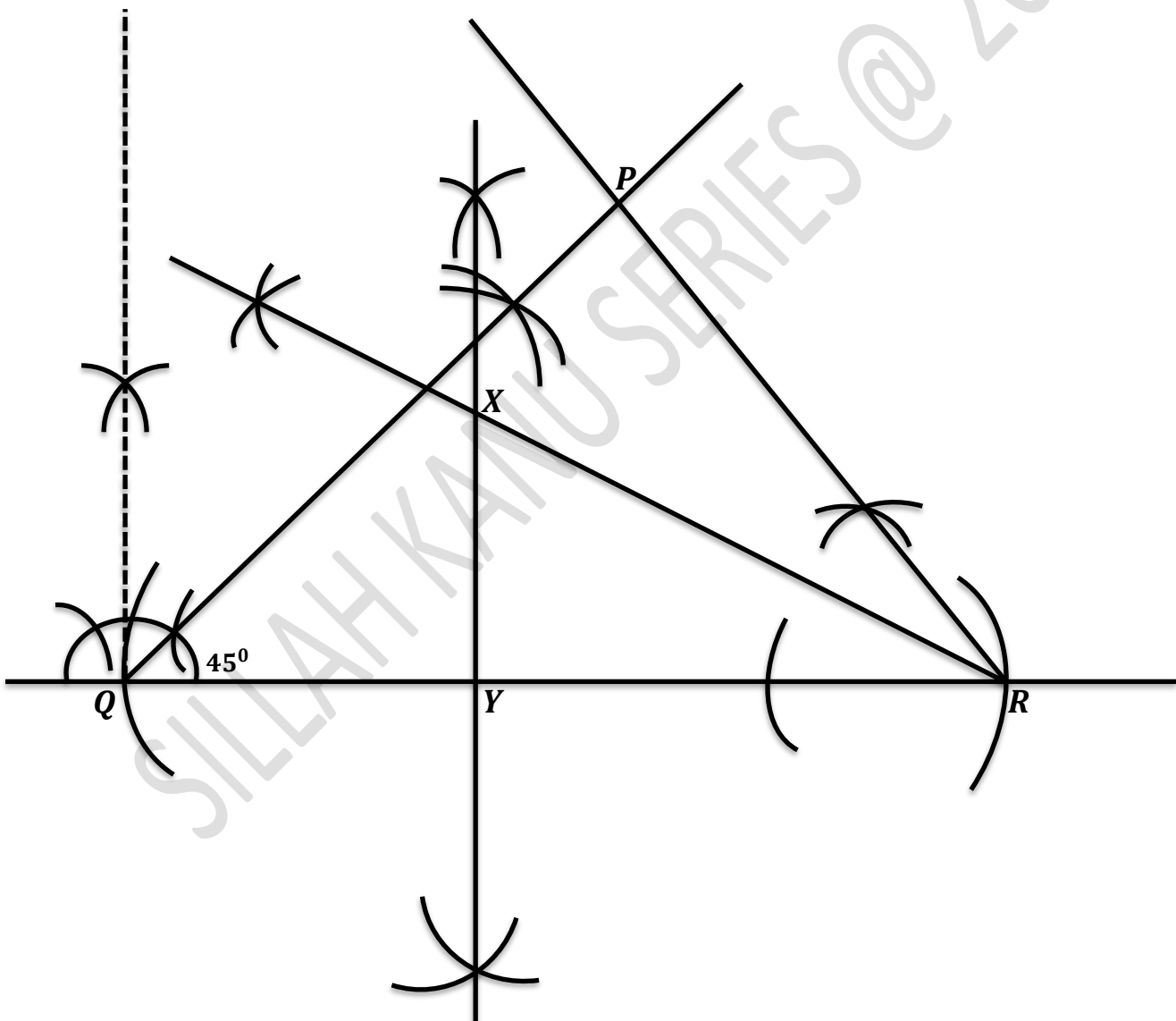
$$PQ = 7\text{ cm}$$

Example 3(a). Using a pair of compasses and a ruler only, construct a triangle PQR with $QR = 8\text{cm}$, angle $PQR = 45^\circ$ and angle

$$QPR = 60^\circ$$

(b). Bisect $\overline{QR}$ and angle QRP . Let the two bisectors meet at X . Let the midpoint of (where the bisector cuts QR) be Y .

(c). Measure $\overline{PQ}$, $\overline{PX}$ and angle RXY



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$$PQ = 7.0\text{cm}, PX = 2.9\text{cm}$$

Angle $RXY = 60^\circ$

Example 4(a). Construct $\triangle XYZ$ with $\overline{XY} = 8\text{cm}$, angle $ZXY = 45^\circ$ and $ZX = 7\text{cm}$.

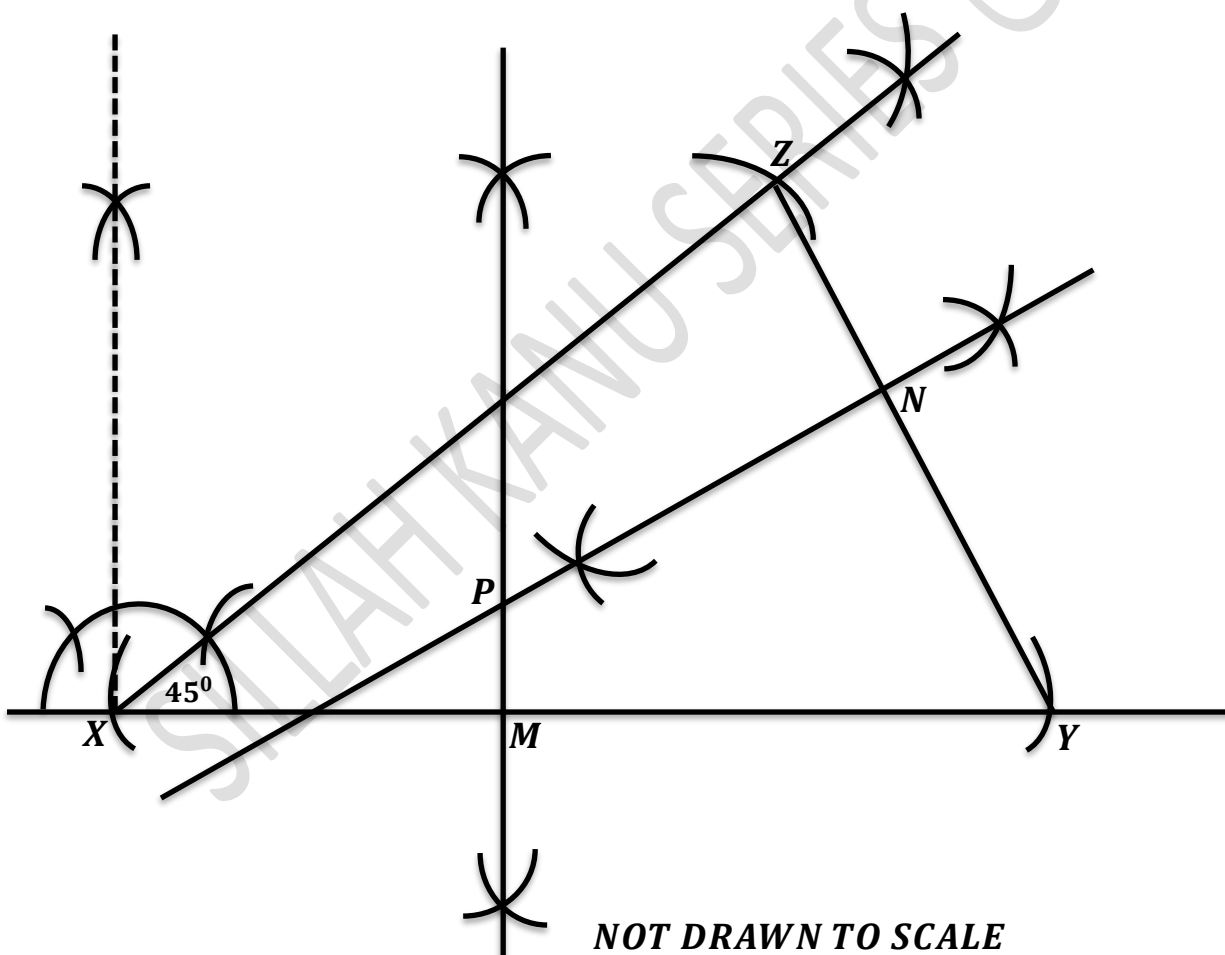
(b). Measure ZY and angle XZY

(c). Bisect $\overline{XY}$ and let the mid – point be M .

(d). Bisect $\overline{ZY}$ and let the mid – point be N .

(e). Measure the angle NPM

Solution



b. $ZY = 6\text{ cm}$. Angle $XZY = 76^\circ$

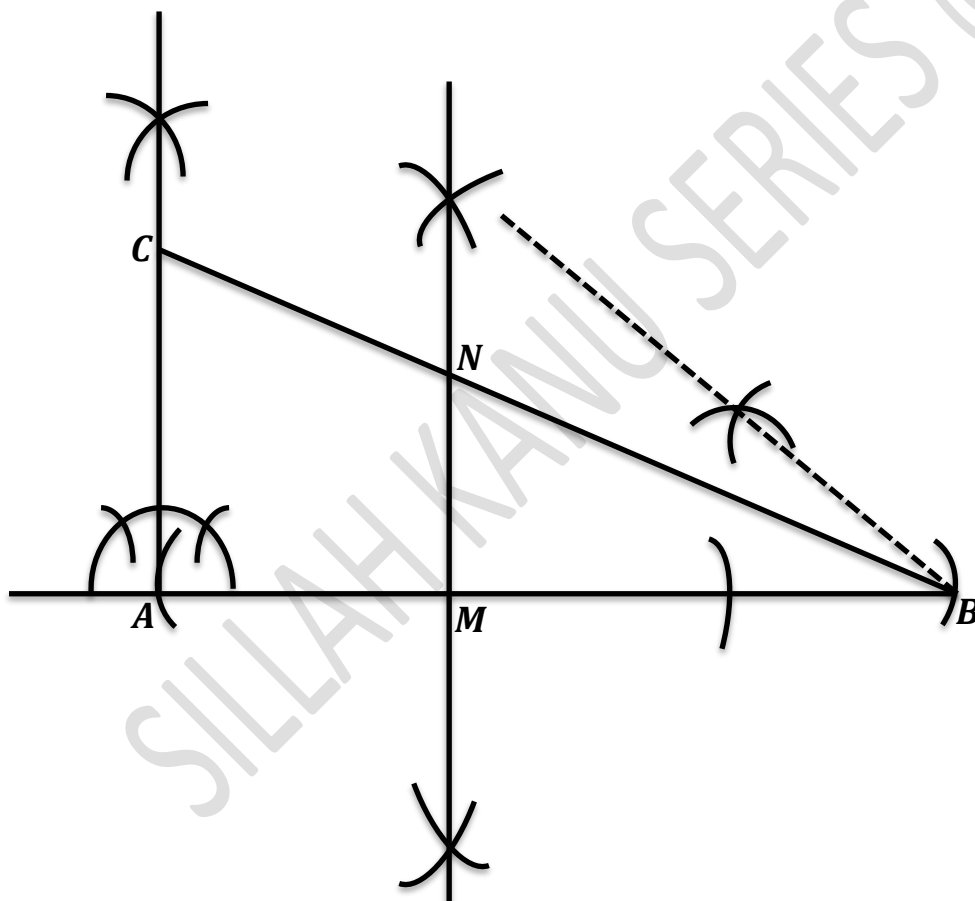
c. Angle $NPM = 122^\circ$

Example 5(a). Using a ruler and a pair of compasses only construct triangle ABC in which $AB = 6\text{cm}$, $\angle BAC = 90^\circ$ and $\angle ABC = 30^\circ$.

(b). Measure AC

(c). Bisect AB and name the point where the line cuts AB and BC respectively.

Solution



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AC = 5.1cm. M is the point where the bisector cuts AB and N is the

point where the bisector cuts BC

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SECTION 9: SKS PAPER 24

Answer all questions

Part 1

Question 1. Construct $\triangle XYZ$ in which $XY = 8.3 \text{ cm}$,
 $YZ = 11.9 \text{ cm}$ and $\widehat{XYZ} = 60^\circ$.

(b). Construct M , the mid – point of XZ

(c). Measure YM

Answer [$YM = 8.8 \text{ cm}$]

Question 2. Construct $\triangle ABC$ in which $AB = 9 \text{ cm}$,
 $BC = 12 \text{ cm}$ and $\widehat{ABC} = 60^\circ$.

(b). Construct the bisector of $\widehat{A}$ and let it meet BC at D

(c). Measure DC

Answer [$DC = 6.6 \text{ cm}$]

Question 3a). Using a ruler and a pair of compasses only, construct a triangle UVW with $VW = 8 \text{ cm}$, $UV = UW = 10 \text{ cm}$

b). Bisect $\widehat{U}$ and draw a perpendicular from U to VW . Let the point of where the bisector meet the perpendicular be X .

c). Measure $\widehat{U}$, $\overline{VX}$ and $\overline{WX}$

Answers: $\widehat{U} = 67^\circ$, $\overline{VX} = 4.8 \text{ cm}$ and $\overline{WX} = 4.8 \text{ cm}$

Question 4a). Using a ruler and a pair of compasses only, construct a triangle ABC with $BC = 6 \text{ cm}$, angle $ABC = 30^\circ$ and angle

$ACB = 45^{\circ}$.

b). Measure the lines AB and AC and angle BAC

c). Construct the perpendicular from A to meet BC at D.

d). Measure AD

Answers: $AB = 4.4\text{ cm}$, $AC = 3.2\text{ cm}$ and angle $BAC = 105^{\circ}$ and

$AD = 2.2\text{ cm}$

Question 5a). Construct an equilateral triangle with sides of length 7.2 cm.

b). Construct the perpendicular from a vertex of the equilateral triangle to the opposite side

c). Measure the length of the perpendicular

Answer: The length of the perpendicular is 6.2 cm.

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5. **Conteh**, A.O (2014) *MATHEMATICS MADE SIMPLE FOR JUNIOR SECONDARY SCHOOLS*, Third edition, Eddie K. Enterprises, Freetown, Sierra Leone.
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